MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project Mansourah Massarah Gold Project

Customer: MA'ADEN GOLD AND BASE METALS (MA'ADEN)

Title: CALCULATION
3210 GRINDING & CLASSIFICATION
CRUSHED ORE STORAGE BIN

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
STAAD	:	Commonly used analysis and design software package named STAAD.Pro
Sec.	:	Section
max.	:	Maximum
ref.	:	Reference
No.	:	Number
cl.	:	Clause
nos.	:	Numbers
Col.	:	Column
approx.	:	Approximate
eqn.	:	Equation
mkd.	:	Marked
Dia.	:	Diameter
Spac.	:	Spacing
Rebar	:	Reinforcement Bar

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1. INTENT

This document is to Prepare for design of **CRUSHED ORE STORAGE BIN (3210)** of approximate 18000T storage capacity. The analysis & design is carried out in accordance with project specifications along with codes & standards for the vertical wall and heavy slab. The foundation raft and the dowel for vertical wall is covered in seperate calculation document no. MD-216-3210-EG-CV-CAL-1140.
The design calculation ensures the safety and serviceability of the complete silo system.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in Excel and STAAD. The silo shell, thick slab and thick slab supporting columns are modeled in STAAD and possible load case are applied. The stresses due to all possible load case combinations are taken out from the analysis of STAAD and used directly as force values for member strength calculation.

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4. INPUT DATA

Refer calculation document no. MD-216-3210-EG-CV-CAL-1140.

5. LOAD

Refer calculation document no. MD-216-3210-EG-CV-CAL-1140.

6. CORRECTION FACTORS FOR LOAD APPLICATION IN STAAD

Refer calculation document no. MD-216-3210-EG-CV-CAL-1140.

7. SUMMARY OF MATERIAL LOADS APPLIED IN STAAD (CONSIDERING CORRECTION FACTORS)

Refer calculation document no. MD-216-3210-EG-CV-CAL-1140.

8. DESIGN REQUIREMENTS

Refer calculation document no. MD-216-3210-EG-CV-CAL-1140.

9. PRIMARY LOAD CASES CONSIDERED IN STAAD:

Refer calculation document no. MD-216-3210-EG-CV-CAL-1140.

10. LOAD COMBINATIONS CONSIDERED IN STAAD

Refer calculation document no. MD-216-3210-EG-CV-CAL-1140.

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11. DESIGN OF VERTICAL WALL

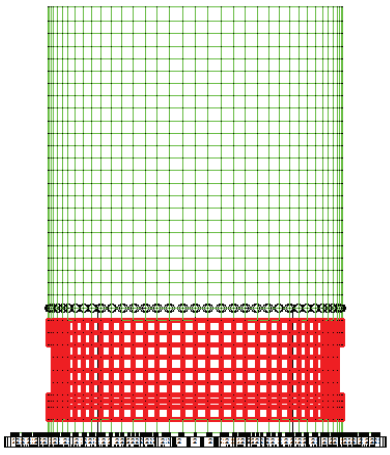
- **Design of Vertical Reinforcement:**

  The verical reinforcement of wall governed by the minimum reinforcement criteria. So, for simplicity the same reinforcement as dowel reinforcement is continued upto Vertical wall Zone 2. For each zone the Vertical reinforcement is calculated. Additional reinforcement for the opening provided as per standrad.

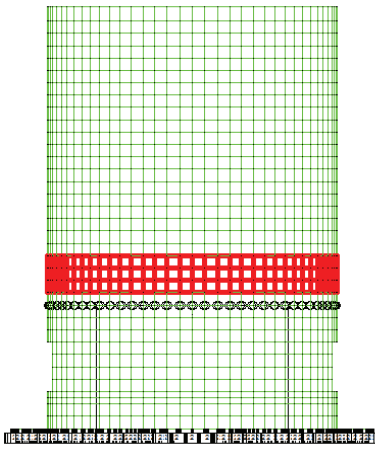
- **Design of Horizontal Reinforcement:**

  For horizontal reinforcement the total verical wall is grouped in seperate zones as per height and for each zone horizontal reinforcement calculated seperately. For concealed beams seperate groups are created and Reinforcement provided accordingly. The additional reinforcement for the opening are given as per standard.

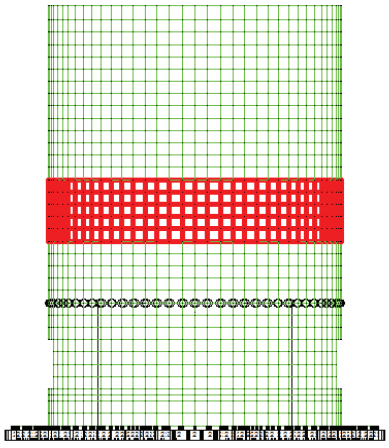
- **STAAD Plate Groups for Vertical Wall:**



TOP_WALL_1



TOP_WALL_2

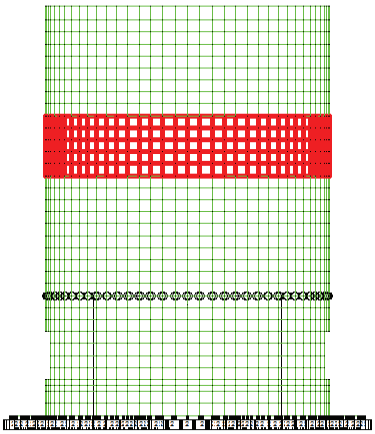


TOP_WALL_3

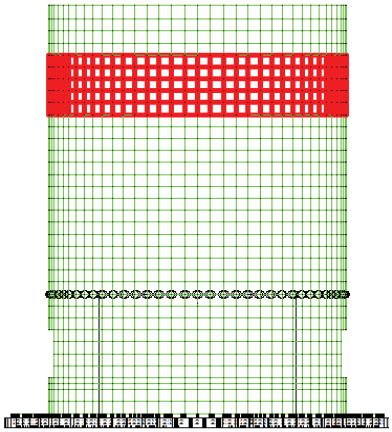
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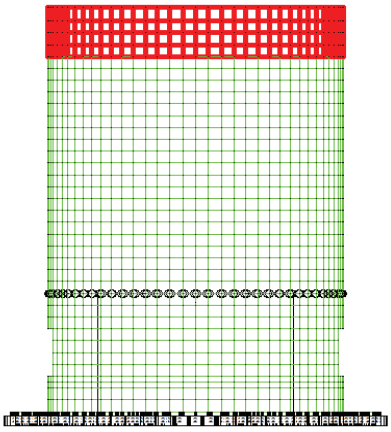
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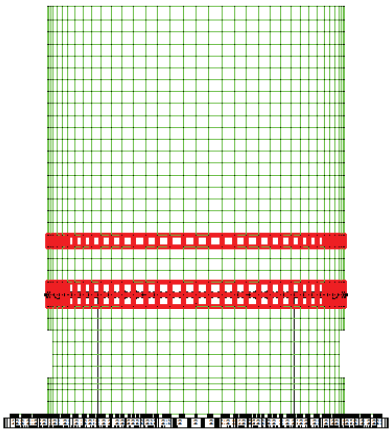
TOP_WALL_4



TOP_WALL_5



TOP_WALL_6



TOP_WALL_BEAM

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• **Vertical Wall: Zone 1**

Design of Wall

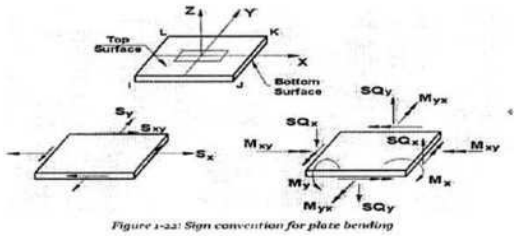
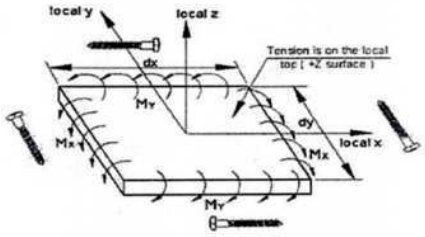


Figure 1-22: Sign convention for plate bending



Sign convention :

Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :

Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :

SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :

+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:

PLATE CENTER STRESS STAAD GROUP : TOP_WALL_1 At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	10283	0.505	0.155	-1.62	-1.575	-1.671	-78.97	-127.6	-190.6	-269.6	-318.2	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qx	10221	-0.499	0.152	-1.58	-1.572	1.667	-86.5	-127.3	0	-86.5	-127.3	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Qy	10133	0.029	0.257	-0.88	-3.242	0.038	-3.934	-461.7	-77.1	-81.04	-538.8	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qy	10701	-0.162	-0.353	-1.31	-5.954	-1.113	8.793	-532.9	-46.94	55.74	-579.8	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Sx	10218	0.05	-0.032	2.712	0.033	-0.033	-221.4	4.175	-9.947	-231.3	14.12	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min Sx	10283	0.438	0.14	-1.78	-1.399	-1.495	-61.44	-119.7	-169.1	-230.5	-288.8	141 0.9X[D] + 1.6X[W1B + H2C + H3C + H5C]
Max Sy	10218	0.083	0.019	1.107	0.036	-0.031	-288.3	-16.37	-24.8	-313.1	-41.16	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Sy	10295	-0.057	0.046	-1.48	-7.298	1.207	67.31	245.9	-13	80.31	258.9	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Max Sxy	10149	-0.127	0.033	0.122	-2.383	1.985	-3.702	31.23	105.7	-109.4	136.9	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sxy	10211	0.127	0.034	0.115	-2.383	-1.986	-2.134	30.74	-105.6	-107.7	136.4	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Mx	10491	-0.006	-0.062	-0.98	-4.315	-0.1	205.8	259.9	-3.858	209.6	263.8	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Mx	10578	0.065	-0.033	1.216	-0.021	0.056	-353.3	-48.49	30.28	-383.6	-78.77	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max My	10379	0.006	-0.028	-1.11	-5.746	-0.909	68.64	386.7	53.65	122.3	440.4	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min My	10701	-0.162	-0.353	-1.31	-5.954	-1.113	8.793	-532.9	-46.94	55.74	-579.8	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Mxy	10149	-0.114	0.149	-0.81	-2.031	1.654	-58.83	-238.2	250.3	-309.1	-488.5	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Mxy	10211	0.116	0.153	-0.83	-2.03	-1.657	-54.04	-239.6	-249.9	-304	-489.6	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]

PLATE CENTER STRESS STAAD GROUP : TOP_WALL_1 At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	10283	0.505	0.155	-1.62	-1.575	-1.671	-78.97	-127.6	-190.6	-269.6	-318.2	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qx	10221	-0.499	0.152	-1.58	-1.572	1.667	-86.5	-127.3	0	-86.5	-127.3	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Qy	10133	0.029	0.257	-0.88	-3.242	0.038	-3.934	-461.7	-77.1	-81.04	-538.8	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qy	10701	-0.162	-0.353	-1.31	-5.954	-1.113	8.793	-532.9	-46.94	55.74	-579.8	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Sx	10218	0.05	-0.032	2.712	0.033	-0.033	-221.4	4.175	-9.947	-231.3	14.12	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min Sx	10283	0.438	0.14	-1.78	-1.399	-1.495	-61.44	-119.7	-169.1	-230.5	-288.8	141 0.9X[D] + 1.6X[W1B + H2C + H3C + H5C]
Max Sy	10218	0.083	0.019	1.107	0.036	-0.031	-288.3	-16.37	-24.8	-313.1	-41.16	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Sy	10295	-0.057	0.046	-1.48	-7.298	1.207	67.31	245.9	-13	80.31	258.9	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Max Sxy	10149	-0.127	0.033	0.122	-2.383	1.985	-3.702	31.23	105.7	-109.4	136.9	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sxy	10211	0.127	0.034	0.115	-2.383	-1.986	-2.134	30.74	-105.6	-107.7	136.4	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Mx	10491	-0.006	-0.062	-0.98	-4.315	-0.1	205.8	259.9	-3.858	209.6	263.8	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Mx	10578	0.065	-0.033	1.216	-0.021	0.056	-353.3	-48.49	30.28	-383.6	-78.77	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max My	10379	0.006	-0.028	-1.11	-5.746	-0.909	68.64	386.7	53.65	122.3	440.4	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min My	10701	-0.162	-0.353	-1.31	-5.954	-1.113	8.793	-532.9	-46.94	55.74	-579.8	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Mxy	10149	-0.114	0.149	-0.81	-2.031	1.654	-58.83	-238.2	250.3	-309.1	-488.5	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Mxy	10211	0.116	0.153	-0.83	-2.03	-1.657	-54.04	-239.6	-249.9	-304	-489.6	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		1425	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

Force Values			Depth		Reinforcement Calculation							
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2
Max Mx	210	top face	50	1359	209.6	0.014	409	3562.5	32	225.84	200	4022.9
Min Mx	45	top face	50	1359	45		87.6		32	225.84	200	4022.9
Max Mxy	-309	bottom face	50	1359	309.1		604		32	225.84	200	4022.9
Min Mxy	-304	bottom face	50	1359	304	*As	594		32	225.84	200	4022.9

Shear Check		Qx	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa	KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.51	719.63	269.6	0.003	1.359	1534	1355	fc'2332	1354.8	4996	> 720	ok	1016	φVc>Vu	not required
	Min Qx	0.50	711.08	86.5			2050			1354.8	4996	> 711	ok	1016	φVc>Vu	not required

Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation							
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2
Max My	440	top face	50	1359	440.4	0.014	861	3562.5	32	225.84	150	5363.9
Min My	-580	bottom face	50	1359	579.8		1135		32	225.84	150	5363.9
Max Mxy	-489	bottom face	50	1359	488.5		956		32	225.84	150	5363.9
Min Mxy	-490	bottom face	50	1359	489.6	*As	958		32	225.84	150	5363.9

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.26	1	366	538.8	0.004	1.359	1350	1355	2332	1349.6	1410	> 366	ok	1012	φVc>Vu	not required
	Min Qy	0.35	1	503	579.8			1367			1354.8	1414	>503	ok	1016	φVc>Vu	not required

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• **Vertical Wall: Zone 2**

Design of Wall

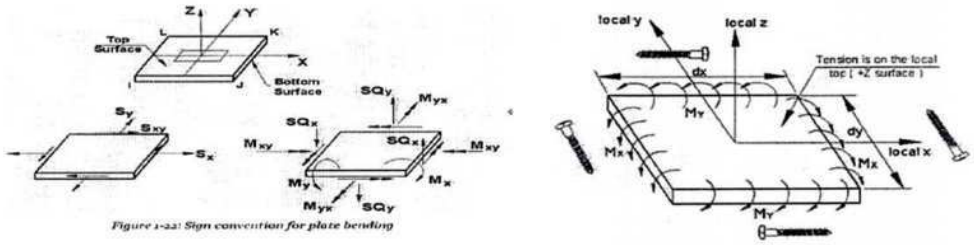


Figure 1-22: Sign convention for plate bending

Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:

PLATE CENTER STRESS STAAD GROUP : TOP_WALL_2 At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	223	0.126	0.27	0.37	-0.322	0.496	-1.642	-120.4	-9.291	-10.93	-129.6	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Qx	229	-0.121	0.261	0.92	-1.823	-0.956	-22.29	-141.4	0	-22.29	-141.4	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Qy	225	0.052	0.415	0.231	-0.186	-0.243	-106.1	-181.2	-2.489	-108.6	-183.7	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Qy	447	-0.008	-0.024	2.248	-0.979	0.115	94.56	234.1	1.828	96.39	235.9	109 1.4X[D] + 1.7X[L + H2B + H3B + H5B]
Max Sx	459	0.002	0.008	2.7	-2.056	-0.027	-14.7	170	2.123	-16.82	172.1	109 1.4X[D] + 1.7X[L + H2B + H3B + H5B]
Min Sx	272	0.001	0.023	-0.21	-0.444	-0.023	6.514	-6.065	0.369	6.883	-6.434	101 1.4X[D]
Max Sy	225	0.049	0.388	0.273	-0.019	-0.149	-104.7	-173.8	-1.831	-106.5	-175.7	141 0.9X[D] + 1.6X[W1B + H2C + H3C + H5C]
Min Sy	213	0.003	0.233	1.174	-4.164	-0.02	-38.64	-75.84	-9.455	-48.1	-85.29	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Max Sxy	221	-0.04	0.24	0.86	-2.233	0.963	24.81	5.356	7.257	32.07	12.61	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sxy	229	-0.121	0.261	0.92	-1.823	-0.956	-22.29	-141.4	-19.4	-41.68	-160.8	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Mx	402	-0.005	-0.002	2.129	-1.909	0.043	104	248.4	-1.998	106	250.4	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Mx	227	-0.018	0.398	0.445	-0.805	-0.905	-113.4	-191.5	-31.57	-145	-223.1	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max My	402	-0.005	-0.002	2.129	-1.909	0.043	104	248.4	-1.998	106	250.4	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min My	227	-0.018	0.398	0.445	-0.805	-0.905	-113.4	-191.5	-31.57	-145	-223.1	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Mxy	265	-0.032	0.244	1.002	-3.334	0.737	-10.65	-31.33	33.27	-43.92	-64.6	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min Mxy	207	0.031	0.246	0.995	-3.331	-0.725	-9.677	-31.54	-34.19	-43.87	-65.74	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]

PLATE CENTER STRESS STAAD GROUP : TOP_WALL_2 At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	223	0.126	0.27	0.37	-0.322	0.496	-1.642	-120.4	-9.291	-10.93	-129.6	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Qx	229	-0.121	0.261	0.92	-1.823	-0.956	-22.29	-141.4	0	-22.29	-141.4	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Qy	225	0.052	0.415	0.231	-0.186	-0.243	-106.1	-181.2	-2.489	-108.6	-183.7	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Qy	447	-0.008	-0.024	2.248	-0.979	0.115	94.56	234.1	1.828	96.39	235.9	109 1.4X[D] + 1.7X[L + H2B + H3B + H5B]
Max Sx	459	0.002	0.008	2.7	-2.056	-0.027	-14.7	170	2.123	-16.82	172.1	109 1.4X[D] + 1.7X[L + H2B + H3B + H5B]
Min Sx	272	0.001	0.023	-0.21	-0.444	-0.023	6.514	-6.065	0.369	6.883	-6.434	101 1.4X[D]
Max Sy	225	0.049	0.388	0.273	-0.019	-0.149	-104.7	-173.8	-1.831	-106.5	-175.7	141 0.9X[D] + 1.6X[W1B + H2C + H3C + H5C]
Min Sy	213	0.003	0.233	1.174	-4.164	-0.02	-38.64	-75.84	-9.455	-48.1	-85.29	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Max Sxy	221	-0.04	0.24	0.86	-2.233	0.963	24.81	5.356	7.257	32.07	12.61	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sxy	229	-0.121	0.261	0.92	-1.823	-0.956	-22.29	-141.4	-19.4	-41.68	-160.8	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Mx	402	-0.005	-0.002	2.129	-1.909	0.043	104	248.4	-1.998	106	250.4	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Mx	227	-0.018	0.398	0.445	-0.805	-0.905	-113.4	-191.5	-31.57	-145	-223.1	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max My	402	-0.005	-0.002	2.129	-1.909	0.043	104	248.4	-1.998	106	250.4	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min My	227	-0.018	0.398	0.445	-0.805	-0.905	-113.4	-191.5	-31.57	-145	-223.1	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Mxy	265	-0.032	0.244	1.002	-3.334	0.737	-10.65	-31.33	33.27	-43.92	-64.6	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min Mxy	207	0.031	0.246	0.995	-3.331	-0.725	-9.677	-31.54	-34.19	-43.87	-65.74	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		1000	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

Force Values			Depth		Reinforcement Calculation								
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2
Max Mx	106	top face	50	938	106		300		25	196.43	150	3273.9
Min Mx	45	top face	50	938	45	0.014	127.1	2500	25	196.43	150	3273.9
Max Mxy	-44	bottom face	50	938	43.92		124		25	196.43	150	3273.9
Min Mxy	-44	bottom face	50	938	43.87	*As	124		25	196.43	150	3273.9

Shear Check		Qx	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa	KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.13	126	10.93	0.003	0.938	1489	943.1	fc'1608	943.07	3453	> 126	ok	707.3	φVc>Vu	not required
	Min Qx	0.12	121	22.29			1171			943.07	3453	> 121	ok	707.3	φVc>Vu	not required

Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation								
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2
Max My	250	top face	50	934	250.4		713		32	321.83	150	5363.9
Min My	-223	bottom face	50	934	223.1	0.014	635	2500	32	321.83	150	5363.9
Max Mxy	-65	bottom face	50	934	64.6		183		32	321.83	150	5363.9
Min Mxy	-66	bottom face	50	934	65.74	*As	186		32	321.83	150	5363.9

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.42	1	415	183.7	0.006	0.934	1005	943.1	1608	943.07	982	> 415	ok	707.3	φVc>Vu	not required
	Min Qy	0.02	1	24	235.9			892.7			892.72	944	>24	ok	669.5	φVc>Vu	not required

Nominal shear strength required by Reinf. Vs= Not Required KN SECTION 11.2

Spacing Required of Strrups, s= Not Required SECTION 11.5.6.2

So, shear links are not required as per design, though single links links for this transition zone additionally.

Diameter of Shear Link Provided = 12 mm
Spacing Provided : Along X-X = 300 mm
Along Y-Y = 350 mm OK

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• **Vertical Wall: Zone 3**

Design of Wall

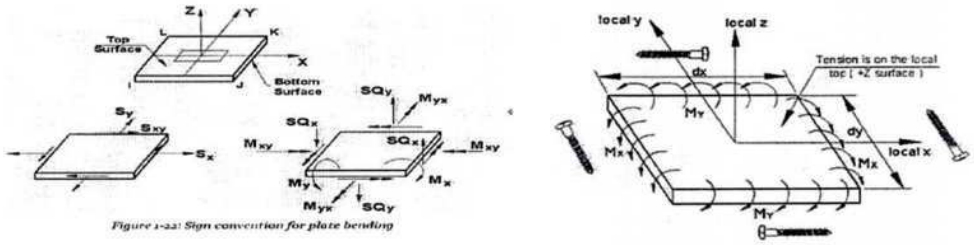


Figure 1-22: Sign convention for plate bending

Sign convention :

Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :

Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :

SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :

+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:

PLATE CENTER STRESS STAAD GROUP : TOP_WALL_3 At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	607	0.032	-0.047	3.022	-2.283	-0.263	23.3	142.4	3.321	26.62	145.7	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Qx	665	-0.032	-0.046	3.028	-2.289	0.265	20	140.3	0	20	140.3	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Qy	613	0	0.003	0.03	-0.731	0.001	-12.73	-0.209	-2.685	-15.41	-2.894	101 1.4X[D]
Min Qy	647	-0.006	-0.069	2.83	-1.953	0.058	73.24	154.5	3.904	77.15	158.4	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Sx	759	0	-0.032	3.171	-2.27	0.003	-31.51	82.86	9.9	-41.41	92.76	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sx	602	0	-0.003	-0.06	-0.445	-0.002	14.5	9.654	0.034	14.53	9.688	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sy	1004	-0.001	-0.002	-0.02	-0.396	0.006	12.06	2.847	-0.034	12.1	2.881	115 1.2X[D] + 0.8X[W1A]
Min Sy	613	0	-0.032	2.93	-2.564	0.002	-27.71	115.5	-9.354	-37.06	124.8	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Max Sxy	665	-0.032	-0.046	3.028	-2.289	0.265	20	140.3	-4.185	24.19	144.5	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sxy	607	0.032	-0.047	3.022	-2.283	-0.263	23.3	142.4	3.321	26.62	145.7	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Mx	602	-0.002	-0.062	2.828	-1.829	0.005	96.1	171.2	-1.84	97.94	173.1	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Mx	935	0.003	-0.015	2.749	-1.78	-0.017	-35.25	24.75	0.328	-35.58	25.08	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Max My	602	-0.002	-0.062	2.828	-1.829	0.005	96.1	171.2	-1.84	97.94	173.1	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min My	1050	0	-0.001	0.009	-0.526	0.002	-5.811	-1.687	-3.528	-9.339	-5.215	101 1.4X[D]
Max Mxy	753	0.024	-0.047	3.035	-2.052	-0.153	14.3	92.97	23.74	38.04	116.7	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Mxy	719	-0.022	-0.046	3.024	-2.06	0.149	12.91	93.88	-21.28	34.19	115.2	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]

PLATE CENTER STRESS STAAD GROUP : TOP_WALL_3 At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	607	0.032	-0.047	3.022	-2.283	-0.263	23.3	142.4	3.321	26.62	145.7	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Qx	665	-0.032	-0.046	3.028	-2.289	0.265	20	140.3	0	20	140.3	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Qy	613	0	0.003	0.03	-0.731	0.001	-12.73	-0.209	-2.685	-15.41	-2.894	101 1.4X[D]
Min Qy	647	-0.006	-0.069	2.83	-1.953	0.058	73.24	154.5	3.904	77.15	158.4	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Sx	759	0	-0.032	3.171	-2.27	0.003	-31.51	82.86	9.9	-41.41	92.76	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sx	602	0	-0.003	-0.06	-0.445	-0.002	14.5	9.654	0.034	14.53	9.688	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sy	1004	-0.001	-0.002	-0.02	-0.396	0.006	12.06	2.847	-0.034	12.1	2.881	115 1.2X[D] + 0.8X[W1A]
Min Sy	613	0	-0.032	2.93	-2.564	0.002	-27.71	115.5	-9.354	-37.06	124.8	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Max Sxy	665	-0.032	-0.046	3.028	-2.289	0.265	20	140.3	-4.185	24.19	144.5	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sxy	607	0.032	-0.047	3.022	-2.283	-0.263	23.3	142.4	3.321	26.62	145.7	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Mx	602	-0.002	-0.062	2.828	-1.829	0.005	96.1	171.2	-1.84	97.94	173.1	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Mx	935	0.003	-0.015	2.749	-1.78	-0.017	-35.25	24.75	0.328	-35.58	25.08	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Max My	602	-0.002	-0.062	2.828	-1.829	0.005	96.1	171.2	-1.84	97.94	173.1	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min My	1050	0	-0.001	0.009	-0.526	0.002	-5.811	-1.687	-3.528	-9.339	-5.215	101 1.4X[D]
Max Mxy	753	0.024	-0.047	3.035	-2.052	-0.153	14.3	92.97	23.74	38.04	116.7	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Mxy	719	-0.022	-0.046	3.024	-2.06	0.149	12.91	93.88	-21.28	34.19	115.2	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		1000	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

Force Values			Depth		Reinforcement Calculation							
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2
Max Mx	98	top face	50	938	97.94		277		25	196.43	175	2806.2
Min Mx	45	top face	50	938	45	0.014	127.1	2500	25	196.43	175	2806.2
Max Mxy	38	top face	50	938	38.04		107		25	196.43	175	2806.2
Min Mxy	34	top face	50	938	34.19	*As	97		25	196.43	175	2806.2

Shear Check		Qx	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa	KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.03	32	26.62	0.003	0.938	941.2	935.1	fc'1608	935.12	3447	> 32	ok	701.3	φVc>Vu	not required
	Min Qx	0.03	32	20			959			935.12	3447	> 32	ok	701.3	φVc>Vu	not required

Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation							
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2
Max My	173	top face	50	934	173.1		492		32	321.83	150	5363.9
Min My	-5	bottom face	50	934	5.215	0.014	15	2500	32	321.83	150	5363.9
Max Mxy	117	top face	50	934	116.7		331		32	321.83	150	5363.9
Min Mxy	115	top face	50	934	115.2	*As	327		32	321.83	150	5363.9

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.00	1	3	2.894	0.006	0.934	933.8	935.1	1608	933.77	975	> 3	ok	700.3	φVc>Vu	not required
	Min Qy	0.07	1	69	158.4			906.9			906.9	955	>69	ok	680.2	φVc>Vu	not required

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• **Vertical Wall: Zone 4**

Design of Wall

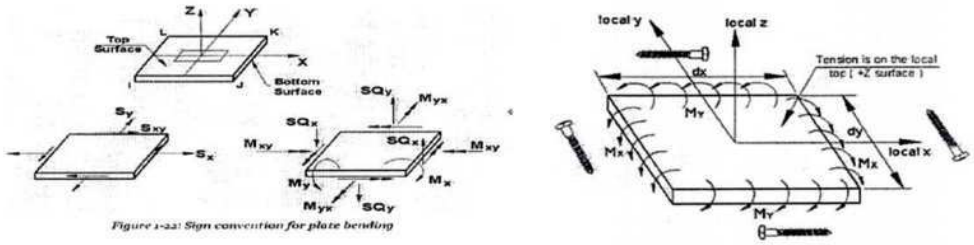


Figure 1-22: Sign convention for plate bending

Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:

PLATE CENTER STRESS STAAD GROUP : TOP_WALL_4 At the face of the column

Load condition		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	L/C
Max Qx	1107	0.014	-0.002	2.624	-1.311	-0.071	17.01	14.75	4.576	21.59	19.32	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Qx	1165	-0.013	-0.001	2.626	-1.312	0.069	15.01	14.4	0	15.01	14.4	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Qy	1361	-0.007	0.007	2.304	-1.008	0.034	3.316	18.24	4.163	7.479	22.4	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Qy	1147	-0.003	-0.01	1.997	-1.091	0.017	6.928	3.232	5.767	12.7	8.999	113 1.4X[D] + 1.7X[L + H2C + H4C + H6C]
Max Sx	1113	0.004	0.002	2.655	-1.369	-0.027	-13.64	10.82	-8.705	-22.35	19.52	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sx	1402	0	-0.001	-0.02	-0.318	0	10.41	0.408	0.585	10.99	0.993	119 1.2X[D] + 1.0X[E1A] + 1.0X[L]
Max Sy	1504	0	0	-0.01	-0.28	0.003	10.67	1.11	0.388	11.06	1.498	115 1.2X[D] + 0.8X[W1A]
Min Sy	1120	-0.002	-0.001	2.462	-1.386	0.02	-22.17	4.535	11.42	-33.59	15.96	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Max Sxy	1110	-0.012	-0.003	2.466	-1.075	0.099	16.21	13.81	-2.16	18.37	15.97	143 0.9X[D] + 1.6X[W1B + H2B + H4B + H6B]
Min Sxy	1164	0.011	-0.003	2.29	-1.076	-0.097	15.94	12.91	1.865	17.8	14.78	137 0.9X[D] + 1.6X[W1B + H1B + H4B + H6B]
Max Mx	1102	0.001	-0.009	2.569	-1.228	-0.006	57.36	19.1	1.065	58.42	20.17	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Mx	1135	0.002	-0.001	2.461	-1.379	-0.007	-32.9	6.978	0.079	-32.98	7.057	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Max My	1502	0	-0.002	1.867	-0.685	-0.001	53.26	29.13	1.277	54.53	30.4	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min My	1543	-0.001	-0.001	-0.01	-0.303	-0.004	-5.751	-2.201	0.947	-6.698	-3.148	120 1.2X[D] + 1.0X[E1B] + 1.0X[L]
Max Mxy	1153	0.005	-0.005	2.62	-1.347	-0.014	-3.733	7.85	17.81	-21.54	25.66	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Mxy	1119	-0.004	-0.005	2.622	-1.351	0.012	-5.893	8.018	-16.9	-22.79	24.91	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]

PLATE CENTER STRESS STAAD GROUP : TOP_WALL_4 At a distance "d" from the face of the column

Load condition		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	L/C
Max Qx	1107	0.014	-0.002	2.624	-1.311	-0.071	17.01	14.75	4.576	21.59	19.32	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Qx	1165	-0.013	-0.001	2.626	-1.312	0.069	15.01	14.4	0	15.01	14.4	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Qy	1361	-0.007	0.007	2.304	-1.008	0.034	3.316	18.24	4.163	7.479	22.4	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Qy	1147	-0.003	-0.01	1.997	-1.091	0.017	6.928	3.232	5.767	12.7	8.999	113 1.4X[D] + 1.7X[L + H2C + H4C + H6C]
Max Sx	1113	0.004	0.002	2.655	-1.369	-0.027	-13.64	10.82	-8.705	-22.35	19.52	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sx	1402	0	-0.001	-0.02	-0.318	0	10.41	0.408	0.585	10.99	0.993	119 1.2X[D] + 1.0X[E1A] + 1.0X[L]
Max Sy	1504	0	0	-0.01	-0.28	0.003	10.67	1.11	0.388	11.06	1.498	115 1.2X[D] + 0.8X[W1A]
Min Sy	1120	-0.002	-0.001	2.462	-1.386	0.02	-22.17	4.535	11.42	-33.59	15.96	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Max Sxy	1110	-0.012	-0.003	2.466	-1.075	0.099	16.21	13.81	-2.16	18.37	15.97	143 0.9X[D] + 1.6X[W1B + H2B + H4B + H6B]
Min Sxy	1164	0.011	-0.003	2.29	-1.076	-0.097	15.94	12.91	1.865	17.8	14.78	137 0.9X[D] + 1.6X[W1B + H1B + H4B + H6B]
Max Mx	1102	0.001	-0.009	2.569	-1.228	-0.006	57.36	19.1	1.065	58.42	20.17	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Mx	1135	0.002	-0.001	2.461	-1.379	-0.007	-32.9	6.978	0.079	-32.98	7.057	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Max My	1502	0	-0.002	1.867	-0.685	-0.001	53.26	29.13	1.277	54.53	30.4	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min My	1543	-0.001	-0.001	-0.01	-0.303	-0.004	-5.751	-2.201	0.947	-6.698	-3.148	120 1.2X[D] + 1.0X[E1B] + 1.0X[L]
Max Mxy	1153	0.005	-0.005	2.62	-1.347	-0.014	-3.733	7.85	17.81	-21.54	25.66	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Mxy	1119	-0.004	-0.005	2.622	-1.351	0.012	-5.893	8.018	-16.9	-22.79	24.91	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		1000	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

Force Values			Depth		Reinforcement Calculation								
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2
Max Mx	58	top face	50	938	58.42		165		25	196.43	175	2806.2
Min Mx	45	top face	50	938	45	0.014	127.1	2500	25	196.43	175	2806.2
Max Mxy	-22	bottom face	50	938	21.54		61		25	196.43	175	2806.2
Min Mxy	-23	bottom face	50	938	22.79	*As	64		25	196.43	175	2806.2

Shear Check		Qx	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa	KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.01	14	21.59	0.003	0.938	916.4	935.1	fc'1608	916.41	3433	> 14	ok	687.3	φVc>Vu	not required
	Min Qx	0.01	13	15.01			926.2			926.16	3440	> 13	ok	694.6	φVc>Vu	not required

Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation								
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2
Max My	30	top face	50	938	30.4		86		25	196.43	150	3273.9
Min My	-3	bottom face	50	938	3.148	0.014	9	2500	25	196.43	150	3273.9
Max Mxy	26	top face	50	938	25.66		72		25	196.43	150	3273.9
Min Mxy	25	top face	50	938	24.91	*As	70		25	196.43	150	3273.9

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.01	1	7	22.4	0.003	0.938	901.4	935.1	1608	901.39	951	> 7	ok	676	φVc>Vu	not required
	Min Qy	0.01	1	10	8.999			937.1			935.12	976	>10	ok	701.3	φVc>Vu	not required

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• **Vertical Wall: Zone 5**

Design of Wall

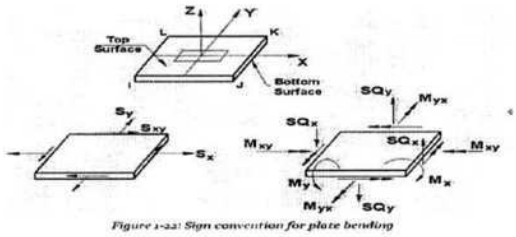
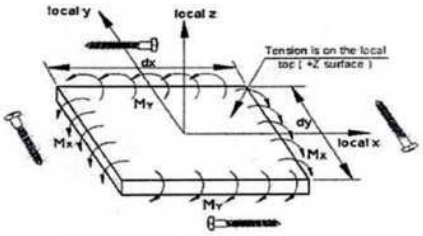


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:

PLATE CENTER STRESS STAAD GROUP : TOP_WALL_5 At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	1660	0.009	-0.004	1.543	-0.581	-0.03	12.51	23.52	-4.111	16.62	27.63	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min Qx	1614	-0.009	-0.003	1.543	-0.581	0.031	12.86	23.64	0	12.86	23.64	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Max Qy	1909	0.001	0.001	-0.01	-0.186	-0.005	6.228	0.897	-0.896	7.124	1.793	119 1.2X[D] + 1.0X[E1A] + 1.0X[L]
Min Qy	1938	0	-0.026	0.851	-0.284	-0.001	-39.68	-22.25	-1.094	-40.78	-23.34	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Sx	1622	-0.005	-0.004	1.675	-0.598	0.018	-15.42	20.23	10.14	-25.56	30.37	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sx	1602	0	0	-0.02	-0.267	0	10.34	0.162	0.513	10.85	0.675	119 1.2X[D] + 1.0X[E1A] + 1.0X[L]
Max Sy	2061	-0.004	-0.011	0.334	-0.11	-0.005	10.79	-23.98	3.451	14.24	-27.43	141 0.9X[D] + 1.6X[W1B + H2C + H3C + H5C]
Min Sy	1645	-0.004	-0.007	1.666	-0.602	0.011	-17.56	17.26	5.741	-23.3	23	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Sxy	1612	-0.008	-0.004	1.456	-0.439	0.047	17.72	22.88	1.75	19.47	24.63	137 0.9X[D] + 1.6X[W1B + H1B + H4B + H6B]
Min Sxy	1662	0.008	-0.004	1.456	-0.44	-0.047	17.23	22.72	-2.021	19.25	24.74	137 0.9X[D] + 1.6X[W1B + H1B + H4B + H6B]
Max Mx	1602	0	-0.006	1.636	-0.564	0	52.92	28.89	1.193	54.12	30.08	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Mx	2035	0	-0.023	0.583	-0.209	-0.001	-48.84	-42.34	-1.049	-49.89	-43.39	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max My	1601	0.003	-0.006	1.638	-0.565	-0.008	51.39	28.93	1.556	52.94	30.49	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min My	2038	0	-0.024	0.581	-0.212	-0.001	-46.41	-43	-1.043	-47.45	-44.04	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Mxy	1653	-0.004	-0.005	1.666	-0.594	0.018	-3.151	21.13	11.4	-14.55	32.53	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Mxy	1619	0.004	-0.005	1.549	-0.595	-0.018	-5.453	19.05	-11.04	-16.49	30.09	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]

PLATE CENTER STRESS STAAD GROUP : TOP_WALL_5 At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	1660	0.009	-0.004	1.543	-0.581	-0.03	12.51	23.52	-4.111	16.62	27.63	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min Qx	1614	-0.009	-0.003	1.543	-0.581	0.031	12.86	23.64	0	12.86	23.64	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Max Qy	1909	0.001	0.001	-0.01	-0.186	-0.005	6.228	0.897	-0.896	7.124	1.793	119 1.2X[D] + 1.0X[E1A] + 1.0X[L]
Min Qy	1938	0	-0.026	0.851	-0.284	-0.001	-39.68	-22.25	-1.094	-40.78	-23.34	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Sx	1622	-0.005	-0.004	1.675	-0.598	0.018	-15.42	20.23	10.14	-25.56	30.37	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sx	1602	0	0	-0.02	-0.267	0	10.34	0.162	0.513	10.85	0.675	119 1.2X[D] + 1.0X[E1A] + 1.0X[L]
Max Sy	2061	-0.004	-0.011	0.334	-0.11	-0.005	10.79	-23.98	3.451	14.24	-27.43	141 0.9X[D] + 1.6X[W1B + H2C + H3C + H5C]
Min Sy	1645	-0.004	-0.007	1.666	-0.602	0.011	-17.56	17.26	5.741	-23.3	23	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Sxy	1612	-0.008	-0.004	1.456	-0.439	0.047	17.72	22.88	1.75	19.47	24.63	137 0.9X[D] + 1.6X[W1B + H1B + H4B + H6B]
Min Sxy	1662	0.008	-0.004	1.456	-0.44	-0.047	17.23	22.72	-2.021	19.25	24.74	137 0.9X[D] + 1.6X[W1B + H1B + H4B + H6B]
Max Mx	1602	0	-0.006	1.636	-0.564	0	52.92	28.89	1.193	54.12	30.08	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Mx	2035	0	-0.023	0.583	-0.209	-0.001	-48.84	-42.34	-1.049	-49.89	-43.39	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max My	1601	0.003	-0.006	1.638	-0.565	-0.008	51.39	28.93	1.556	52.94	30.49	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min My	2038	0	-0.024	0.581	-0.212	-0.001	-46.41	-43	-1.043	-47.45	-44.04	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Mxy	1653	-0.004	-0.005	1.666	-0.594	0.018	-3.151	21.13	11.4	-14.55	32.53	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Mxy	1619	0.004	-0.005	1.549	-0.595	-0.018	-5.453	19.05	-11.04	-16.49	30.09	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		1000	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

Force Values			Depth		Reinforcement Calculation							
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2
Max Mx	54	top face	50	938	54.12		153		25	196.43	175	2806.2
Min Mx	45	top face	50	938	45	0.014	127.1	2500	25	196.43	175	2806.2
Max Mxy	-15	bottom face	50	938	14.55		41		25	196.43	175	2806.2
Min Mxy	-16	bottom face	50	938	16.49	*As	47		25	196.43	175	2806.2

Shear Check		Qx	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa	KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.01	9	16.62	0.003	0.938	911.6	935.1	fc'1608	911.64	3429	> 9	ok	683.7	φVc>Vu	not required
	Min Qx	0.01	9	12.86			918.7			918.72	3434	> 9	ok	689	φVc>Vu	not required

Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation							
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2
Max My	30	top face	50	938	30.49		86		25	196.43	150	3273.9
Min My	-44	bottom face	50	938	44.04	0.014	124	2500	25	196.43	150	3273.9
Max Mxy	33	top face	50	938	32.53		92		25	196.43	150	3273.9
Min Mxy	30	top face	50	938	30.09	*As	85		25	196.43	150	3273.9

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.00	1	1	1.793	0.003	0.938	912.4	935.1	1608	912.36	959	> 1	ok	684.3	φVc>Vu	not required
	Min Qy	0.03	1	26	23.34			937.2			935.12	976	>26	ok	701.3	φVc>Vu	not required

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• **Vertical Wall: Zone 6**

Design of Wall

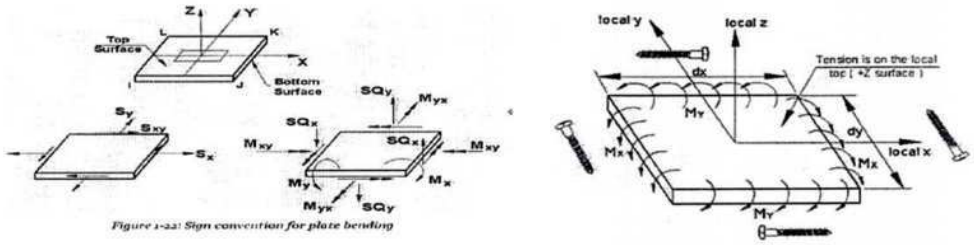


Figure 1-22: Sign convention for plate bending

Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:

PLATE CENTER STRESS STAAD GROUP : TOP_WALL_6 At the face of the column

Load condition		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	L/C
Max Qx	2417	0.018	0.009	-0.16	-0.055	-0.001	1.043	-4.16	-7.702	8.745	-11.86	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min Qx	2453	-0.018	0.009	-0.15	-0.055	0.001	-6.758	-4.399	0	-6.758	-4.399	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Max Qy	2335	0	0.02	0.003	-0.092	0	-53.15	-26.12	-1.091	-54.24	-27.21	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Qy	2138	0	-0.006	0.344	-0.164	-0.001	-51.38	-54.03	-1	-52.38	-55.03	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Sx	2135	0	-0.005	0.345	-0.161	-0.001	-53	-53.45	-1.05	-54.05	-54.5	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sx	2403	0.004	0.008	-0.19	-0.023	0	44.93	-3.143	-1.14	46.07	-4.283	131 0.9X[D] + 1.6X[W1A + H2B + H4B + H6B]
Max Sy	2459	-0.014	0.009	-0.18	-0.023	-0.001	18.46	-3.553	5.611	24.07	-9.164	143 0.9X[D] + 1.6X[W1B + H2B + H4B + H6B]
Min Sy	2136	0	-0.006	0.344	-0.164	0.001	-51.38	-54.03	1.169	-52.54	-55.2	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Sxy	2124	-0.005	-0.004	0.308	-0.09	0.018	-27.35	-46.42	7.767	-35.12	-54.18	131 0.9X[D] + 1.6X[W1A + H2B + H4B + H6B]
Min Sxy	2119	0.007	-0.004	0.278	-0.089	-0.018	-12.82	-41.62	-7.642	-20.46	-49.26	125 0.9X[D] + 1.6X[W1A + H1B + H4B + H6B]
Max Mx	2402	-0.001	0.008	-0.17	-0.055	0	55.12	-3.309	0.617	55.74	-3.926	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min Mx	2235	0	0.015	0.154	-0.127	-0.001	-54.06	-46.01	-1.045	-55.11	-47.06	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max My	2101	0	0	-0.01	-0.131	0	11.55	1.779	0.137	11.69	1.916	119 1.2X[D] + 1.0X[E1A] + 1.0X[L]
Min My	2138	0	-0.006	0.344	-0.164	-0.001	-51.38	-54.03	-1	-52.38	-55.03	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Mxy	2353	-0.007	0.018	-0.02	-0.092	0.006	-9.562	-20.7	9.847	-19.41	-30.55	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min Mxy	2319	0.007	0.018	-0.02	-0.092	-0.005	-11.03	-20.73	-9.701	-20.73	-30.43	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]

PLATE CENTER STRESS STAAD GROUP : TOP_WALL_6 At a distance "d" from the face of the column

Load condition		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	L/C
Max Qx	2417	0.018	0.009	-0.16	-0.055	-0.001	1.043	-4.16	-7.702	8.745	-11.86	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min Qx	2453	-0.018	0.009	-0.15	-0.055	0.001	-6.758	-4.399	0	-6.758	-4.399	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Max Qy	2335	0	0.02	0.003	-0.092	0	-53.15	-26.12	-1.091	-54.24	-27.21	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Qy	2138	0	-0.006	0.344	-0.164	-0.001	-51.38	-54.03	-1	-52.38	-55.03	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Sx	2135	0	-0.005	0.345	-0.161	-0.001	-53	-53.45	-1.05	-54.05	-54.5	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sx	2403	0.004	0.008	-0.19	-0.023	0	44.93	-3.143	-1.14	46.07	-4.283	131 0.9X[D] + 1.6X[W1A + H2B + H4B + H6B]
Max Sy	2459	-0.014	0.009	-0.18	-0.023	-0.001	18.46	-3.553	5.611	24.07	-9.164	143 0.9X[D] + 1.6X[W1B + H2B + H4B + H6B]
Min Sy	2136	0	-0.006	0.344	-0.164	0.001	-51.38	-54.03	1.169	-52.54	-55.2	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Sxy	2124	-0.005	-0.004	0.308	-0.09	0.018	-27.35	-46.42	7.767	-35.12	-54.18	131 0.9X[D] + 1.6X[W1A + H2B + H4B + H6B]
Min Sxy	2119	0.007	-0.004	0.278	-0.089	-0.018	-12.82	-41.62	-7.642	-20.46	-49.26	125 0.9X[D] + 1.6X[W1A + H1B + H4B + H6B]
Max Mx	2402	-0.001	0.008	-0.17	-0.055	0	55.12	-3.309	0.617	55.74	-3.926	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min Mx	2235	0	0.015	0.154	-0.127	-0.001	-54.06	-46.01	-1.045	-55.11	-47.06	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max My	2101	0	0	-0.01	-0.131	0	11.55	1.779	0.137	11.69	1.916	119 1.2X[D] + 1.0X[E1A] + 1.0X[L]
Min My	2138	0	-0.006	0.344	-0.164	-0.001	-51.38	-54.03	-1	-52.38	-55.03	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Mxy	2353	-0.007	0.018	-0.02	-0.092	0.006	-9.562	-20.7	9.847	-19.41	-30.55	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min Mxy	2319	0.007	0.018	-0.02	-0.092	-0.005	-11.03	-20.73	-9.701	-20.73	-30.43	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		1000	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

Force Values			Depth		Reinforcement Calculation								
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2
Max Mx	56	top face	50	938	55.74		157		25	196.43	175	2806.2
Min Mx	45	top face	50	938	45	0.014	127.1	2500	25	196.43	175	2806.2
Max Mxy	-19	bottom face	50	938	19.41		55		25	196.43	175	2806.2
Min Mxy	-21	bottom face	50	938	20.73	*As	59		25	196.43	175	2806.2

Shear Check		Qx	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa	KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.02	18	8.745	0.003	0.938	979.5	935.1	fc'1608	935.12	3447	> 18	ok	701.3	φVc>Vu	not required
	Min Qx	0.02	18	6.758			1007			935.12	3447	> 18	ok	701.3	φVc>Vu	not required

Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation								
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2
Max My	2	top face	50	938	1.916		5		25	196.43	150	3273.9
Min My	-55	bottom face	50	938	55.03	0.014	155	2500	25	196.43	150	3273.9
Max Mxy	-31	bottom face	50	938	30.55		86		25	196.43	150	3273.9
Min Mxy	-30	bottom face	50	938	30.43	*As	86		25	196.43	150	3273.9

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.02	1	20	27.21	0.003	0.938	920.3	935.1	1608	920.29	965	> 20	ok	690.2	φVc>Vu	not required
	Min Qy	0.01	1	6	55.03			892.3			892.29	944	>6	ok	669.2	φVc>Vu	not required

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• **Vertical Wall: Conceal Beam**

Design of Wall

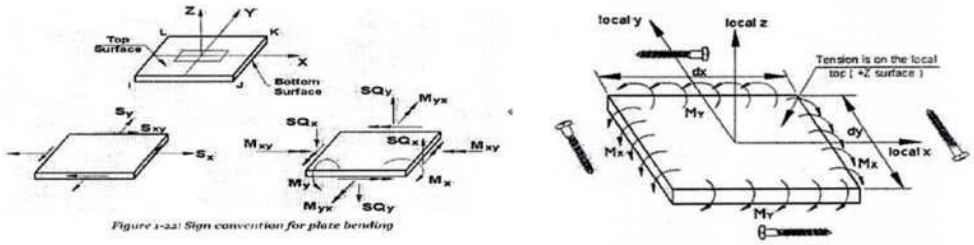


Figure 1-22: Sign convention for plate bending

Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:

PLATE CENTER STRESS STAAD GROUP : TOP_WALL_ BEAM At the face of the column

Load condition		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	L/C
Max Qx	125	0.666	0.984	-0.02	0.669	-0.18	-165.9	-912.6	-279.5	-445.4	-1192	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Qx	127	-0.648	0.959	0.174	-0.382	-1.557	-180	-906.7	0	-180	-906.7	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Qy	125	0.666	0.984	-0.02	0.669	-0.18	-165.9	-912.6	-279.5	-445.4	-1192	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Qy	10749	-0.011	-0.38	0.053	-3.385	-0.733	-82.04	-344.2	107.3	-189.3	-451.4	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Sx	559	0.001	-0.023	3.014	-2.898	-0.006	-29.03	158.4	8.333	-37.37	166.7	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sx	10721	0.005	-0.107	-0.45	-0.825	-0.32	-138.6	-333.3	-40.11	-178.7	-373.4	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Sy	125	0.633	0.926	0.039	0.733	-0.066	-156	-863.3	-265.1	-421.1	-1128	141 0.9X[D] + 1.6X[W1B + H2C + H3C + H5C]
Min Sy	113	0.019	0.443	0.331	-4.609	-0.018	-73.17	-423.2	-16.13	-89.3	-439.3	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Max Sxy	10741	0.206	-0.187	-0.29	-1.825	1.547	-168.1	-345.7	228.9	-397	-574.6	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sxy	10747	0.547	0.444	0.154	-1.669	-1.694	263.6	736.5	798.3	1062	1535	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Mx	10747	0.547	0.443	0.152	-1.663	-1.688	265.7	742.9	800.4	1066	1543	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Mx	10769	-0.1	-0.104	0.786	-0.135	-0.185	-243.9	-112	-85.8	-329.7	-197.9	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max My	10747	0.547	0.443	0.152	-1.663	-1.688	265.7	742.9	800.4	1066	1543	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min My	125	0.666	0.984	-0.02	0.669	-0.18	-165.9	-912.6	-279.5	-445.4	-1192	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Mxy	10747	0.547	0.443	0.152	-1.663	-1.688	265.7	742.9	800.4	1066	1543	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Mxy	10745	-0.63	0.559	1.06	-0.179	0.116	223.6	564.2	-742.6	966.2	1307	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]

PLATE CENTER STRESS STAAD GROUP : TOP_WALL_ BEAM At a distance "d" from the face of the column

Load condition		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	L/C
Max Qx	125	0.666	0.984	-0.02	0.669	-0.18	-165.9	-912.6	-279.5	-445.4	-1192	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Qx	127	-0.648	0.959	0.174	-0.382	-1.557	-180	-906.7	0	-180	-906.7	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Qy	125	0.666	0.984	-0.02	0.669	-0.18	-165.9	-912.6	-279.5	-445.4	-1192	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Qy	10749	-0.011	-0.38	0.053	-3.385	-0.733	-82.04	-344.2	107.3	-189.3	-451.4	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Sx	559	0.001	-0.023	3.014	-2.898	-0.006	-29.03	158.4	8.333	-37.37	166.7	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sx	10721	0.005	-0.107	-0.45	-0.825	-0.32	-138.6	-333.3	-40.11	-178.7	-373.4	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Sy	125	0.633	0.926	0.039	0.733	-0.066	-156	-863.3	-265.1	-421.1	-1128	141 0.9X[D] + 1.6X[W1B + H2C + H3C + H5C]
Min Sy	113	0.019	0.443	0.331	-4.609	-0.018	-73.17	-423.2	-16.13	-89.3	-439.3	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Max Sxy	10741	0.206	-0.187	-0.29	-1.825	1.547	-168.1	-345.7	228.9	-397	-574.6	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sxy	10747	0.547	0.444	0.154	-1.669	-1.694	263.6	736.5	798.3	1062	1535	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Mx	10747	0.547	0.443	0.152	-1.663	-1.688	265.7	742.9	800.4	1066	1543	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Mx	10769	-0.1	-0.104	0.786	-0.135	-0.185	-243.9	-112	-85.8	-329.7	-197.9	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max My	10747	0.547	0.443	0.152	-1.663	-1.688	265.7	742.9	800.4	1066	1543	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min My	125	0.666	0.984	-0.02	0.669	-0.18	-165.9	-912.6	-279.5	-445.4	-1192	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Mxy	10747	0.547	0.443	0.152	-1.663	-1.688	265.7	742.9	800.4	1066	1543	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Mxy	10745	-0.63	0.559	1.06	-0.179	0.116	223.6	564.2	-742.6	966.2	1307	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		1000	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

Force Values			Depth		Reinforcement Calculation								
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2
Max Mx	1066	top face	50	934	1066		3092		32	260.22	140	5747
Min Mx	45	top face	50	934	45	0.014	127.6	2500	32	321.83	140	5747
Max Mxy	1066	top face	50	934	1066		3092		32	260.22	140	5747
Min Mxy	966	top face	50	934	966.2	*As	2796		32	287.79	140	5747

Shear Check		Qx	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa	KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.67	666	445.4	0.006	0.934	1021	981.8	fc'1602	981.8	3472	> 666	ok	736.3	φVc>Vu	not required
	Min Qx	0.65	648	180			1213			981.8	3472	> 648	ok	736.3	φVc>Vu	not required

Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation								
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2
Max My	1543	top face	50	934	1543		4526		32	177.77	140	5747
Min My	-1192	bottom face	50	934	1192	0.014	3467	2500	32	232.05	140	5747
Max Mxy	1543	top face	50	934	1543		4526		32	177.77	140	5747
Min Mxy	1307	top face	50	934	1307	*As	3811		32	211.12	140	5747

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.98	1	984	1192	0.006	0.934	959.4	981.8	1602	959.42	993	> 984	ok	719.6	φVc<Vu	required
	Min Qy	0.38	1	380	451.4			960.9			960.91	994	>380	ok	720.7	φVc>Vu	not required

Nominal shear strength required by Reinf. Vs=		352.6	KN	SECTION 11.2
Spacing Required of Strrups, s=		160	mm	SECTION 11.5.6.2
	or	467	mm	SECTION 11.5.4.1
	or	750	mm	
Hence, spacing required =		160	mm	
Diameter of Shear Link Provided	=	16	mm	
Spacing Provided : Along X-X	=	140	mm	
Along Y-Y	=	140	mm	OK

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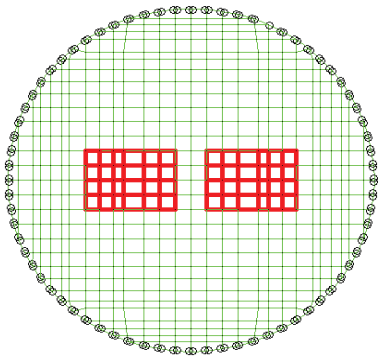
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12. DESIGN OF HEAVY SLAB

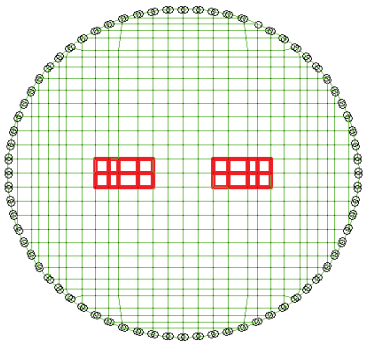
Based on the maximum moments and shear from the STAAD analysis, the design of heavy slab is carried out for both directions X & Y and top & bottom reinforcements.

The plate elements of STAAD model devided in several groups and individually for each group the reinforements calculated for maximum moment and shear of that group.

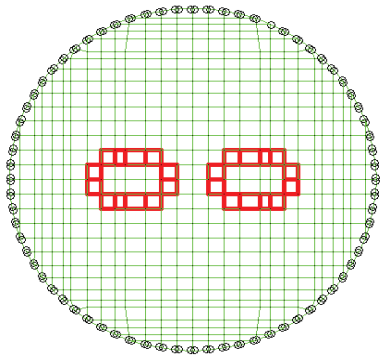
• STAAD Plate Groups for Heavy Slab:



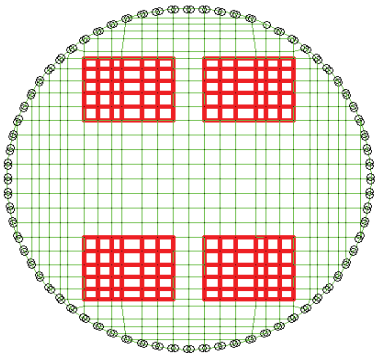
HEAVY_SLAB_1
(For Ttotal)



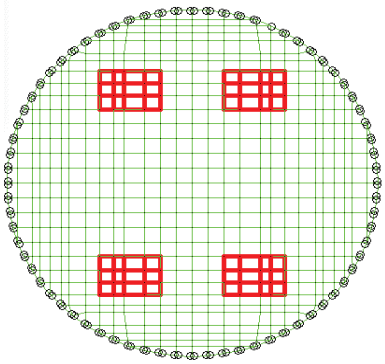
HEAVY_SLAB_1A
(Main Reinforcement)



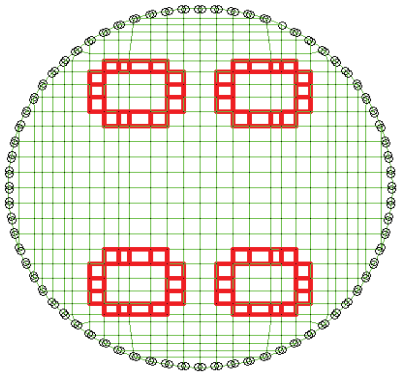
HEAVY_SLAB_1B
(Main Reinforcement & Additional Shear Links)



HEAVY_SLAB_2
(For Ttotal)



HEAVY_SLAB_2A
(Main Reinforcement)

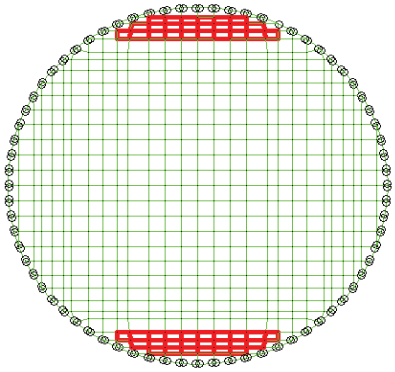


HEAVY_SLAB_2B
(Main Reinforcement & Additional Shear Links)

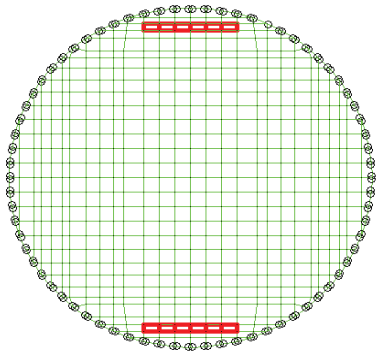
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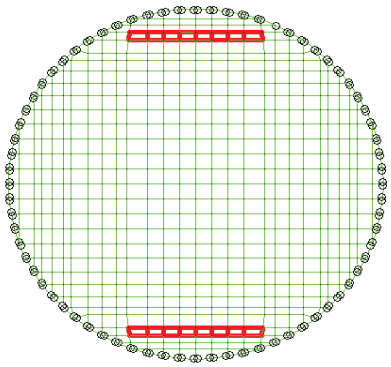
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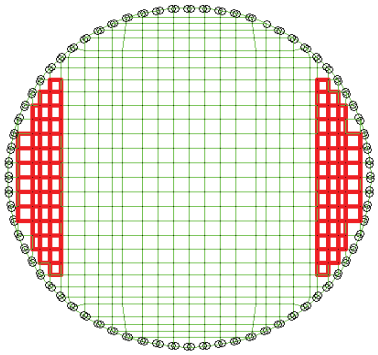
HEAVY_SLAB_3
(For Ttotal)



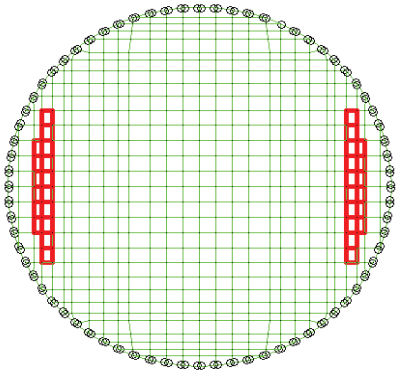
HEAVY_SLAB_3A
(Main Reinforcement)



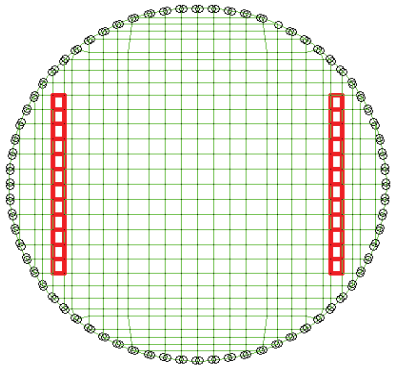
HEAVY_SLAB_3B
(Main Reinforcement & Additional Shear Links)



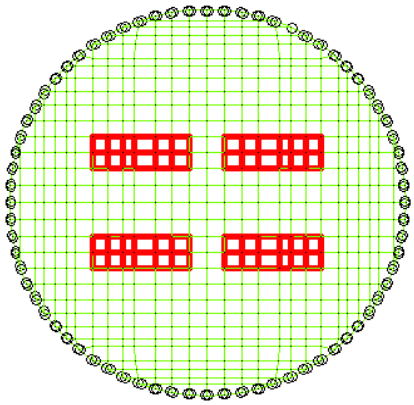
HEAVY_SLAB_4
(For Ttotal)



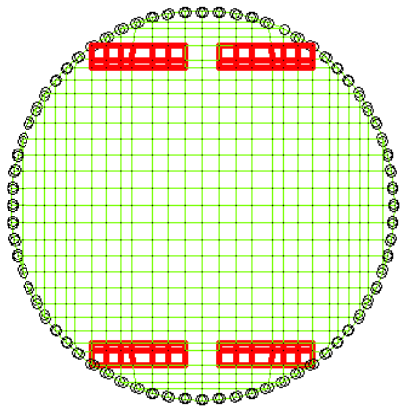
HEAVY_SLAB_4A
(Main Reinforcement)



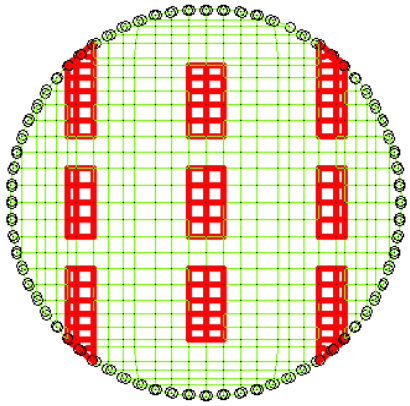
HEAVY_SLAB_4B
(Main Reinforcement & Additional Shear Links)



CONCEAL BEAM_CB2



CONCEAL BEAM_CB3



CONCEAL BEAM_CB4

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• **Area Heavy Slab (Part 1):** Heavy_Slab_1A

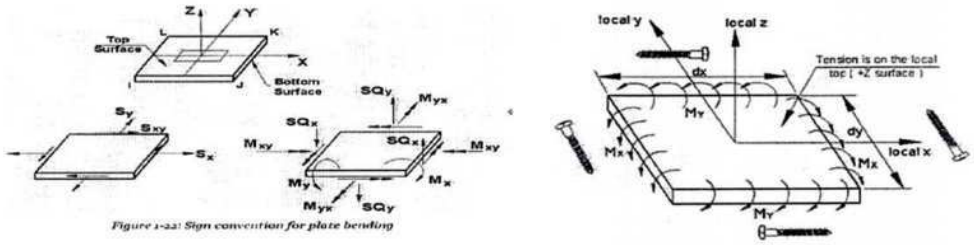


Figure 1-22: Sign convention for plate bending

Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:

PLATE CENTER STRESS STAAD GROUP : Heavy_Slab_1A At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	12546	0.664	-0.126	0.426	0.421	0.001	-1586	-1560	204.1	-1790	-1764	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Qx	12266	-0.661	0.107	0.398	0.401	0.004	-1565	-1582	0	-1565	-1582	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Qy	12548	0.663	0.11	0.423	0.421	-0.004	-1588	-1572	-168	-1756	-1740	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Qy	12257	-0.66	-0.13	0.426	0.426	0	-1555	-1572	-183	-1738	-1755	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Sx	12546	0.411	-0.077	0.556	0.556	-0.001	-280.8	-275.4	124.3	-405.1	-399.7	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sx	12207	0	0.001	0.034	0.023	0.008	-51.82	55.72	-15.91	-67.73	71.63	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sy	12206	0.088	-0.054	0.533	0.566	-0.004	-985.8	-419.3	87.81	-1074	-507.1	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sy	12263	-0.043	0	0.038	0.023	0.005	-15.08	87.13	-1.607	-16.69	88.73	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sxy	12207	0.087	0.035	0.499	0.538	0.013	-987.5	-433.6	-105	-1092	-538.6	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min Sxy	12708	-0.088	0.036	0.499	0.536	-0.012	-995.3	-429.9	103.7	-1099	-533.5	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Max Mx	12546	0.068	-0.014	0.051	0.031	-0.003	102.5	98.47	20.16	122.7	118.6	101 1.4X[D]
Min Mx	12689	0.021	-0.062	0.417	0.423	0.002	-2933	-1573	-51.92	-2984	-1625	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max My	12258	-0.044	-0.006	0.05	0.03	0.003	-7.863	100.2	-3.894	-11.76	104	101 1.4X[D]
Min My	12207	0.209	0.075	0.407	0.427	0.006	-2604	-1744	-157.9	-2762	-1902	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Mxy	12546	0.664	-0.126	0.426	0.421	0.001	-1586	-1560	204.1	-1790	-1764	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Mxy	12257	-0.66	-0.13	0.402	0.403	0	-1556	-1572	-183.1	-1739	-1755	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]

PLATE CENTER STRESS STAAD GROUP : Heavy_Slab_1A At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	12546	0.664	-0.126	0.426	0.421	0.001	-1586	-1560	204.1	-1790	-1764	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Qx	12266	-0.661	0.107	0.398	0.401	0.004	-1565	-1582	0	-1565	-1582	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Qy	12548	0.663	0.11	0.423	0.421	-0.004	-1588	-1572	-168	-1756	-1740	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Qy	12257	-0.66	-0.13	0.426	0.426	0	-1555	-1572	-183	-1738	-1755	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Sx	12546	0.411	-0.077	0.556	0.556	-0.001	-280.8	-275.4	124.3	-405.1	-399.7	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sx	12207	0	0.001	0.034	0.023	0.008	-51.82	55.72	-15.91	-67.73	71.63	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sy	12206	0.088	-0.054	0.533	0.566	-0.004	-985.8	-419.3	87.81	-1074	-507.1	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sy	12263	-0.043	0	0.038	0.023	0.005	-15.08	87.13	-1.607	-16.69	88.73	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sxy	12207	0.087	0.035	0.499	0.538	0.013	-987.5	-433.6	-105	-1092	-538.6	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min Sxy	12708	-0.088	0.036	0.499	0.536	-0.012	-995.3	-429.9	103.7	-1099	-533.5	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Max Mx	12546	0.068	-0.014	0.051	0.031	-0.003	102.5	98.47	20.16	122.7	118.6	101 1.4X[D]
Min Mx	12689	0.021	-0.062	0.417	0.423	0.002	-2933	-1573	-51.92	-2984	-1625	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max My	12258	-0.044	-0.006	0.05	0.03	0.003	-7.863	100.2	-3.894	-11.76	104	101 1.4X[D]
Min My	12207	0.209	0.075	0.407	0.427	0.006	-2604	-1744	-157.9	-2762	-1902	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Mxy	12546	0.664	-0.126	0.426	0.421	0.001	-1586	-1560	204.1	-1790	-1764	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Mxy	12257	-0.66	-0.13	0.402	0.403	0	-1556	-1572	-183.1	-1739	-1755	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	40	mm
	top face	40	mm
Thicknes of slab =		1500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

Force Values			Depth		Reinforcement Calculation							
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2
Max Mx	123	top face	40	1444	122.7	0.014	225	2700	32	297.99	150	5363.9
Min Mx	45	top face	40	1444	45		82.5		32	297.99	150	5363.9
Max Mxy	-1790	bottom face	40	1444	1790		3334		32	241.33	150	5363.9
Min Mxy	-1739	bottom face	40	1444	1739	*As	3236		32	248.6	150	5363.9

Shear Check		Qx	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa	KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.66	996	1790	0.004	1.444	1440	1458	fc'2477	1440.1	5309	> 996	ok	1080	φVc>Vu	not required
	Min Qx	0.66	991.5	1565			1450			1450.3	5316	> 992	ok	1088	φVc>Vu	not required

Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation							
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2
Max My	104	top face	40	1444	104	0.014	191	2700	32	297.99	150	5363.9
Min My	-1902	bottom face	40	1444	1902		3545		32	226.95	150	5363.9
Max Mxy	-1764	bottom face	40	1444	1764		3284		32	245	150	5363.9
Min Mxy	-1755	bottom face	40	1444	1755	*As	3267		32	246.25	150	5363.9

Shear Check		Qx	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa	KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.11	165	1740	0.004	1.444	1379	1458	2477	1379.3	1457	> 165	ok	1035	φVc>Vu	not required
	Min Qy	0.13	195	1755			1381			1381.5	1459	>195	ok	1036	φVc>Vu	not required

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• **Area Raft Other (Part 1):** Heavy_Slab_1B

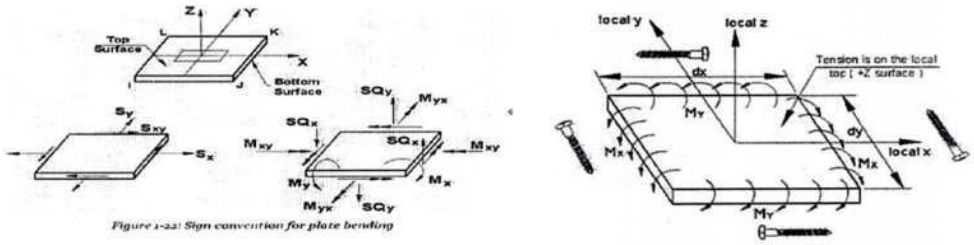


Figure 1-22: Sign convention for plate bending

Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:

PLATE CENTER STRESS STAAD GROUP : Heavy_Slab_1B At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	12545	0.893	-0.251	0.408	0.398	0.003	-1779	-932.4	338.6	-2118	-1271	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qx	12267	-0.889	0.235	0.398	0.4	0.008	-1768	-976.3	0	-1768	-976.3	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Qy	12547	0.697	0.293	0.427	0.418	-0.003	-46.65	-1802	-229.9	-276.5	-2032	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Qy	12256	-0.693	-0.313	0.406	0.398	-0.001	-24.36	-1798	-237.6	-262	-2036	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Sx	12545	0.552	-0.151	0.564	0.553	0	-399.7	105	204.4	-604.1	309.4	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sx	12194	0.014	0.009	0.03	0.023	0.01	-10.24	29.44	-18.21	-28.45	47.64	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sy	12193	0.175	-0.116	0.517	0.569	-0.006	-603.8	-562.7	103.8	-707.5	-666.4	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sy	12208	0.01	0.005	0.033	0.02	0.011	-65.93	80.41	-36.08	-102	116.5	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sxy	12208	0.175	0.085	0.505	0.531	0.026	-1072	-204.6	-222.4	-1294	-427	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min Sxy	12709	-0.175	0.085	0.505	0.53	-0.025	-1078	-201.1	221.8	-1299	-422.9	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Max Mx	12256	-0.421	-0.187	0.529	0.532	0.002	662.8	-415	-151.5	814.3	-566.5	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min Mx	12688	-0.028	-0.108	0.425	0.422	0.008	-3051	-1305	-148.3	-3199	-1453	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max My	12252	-0.092	-0.024	0.055	0.03	0.002	81	162.4	-32.62	113.6	195	101 1.4X[D]
Min My	12194	0.353	0.174	0.403	0.425	0.009	-1932	-1928	-179.9	-2112	-2108	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Mxy	12545	0.893	-0.251	0.432	0.421	0.003	-1780	-932.6	338.7	-2118	-1271	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Mxy	12208	0.363	0.169	0.407	0.427	0.013	-2522	-1347	-319	-2841	-1666	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]

PLATE CENTER STRESS STAAD GROUP : Heavy_Slab_1B At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	12545	0.893	-0.251	0.408	0.398	0.003	-1779	-932.4	338.6	-2118	-1271	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qx	12267	-0.889	0.235	0.398	0.4	0.008	-1768	-976.3	0	-1768	-976.3	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Qy	12547	0.697	0.293	0.427	0.418	-0.003	-46.65	-1802	-229.9	-276.5	-2032	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Qy	12256	-0.693	-0.313	0.406	0.398	-0.001	-24.36	-1798	-237.6	-262	-2036	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Sx	12545	0.552	-0.151	0.564	0.553	0	-399.7	105	204.4	-604.1	309.4	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sx	12194	0.014	0.009	0.03	0.023	0.01	-10.24	29.44	-18.21	-28.45	47.64	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sy	12193	0.175	-0.116	0.517	0.569	-0.006	-603.8	-562.7	103.8	-707.5	-666.4	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sy	12208	0.01	0.005	0.033	0.02	0.011	-65.93	80.41	-36.08	-102	116.5	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sxy	12208	0.175	0.085	0.505	0.531	0.026	-1072	-204.6	-222.4	-1294	-427	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min Sxy	12709	-0.175	0.085	0.505	0.53	-0.025	-1078	-201.1	221.8	-1299	-422.9	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Max Mx	12256	-0.421	-0.187	0.529	0.532	0.002	662.8	-415	-151.5	814.3	-566.5	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min Mx	12688	-0.028	-0.108	0.425	0.422	0.008	-3051	-1305	-148.3	-3199	-1453	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max My	12252	-0.092	-0.024	0.055	0.03	0.002	81	162.4	-32.62	113.6	195	101 1.4X[D]
Min My	12194	0.353	0.174	0.403	0.425	0.009	-1932	-1928	-179.9	-2112	-2108	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Mxy	12545	0.893	-0.251	0.432	0.421	0.003	-1780	-932.6	338.7	-2118	-1271	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Mxy	12208	0.363	0.169	0.407	0.427	0.013	-2722	-1347	-319	-3041	-1666	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	40	mm
	top face	40	mm
Thicknes of slab =		1500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

Force Values			Depth		Reinforcement Calculation							
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2
Max Mx	814	top face	40	1444	814.3	0.014	1503	2700	32	297.99	150	5363.9
Min Mx	45	top face	40	1444	45		82.5		32	297.99	150	5363.9
Max Mxy	-2118	bottom face	40	1444	2118		3958		32	203.3	150	5363.9
Min Mxy	-2841	bottom face	40	1444	2841	*As	5343		32	150.57	150	5363.9

Shear Check		Qx	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa	KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.89	1339.5	2118	0.004	1.444	1450	1458	fc'2477	1450.1	5316	> 1340	ok	1088	φVc<Vu	required
	Min Qx	0.89	1333.5	1768			1466			1458	5322	> 1334	ok	1094	φVc<Vu	required

Nominal shear strength required by Reinf. Vs= 335.9 KN SECTION 11.2

Spacing Required of Strrups, s= 242 mm SECTION 11.5.6.2
or 722 mm SECTION 11.5.4.1
or 1125 mm

Hence, spacing required = 242 mm

Diameter of Shear Link Provided = 16 mm
Spacing Provided : Along X-X = 150 mm
Along Y-Y = 150 mm OK

Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation							
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2
Max My	195	top face	40	1444	195	0.014	358	2700	32	297.99	150	5363.9
Min My	-2108	bottom face	40	1444	2108		3938		32	204.32	150	5363.9
Max Mxy	-1271	bottom face	40	1444	1271		2356		32	297.99	150	5363.9
Min Mxy	-1666	bottom face	40	1444	1666	*As	3099		32	259.6	150	5363.9

Shear Check		Qx	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa	KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.29	440	2032	0.004	1.444	1395	1458	2477	1395.3	1469	> 440	ok	1046	φVc>Vu	not required
	Min Qy	0.31	470	2036			1397			1397.2	1471	>470	ok	1048	φVc>Vu	not required

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• **Area Raft Other (Part 2):** Heavy_Slab_2A

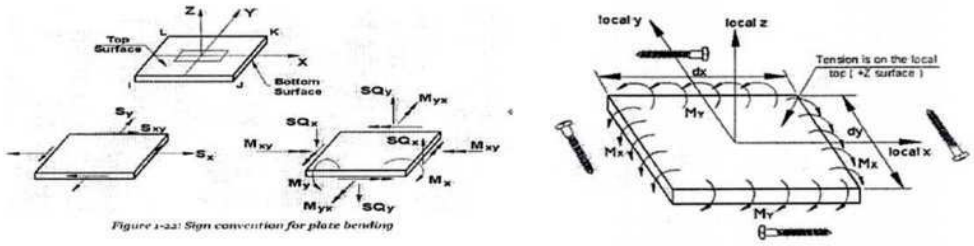


Figure 1-22: Sign convention for plate bending

Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:

PLATE CENTER STRESS STAAD GROUP : Heavy_Slab_2A At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	12527	0.558	0.286	0.458	0.417	-0.011	-1776	-1972	-571.1	-2347	-2543	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Qx	12445	-0.542	-0.297	0.409	0.389	0.013	-1789	-2081	0	-1789	-2081	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Qy	12527	0.558	0.286	0.436	0.394	-0.011	-1776	-1972	-571.1	-2347	-2544	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qy	12639	0.542	-0.297	0.41	0.388	-0.012	-1794	-2077	633.6	-2427	-2711	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Sx	12201	0.124	0.141	0.636	0.516	0.002	-1089	-617.4	15.52	-1105	-632.9	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sx	12475	-0.042	-0.009	0.018	0.011	0.001	26.64	-68.39	-45.69	72.33	-114.1	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sy	12230	-0.277	-0.046	0.573	0.56	0.05	-419.4	-833.7	116.9	-536.3	-950.6	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sy	12472	0.002	0.001	0.019	0.009	0.004	-74.48	-79.86	-44.43	-118.9	-124.3	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sxy	12225	-0.149	-0.064	0.593	0.519	0.059	-826.7	-816.2	126.6	-953.3	-942.7	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sxy	12616	0.159	-0.033	0.586	0.513	-0.059	-905.2	-865.2	-128.9	-1034	-994.1	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Mx	12527	0.063	0.047	0.068	0.031	-0.009	64.03	27.99	-66.27	130.3	94.26	101 1.4X[D]
Min Mx	12632	-0.019	0.138	0.483	0.388	-0.002	-2883	-1936	-100.2	-2983	-2036	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max My	12633	0.037	0.03	0.073	0.028	-0.009	-33.07	31.08	-45.26	-78.33	76.34	101 1.4X[D]
Min My	12460	-0.432	-0.072	0.432	0.411	0.011	-1743	-2407	-393.8	-2137	-2801	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Mxy	12639	0.542	-0.297	0.434	0.411	-0.011	-1794	-2078	633.7	-2427	-2711	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Mxy	12445	-0.542	-0.297	0.409	0.389	0.013	-1789	-2081	-628.5	-2418	-2710	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]

PLATE CENTER STRESS STAAD GROUP : Heavy_Slab_2A At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	12527	0.558	0.286	0.458	0.417	-0.011	-1776	-1972	-571.1	-2347	-2543	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Qx	12445	-0.542	-0.297	0.409	0.389	0.013	-1789	-2081	0	-1789	-2081	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Qy	12527	0.558	0.286	0.436	0.394	-0.011	-1776	-1972	-571.1	-2347	-2544	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qy	12639	0.542	-0.297	0.41	0.388	-0.012	-1794	-2077	633.6	-2427	-2711	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Sx	12201	0.124	0.141	0.636	0.516	0.002	-1089	-617.4	15.52	-1105	-632.9	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sx	12475	-0.042	-0.009	0.018	0.011	0.001	26.64	-68.39	-45.69	72.33	-114.1	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sy	12230	-0.277	-0.046	0.573	0.56	0.05	-419.4	-833.7	116.9	-536.3	-950.6	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sy	12472	0.002	0.001	0.019	0.009	0.004	-74.48	-79.86	-44.43	-118.9	-124.3	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sxy	12225	-0.149	-0.064	0.593	0.519	0.059	-826.7	-816.2	126.6	-953.3	-942.7	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sxy	12616	0.159	-0.033	0.586	0.513	-0.059	-905.2	-865.2	-128.9	-1034	-994.1	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Mx	12527	0.063	0.047	0.068	0.031	-0.009	64.03	27.99	-66.27	130.3	94.26	101 1.4X[D]
Min Mx	12632	-0.019	0.138	0.483	0.388	-0.002	-2883	-1936	-100.2	-2983	-2036	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max My	12633	0.037	0.03	0.073	0.028	-0.009	-33.07	31.08	-45.26	-78.33	76.34	101 1.4X[D]
Min My	12460	-0.432	-0.072	0.432	0.411	0.011	-1743	-2407	-393.8	-2137	-2801	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Mxy	12639	0.542	-0.297	0.434	0.411	-0.011	-1794	-2078	633.7	-2427	-2711	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Mxy	12445	-0.542	-0.297	0.409	0.389	0.013	-1789	-2081	-628.5	-2418	-2710	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	40	mm
	top face	40	mm
Thicknes of slab =		1500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

Force Values			Depth		Reinforcement Calculation							
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2
Max Mx	130	top face	40	1444	130.3		239		32	297.99	150	5363.9
Min Mx	45	top face	40	1444	45	0.014	82.5	2700	32	297.99	150	5363.9
Max Mxy	-2427	bottom face	40	1444	2427		4548		32	176.9	150	5363.9
Min Mxy	-2418	bottom face	40	1444	2418	*As	4529		32	177.63	150	5363.9

Shear Check		Qx	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa	KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.56	837	2347	0.004	1.444	1414	1458	fc'2477	1413.8	5289	> 837	ok	1060	φVc>Vu	not required
	Min Qx	0.54	813	1789			1427			1426.7	5299	> 813	ok	1070	φVc>Vu	not required

Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation							
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2
Max My	76	top face	40	1444	76.34		140		32	297.99	150	5363.9
Min My	-2801	bottom face	40	1444	2801	0.014	5267	2700	32	152.76	150	5363.9
Max Mxy	-2711	bottom face	40	1444	2711		5094		32	157.95	150	5363.9
Min Mxy	-2710	bottom face	40	1444	2710	*As	5091		32	158.03	150	5363.9

Shear Check		Qx	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa	KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.29	429	2544	0.004	1.444	1389	1458	2477	1389.1	1465	> 429	ok	1042	φVc>Vu	not required
	Min Qy	0.30	446	2711			1388			1388.5	1464	>446	ok	1041	φVc>Vu	not required

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• **Area Raft Other (Part 2):** Heavy_Slab_2B

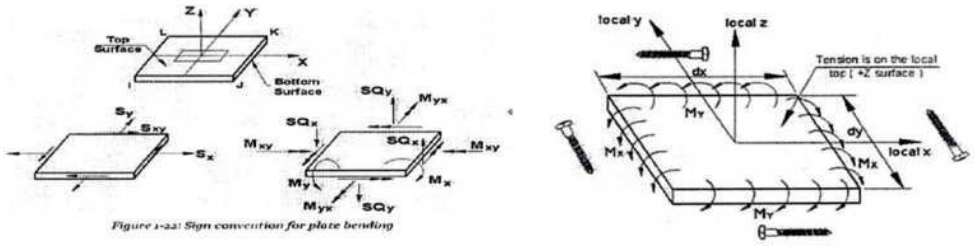


Figure 1-22: Sign convention for plate bending

Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:

PLATE CENTER STRESS STAAD GROUP : Heavy_Slab_2B At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	12529	0.84	0.4	0.433	0.393	-0.004	-1910	-1172	-630.4	-2540	-1802	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qx	12427	-0.83	-0.412	0.408	0.391	0.013	-1918	-1267	0	-1918	-1267	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Qy	12526	0.618	0.571	0.429	0.408	-0.007	-454.7	-2141	-585.4	-1040	-2726	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qy	12638	0.601	-0.591	0.409	0.387	-0.009	-500.9	-2245	641	-1142	-2886	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Sx	12187	0.165	0.055	0.652	0.523	0	-863.9	-788.4	68.02	-931.9	-856.5	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sx	12480	-0.003	0.007	0.014	0.009	0.003	-68.13	-80.36	-46.37	-114.5	-126.7	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sy	12231	-0.339	-0.124	0.566	0.584	0.032	187.7	-886.1	10.67	198.4	-896.8	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sy	12481	-0.01	0.011	0.014	0.009	0.003	-65.32	-77.85	-52.24	-117.6	-130.1	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sxy	12216	-0.128	-0.154	0.57	0.52	0.077	-776.3	-726.5	91.14	-867.4	-817.7	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sxy	12604	0.149	-0.108	0.56	0.516	-0.075	-872.8	-809.6	-99.01	-971.8	-908.6	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Mx	12237	-0.383	0.394	0.575	0.559	0.02	417.1	-706.6	383.5	800.6	-1090	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Mx	12531	-0.062	0.185	0.473	0.4	0.009	-3052	-1514	-94.39	-3146	-1609	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max My	12529	0.091	0.051	0.066	0.03	-0.006	56.87	125.7	-71.06	127.9	196.7	101 1.4X[D]
Min My	12459	-0.465	-0.118	0.432	0.409	0.01	-741.7	-2513	-304.1	-1046	-2817	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Mxy	12637	0.83	-0.412	0.432	0.413	-0.012	-1924	-1262	686	-2610	-1948	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Mxy	12427	-0.83	-0.412	0.408	0.391	0.013	-1918	-1267	-679.4	-2597	-1947	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]

PLATE CENTER STRESS STAAD GROUP : Heavy_Slab_2B At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	12529	0.84	0.4	0.433	0.393	-0.004	-1910	-1172	-630.4	-2540	-1802	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qx	12427	-0.83	-0.412	0.408	0.391	0.013	-1918	-1267	0	-1918	-1267	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Qy	12526	0.618	0.571	0.429	0.408	-0.007	-454.7	-2141	-585.4	-1040	-2726	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qy	12638	0.601	-0.591	0.409	0.387	-0.009	-500.9	-2245	641	-1142	-2886	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Sx	12187	0.165	0.055	0.652	0.523	0	-863.9	-788.4	68.02	-931.9	-856.5	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sx	12480	-0.003	0.007	0.014	0.009	0.003	-68.13	-80.36	-46.37	-114.5	-126.7	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sy	12231	-0.339	-0.124	0.566	0.584	0.032	187.7	-886.1	10.67	198.4	-896.8	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sy	12481	-0.01	0.011	0.014	0.009	0.003	-65.32	-77.85	-52.24	-117.6	-130.1	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sxy	12216	-0.128	-0.154	0.57	0.52	0.077	-776.3	-726.5	91.14	-867.4	-817.7	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sxy	12604	0.149	-0.108	0.56	0.516	-0.075	-872.8	-809.6	-99.01	-971.8	-908.6	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Mx	12237	-0.383	0.394	0.575	0.559	0.02	417.1	-706.6	383.5	800.6	-1090	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Mx	12531	-0.062	0.185	0.473	0.4	0.009	-3052	-1514	-94.39	-3146	-1609	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max My	12529	0.091	0.051	0.066	0.03	-0.006	56.87	125.7	-71.06	127.9	196.7	101 1.4X[D]
Min My	12459	-0.465	-0.118	0.432	0.409	0.01	-741.7	-2593	-304.1	-1046	-2897	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Mxy	12637	0.83	-0.412	0.432	0.413	-0.012	-1924	-1262	686	-2610	-1948	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Mxy	12427	-0.83	-0.412	0.408	0.391	0.013	-1918	-1267	-679.4	-2597	-1947	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	40	mm
	top face	40	mm
Thicknes of slab =		1500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

Force Values			Depth		Reinforcement Calculation							
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2
Max Mx	801	top face	40	1444	800.6	0.014	1477	2700	32	297.99	150	5363.9
Min Mx	45	top face	40	1444	45		82.5		32	297.99	150	5363.9
Max Mxy	-2610	bottom face	40	1444	2610		4899		32	164.24	150	5363.9
Min Mxy	-2597	bottom face	40	1444	2597	*As	4874		32	165.07	150	5363.9

Shear Check		Qx	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa	KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.84	1260	2540	0.004	1.444	1432	1458	fc'2477	1432.2	5303	> 1260	ok	1074	φVc<Vu	required
	Min Qx	0.83	1245	1918			1452			1452.3	5318	> 1245	ok	1089	φVc<Vu	required

Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation							
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2
Max My	197	top face	40	1444	196.7	0.014	361	2700	32	297.99	150	5363.9
Min My	-2817	bottom face	40	1444	2817		5298		32	151.87	150	5363.9
Max Mxy	-1948	bottom face	40	1444	1948		3634		32	221.42	150	5363.9
Min Mxy	-1947	bottom face	40	1444	1947	*As	3631		32	221.58	150	5363.9

Shear Check		Qx	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa	KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.57	857	2726	0.004	1.444	1408	1458	2477	1408.2	1479	> 857	ok	1056	φVc>Vu	not required
	Min Qy	0.59	887	2886			1407			1407.3	1478	>887	ok	1055	φVc>Vu	not required

Nominal shear strength required by Reinf. Vs= Not Required KN SECTION 11.2

Spacing Required of Strrups, s= Not Required mm SECTION 11.5.6.2

Though, not required as design of of view below mentioned stirrups provided.

Diameter of Shear Link Provided = 16 mm

Spacing Provided : Along X-X = 150 mm

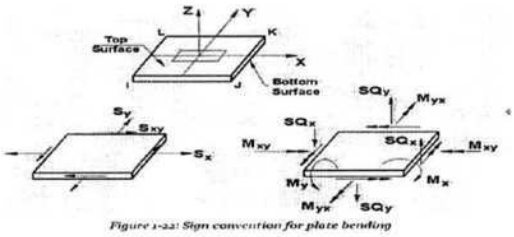
Along Y-Y = 150 mm OK

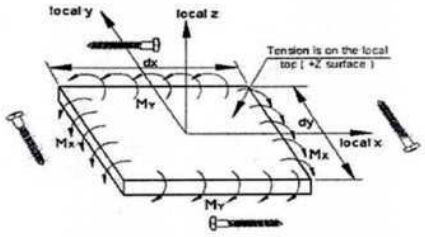
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• **Area Raft Other (Part 3):** Heavy_Slab_3A





Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:

PLATE CENTER STRESS STAAD GROUP : Heavy_Slab_3A At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	12517	0.274	-0.238	0.5	0.463	0.012	-837.9	-754.2	-578.5	-1416	-1333	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qx	12508	-0.248	0.245	0.4	0.383	0	-1025	-781.7	0	-1025	-781.7	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Qy	12509	-0.04	0.537	0.424	0.434	-0.012	-1484	-687.4	-399.3	-1883	-1087	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Qy	12518	0.097	-0.55	0.505	0.471	0.015	-1307	-731.2	-417.6	-1724	-1149	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Sx	12344	-0.104	-0.233	0.659	0.637	0.018	-217.3	-242.4	258.6	-476	-501.1	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sx	12509	-0.011	0.016	-0	0.009	0.003	4.697	-11.33	-59.12	63.81	-70.45	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sy	12518	0.126	-0.246	0.651	0.638	-0.011	-272.6	-335.3	-308.7	-581.3	-644	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sy	12507	-0.016	-0.036	0.002	0.007	0.001	102.7	-9.68	-23.97	126.7	-33.65	115 1.2X[D] + 0.8X[W1A]
Max Sxy	12578	0.045	0.206	0.485	0.509	0.025	-461	-243.4	268.8	-729.8	-512.1	131 0.9X[D] + 1.6X[W1A + H2B + H4B + H6B]
Min Sxy	12509	-0.053	0.225	0.514	0.543	-0.023	-489.7	-259.5	-315.9	-805.6	-575.4	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Mx	12350	-0.113	0.185	0.601	0.6	0.015	449.5	-258.5	157.7	607.1	-416.1	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min Mx	12578	0.04	0.537	0.424	0.434	0.014	-1485	-686.4	400.8	-1885	-1087	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max My	12395	0.02	0.047	0.069	0.041	-0.003	166.3	-0.706	-26.27	192.6	-26.97	114 1.2X[D] + 1.0X[L]
Min My	12507	-0.155	-0.048	0.403	0.435	-0.001	-456	-870.8	-259.3	-715.3	-1130	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Mxy	12577	0.248	0.244	0.4	0.383	0.002	-1025	-781.1	523.1	-1549	-1304	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Mxy	12517	0.274	-0.238	0.522	0.486	0.012	-838.6	-753.9	-578.9	-1417	-1333	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]

PLATE CENTER STRESS STAAD GROUP : Heavy_Slab_3A At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	12517	0.274	-0.238	0.5	0.463	0.012	-837.9	-754.2	-578.5	-1416	-1333	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qx	12508	-0.248	0.245	0.4	0.383	0	-1025	-781.7	0	-1025	-781.7	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Qy	12509	-0.04	0.537	0.424	0.434	-0.012	-1484	-687.4	-399.3	-1883	-1087	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Qy	12518	0.097	-0.55	0.505	0.471	0.015	-1307	-731.2	-417.6	-1724	-1149	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Sx	12344	-0.104	-0.233	0.659	0.637	0.018	-217.3	-242.4	258.6	-476	-501.1	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sx	12509	-0.011	0.016	-0	0.009	0.003	4.697	-11.33	-59.12	63.81	-70.45	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sy	12518	0.126	-0.246	0.651	0.638	-0.011	-272.6	-335.3	-308.7	-581.3	-644	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sy	12507	-0.016	-0.036	0.002	0.007	0.001	102.7	-9.68	-23.97	126.7	-33.65	115 1.2X[D] + 0.8X[W1A]
Max Sxy	12578	0.045	0.206	0.485	0.509	0.025	-461	-243.4	268.8	-729.8	-512.1	131 0.9X[D] + 1.6X[W1A + H2B + H4B + H6B]
Min Sxy	12509	-0.053	0.225	0.514	0.543	-0.023	-489.7	-259.5	-315.9	-805.6	-575.4	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Mx	12350	-0.113	0.185	0.601	0.6	0.015	449.5	-258.5	157.7	607.1	-416.1	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min Mx	12578	0.04	0.537	0.424	0.434	0.014	-1485	-686.4	400.8	-1885	-1087	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max My	12395	0.02	0.047	0.069	0.041	-0.003	166.3	-0.706	-26.27	192.6	-26.97	114 1.2X[D] + 1.0X[L]
Min My	12507	-0.155	-0.048	0.403	0.435	-0.001	-456	-870.8	-259.3	-715.3	-1130	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Mxy	12577	0.248	0.244	0.4	0.383	0.002	-1025	-781.1	523.1	-1549	-1304	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Mxy	12517	0.274	-0.238	0.522	0.486	0.012	-838.6	-753.9	-578.9	-1417	-1333	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	40	mm
	top face	40	mm
Thicknes of slab =		1500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

Force Values			Depth		Reinforcement Calculation							
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2
Max Mx	607	top face	40	1444	607.1	0.014	1118	2700	32	297.99	150	5363.9
Min Mx	45	top face	40	1444	45		82.5		32	297.99	150	5363.9
Max Mxy	-1549	bottom face	40	1444	1549		2877		32	279.61	150	5363.9
Min Mxy	-1417	bottom face	40	1444	1417	*As	2631		32	297.99	150	5363.9

Shear Check		Qx	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa	KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.27	411	1416	0.004	1.444	1405	1458	fc'2477	1405.1	5282	> 411	ok	1054	φVc>Vu	not required
	Min Qx	0.25	372	1025			1415			1414.7	5290	> 372	ok	1061	φVc>Vu	not required

Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation							
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2
Max My	-27	bottom face	40	1444	26.97	0.014	49	2700	32	297.99	150	5363.9
Min My	-1130	bottom face	40	1444	1130		2092		32	297.99	150	5363.9
Max Mxy	-1304	bottom face	40	1444	1304		2418		32	297.99	150	5363.9
Min Mxy	-1333	bottom face	40	1444	1333	*As	2472		32	297.99	150	5363.9

Shear Check		Qx	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa	KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.54	806	1087	0.004	1.444	1464	1458	2477	1458	1516	> 806	ok	1094	φVc>Vu	not required
	Min Qy	0.55	825	1149			1461			1458	1516	>825	ok	1094	φVc>Vu	not required

Nominal shear strength required by Reinf. Vs= Not Required KN SECTION 11.2

Spacing Required of Strrups, s= Not Required mm SECTION 11.5.6.2

Though, Not required as per design, provided below mentioned stirrups.

Diameter of Shear Link Provided = 16 mm

Spacing Provided : Along X-X = 150 mm

Along Y-Y = 150 mm OK

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• **Area Raft Other (Part 3):** Heavy_Slab_3B

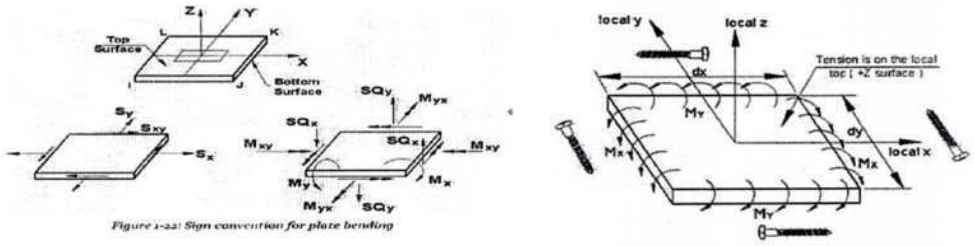


Figure 1-22: Sign convention for plate bending

Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:

PLATE CENTER STRESS STAAD GROUP : Heavy_Slab_3B At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	12341	0.58	-0.813	0.458	0.452	0.034	-999.3	-382.2	-366.5	-1366	-748.7	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Qx	12505	-0.522	0.097	0.409	0.395	0.01	-1151	-892.9	0	-1151	-892.9	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Qy	12493	0.077	0.625	0.431	0.4	0.003	-1785	-986.7	-253.7	-2038	-1240	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Qy	12341	0.58	-0.813	0.458	0.452	0.034	-999.3	-382.2	-366.5	-1366	-748.7	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Sx	12351	-0.382	0.051	0.617	0.625	0.013	22.98	-230.2	374.8	397.8	-605	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sx	12493	-0.004	0.026	0.001	0.008	0.003	-24.79	-27.74	-55.96	-80.75	-83.7	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sy	12345	-0.192	-0.127	0.604	0.625	0.023	-298.8	-368.5	244	-542.7	-612.5	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sy	12506	-0.026	-0.071	0.005	0.006	0	136.2	6.847	-29.98	166.2	36.83	115 1.2X[D] + 0.8X[W1A]
Max Sxy	12341	0.58	-0.813	0.458	0.452	0.034	-999.3	-382.2	-366.5	-1366	-748.7	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Sxy	12592	0.193	-0.171	0.548	0.568	-0.015	-353.1	-370.8	-225.8	-578.9	-596.6	131 0.9X[D] + 1.6X[W1A + H2B + H4B + H6B]
Max Mx	12352	-0.175	0.455	0.586	0.592	0.011	704.2	-205.9	221.9	926.2	-427.8	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min Mx	12574	-0.077	0.625	0.432	0.4	-0.001	-1786	-985.4	255	-2041	-1240	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max My	12396	0.035	0.097	0.078	0.043	-0.004	232.2	21.73	-35.44	267.6	57.16	101 1.4X[D]
Min My	12591	0.01	-0.631	0.425	0.401	0.012	-1679	-1032	-131.8	-1810	-1164	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Mxy	12572	0.522	0.097	0.409	0.395	-0.008	-1152	-892.1	627.1	-1779	-1519	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Mxy	12519	0.565	-0.083	0.5	0.494	0.019	-980	-842.7	-661.8	-1642	-1505	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]

PLATE CENTER STRESS STAAD GROUP : Heavy_Slab_3B At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	12341	0.58	-0.813	0.458	0.452	0.034	-999.3	-382.2	-366.5	-1366	-748.7	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Qx	12505	-0.522	0.097	0.409	0.395	0.01	-1151	-892.9	0	-1151	-892.9	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Qy	12493	0.077	0.625	0.431	0.4	0.003	-1785	-986.7	-253.7	-2038	-1240	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Qy	12341	0.58	-0.813	0.458	0.452	0.034	-999.3	-382.2	-366.5	-1366	-748.7	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Sx	12351	-0.382	0.051	0.617	0.625	0.013	22.98	-230.2	374.8	397.8	-605	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sx	12493	-0.004	0.026	0.001	0.008	0.003	-24.79	-27.74	-55.96	-80.75	-83.7	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sy	12345	-0.192	-0.127	0.604	0.625	0.023	-298.8	-368.5	244	-542.7	-612.5	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sy	12506	-0.026	-0.071	0.005	0.006	0	136.2	6.847	-29.98	166.2	36.83	115 1.2X[D] + 0.8X[W1A]
Max Sxy	12341	0.58	-0.813	0.458	0.452	0.034	-999.3	-382.2	-366.5	-1366	-748.7	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Sxy	12592	0.193	-0.171	0.548	0.568	-0.015	-353.1	-370.8	-225.8	-578.9	-596.6	131 0.9X[D] + 1.6X[W1A + H2B + H4B + H6B]
Max Mx	12352	-0.175	0.455	0.586	0.592	0.011	704.2	-205.9	221.9	926.2	-427.8	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min Mx	12574	-0.077	0.625	0.432	0.4	-0.001	-1786	-985.4	255	-2041	-1240	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max My	12396	0.035	0.097	0.078	0.043	-0.004	232.2	21.73	-35.44	267.6	57.16	101 1.4X[D]
Min My	12591	0.01	-0.631	0.425	0.401	0.012	-1679	-1032	-131.8	-1810	-1164	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Mxy	12572	0.522	0.097	0.409	0.395	-0.008	-1152	-892.1	627.1	-1779	-1519	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Mxy	12519	0.565	-0.083	0.5	0.494	0.019	-980	-842.7	-661.8	-1642	-1505	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	40	mm
	top face	40	mm
Thicknes of slab =		1500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

Force Values			Depth		Reinforcement Calculation								
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2
Max Mx	926	top face	40	1444	926.2		1711		32	297.99	150	5363.9
Min Mx	45	top face	40	1444	45	0.014	82.5	2700	32	297.99	150	5363.9
Max Mxy	-1779	bottom face	40	1444	1779		3313		32	242.87	150	5363.9
Min Mxy	-1642	bottom face	40	1444	1642	*As	3053		32	263.49	150	5363.9

Shear Check		Qx	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa	KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.58	870	1366	0.004	1.444	1451	1458	fc'2477	1450.7	5317	> 870	ok	1088	φVc>Vu	not required
	Min Qx	0.52	783	1151			1456			1456.4	5321	> 783	ok	1092	φVc>Vu	not required

Nominal shear strength required by Reinf. Vs= Not Required KN SECTION 11.2

Spacing Required of Strrups, s= Not Required mm SECTION 11.5.6.2
or 722 mm SECTION 11.5.4.1
or 1125 mm

Hence, spacing required = 722 mm

Diameter of Shear Link Provided = 16 mm
Spacing Provided : Along X-X = 150 mm
Along Y-Y = 150 mm OK

Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation								
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2
Max My	57	top face	40	1444	57.16		105		32	297.99	150	5363.9
Min My	-1164	bottom face	40	1444	1164	0.014	2154	2700	32	297.99	150	5363.9
Max Mxy	-1519	bottom face	40	1444	1519		2822		32	285.09	150	5363.9
Min Mxy	-1505	bottom face	40	1444	1505	*As	2795		32	287.91	150	5363.9

Shear Check		Qx	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa	KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.63	938	1240	0.004	1.444	1466	1458	2477	1458	1516	> 938	ok	1094	φVc>Vu	not required
	Min Qy	0.81	1220	748.7			1581			1458	1516	>1220	ok	1094	φVc<Vu	required

Nominal shear strength required by Reinf. Vs= 168.0 KN SECTION 11.2

Spacing Required of Strrups, s= 484 mm SECTION 11.5.6.2
or 722 mm SECTION 11.5.4.1
or 1125 mm

Hence, spacing required = 484 mm

Diameter of Shear Link Provided = 16 mm
Spacing Provided : Along X-X = 150 mm
Along Y-Y = 150 mm OK

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• **Area Raft Other (Part 4):** Heavy_Slab_4A

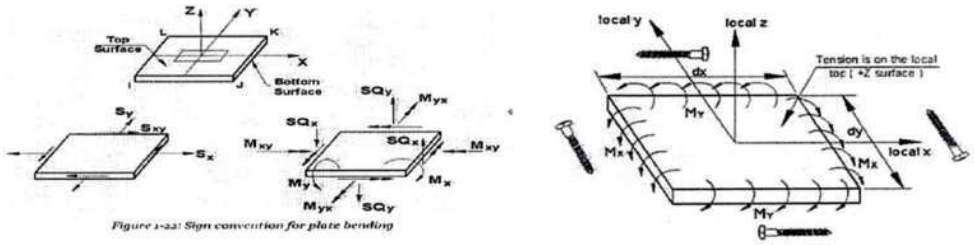


Figure 1-22: Sign convention for plate bending

Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:

PLATE CENTER STRESS STAAD GROUP : Heavy_Slab_4A At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	12677	0.456	-0.343	0.462	0.37	-0.003	-851	-468.7	160.8	-1012	-629.5	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qx	12298	-0.455	-0.344	0.468	0.369	0.012	-852.5	-464.1	0	-852.5	-464.1	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Qy	12736	0.12	0.463	0.382	0.341	-0.013	-831.9	-1329	-325.4	-1157	-1654	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qy	12682	0.127	-0.476	0.393	0.347	0.006	-827	-1299	340.2	-1167	-1639	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Sx	12288	-0.063	0.171	0.695	0.628	0.047	81.86	-78.03	468.9	550.7	-546.9	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sx	12315	-0.018	0.01	0.012	-0.016	0.014	21.61	-23.03	2.751	24.36	-25.78	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sy	12537	0.061	0.172	0.69	0.629	-0.043	86.81	-87.83	-478.3	565.1	-566.1	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sy	12315	-0.018	0.01	0.012	-0.016	0.014	21.61	-23.03	2.751	24.36	-25.78	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sxy	12288	-0.063	0.172	0.667	0.599	0.048	82.56	-78.13	469.5	552.1	-547.6	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min Sxy	12537	0.061	0.172	0.662	0.6	-0.044	87.32	-87.57	-478.5	565.9	-566.1	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Max Mx	12537	0.03	0.042	0.08	0.052	-0.022	88.2	80.42	-106	194.2	186.4	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Min Mx	12735	-0.057	0.323	0.412	0.331	0.003	-895.7	-1293	-257.2	-1153	-1550	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max My	12293	-0.395	0.074	0.596	0.567	0.007	-20.29	319.6	345.7	-366	665.3	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min My	12316	0.107	0.172	0.368	0.347	0.014	-844.1	-1893	112.3	-956.4	-2006	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Mxy	12288	-0.063	0.172	0.667	0.599	0.048	82.56	-78.13	469.5	552.1	-547.6	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min Mxy	12537	-0.086	0.147	0.536	0.461	-0.024	-461.6	-962.1	-484.3	-945.9	-1446	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]

PLATE CENTER STRESS STAAD GROUP : Heavy_Slab_4A At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	12677	0.456	-0.343	0.462	0.37	-0.003	-851	-468.7	160.8	-1012	-629.5	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qx	12298	-0.455	-0.344	0.468	0.369	0.012	-852.5	-464.1	0	-852.5	-464.1	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Qy	12736	0.12	0.463	0.382	0.341	-0.013	-831.9	-1329	-325.4	-1157	-1654	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qy	12682	0.127	-0.476	0.393	0.347	0.006	-827	-1299	340.2	-1167	-1639	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Sx	12288	-0.063	0.171	0.695	0.628	0.047	81.86	-78.03	468.9	550.7	-546.9	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sx	12315	-0.018	0.01	0.012	-0.016	0.014	21.61	-23.03	2.751	24.36	-25.78	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sy	12537	0.061	0.172	0.69	0.629	-0.043	86.81	-87.83	-478.3	565.1	-566.1	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sy	12315	-0.018	0.01	0.012	-0.016	0.014	21.61	-23.03	2.751	24.36	-25.78	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sxy	12288	-0.063	0.172	0.667	0.599	0.048	82.56	-78.13	469.5	552.1	-547.6	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min Sxy	12537	0.061	0.172	0.662	0.6	-0.044	87.32	-87.57	-478.5	565.9	-566.1	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Max Mx	12537	0.03	0.042	0.08	0.052	-0.022	88.2	80.42	-106	194.2	186.4	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Min Mx	12735	-0.057	0.323	0.412	0.331	0.003	-895.7	-1293	-257.2	-1153	-1550	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max My	12293	-0.395	0.074	0.596	0.567	0.007	-20.29	319.6	345.7	-366	665.3	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min My	12316	0.107	0.172	0.368	0.347	0.014	-844.1	-1893	112.3	-956.4	-2006	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Mxy	12288	-0.063	0.172	0.667	0.599	0.048	82.56	-78.13	469.5	552.1	-547.6	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min Mxy	12537	-0.086	0.147	0.536	0.461	-0.024	-461.6	-962.1	-484.3	-945.9	-1446	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	40	mm
	top face	40	mm
Thicknes of slab =		1500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

Force Values			Depth		Reinforcement Calculation							
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2
Max Mx	194	top face	40	1444	194.2	0.014	356	2700	32	297.99	150	5363.9
Min Mx	45	top face	40	1444	45		82.5		32	297.99	150	5363.9
Max Mxy	552	top face	40	1444	552.1		1016		32	297.99	150	5363.9
Min Mxy	-946	bottom face	40	1444	945.9	*As	1748		32	297.99	150	5363.9

Shear Check		Qx	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa	KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.46	684	1012	0.004	1.444	1456	1458	fc'2477	1455.9	5321	> 684	ok	1092	φVc>Vu	not required
	Min Qx	0.46	682.5	852.5			1472			1458	5322	> 683	ok	1094	φVc>Vu	not required

Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation							
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2
Max My	665	top face	40	1444	665.3	0.014	1226	2700	32	297.99	150	5363.9
Min My	-2006	bottom face	40	1444	2006		3743		32	214.98	150	5363.9
Max Mxy	-548	bottom face	40	1444	547.6		1008		32	297.99	150	5363.9
Min Mxy	-1446	bottom face	40	1444	1446	*As	2685		32	297.99	150	5363.9

Shear Check		Qx	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa	KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.46	695	1654	0.004	1.444	1422	1458	2477	1422.1	1489	> 695	ok	1067	φVc>Vu	not required
	Min Qy	0.48	714	1639			1424			1424.2	1491	>714	ok	1068	φVc>Vu	not required

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• **Area Raft Other (Part 4):** Heavy_Slab_4B

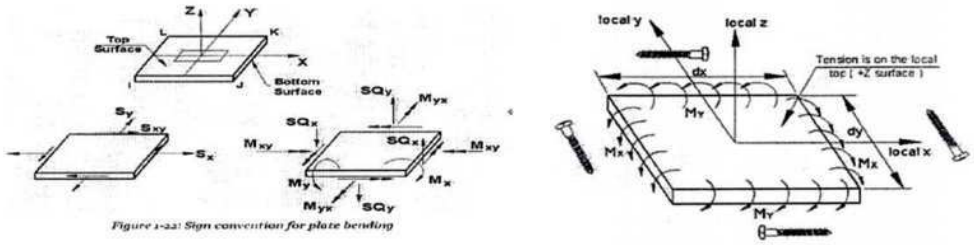


Figure 1-22: Sign convention for plate bending

Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:

PLATE CENTER STRESS STAAD GROUP : Heavy_Slab_4B At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	12676	1.166	-0.795	0.463	0.382	0.01	-465.2	70.93	286.5	-751.7	357.4	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qx	12299	-1.162	-0.794	0.468	0.384	-0.001	-462.9	69.54	0	-462.9	69.54	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Qy	12737	0.193	0.818	0.38	0.368	-0.019	-525.5	-1478	-331.4	-857	-1809	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qy	12683	0.199	-0.836	0.391	0.373	0.014	-516.9	-1444	346.4	-863.4	-1791	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Sx	12289	-0.092	0.502	0.67	0.6	0.016	238.2	-195.3	506.5	744.7	-701.8	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sx	12317	-0.011	0.017	0.017	0.012	0.014	60.5	-22.72	-1.549	62.05	-24.27	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sy	12284	0.18	0.165	0.665	0.66	0.03	53.87	-510.5	345.4	399.3	-855.9	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sy	12323	-0.034	0.072	0.021	0.011	0.018	84.74	21.29	5.084	89.82	26.38	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sxy	12328	-0.879	0.54	0.505	0.513	0.043	301.1	540.4	52.49	353.6	592.9	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min Sxy	12748	0.88	0.54	0.505	0.514	-0.041	299.5	542.3	-55.02	354.5	597.4	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Max Mx	12665	0.863	0.468	0.587	0.546	0.014	340.2	674.1	-450.8	791	1125	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min Mx	12721	-0.111	0.238	0.374	0.388	-0.011	-784.7	-2028	-64.47	-849.2	-2092	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max My	12292	-0.862	0.465	0.59	0.545	-0.01	339.6	676.9	440	779.6	1117	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min My	12317	0.11	0.237	0.375	0.388	0.017	-778.9	-2030	51.09	-830	-2081	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Mxy	12752	1.049	-0.6	0.505	0.419	-0.037	-427.5	34.36	590	-1018	624.4	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Mxy	12536	-0.012	0.633	0.512	0.451	0.007	-329.2	-1224	-613.9	-943.1	-1838	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]

PLATE CENTER STRESS STAAD GROUP : Heavy_Slab_4B At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	12676	1.166	-0.795	0.463	0.382	0.01	-465.2	70.93	286.5	-751.7	357.4	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qx	12299	-1.162	-0.794	0.468	0.384	-0.001	-462.9	69.54	0	-462.9	69.54	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Qy	12737	0.193	0.818	0.38	0.368	-0.019	-525.5	-1478	-331.4	-857	-1809	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qy	12683	0.199	-0.836	0.391	0.373	0.014	-516.9	-1444	346.4	-863.4	-1791	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Sx	12289	-0.092	0.502	0.67	0.6	0.016	238.2	-195.3	506.5	744.7	-701.8	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sx	12317	-0.011	0.017	0.017	0.012	0.014	60.5	-22.72	-1.549	62.05	-24.27	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sy	12284	0.18	0.165	0.665	0.66	0.03	53.87	-510.5	345.4	399.3	-855.9	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sy	12323	-0.034	0.072	0.021	0.011	0.018	84.74	21.29	5.084	89.82	26.38	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sxy	12328	-0.879	0.54	0.505	0.513	0.043	301.1	540.4	52.49	353.6	592.9	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min Sxy	12748	0.88	0.54	0.505	0.514	-0.041	299.5	542.3	-55.02	354.5	597.4	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Max Mx	12665	0.863	0.468	0.587	0.546	0.014	340.2	674.1	-450.8	791	1125	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min Mx	12721	-0.111	0.238	0.374	0.388	-0.011	-784.7	-2028	-64.47	-849.2	-2092	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max My	12292	-0.862	0.465	0.59	0.545	-0.01	339.6	676.9	440	779.6	1117	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min My	12317	0.11	0.237	0.375	0.388	0.017	-778.9	-2030	51.09	-830	-2081	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Mxy	12752	1.049	-0.6	0.505	0.419	-0.037	-427.5	34.36	590	-1018	624.4	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Mxy	12536	-0.012	0.633	0.512	0.451	0.007	-329.2	-1224	-613.9	-943.1	-1838	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	40	mm
	top face	40	mm
Thicknes of slab =		1500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

Force Values			Depth		Reinforcement Calculation								
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2
Max Mx	791	top face	40	1444	791		1460		32	297.99	150	5363.9
Min Mx	45	top face	40	1444	45	0.014	82.5	2700	32	297.99	150	5363.9
Max Mxy	-1018	bottom face	40	1444	1018		1881		32	297.99	150	5363.9
Min Mxy	-943	bottom face	40	1444	943.1	*As	1743		32	297.99	150	5363.9

Shear Check		Qx	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$			Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa	KN	KN-m/m		m^2	KN	KN	KN	KN	KN				KN		
Load condition	Max Qx	1.17	1749	751.7	0.004	1.444	1673	1458	fc'2477	1458	5322	> 1749		ok	1094	φVc<Vu	required
	Min Qx	1.16	1743	462.9			1863			1458	5322	> 1743		ok	1094	φVc<Vu	required

Nominal shear strength required by Reinf. Vs= 874.0 KN SECTION 11.2

Spacing Required of Strrups, s= 93 mm SECTION 11.5.6.2
or 722 mm SECTION 11.5.4.1
or 1125 mm

Hence, spacing required = 93 mm

Diameter of Shear Link Provided = 16 mm
Spacing Provided : Along X-X = 150 mm
Along Y-Y = 150 mm DECREASE SPACING

Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation								
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2
Max My	1117	top face	40	1444	1117		2067		32	297.99	150	5363.9
Min My	-2081	bottom face	40	1444	2081	0.014	3887	2700	32	207.01	150	5363.9
Max Mxy	624	top face	40	1444	624.4		1150		32	297.99	150	5363.9
Min Mxy	-1838	bottom face	40	1444	1838	*As	3425		32	234.92	150	5363.9

Shear Check		Qx	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$			Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa	KN	KN-m/m		m^2	KN	KN	KN	KN	KN				KN		
Load condition	Max Qy	0.82	1227	1809	0.004	1.444	1456	1458	2477	1456.2	1515	> 1227		ok	1092	φVc<Vu	required
	Min Qy	0.84	1254	1791			1459			1458	1516	>1254		ok	1094	φVc<Vu	required

Nominal shear strength required by Reinf. Vs= 215.8 KN SECTION 11.2

Spacing Required of Strrups, s= 377 mm SECTION 11.5.6.2
or 722 mm SECTION 11.5.4.1
or 1125 mm

Hence, spacing required = 377 mm

Diameter of Shear Link Provided = 16 mm
Spacing Provided : Along X-X = 150 mm
Along Y-Y = 150 mm OK

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• **Heavy Slab Concealed Beam (CB2):** Suport Portion

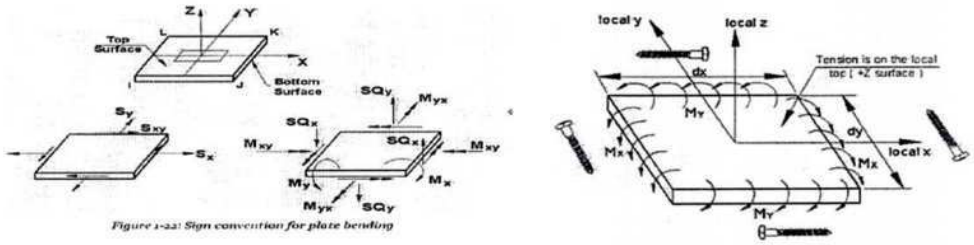


Figure 1-22 Sign convention for plate bending

Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:

PLATE CENTER STRESS STAAD GROUP : CB2_MOMENT At the face of the column

Load condition		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	L/C
Max Qx	12541	2.087	-0.151	0.417	0.399	0.003	-279.1	761.3	376	-655.1	1137	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qx	12271	-2.08	0.136	0.403	0.395	0.008	-294.2	707.6	0	-294.2	707.6	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Qy	12540	2.062	0.363	0.422	0.4	0	-370.7	698.3	-562.3	-933	1261	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qy	12413	-2.054	-0.376	0.429	0.416	0.009	-395.6	624.1	-598.8	-994.4	1223	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Sx	12190	0.69	0.104	0.608	0.546	-0.028	-737.6	343.4	8.284	-745.9	351.7	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sx	12413	-0.188	-0.035	0.031	0.018	0.001	192.2	259.2	-56.52	248.7	315.7	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sy	12191	0.66	-0.044	0.575	0.555	-0.032	-698.9	360.5	285.1	-983.9	645.6	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sy	12390	0.083	-0.024	0.037	0.012	0.012	-50.16	113.6	-21.47	-71.63	135.1	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sxy	12196	0.655	0.031	0.516	0.522	0.041	-704.1	283.6	-302.6	-1007	586.2	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min Sxy	12726	-0.656	0.031	0.516	0.522	-0.039	-707.6	286.6	301.4	-1009	588	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Max Mx	12248	-1.262	-0.105	0.569	0.554	0.003	524.6	1143	-216.8	741.4	1360	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Mx	12670	-0.091	0.11	0.456	0.41	0.014	-3173	-1139	-119.5	-3292	-1259	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max My	12541	1.268	-0.104	0.541	0.525	-0.004	516.1	1147	229.5	745.6	1377	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min My	12392	0.093	-0.118	0.426	0.42	0.022	-3143	-1233	-154.7	-3298	-1388	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Mxy	12243	-2.037	0.349	0.447	0.427	-0.003	-330.7	691.3	610.8	-941.5	1302	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Mxy	12413	-2.054	-0.376	0.405	0.392	0.009	-395.4	624.4	-598.8	-994.3	1223	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]

PLATE CENTER STRESS STAAD GROUP : CB2_SHEAR At a distance "d" from the face of the column

Load condition		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	L/C
Max Qx	12543	1.129	-0.12	0.417	0.396	0.003	-1991	-402.9	83.78	-2074	-486.7	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qx	12268	-1.123	0.105	0.4	0.397	0.011	-1987	-466.7	0	-1987	-466.7	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Qy	12542	1.107	0.271	0.426	0.394	0.001	-2034	-496.9	-362.2	-2396	-859.1	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qy	12412	-1.1	-0.284	0.428	0.417	0.012	-2037	-576.9	-401.7	-2439	-978.6	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Sx	12203	0.276	0.126	0.602	0.541	-0.02	-1190	26.54	74.21	-1264	100.7	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sx	12412	-0.108	-0.034	0.031	0.017	0.003	38.57	155.7	-42.53	81.11	198.2	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sy	12204	0.278	-0.041	0.576	0.551	-0.022	-1160	66.04	177.8	-1338	243.9	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sy	12391	0.017	-0.032	0.034	0.014	0.01	-94.62	80.36	-29.72	-124.3	110.1	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sxy	12209	0.277	0.027	0.518	0.522	0.032	-1165	-5.758	-200.4	-1366	-206.2	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min Sxy	12710	-0.277	0.027	0.518	0.521	-0.03	-1170	-2.548	200	-1370	-202.6	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Max Mx	12543	0.116	-0.009	0.06	0.03	-0.002	59.5	212.6	5.324	64.83	217.9	101 1.4X[D]
Min Mx	12670	-0.091	0.11	0.456	0.41	0.014	-3173	-1139	-119.5	-3292	-1259	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max My	12543	0.7	-0.063	0.545	0.521	0	-535.5	420.3	51.06	-586.5	471.3	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min My	12392	0.093	-0.118	0.426	0.42	0.022	-3143	-1233	-154.7	-3298	-1388	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Mxy	12634	1.101	-0.284	0.429	0.416	-0.011	-2045	-569.8	410	-2455	-979.8	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Mxy	12412	-1.1	-0.284	0.404	0.394	0.013	-2037	-576.6	-401.7	-2439	-978.3	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	40	mm
	top face	40	mm
Thicknes of slab =		1500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

Force Values			Depth		Reinforcement Calculation								
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided	
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	
Max Mx	741	top face	40	1444	741.4		1367		32	297.99	140	5747	
Min Mx	45	top face	40	1444	45	0.014	82.5	2700	32	297.99	140	5747	Provdie (D32+D25) for Top and Bottom Face
Max Mxy	-942	bottom face	40	1444	941.5		1740		32	297.99	140	5747	
Min Mxy	-994	bottom face	40	1444	994.3	*As	1838		32	297.99	140	5747	

Shear Check		Qx	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$			Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa	KN	KN-m/m		m^2	KN	KN	KN	KN	KN				KN		
Load condition	Max Qx	1.13	1693.5	2074	0.004	1.444	1482	1465	fc'2477	1464.6	5327	> 1694	ok	1098	φVc<Vu	required	
	Min Qx	1.12	1684.5	1987			1486			1464.6	5327	> 1685	ok	1098	φVc<Vu	required	

Nominal shear strength required by Reinf. Vs= 793.4 KN SECTION 11.2

Spacing Required of Strrups, s= 110 mm SECTION 11.5.6.2
or 722 mm SECTION 11.5.4.1
or 1125 mm

Hence, spacing required = 110 mm

Diameter of Shear Link Provided = 16 mm
Spacing Provided : Along X-X = 100 mm
Along Y-Y = 140 mm OK

Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation								
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided	
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	
Max My	1377	top face	40	1444	1377		2554		32	297.99	150	5363.9	
Min My	-1388	bottom face	40	1444	1388	0.014	2575	2700	32	297.99	150	5363.9	
Max Mxy	1302	top face	40	1444	1302		2414		32	297.99	150	5363.9	
Min Mxy	1223	top face	40	1444	1223	*As	2266		32	297.99	150	5363.9	

Shear Check		Qx	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$			Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa	KN	KN-m/m		m^2	KN	KN	KN	KN	KN				KN		
Load condition	Max Qy	0.27	407	859.1	0.004	1.444	1434	1465	2477	1433.6	1498	> 407	ok	1075	φVc>Vu	not required	
	Min Qy	0.28	426	978.6			1428			1428.3	1494	>426	ok	1071	φVc>Vu	not required	

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• **Heavy Slab Concealed Beam (CB2):** Span Portion

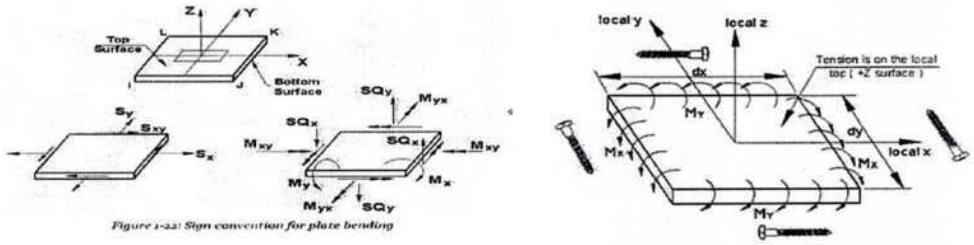


Figure 1-22: Sign convention for plate bending

Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:

PLATE CENTER STRESS STAAD GROUP : CB2_Middle At the face of the column

Load condition		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	L/C
Max Qx	12647	0.435	-0.007	0.417	0.395	0.007	-2902	-919.4	-66.34	-2969	-985.7	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qx	12265	-0.432	-0.006	0.421	0.423	0.015	-2884	-989.2	0	-2884	-989.2	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Qy	12646	0.42	0.115	0.43	0.389	0.006	-2914	-1003	-227.8	-3142	-1231	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qy	12411	-0.414	-0.124	0.427	0.419	0.017	-2897	-1091	-267.2	-3164	-1358	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Sx	12224	0.014	0.085	0.596	0.539	-0.011	-1318	-63.06	105.6	-1423	-168.6	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sx	12411	-0.053	-0.022	0.032	0.016	0.006	-54.61	99.16	-36.55	-91.16	135.7	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sy	12245	0.01	-0.002	0.577	0.549	-0.014	-1299	-8.37	124.2	-1424	-132.5	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sy	12392	-0.014	-0.022	0.033	0.015	0.008	-95.96	76.27	-31.52	-127.5	107.8	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sxy	12262	0.01	-0.013	0.519	0.521	0.024	-1309	-75.71	-148.8	-1457	-224.5	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min Sxy	12692	-0.01	-0.013	0.52	0.52	-0.024	-1313	-72.47	149.3	-1463	-221.8	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Max Mx	12647	0.045	0.003	0.052	0.027	-0.002	-33.05	136.9	-11.88	-44.94	148.8	115 1.2X[D] + 0.8X[W1A]
Min Mx	12670	-0.091	0.11	0.456	0.41	0.014	-3173	-1139	-119.5	-3292	-1259	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max My	12246	-0.054	0.004	0.06	0.03	0.002	-45.06	158.1	13.63	-58.69	171.7	101 1.4X[D]
Min My	12392	0.093	-0.118	0.426	0.42	0.022	-3143	-1233	-154.7	-3298	-1388	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Mxy	12653	0.415	-0.123	0.428	0.417	-0.015	-2908	-1083	272	-3180	-1355	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Mxy	12411	-0.414	-0.124	0.427	0.419	0.017	-2897	-1091	-267.2	-3164	-1358	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]

PLATE CENTER STRESS STAAD GROUP : CB2_Middle At a distance "d" from the face of the column

Load condition		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	L/C
Max Qx	12647	0.435	-0.007	0.417	0.395	0.007	-2902	-919.4	-66.34	-2969	-985.7	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qx	12265	-0.432	-0.006	0.421	0.423	0.015	-2884	-989.2	0	-2884	-989.2	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Qy	12646	0.42	0.115	0.43	0.389	0.006	-2914	-1003	-227.8	-3142	-1231	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qy	12411	-0.414	-0.124	0.427	0.419	0.017	-2897	-1091	-267.2	-3164	-1358	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Sx	12224	0.014	0.085	0.596	0.539	-0.011	-1318	-63.06	105.6	-1423	-168.6	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sx	12411	-0.053	-0.022	0.032	0.016	0.006	-54.61	99.16	-36.55	-91.16	135.7	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sy	12245	0.01	-0.002	0.577	0.549	-0.014	-1299	-8.37	124.2	-1424	-132.5	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sy	12392	-0.014	-0.022	0.033	0.015	0.008	-95.96	76.27	-31.52	-127.5	107.8	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sxy	12262	0.01	-0.013	0.519	0.521	0.024	-1309	-75.71	-148.8	-1457	-224.5	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min Sxy	12692	-0.01	-0.013	0.52	0.52	-0.024	-1313	-72.47	149.3	-1463	-221.8	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Max Mx	12647	0.045	0.003	0.052	0.027	-0.002	-33.05	136.9	-11.88	-44.94	148.8	115 1.2X[D] + 0.8X[W1A]
Min Mx	12670	-0.091	0.11	0.456	0.41	0.014	-3173	-1139	-119.5	-3292	-1259	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max My	12246	-0.054	0.004	0.06	0.03	0.002	-45.06	158.1	13.63	-58.69	171.7	101 1.4X[D]
Min My	12392	0.093	-0.118	0.426	0.42	0.022	-3143	-1233	-154.7	-3298	-1388	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Mxy	12653	0.415	-0.123	0.428	0.417	-0.015	-2908	-1083	272	-3180	-1355	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Mxy	12411	-0.414	-0.124	0.427	0.419	0.017	-2897	-1091	-267.2	-3164	-1358	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	40	mm
	top face	40	mm
Thicknes of slab =		1500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

Force Values			Depth		Reinforcement Calculation								
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided	
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	
Max Mx	-45	bottom face	40	1440	44.94		83		40.61	479.87	150	8637.7	
Min Mx	45	top face	40	1440	45	0.014	82.7	2700	40.61	479.87	150	8637.7	Provdie (D32+D25) for Top and Bottom Face
Max Mxy	-3180	bottom face	40	1440	3180		6021		40.61	215.19	150	8637.7	
Min Mxy	-3164	bottom face	40	1440	3164	*As	5990		40.61	216.31	150	8637.7	

Shear Check		Qx	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa	KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.44	652.5	2969	0.006	1.44	1409	1510	fc'2470	1409.2	5273	> 653	ok	1057	φVc>Vu	not required
	Min Qx	0.43	648	2884			1410			1410.3	5274	> 648	ok	1058	φVc>Vu	not required

Nominal shear strength required by Reinf. Vs= Not Required SECTION 11.2

Spacing Required of Strrups, s= Not Required SECTION 11.5.6.2

Though, Link Provided

Diameter of Shear Link Provided = 16 mm

Spacing Provided : Along X-X = 175 mm

Along Y-Y = 140 mm OK

Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation								
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided	
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	
Max My	172	top face	40	1444	171.7		315		32	297.99	150	5363.9	
Min My	-1388	bottom face	40	1444	1388	0.014	2575	2700	32	297.99	150	5363.9	
Max Mxy	-1355	bottom face	40	1444	1355		2513		32	297.99	150	5363.9	
Min Mxy	-1358	bottom face	40	1444	1358	*As	2519		32	297.99	150	5363.9	

Shear Check		Qx	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa	KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.12	173	1231	0.004	1.444	1392	1510	2470	1392.4	1466	> 173	ok	1044	φVc>Vu	not required
	Min Qy	0.12	186	1358			1392			1391.7	1465	>186	ok	1044	φVc>Vu	not required

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• **Heavy Slab Concealed Beam (CB3):** Suport Portion

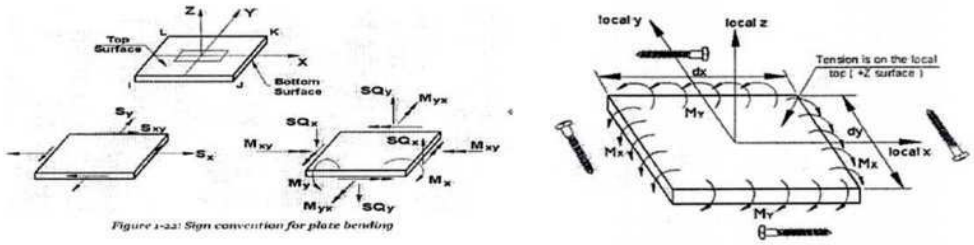


Figure 1-22: Sign convention for plate bending

Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:

PLATE CENTER STRESS STAAD GROUP : CB3_Moment At the face of the column

Load condition		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	L/C
Max Qx	12521	0.951	-0.023	0.431	0.449	-0.004	-1018	-1164	-186.2	-1205	-1350	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qx	12499	-0.9	0.024	0.405	0.387	0.008	-1141	-1279	0	-1141	-1279	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Qy	12486	0.609	0.804	0.437	0.435	0.07	-1177	-1016	325	-1502	-1341	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Qy	12338	0.198	-1.102	0.448	0.316	0.052	-1596	-402	-260	-1856	-662.1	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Sx	12353	-0.528	-0.106	0.586	0.621	0.019	36.41	-112.2	214.3	250.7	-326.5	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sx	12488	0.015	0.038	0.002	0.008	0.004	-44.22	-43.05	-41.08	-85.3	-84.13	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sy	12353	-0.528	-0.106	0.586	0.621	0.019	36.41	-112.2	214.3	250.7	-326.5	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sy	12336	0.028	-0.01	0.027	0.002	0.015	-93.07	-49.32	-6.507	-99.58	-55.83	118 1.2X[D] + 1.6X[W1B] + 1.0X[L]
Max Sxy	12486	0.312	0.419	0.537	0.548	0.092	-452.2	-418.6	37.84	-490	-456.4	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sxy	12600	0.002	-0.175	0.524	0.433	-0.091	-1015	-718.3	-15.7	-1031	-734	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Mx	12521	0.099	0.011	0.068	0.046	-0.007	125.4	23.22	-18.2	143.6	41.42	101 1.4X[D]
Min Mx	12600	-0.095	-0.44	0.409	0.33	-0.062	-2308	-1726	-10.34	-2318	-1736	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max My	12520	0.094	-0.007	0.072	0.047	-0.006	123.3	24.38	-38.39	161.7	62.77	101 1.4X[D]
Min My	12495	0.127	0.467	0.421	0.403	-0.001	-2211	-1836	-155.2	-2366	-1991	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Mxy	12597	-0.717	-0.519	0.315	0.338	0.074	-1343	-925.2	451.7	-1795	-1377	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Mxy	12335	0.649	-0.346	0.382	0.415	-0.071	-1287	-752.1	-459.8	-1747	-1212	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]

PLATE CENTER STRESS STAAD GROUP : CB3_Shear At a distance "d" from the face of the column

Load condition		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	L/C
Max Qx	12598	0.49	-0.371	0.415	0.436	-0.024	-1774	-1564	-232.5	-2006	-1796	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qx	12498	-0.448	0.373	0.405	0.393	0.008	-1852	-1641	0	-1852	-1641	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Qy	12488	0.38	0.703	0.375	0.401	0.006	-1727	-1234	70.29	-1797	-1305	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Qy	12338	0.198	-1.102	0.448	0.316	0.052	-1596	-402	-260	-1856	-662.1	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Sx	12197	0.046	-0.268	0.58	0.407	0.035	-945.6	-486.1	-58.08	-1004	-544.2	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sx	12488	0.015	0.038	0.002	0.008	0.004	-44.22	-43.05	-41.08	-85.3	-84.13	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sy	12346	-0.236	-0.136	0.568	0.611	0.038	-386.1	-425.1	186.1	-572.1	-611.2	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sy	12336	0.028	-0.01	0.027	0.002	0.015	-93.07	-49.32	-6.507	-99.58	-55.83	118 1.2X[D] + 1.6X[W1B] + 1.0X[L]
Max Sxy	12210	-0.023	-0.258	0.54	0.439	0.09	-898.2	-569.3	-8.033	-906.3	-577.3	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sxy	12600	0.002	-0.175	0.524	0.433	-0.091	-1015	-718.3	-15.7	-1031	-734	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Mx	12593	0.054	-0.006	0.068	0.047	-0.01	43.76	-28.7	-35.01	78.77	-63.7	101 1.4X[D]
Min Mx	12600	-0.095	-0.44	0.409	0.33	-0.062	-2308	-1726	-10.34	-2318	-1736	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max My	12596	-0.014	-0.007	0.036	0.007	-0.018	-67.71	-12.58	11.96	-79.67	-24.53	115 1.2X[D] + 0.8X[W1A]
Min My	12495	0.127	0.467	0.421	0.403	-0.001	-2211	-1836	-155.2	-2366	-1991	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Mxy	12570	0.357	0.366	0.429	0.417	-0.004	-1776	-1314	401.5	-2177	-1715	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Mxy	12501	-0.357	0.366	0.403	0.393	0.006	-1774	-1315	-399.3	-2173	-1714	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	40	mm
	top face	40	mm
Thicknes of slab =		1500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

Force Values			Depth		Reinforcement Calculation							
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2
Max Mx	144	top face	40	1444	143.6	0.014	263	2700	32	297.99	150	5363.9
Min Mx	45	top face	40	1444	45		82.5		32	297.99	150	5363.9
Max Mxy	-1795	bottom face	40	1444	1795		3343		32	240.64	150	5363.9
Min Mxy	-1747	bottom face	40	1444	1747	*As	3252		32	247.42	150	5363.9

Shear Check		Qx	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa	KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.49	735	2006	0.004	1.444	1415	1458	fc'2477	1415.1	5290	> 735	ok	1061	φVc>Vu	not required
	Min Qx	0.45	672	1852			1415			1414.6	5290	> 672	ok	1061	φVc>Vu	not required

Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation							
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2
Max My	63	top face	40	1440	62.77	0.014	115	2700	40.61	479.92	120	10798
Min My	-1991	bottom face	40	1440	1991		3727		40.61	347.68	120	10798
Max Mxy	-1377	bottom face	40	1440	1377		2562		40.61	479.92	120	10798
Min Mxy	-1212	bottom face	40	1440	1212	*As	2252		40.61	479.92	120	10798

Provdie (D32+D25) for Top and Bottom Face

Shear Check		Qx	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa	KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.70	1055	1305	0.008	1.44	1473	1458	2477	1458	1516	> 1055	ok	1094	φVc>Vu	not required
	Min Qy	1.10	1653	662.1			1696			1458	1516	check	check	1094	φVc<Vu	required

Nominal shear strength required by Reinf. Vs=			746.0	KN	SECTION 11.2
Spacing Required of Strrups, s=			163	mm	SECTION 11.5.6.2
		or	720	mm	SECTION 11.5.4.1
		or	1125	mm	
Hence, spacing required =			163	mm	
Diameter of Shear Link Provided		=	16	mm	
Spacing Provided :	Along X-X	=	140	mm	
	Along Y-Y	=	100	mm	OK

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• **Heavy Slab Concealed Beam (CB3):** Span Portion

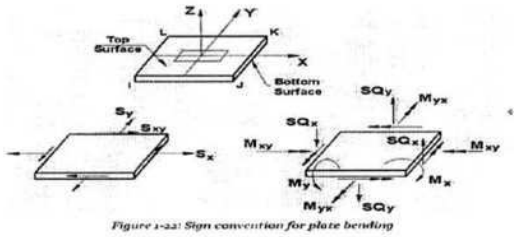
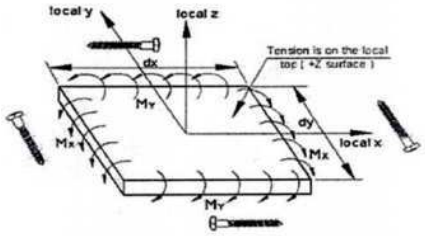


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:

PLATE CENTER STRESS STAAD GROUP : CB3_Middle At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	12598	0.49	-0.371	0.415	0.436	-0.024	-1774	-1564	-232.5	-2006	-1796	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qx	12498	-0.448	0.373	0.405	0.393	0.008	-1852	-1641	0	-1852	-1641	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Qy	12488	0.38	0.703	0.375	0.401	0.006	-1727	-1234	70.29	-1797	-1305	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Qy	12338	0.198	-1.102	0.448	0.316	0.052	-1596	-402	-260	-1856	-662.1	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Sx	12197	0.046	-0.268	0.58	0.407	0.035	-945.6	-486.1	-58.08	-1004	-544.2	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sx	12488	0.015	0.038	0.002	0.008	0.004	-44.22	-43.05	-41.08	-85.3	-84.13	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sy	12346	-0.236	-0.136	0.568	0.611	0.038	-386.1	-425.1	186.09	-572.1	-611.2	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sy	12336	0.028	-0.01	0.027	0.002	0.015	-93.07	-49.32	-6.507	-99.58	-55.83	118 1.2X[D] + 1.6X[W1B] + 1.0X[L]
Max Sxy	12210	-0.023	-0.258	0.54	0.439	0.09	-898.2	-569.3	-8.033	-906.3	-577.3	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sxy	12600	0.002	-0.175	0.524	0.433	-0.091	-1015	-718.3	-15.7	-1031	-734	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Mx	12593	0.054	-0.006	0.068	0.047	-0.01	43.76	-28.7	-35.01	78.77	-63.7	101 1.4X[D]
Min Mx	12600	-0.095	-0.44	0.409	0.33	-0.062	-2308	-1726	-10.34	-2318	-1736	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max My	12596	-0.014	-0.007	0.036	0.007	-0.018	-67.71	-12.58	11.958	-79.67	-24.53	115 1.2X[D] + 0.8X[W1A]
Min My	12495	0.127	0.467	0.421	0.403	-0.001	-2211	-1836	-155.2	-2366	-1991	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Mxy	12570	0.357	0.366	0.429	0.417	-0.004	-1776	-1314	401.45	-2177	-1715	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Mxy	12501	-0.357	0.366	0.403	0.393	0.006	-1774	-1315	-399.3	-2173	-1714	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]

PLATE CENTER STRESS STAAD GROUP : CB3_Middle At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	12598	0.49	-0.371	0.415	0.436	-0.024	-1774	-1564	-232.5	-2006	-1796	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qx	12498	-0.448	0.373	0.405	0.393	0.008	-1852	-1641	0	-1852	-1641	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Qy	12488	0.38	0.703	0.375	0.401	0.006	-1727	-1234	70.29	-1797	-1305	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Qy	12338	0.198	-0.9	0.448	0.316	0.052	-1596	-402	-260	-1856	-662.1	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Sx	12197	0.046	-0.268	0.58	0.407	0.035	-945.6	-486.1	-58.08	-1004	-544.2	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sx	12488	0.015	0.038	0.002	0.008	0.004	-44.22	-43.05	-41.08	-85.3	-84.13	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sy	12346	-0.236	-0.136	0.568	0.611	0.038	-386.1	-425.1	186.09	-572.1	-611.2	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sy	12336	0.028	-0.01	0.027	0.002	0.015	-93.07	-49.32	-6.507	-99.58	-55.83	118 1.2X[D] + 1.6X[W1B] + 1.0X[L]
Max Sxy	12210	-0.023	-0.258	0.54	0.439	0.09	-898.2	-569.3	-8.033	-906.3	-577.3	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sxy	12600	0.002	-0.175	0.524	0.433	-0.091	-1015	-718.3	-15.7	-1031	-734	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Mx	12593	0.054	-0.006	0.068	0.047	-0.01	43.76	-28.7	-35.01	78.77	-63.7	101 1.4X[D]
Min Mx	12600	-0.095	-0.44	0.409	0.33	-0.062	-2308	-1726	-10.34	-2318	-1736	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max My	12596	-0.014	-0.007	0.036	0.007	-0.018	-67.71	-12.58	11.958	-79.67	-24.53	115 1.2X[D] + 0.8X[W1A]
Min My	12495	0.127	0.467	0.421	0.403	-0.001	-2211	-1836	-155.2	-2366	-1991	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Mxy	12570	0.357	0.366	0.429	0.417	-0.004	-1776	-1314	401.45	-2177	-1715	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Mxy	12501	-0.357	0.366	0.403	0.393	0.006	-1774	-1315	-399.3	-2173	-1714	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	40	mm
	top face	40	mm
Thicknes of slab =		1500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

Force Values			Depth		Reinforcement Calculation								
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2
Max Mx	79	top face	40	1444	78.77	0.014	144	2700	32	297.99	150	5363.9
Min Mx	45	top face	40	1444	45		82.5		32	297.99	150	5363.9
Max Mxy	-2177	bottom face	40	1444	2177		4069		32	197.72	150	5363.9
Min Mxy	-2173	bottom face	40	1444	2173	*As	4062		32	198.08	150	5363.9

Shear Check		Qx	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa	KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.49	735	2006	0.004	1.444	1415.1	1458	fc'2477	1415.1	5290	> 735	ok	1061	φVc>Vu	not required
	Min Qx	0.45	672	1852			1414.6			1414.6	5290	> 672	ok	1061	φVc>Vu	not required

Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation								
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2
Max My	-25	bottom face	40	1440	24.53	0.014	45	2700	40.61	479.87	140	9254.6
Min My	-1991	bottom face	40	1440	1991		3727		40.61	347.64	140	9254.6
Max Mxy	-1715	bottom face	40	1440	1715		3202		40.61	404.68	140	9254.6
Min Mxy	-1714	bottom face	40	1440	1714	*As	3200		40.61	404.9	140	9254.6

Provdie (D32+D25) for Top and Bottom Face

Shear Check		Qx	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa	KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.70	1055	1305	0.006	1.44	1473.3	1458	2477	1458	1516	> 1055	ok	1094	φVc>Vu	not required
	Min Qy	0.90	1350	662.1			1635.3			1458	1516	>1350	ok	1094	φVc<Vu	required

Nominal shear strength required by Reinf. Vs=			342.0	KN	SECTION 11.2
Spacing Required of Strrups, s=			203	mm	SECTION 11.5.6.2
		or	720	mm	SECTION 11.5.4.1
		or	1125	mm	
Hence, spacing required =			203	mm	
Diameter of Shear Link Provided		=	16	mm	
Spacing Provided :	Along X-X	=	140	mm	
	Along Y-Y	=	175	mm	OK

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• **Heavy Slab Concealed Beam (CB4):** Suport Portion

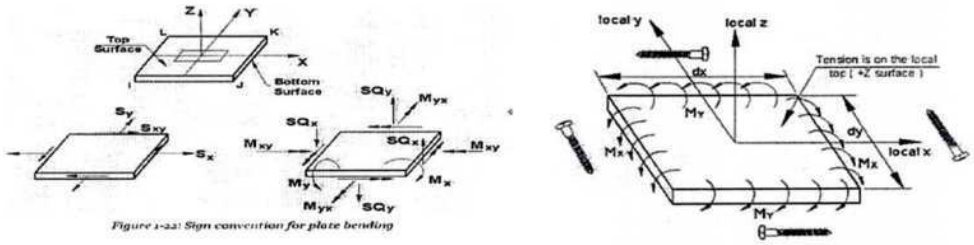


Figure 1-22 Sign convention for plate bending

Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:

PLATE CENTER STRESS STAAD GROUP : CB4_Moment At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	12399	0.898	-1.5	0.455	0.452	0.011	532.5	-1254	133.8	666.4	-1388	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Qx	12458	-0.852	1.468	0.432	0.388	0.014	382.5	-1408	0	382.5	-1408	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Qy	12403	0.429	1.946	0.423	0.416	0.001	1750	-1109	-529	2279	-1638	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qy	12422	0.42	-1.978	0.405	0.395	-0.005	1701	-1191	571.3	2272	-1762	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Sx	12275	0.564	-0.403	0.669	0.87	-0.075	-295.8	-374.4	56.25	-352.1	-430.6	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sx	12478	0.029	0.027	0.011	0.012	0.003	-19.51	-58.45	-32.19	-51.7	-90.64	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sy	12275	0.564	-0.403	0.669	0.87	-0.075	-295.8	-374.4	56.25	-352.1	-430.6	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sy	12432	0.041	-0.096	0.037	0.009	0.011	68.97	31.41	-30.87	99.84	62.28	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sxy	12477	0.264	0.44	0.571	0.772	0.136	-241	-492.1	69.96	-311	-562.1	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sxy	12563	-0.265	0.44	0.571	0.772	-0.134	-241.4	-492	-70.19	-311.6	-562.2	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Mx	12254	-0.437	-1.726	0.414	0.391	-0.004	2062	-900.8	-420	2482	-1321	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Mx	12612	-0.527	-0.276	0.507	0.458	0.014	-1478	-1836	21.18	-1499	-1857	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max My	12254	-0.04	-0.158	0.053	0.035	0.002	455.1	157.5	-45.83	501	203.4	101 1.4X[D]
Min My	12456	-0.205	-0.128	0.433	0.405	0.005	89.85	-2769	-110.8	200.6	-2880	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Mxy	12241	-0.406	1.938	0.446	0.441	-0.003	1772	-1108	590.4	2363	-1698	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Mxy	12425	-0.42	-1.977	0.405	0.395	0.006	1703	-1192	-562.9	2265	-1755	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]

PLATE CENTER STRESS STAAD GROUP : CB4_Shear At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	12466	0.533	0.083	0.462	0.485	0.01	-914	-1960	-15.15	-929.2	-1975	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Qx	12623	-0.549	-0.082	0.49	0.5	0.006	-948	-1861	0	-948	-1861	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Qy	12402	0.321	0.848	0.426	0.417	-0.002	604.2	-2324	-237.2	841.4	-2561	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qy	12423	0.303	-0.876	0.408	0.387	-0.003	532.7	-2434	279.3	812	-2713	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Sx	12276	0.269	-0.15	0.664	0.634	0.023	-338	-612.7	191.8	-529.8	-804.5	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sx	12457	-0.021	0.009	0.018	0.015	0.001	188.2	-87.52	-5.666	193.9	-93.18	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sy	12277	0.27	0.025	0.66	0.671	0	-251.7	-697.1	229.4	-481.2	-926.6	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sy	12440	0.039	-0.055	0.034	0.011	0.008	6.328	-42.62	-37.75	44.08	-80.37	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sxy	12276	0.511	-0.355	0.521	0.483	0.031	-1047	-1556	-50.83	-1097	-1607	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Sxy	12722	-0.028	0.174	0.443	0.522	-0.02	-7.244	-787.9	16.84	-24.08	-804.8	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Max Mx	12255	-0.201	-0.299	0.528	0.534	0.001	1380	-556.1	-68.53	1448	-624.6	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min Mx	12612	-0.527	-0.276	0.507	0.458	0.014	-1478	-1836	21.18	-1499	-1857	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max My	12255	-0.032	-0.049	0.05	0.033	0.001	366.9	56.31	-13.05	380	69.36	101 1.4X[D]
Min My	12456	-0.205	-0.128	0.433	0.405	0.005	89.85	-2769	-110.8	200.6	-2880	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Mxy	12238	-0.283	0.84	0.449	0.443	0.002	628.5	-2315	297	925.4	-2612	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Mxy	12628	-0.188	0.273	0.628	0.579	-0.008	-73.03	-631	-290.9	-363.9	-921.9	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	40	mm
	top face	40	mm
Thicknes of slab =		1500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

Force Values			Depth		Reinforcement Calculation								
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided	
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	
Max Mx	2482	top face	40	1440	2482		4667		40.61	277.61	150	8637.7	
Min Mx	45	top face	40	1440	45	0.014	82.7	2700	40.61	479.87	150	8637.7	Provdie (D32+D25) for Top and Bottom Face
Max Mxy	2363	top face	40	1440	2363		4438		40.61	291.94	150	8637.7	
Min Mxy	2265	top face	40	1440	2265	*As	4251		40.61	304.76	150	8637.7	

Shear Check		Qx	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa	KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.53	799.5	929.2	0.006	1.44	1545	1510	fc'2470	1509.6	5348	> 800	ok	1132	φVc>Vu	not required
	Min Qx	0.55	823.5	948			1546			1509.6	5348	> 824	ok	1132	φVc>Vu	not required

Nominal shear strength required by Reinf. Vs= Not Required KN SECTION 11.2

Spacing Required of Strrups, s= Not Required mm SECTION 11.5.6.2
or 720 mm SECTION 11.5.4.1
or 1125 mm

Hence, spacing required = 720 mm

Diameter of Shear Link Provided = 16 mm
Spacing Provided : Along X-X = 140 mm
Along Y-Y = 100 mm OK

Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation								
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided	
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	
Max My	203	top face	40	1440	203.4		374		40.61	479.87	140	9254.6	
Min My	-2880	bottom face	40	1440	2880	0.014	5437	2700	40.61	238.31	140	9254.6	Provdie (D32+D25) for Top and Bottom Face
Max Mxy	-1698	bottom face	40	1440	1698		3169		40.61	408.84	140	9254.6	
Min Mxy	-1755	bottom face	40	1440	1755	*As	3278		40.61	395.27	140	9254.6	

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.85	1	1272	2561	0.006	1.44	1468	1510	2470	1467.8	1522	> 1272	ok	1101	φVc<Vu	required
	Min Qy	0.88	1	1314	2713			1465			1465.2	1520	>1314	ok	1099	φVc<Vu	required

Nominal shear strength required by Reinf. Vs= 286.8 KN SECTION 11.2

Spacing Required of Strrups, s= 303 mm SECTION 11.5.6.2
or 720 mm SECTION 11.5.4.1
or 1125 mm

Hence, spacing required = 303 mm

Diameter of Shear Link Provided = 16 mm
Spacing Provided : Along X-X = 100 mm
Along Y-Y = 140 mm OK

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• **Heavy Slab Concealed Beam (CB4):** Span Portion

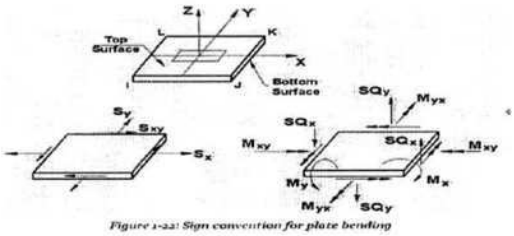
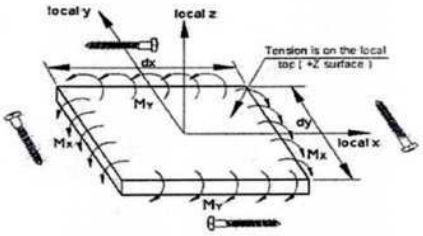


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:

PLATE CENTER STRESS STAAD GROUP : CB4_Middle At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	12466	0.533	0.083	0.462	0.485	0.01	-914	-1960	-15.15	-929.2	-1975	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Qx	12623	-0.549	-0.082	0.49	0.5	0.006	-948	-1861	0	-948	-1861	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Qy	12401	0.139	0.128	0.54	0.55	-0.009	784.8	-1098	-54.53	839.4	-1153	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min Qy	12424	0.126	-0.155	0.515	0.499	0	658.8	-1178	79.64	738.5	-1257	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Max Sx	12277	0.27	0.025	0.66	0.671	0	-251.7	-697.1	229.4	-481.2	-926.6	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sx	12456	-0.016	-0.041	0.022	0.013	0	190.3	-74.49	-16.05	206.4	-90.53	115 1.2X[D] + 0.8X[W1A]
Max Sy	12277	0.27	0.025	0.66	0.671	0	-251.7	-697.1	229.4	-481.2	-926.6	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sy	12465	0.035	-0.011	0.028	0.011	0.005	-14.76	-72.48	-41.51	-56.28	-114	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sxy	12622	-0.384	0.006	0.494	0.454	0.022	-1154	-1637	-26.08	-1180	-1663	140 0.9X[D] + 1.6X[W1B + H2B + H3B + H5B]
Min Sxy	12623	-0.03	0.019	0.086	0.059	-0.016	25.47	-22.5	-74.59	100.1	-97.09	101 1.4X[D]
Max Mx	12235	-0.123	0.126	0.54	0.551	0.01	793.5	-1093	69.04	862.5	-1162	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min Mx	12622	-0.496	-0.001	0.51	0.449	0.015	-1450	-2037	-35.33	-1485	-2072	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max My	12623	-0.025	0.017	0.075	0.054	-0.013	28.57	-19.53	-66.36	94.93	-85.89	115 1.2X[D] + 0.8X[W1A]
Min My	12456	-0.205	-0.128	0.433	0.405	0.005	89.85	-2729	-110.8	200.6	-2840	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Mxy	12277	0.271	0.025	0.633	0.642	0.002	-251.4	-697.9	230	-481.4	-928	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min Mxy	12623	-0.28	0.025	0.626	0.639	0.004	-248.3	-725.6	-241.1	-489.4	-966.7	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]

PLATE CENTER STRESS STAAD GROUP : CB4_Middle At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	12466	0.533	0.083	0.462	0.485	0.01	-914	-1960	-15.15	-929.2	-1975	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Qx	12623	-0.549	-0.082	0.49	0.5	0.006	-948	-1861	0	-948	-1861	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Qy	12401	0.139	0.128	0.54	0.55	-0.009	784.8	-1098	-54.53	839.4	-1153	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min Qy	12424	0.126	-0.155	0.515	0.499	0	658.8	-1178	79.64	738.5	-1257	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Max Sx	12277	0.27	0.025	0.66	0.671	0	-251.7	-697.1	229.4	-481.2	-926.6	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sx	12456	-0.016	-0.041	0.022	0.013	0	190.3	-74.49	-16.05	206.4	-90.53	115 1.2X[D] + 0.8X[W1A]
Max Sy	12277	0.27	0.025	0.66	0.671	0	-251.7	-697.1	229.4	-481.2	-926.6	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Sy	12465	0.035	-0.011	0.028	0.011	0.005	-14.76	-72.48	-41.51	-56.28	-114	117 1.2X[D] + 1.6X[W1A] + 1.0X[L]
Max Sxy	12622	-0.384	0.006	0.494	0.454	0.022	-1154	-1637	-26.08	-1180	-1663	140 0.9X[D] + 1.6X[W1B + H2B + H3B + H5B]
Min Sxy	12623	-0.03	0.019	0.086	0.059	-0.016	25.47	-22.5	-74.59	100.1	-97.09	101 1.4X[D]
Max Mx	12235	-0.123	0.126	0.54	0.551	0.01	793.5	-1093	69.04	862.5	-1162	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min Mx	12622	-0.496	-0.001	0.51	0.449	0.015	-1450	-2037	-35.33	-1485	-2072	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max My	12623	-0.025	0.017	0.075	0.054	-0.013	28.57	-19.53	-66.36	94.93	-85.89	115 1.2X[D] + 0.8X[W1A]
Min My	12456	-0.205	-0.128	0.433	0.405	0.005	89.85	-2729	-110.8	200.6	-2840	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Mxy	12277	0.271	0.025	0.633	0.642	0.002	-251.4	-697.9	230	-481.4	-928	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min Mxy	12623	-0.28	0.025	0.626	0.639	0.004	-248.3	-725.6	-241.1	-489.4	-966.7	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	40	mm
	top face	40	mm
Thicknes of slab =		1500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

Force Values			Depth		Reinforcement Calculation								
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided	
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	
Max Mx	863	top face	40	1444	862.5		1593		32	297.99	150	5363.9	
Min Mx	45	top face	40	1444	45	0.014	82.5	2700	32	297.99	150	5363.9	Provide (D32+D25) for Top and Bottom Face
Max Mxy	-481	bottom face	40	1444	481.4		886		32	297.99	150	5363.9	
Min Mxy	-489	bottom face	40	1444	489.4	*As	901		32	297.99	150	5363.9	

Shear Check		Qx	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$			Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa	KN	KN-m/m		m^2	KN	KN	KN	KN	KN				KN		
Load condition	Max Qx	0.53	799.5	929.2	0.004	1.444	1480	1458	fc'2477	1458	5322	> 800		ok	1094	φVc>Vu	not required
	Min Qx	0.55	823.5	948			1481			1458	5322	> 824		ok	1094	φVc>Vu	not required

Nominal shear strength required by Reinf. Vs= Not Required SECTION 11.2

Spacing Required of Strrups, s= Not Required SECTION 11.5.6.2

or 722 mm SECTION 11.5.4.1

or 1125 mm

Hence, spacing required = 722 mm

Though, Link Provided

Diameter of Shear Link Provided = 16 mm

Spacing Provided : Along X-X = 175 mm

Along Y-Y = 140 mm OK

Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation								
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided	
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	
Max My	-86	bottom face	40	1444	85.89		157		32	297.99	150	5363.9	
Min My	-2840	bottom face	40	1444	2840	0.014	5342	2700	32	150.6	150	5363.9	
Max Mxy	-928	bottom face	40	1444	928		1714		32	297.99	150	5363.9	
Min Mxy	-967	bottom face	40	1444	966.7	*As	1787		32	297.99	150	5363.9	

Shear Check		Qx	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$			Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa	KN	KN-m/m		m^2	KN	KN	KN	KN	KN				KN		
Load condition	Max Qy	0.13	192	1153	0.004	1.444	1389	1458	2477	1388.8	1464	> 192		ok	1042	φVc>Vu	not required
	Min Qy	0.16	233	1257			1391			1391.2	1466	>233		ok	1043	φVc>Vu	not required

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REFERENCES

- **Bulk Solid Load**

[1] Eurocode 1 - Actions on structures - Part4 : Silos and Tanks; EN 1991-4 : 2007

[2] Standard Practice for Design and Construction of Concrete Silos and Stacking Tubes for storing Granular Materials; ACI 313-98

- **Wind Load**

[3] Saudi Building Code Requirements : Structural - Loading and Forces; SBC 301-2007

- **Seismic Load**

[4] Saudi Building Code Requirements : Structural - Loading and Forces; SBC 301-2007

- **Structural Design**

[5] Saudi concrete structures Code; SBC 304-2018

[6] Building Code Requirements for Structural Concrete; ACI 318-14

[7] Design and Construction of Silos and Bunkers by Sargis S. Safarian & Ernest C. Harris

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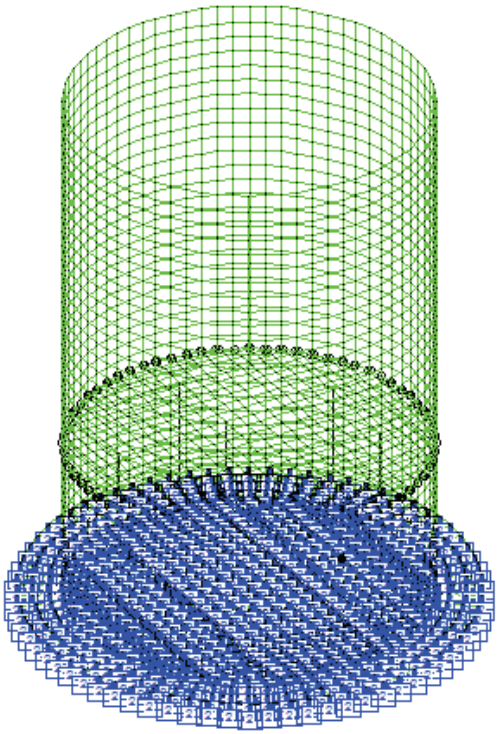
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ATTACHMENT 1 STAAD MODEL SCHEMATIC

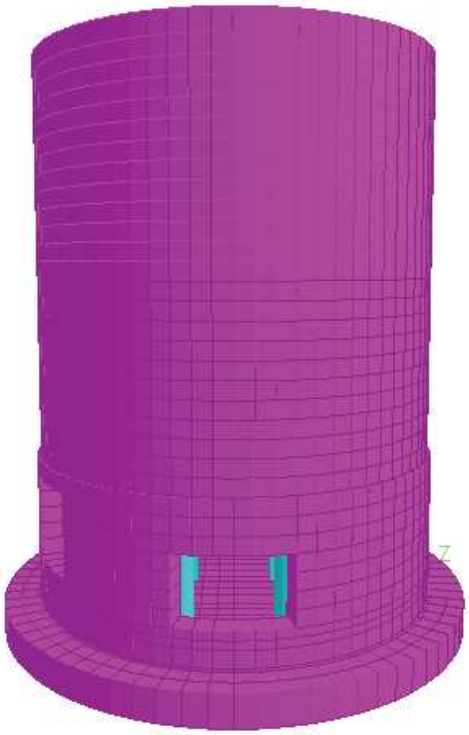
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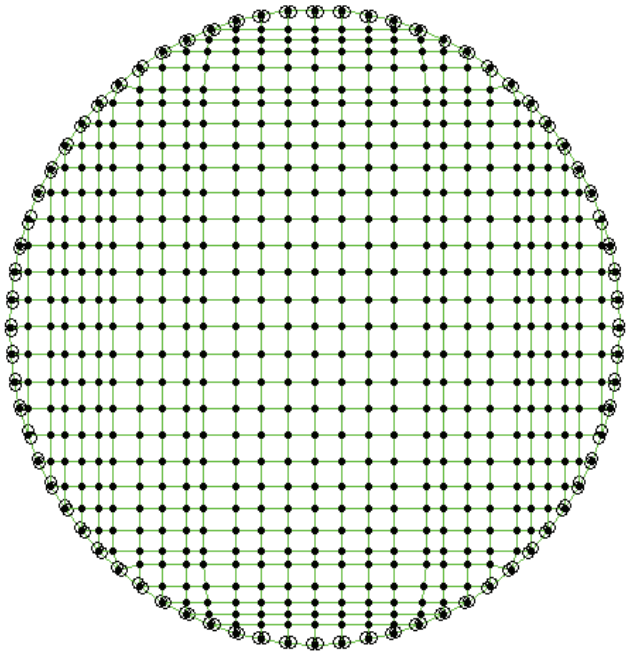
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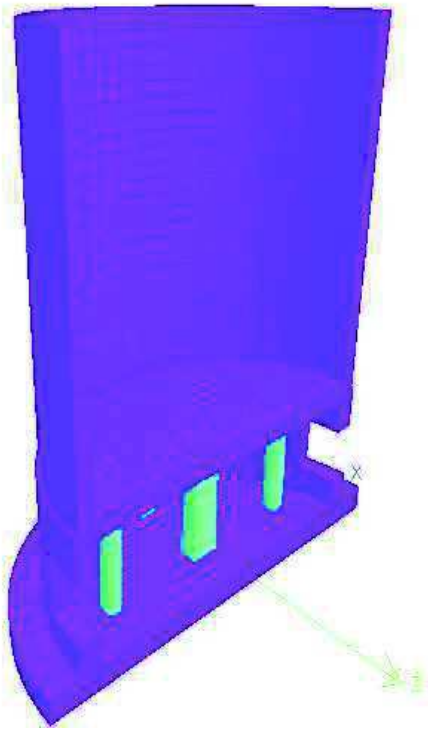
Modeling Mode View



STAAD Rendered View



Heavy Slab Plan



Cross-Sectional View

ATTACHMENT 2 STAAD OUTPUT

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*           Version  20.07.11.70                  *
*           Proprietary Program of                *
*           Bentley Systems, Inc.                 *
*           Date=    JUN 24, 2020                 *
*           Time=    16:25:55                     *
*                                                *
*      USER ID: LNT ECC DIV                      *
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    4. ENGINEER DATE 22-06-2020
    5. JOB NAME MANSOURAH MASSARAH GOLD PROJECT
    6. JOB CLIENT MA'ADEN GOLD AND BASE METALS (MA'ADEN)
    7. ENGINEER NAME SVK
    8. CHECKER NAME UR
    9. JOB REF CRUSHED ORE STORAGE BIN
   10. JOB NO O19045
   11. JOB REV R1
   12. JOB PART 450
   13. APPROVED NAME AG
   14. CHECKER DATE 22-06-2020
   15. APPROVED DATE 22-06-2020
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   19. *
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ELEMENT PROPERTY
101 TO 172 201 TO 272 301 TO 372 401 TO 472 501 TO 572 601 TO 672 701 TO 772 -
801 TO 872 901 TO 972 1001 TO 1072 1101 TO 1172 1201 TO 1272 1301 TO 1372 -
1401 TO 1472 1501 TO 1572 1601 TO 1672 1701 TO 1772 1801 TO 1872 -
1901 TO 1972 2001 TO 2072 2101 TO 2172 2201 TO 2272 2301 TO 2372 -
2401 TO 2472 THICKNESS 1
10879 TO 10981 11198 TO 11203 11205 11206 11210 TO 11222 11225 TO 11253 11255 -
11256 TO 11259 11371 TO 11399 11487 TO 11501 11503 11511 TO 12067 -
12885 TO 12956 THICKNESS 1.85
10001 TO 10288 10295 TO 10354 10367 TO 10380 10382 10384 10386 10388 TO 10404 -
10406 10408 10410 10412 TO 10426 10439 TO 10452 10454 10456 10458 -
10460 TO 10476 10478 10480 10482 10484 TO 10498 10511 TO 10524 10526 10528 -
10530 10532 TO 10548 10550 10552 10554 10556 TO 10570 10577 TO 10596 10598 -
10600 10602 10604 TO 10620 10622 10624 10626 10628 TO 10668 10670 10672 -
10674 10676 TO 10692 10694 10696 10698 10700 TO 10792 THICKNESS 1.425
12172 TO 12787 THICKNESS 1.5
*
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185E+007
POISSON 0.17
DENSITY 24
ALPHA 1E-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 35000
END DEFINE MATERIAL
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*
*
START GROUP DEFINITION
ELEMENT
*
_OPNG1 134 136 138 140 234 236 238 240 334 336 338 340 434 436 438 440
*
*
*
_THICK_SLAB 12172 TO 12787
*
_PL_NEAR_OPNG 10007 10008 10019 10029 10043 10053 10065 10066 10079 10080 -
10091 10101 10115 10125 10137 10138 10151 10152 10163 10173 10187 10197 -
10209 10210 10223 10224 10235 10245 10259 10269 10281 10282 10295 10296 -
10307 10317 10331 10341 10353 10354 10367 10368 10379 10389 10403 10413 -
10425 10426 10439 10440 10451 10461 10475 10485 10497 10498 10511 10512 -
10523 10533 10547 10557 10569 10570 10583 10584 10595 10605 10619 10629 -
10641 10642 10655 10656 10667 10677 10691 10701 10713 10714 10727 10728 -
10739 10749 10763 10773 10785 10786
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*
_PL_OTHER 10001 TO 10006 10009 TO 10018 10020 TO 10028 10030 TO 10042 10044 -
10045 TO 10052 10054 TO 10064 10067 TO 10078 10081 TO 10090 10092 TO 10100 -
10102 TO 10114 10116 TO 10124 10126 TO 10136 10139 TO 10150 10153 TO 10162 -
10164 TO 10172 10174 TO 10186 10188 TO 10196 10198 TO 10208 10211 TO 10222 -
10225 TO 10234 10236 TO 10244 10246 TO 10258 10260 TO 10268 10270 TO 10280 -
10283 TO 10288 10297 TO 10306 10308 TO 10316 10318 TO 10330 10332 TO 10340 -
10342 TO 10352 10369 TO 10378 10380 10382 10384 10386 10388 10390 TO 10402 -
10404 10406 10408 10410 10412 10414 TO 10424 10441 TO 10450 10452 10454 -
10456 10458 10460 10462 TO 10474 10476 10478 10480 10482 10484 -
10486 TO 10496 10513 TO 10522 10524 10526 10528 10530 10532 10534 TO 10546 -
10548 10550 10552 10554 10556 10558 TO 10568 10585 TO 10594 10596 10598 -
10600 10602 10604 10606 TO 10618 10620 10622 10624 10626 10628 -
10630 TO 10640 10657 TO 10666 10668 10670 10672 10674 10676 10678 TO 10690 -
10692 10694 10696 10698 10700 10702 TO 10712 10729 TO 10738 10740 10742 -
10744 10746 10748 10750 TO 10762 10764 10766 10768 10770 10772 -
10774 TO 10784
_PL_BEAM 10577 TO 10582 10643 TO 10654 10715 TO 10726 10741 10743 10745 10747 -
10765 10767 10769 10771 10787 TO 10792
_PL_BTTM_WALL 10145 TO 10150 10155 TO 10160 10162 10164 TO 10172 10174 10176 -
10177 TO 10184 10186 10188 TO 10196 10198 10200 TO 10206 10211 TO 10222 10227 -
10228 TO 10232 10234 10236 TO 10244 10246 10248 TO 10256 10258 10260 TO 10268 -
10270 10272 TO 10278 10283 TO 10288 10299 TO 10304 10306 10308 TO 10316 -
10318 10320 TO 10328 10330 10332 TO 10340 10342 10344 TO 10350 -
10371 TO 10376 10378 10380 10382 10384 10386 10388 10390 10392 TO 10400 -
10402 10404 10406 10408 10410 10412 10414 10416 TO 10422 10443 TO 10448 -
10450 10452 10454 10456 10458 10460 10462 10464 TO 10472 10474 10476 10478 -
10480 10482 10484 10486 10488 TO 10494 10515 TO 10520 10522 10524 10526 -
10528 10530 10532 10534 10536 TO 10544 10546 10548 10550 10552 10554 10556 -
10558 10560 TO 10566 10587 TO 10592 10594 10596 10598 10600 10602 10604 -
10606 10608 TO 10616 10618 10620 10622 10624 10626 10628 10630 -
10632 TO 10638 10659 TO 10664 10666 10668 10670 10672 10674 10676 10678 -
10680 TO 10688 10690 10692 10694 10696 10698 10700 10702 10704 TO 10710 -
10731 TO 10736 10738 10740 10742 10744 10746 10748 10750 10752 TO 10760 -
10762 10764 10766 10768 10770 10772 10774 10776 TO 10782
_RAFT_COL_STRIP 10883 TO 10886 10889 10890 10896 TO 10899 10902 10903 10920 -
10921 10925 10926 10947 10948 10950 10951 10955 10956 10960 10961 10977 -
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11251 TO 11253 11257 TO 11259 11392 TO 11395 11666 TO 11670 11676 TO 11679 -
11683 TO 11686 11689 11690 11693 TO 11696 11702 11705 11706 11711 TO 11715 -
11729 TO 11732 11738 11764 11779 11780 11782 11783 11785 11786 -
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11915 11916 11922 11923 11944 TO 11948 11953 TO 11957 11962 TO 11966 12005 -
12006 TO 12008 12016 TO 12022 12027 TO 12029 12031 TO 12033 12035 TO 12037
*_RAFT_CANTI 11519 TO 11662
_RAFT_BELOW_WALL 11198 11202 11203 11205 11206 11213 11222 11230 11231 11242 -
11248 11371 TO 11376 11380 11396 TO 11399 11487 TO 11501 11503 -
11511 TO 11518 11663 11664 11673 11697 11698 11717 11735 11748 11757 11766 -
11769 11771 11790 11792 11794 11796 11819 11833 TO 11844 11847 11856 11860 -
11861 11864 TO 11866 11868 11869 11877 11889 11890 11905 11942 -
12045 TO 12067
*_RAFT_OTHER 10879 TO 10881 10891 TO 10894 10904 TO 10907 10917 10918 10922 -
*10923 10927 TO 10936 10940 TO 10942 10952 10953 10957 10958 10965 TO 10972 -
*11199 TO 11201 11225 11232 11237 11243 11249 11255 11377 TO 11379 -
*11381 TO 11383 11672 11681 11691 11704 11708 11709 11723 11724 11736 11739 -
*11740 TO 11747 11749 TO 11754 11758 TO 11762 11765 11767 11768 11770 11772 -
*11773 TO 11777 11804 11805 11810 11811 11845 11846 11848 11849 11854 11855 -
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*11892 TO 11896 11898 TO 11904 11912 11913 11920 11925 TO 11937 11940 11941 -
*11950 11951 11959 11960 11968 TO 11980 11994 TO 11998 12010 TO 12014 12024 -
*12025 TO 12026 12042 TO 12044
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*_RAFT_BELOW_COL 10882 10883 10886 TO 10889 10895 TO 10903 10908 TO 10916 -
*10919 10920 10924 TO 10926 10937 TO 10939 10943 TO 10950 10954 10955 10959 -
*10960 TO 10964 10973 TO 10980 11210 TO 11212 11214 TO 11217 11220 11221 11226
*11227 11233 TO 11236 11238 TO 11241 11244 TO 11247 11250 TO 11253 11256 -
*11257 11384 TO 11394 11665 11666 11670 11671 11674 TO 11676 11679 11680 -
*11682 11683 11686 TO 11689 11692 11693 11696 11699 TO 11703 11705 TO 11707 -
*11710 TO 11716 11718 TO 11722 11725 TO 11729 11732 TO 11734 11737 11738 -
*11755 11756 11763 11764 11778 11779 11781 11782 11784 TO 11789 11791 11793 -
*11795 11797 TO 11803 11806 TO 11809 11812 TO 11818 11820 TO 11832 -
*11850 TO 11853 11857 11858 11862 11863 11872 11873 11878 11883 11897 11906 -
*11907 TO 11911 11914 TO 11919 11921 TO 11924 11938 11939 11943 11944 11948 -
*11949 11952 11953 11957 11958 11961 TO 11967 11981 TO 11993 11999 TO 12009 -
*12015 TO 12019 12022 12023 12029 12030 12033 TO 12041
_RAFT_CANT_MOM 11520 11522 11524 11526 11528 11530 11532 11534 11536 11538 -
11540 11542 11544 11546 11548 11550 11552 11554 11556 11558 11560 11562 -
11564 11566 11568 11570 11572 11574 11576 11578 11580 11582 11584 11586 -
11588 11590 11592 11594 11596 11598 11600 11602 11604 11606 11608 11610 -
11612 11614 11616 11618 11620 11622 11624 11626 11628 11630 11632 11634 -
11636 11638 11640 11642 11644 11646 11648 11650 11652 11654 11656 11658 -
11660 11662 12885 TO 12956
*_RAFT_COL_SHEAR 10882 10887 10888 10895 10900 10901 10908 TO 10916 10919 -
*10924 10937 TO 10939 10943 TO 10946 10949 10954 10959 10962 TO 10964 10973 -
*10974 TO 10976 10979 11210 TO 11212 11214 11221 11226 11233 11238 TO 11241 -
*11244 TO 11247 11250 11256 11384 TO 11391 11665 11671 11674 11675 11680 -
*11682 11687 11688 11692 11699 TO 11701 11703 11707 11710 11716 -
*11718 TO 11722 11725 TO 11728 11733 11734 11737 11755 11756 11763 11778 -
*11781 11784 11787 TO 11789 11791 11793 11795 11797 11798 11803 11806 11807 -
*11809 11812 11818 11822 TO 11828 11830 11832 11850 11853 11857 11858 11862 -
*11863 11872 11873 11878 11883 11897 11906 TO 11911 11914 11917 TO 11919 -
*11921 11924 11938 11939 11943 11949 11952 11958 11961 11967 11981 -
*11987 TO 11993 11999 12009 12015 12023 12030 12034 12038 TO 12041
*
_RAFT_CANT_SHEAR 12885 TO 12956
*
_RAFT_COL1_MOM 10881 TO 10883 10886 TO 10889 10894 TO 10903 10907 TO 10916 -
11210 TO 11212 11214 TO 11217 11220 11221 11226 11227 11233 TO 11236 11238 -
11239 TO 11241 11244 TO 11247 11250 TO 11253 11256 11257 11665 11666 11670 -
11671 11701 11710 TO 11716 11718 TO 11722 11743 TO 11745 11812 TO 11818 -
11900 TO 11902 11906 TO 11911 11943 11944 11948 11949 11952 11953 11957 -
11958 11961 TO 11967 11981 TO 11994 11999 TO 12010 12015 TO 12019 -
12022 TO 12024 12029 12030 12033 TO 12041
_RAFT_COL2_MOM 10943 TO 10950 10954 10955 10959 TO 10964 10973 TO 10980 11682 -
11683 11686 TO 11689 11692 11693 11696 11702 11703 11705 TO 11707 -
11725 TO 11727 11806 TO 11809 11820 TO 11832 11914 TO 11919
_RAFT_COL3_MOM 10919 10920 10924 TO 10926 10937 TO 10939 11384 TO 11394 11674 -
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11795 11797 TO 11803 11850 TO 11853 11857 11858 11862 11863 11872 11873 -
11878 11883 11897 11921 TO 11924 11938 11939
_RAFT_COL1_SHEAR 10881 10882 10887 10888 10894 10895 10900 10901 -
10907 TO 10916 11210 TO 11212 11214 11221 11226 11233 11238 TO 11241 11244 -
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12030 12034 12038 TO 12041
_RAFT_COL2_SHEAR 10943 TO 10946 10949 10954 10959 10962 TO 10964 -
10973 TO 10976 10979 11682 11687 11688 11692 11703 11707 11725 TO 11727 -
11806 11807 11809 11822 TO 11828 11830 11832 11914 11917 TO 11919
_RAFT_COL3_SHEAR 10919 10924 10937 TO 10939 11386 11387 11389 TO 11391 11674 -
11675 11680 11728 11733 11734 11737 11755 11756 11763 11778 11781 11784 -
11787 11788 11795 11797 11803 11850 11853 11857 11872 11873 11878 11883 -
11897 11921 11924 11938 11939
_RAFT_OTHR_1 10929 TO 10931 10934 TO 10936 10952 10953 10957 10958 -
10965 TO 10972 11672 11691 11708 11709 11723 11724 11810 11811 11912 11913 -
11926 TO 11935 11950 11951 11968 TO 11975
_RAFT_OTHR_2 10893 10905 10906 10922 10923 10927 10928 10932 10933 11741 -
11742 11751 TO 11754 11760 TO 11762 11884 TO 11886 11893 TO 11896 11898 -
11899 11936 11940 11941 11976 TO 11978 11995 11996 12011
_RAFT_OTHER_3 10940 TO 10942 11681 11704 11736 11739 11740 11804 11805 11920 -
11937
_THICK_SLAB_EFFC 12186 TO 12189 12192 TO 12195 12199 TO 12202 12205 TO 12208 -
12220 TO 12223 12225 TO 12228 12230 12231 12233 12234 12236 12237 12239 -
12240 12250 TO 12253 12256 TO 12261 12263 12264 12266 12267 12269 12270 -
12278 12283 12284 12287 TO 12289 12292 TO 12299 12303 TO 12306 -
12308 TO 12311 12314 TO 12317 12320 TO 12323 12325 TO 12328 12336 TO 12338 -
12341 12342 12344 TO 12347 12349 12351 12353 12385 TO 12387 12416 12419 -
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12467 12471 TO 12476 12488 12490 12492 TO 12494 12497 12501 12502 12504 -
12505 12508 12509 12513 12517 TO 12520 12523 TO 12533 12536 TO 12538 12544 -
12545 TO 12550 12566 12568 TO 12575 12577 12578 12583 12590 TO 12596 12613 -
12614 TO 12621 12624 12626 12627 12630 TO 12633 12636 TO 12641 12648 TO 12651 -
12654 TO 12658 12660 12663 TO 12665 12676 TO 12683 12686 TO 12691 -
12694 TO 12697 12700 TO 12703 12706 TO 12709 12712 TO 12715 12718 TO 12721 -
12724 12725 12728 TO 12731 12733 TO 12737 12748 TO 12750 12752 TO 12754 -
12756 TO 12758 12760 TO 12762
_TOP_WALL_1 10073 TO 10288 10295 TO 10354 10367 TO 10380 10382 10384 10386 -
10388 TO 10404 10406 10408 10410 10412 TO 10426 10439 TO 10452 10454 10456 -
10458 10460 TO 10476 10478 10480 10482 10484 TO 10498 10511 TO 10524 10526 -
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10598 10600 10602 10604 TO 10620 10622 10624 10626 10628 TO 10668 10670 -
10672 10674 10676 TO 10692 10694 10696 10698 10700 TO 10720
_TOP_WALL_2 201 TO 272 301 TO 372 401 TO 472
_TOP_WALL_3 601 TO 672 701 TO 772 801 TO 872 901 TO 972 1001 TO 1072
_TOP_WALL_4 1101 TO 1172 1201 TO 1272 1301 TO 1372 1401 TO 1472 1501 TO 1572
_TOP_WALL_5 1601 TO 1672 1701 TO 1772 1801 TO 1872 1901 TO 1972 2001 TO 2072
_TOP_WALL_6 2101 TO 2172 2201 TO 2272 2301 TO 2372 2401 TO 2472
_TOP_WALL_BEAM 101 TO 172 501 TO 572 10721 TO 10792
_CB2 12190 12191 12196 12203 12204 12209 12224 12229 12242 12243 -
12245 TO 12248 12262 12265 12268 12271 12390 TO 12392 12411 TO 12413 12540 -
12541 TO 12543 12551 12552 12634 12635 12646 12647 12652 12653 12668 TO 12673 -
12692 12693 12710 12711 12726 12727
_CB3 12184 12197 12210 TO 12213 12335 12336 12338 12342 12346 12353 12485 -
12486 TO 12488 12494 TO 12499 12501 12502 12520 12521 12565 TO 12571 12593 -
12594 TO 12602 12643 TO 12645 12699 12717
_CB4 12172 TO 12176 12179 TO 12182 12219 12232 12235 12238 12241 12254 12255 -
12272 12273 12275 TO 12277 12285 12290 12307 12312 12318 12324 -
12364 TO 12367 12369 12370 12376 12379 12380 12399 TO 12403 12406 TO 12409 -
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12422 TO 12425 12432 12433 12439 12440 12447 TO 12449 12456 TO 12458 12465 -
12466 12469 12470 12477 12478 12534 12535 12563 12564 12608 12609 12611 -
12612 12622 12623 12628 12629 12684 12685 12704 12705 12722 12723 12738 -
12739 12742 TO 12746 12755 12759 12763 12764 12776 12777
_HEAVY_SLAB_REST 12185 TO 12189 12192 TO 12195 12198 TO 12202 12205 TO 12208 -
12215 TO 12218 12220 TO 12223 12225 TO 12228 12230 12231 12233 12234 12236 -
12237 12239 12240 12250 TO 12253 12256 TO 12261 12263 12264 12266 12267 -
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12308 TO 12311 12313 TO 12317 12319 TO 12323 12325 TO 12328 12330 TO 12334 -
12337 12339 TO 12341 12343 TO 12345 12347 TO 12352 12355 TO 12363 12368 -
12371 TO 12375 12377 12378 12381 TO 12387 12393 TO 12396 12416 TO 12421 -
12426 TO 12431 12434 TO 12438 12441 TO 12446 12452 TO 12455 12459 TO 12464 -
12467 12468 12471 TO 12476 12479 TO 12484 12489 TO 12493 12504 TO 12519 -
12522 TO 12533 12536 TO 12539 12544 TO 12550 12553 TO 12562 12572 TO 12592 -
12603 TO 12607 12610 12613 TO 12621 12624 TO 12627 12630 TO 12633 -
12636 TO 12642 12648 TO 12651 12654 TO 12665 12676 TO 12683 12686 TO 12691 -
12694 TO 12698 12700 TO 12703 12706 TO 12709 12712 TO 12716 12718 TO 12721 -
12724 12725 12728 TO 12737 12748 TO 12750 12752 TO 12754 12756 TO 12758 -
12760 TO 12762 12765 TO 12775 12778 TO 12787
_TOP_WALL_BEAM_1 101 TO 172 10721 TO 10792
_TOP_WALL_BEAM_2 501 TO 572
_TOP_WALL_BEAM_3 2401 TO 2472
_HEAVY_SLAB_1 12192 TO 12195 12205 TO 12208 12250 TO 12253 12256 TO 12261 -
12263 12264 12266 12267 12269 12270 12416 12544 TO 12550 12648 TO 12651 -
12686 TO 12691 12706 TO 12709 12724 12725
_HEAVY_SLAB_2 12185 TO 12189 12198 TO 12202 12215 TO 12218 12220 TO 12223 -
12225 TO 12228 12230 12231 12233 12234 12236 12237 12239 12240 -
12426 TO 12431 12441 TO 12446 12459 TO 12464 12471 TO 12476 12479 TO 12484 -
12522 TO 12533 12603 TO 12607 12613 TO 12621 12630 TO 12633 12636 TO 12642 -
12654 TO 12661 12694 TO 12698 12712 TO 12716 12728 TO 12732
_HEAVY_SLAB_3 12337 12340 12341 12343 TO 12345 12347 TO 12352 12394 TO 12396 -
12419 12420 12452 TO 12454 12490 12492 12493 12504 TO 12509 12511 12513 -
12515 12517 TO 12519 12572 TO 12575 12577 TO 12579 12582 12583 12587 12590 -
12591 TO 12592
_HEAVY_SLAB_4 12278 12283 12284 12287 TO 12289 12292 TO 12294 12296 TO 12299 -
12303 TO 12306 12308 TO 12311 12314 TO 12317 12320 TO 12323 12325 TO 12328 -
12385 TO 12387 12421 12434 TO 12436 12438 12467 12536 TO 12538 12624 12626 -
12627 12663 TO 12665 12676 TO 12683 12700 TO 12703 12718 TO 12721 -
12733 TO 12737 12748 TO 12750 12752 TO 12754 12756 TO 12758 12760 TO 12762
_HEAVY_SLAB_1A 12206 12207 12257 TO 12260 12263 12266 12546 12548 12649 12650 -
12689 12690 12707 12708
_HEAVY_SLAB_2A 12199 TO 12201 12220 TO 12222 12225 TO 12227 12230 12233 12236 -
12442 TO 12445 12460 TO 12463 12472 TO 12475 12525 12527 12614 TO 12620 -
12631 TO 12633 12639 12655 TO 12658 12660 12695 TO 12697 12713 TO 12715
_HEAVY_SLAB_3A 12344 12349 12350 12395 12453 12507 TO 12509 12517 12518 12577 -
12578
_HEAVY_SLAB_4A 12288 12293 12297 12298 12304 12305 12309 12310 12315 12316 -
12321 12322 12326 12327 12386 12435 12537 12664 12677 12678 12681 12682 -
12701 12702 12719 12720 12735 12736 12749 12750 12753 12757
_HEAVY_SLAB_1B 12193 12194 12205 12208 12250 TO 12252 12256 12261 12264 12267 -
12269 12544 12545 12547 12550 12648 12651 12687 12688 12691 12706 12709 -
12724
_HEAVY_SLAB_2B 12186 TO 12188 12198 12202 12215 TO 12217 12223 12228 12231 -
12234 12237 12239 12427 TO 12430 12441 12446 12459 12464 12471 12476 12480 -
12481 TO 12483 12523 12524 12526 12529 TO 12532 12603 TO 12606 12613 12621 -
12630 12637 12638 12640 12641 12654 12659 12661 12694 12698 12712 12716 -
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12729 TO 12731
_HEAVY_SLAB_3B 12341 12345 12351 12352 12396 12452 12493 12504 TO 12506 12519 -
12572 TO 12574 12591 12592
_HEAVY_SLAB_4B 12284 12289 12292 12299 12306 12311 12317 12323 12328 12387 -
12434 12438 12536 12627 12665 12676 12683 12703 12721 12737 12748 12752 -
12756 12760
_CB2_MOM 12190 12191 12196 12203 12204 12209 12224 12229 12242 12243 12245 -
12246 TO 12248 12262 12265 12268 12271 12390 TO 12392 12411 TO 12413 12540 -
12541 TO 12543 12551 12552 12634 12635 12646 12647 12652 12653 12668 TO 12673 -
12692 12693 12710 12711 12726 12727
_CB2_SHEAR 12203 12204 12209 12224 12229 12242 12245 TO 12247 12262 12265 -
12268 12391 12392 12411 12412 12542 12543 12552 12634 12646 12647 12652 -
12653 12669 TO 12672 12692 12693 12710 12711
_CB3_MOM 12184 12197 12210 TO 12213 12335 12336 12338 12342 12346 12353 12485 -
12486 TO 12488 12494 TO 12499 12501 12502 12520 12521 12565 TO 12571 12593 -
12594 TO 12602 12643 TO 12645 12699 12717
_CB3_SHEAR 12197 12210 TO 12212 12336 12338 12342 12346 12487 12488 12494 -
12495 TO 12498 12501 12566 12568 TO 12570 12593 TO 12596 12598 TO 12601 12644 -
12645 12699 12717
_CB4_MOM 12172 TO 12176 12179 TO 12182 12219 12232 12235 12238 12241 12254 -
12255 12272 12273 12275 TO 12277 12285 12290 12307 12312 12318 12324 12399 -
12400 TO 12403 12406 TO 12409 12422 TO 12425 12432 12433 12439 12440 12447 -
12448 TO 12449 12456 TO 12458 12465 12466 12469 12470 12477 12478 12534 12535 -
12563 12608 12609 12611 12612 12622 12623 12628 12629 12684 12685 12704 -
12705 12722 12723 12738 12739 12742 TO 12746 12755 12759 12763 12764
_CB4_SHEAR 12173 TO 12175 12180 12181 12232 12235 12238 12255 12272 12276 -
12277 12285 12312 12318 12400 TO 12402 12407 12408 12423 12424 12439 12440 -
12447 12448 12456 12457 12465 12466 12469 12470 12611 12612 12622 12623 -
12628 12629 12704 12705 12722 12723 12743 TO 12745 12759 12763 12764
_CB2_MIDDLE 12224 12229 12245 12246 12262 12265 12392 12411 12646 12647 12652 -
12653 12670 12671 12692 12693
_CB3_MIDDLE 12197 12210 TO 12212 12336 12338 12342 12346 12487 12488 12494 -
12495 TO 12498 12501 12566 12568 TO 12570 12593 TO 12596 12598 TO 12601 12644 -
12645 12699 12717
_CB4_MIDDLE 12174 12235 12277 12401 12424 12456 12465 12466 12622 12623 12744 -
12763
END GROUP DEFINITION
*
MEMBER PROPERTY AMERICAN
10802 10803 12870 12871 12878 12879 PRIS YD 2 ZD 1.25
10799 10808 12869 12872 12877 12880 PRIS YD 1.5 ZD 1
MEMBER PROPERTY AMERICAN
12070 12071 12074 12075 12873 TO 12876 12881 TO 12884 PRIS YD 1.4
*
*SUPPORTS
*5001 TO 5072 11962 TO 11977 12006 12008 12009 12011 12013 TO 12351 -
*12353 TO 12365 12367 TO 12391 12393 TO 12403 12405 TO 12417 12419 TO 12443 -
*12445 TO 12455 12457 TO 12470 12472 TO 12561 12563 TO 12703 -
*12704 DIRECT Y SUBGRADE 54000
*
ELEMENT RELEASE
12281 12282 12330 TO 12332 12355 TO 12359 12362 TO 12365 12367 TO 12374 12393 -
12417 12418 12539 12553 TO 12557 12559 12586 12588 12589 12597 12609 12610 -
12625 12662 12765 TO 12787 J1 MX MY MZ
12275 12279 TO 12282 12286 12295 12301 12302 12313 12319 12330 TO 12332 12334 -
12335 12339 12355 TO 12366 12377 TO 12382 12384 12393 12417 12418 -
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12767 J2 MX MY MZ
12286 12301 12302 12313 12319 12360 12369 TO 12373 12375 TO 12384 12437 12455 -
12468 12477 12486 12489 12491 12510 12512 12514 12516 12584 12585 12765 -
12766 12768 TO 12787 J3 MX MY MZ
12374 12455 12512 12514 12516 12539 12553 TO 12564 12567 12576 12580 12581 -
12584 12585 J4 MX MY MZ
*
CONSTANTS
MATERIAL CONCRETE ALL
*
SUPPORTS
10879 TO 10981 11198 TO 11203 11205 11206 11210 TO 11222 11225 TO 11253 11255 -
11256 TO 11259 11371 TO 11399 11487 TO 11501 11503 11511 TO 12067 -
12885 TO 12956 PLATE MAT DIRECT Y SUBGRADE 54000
*
*
DEFINE IBC 2003
SDS 0.067 SD1 0.013 S1 0.02 IE 1 RX 2.5 RZ 2.5 SCLASS 2 PX 0.77 PZ 0.77
SELFWEIGHT 1
JOINT WEIGHT
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* FOR STORE MATERIAL
2001 TO 2072 WEIGHT 17.2
1901 TO 1972 WEIGHT 72.6
1801 TO 1872 WEIGHT 110.8
1701 TO 1772 WEIGHT 110.8
1601 TO 1672 WEIGHT 110.8
1501 TO 1572 WEIGHT 110.8
1401 TO 1472 WEIGHT 110.8
1301 TO 1372 WEIGHT 110.8
1201 TO 1272 WEIGHT 110.8
1101 TO 1172 WEIGHT 110.8
1001 TO 1072 WEIGHT 110.8
901 TO 972 WEIGHT 110.8
801 TO 872 WEIGHT 110.8
701 TO 772 WEIGHT 110.8
601 TO 672 WEIGHT 110.8
501 TO 572 WEIGHT 110.8
401 TO 472 WEIGHT 110.8
301 TO 372 WEIGHT 110.8
201 TO 272 WEIGHT 110.8
101 TO 172 WEIGHT 55.4
*
* FOR DEAD LOAD
*
******** STRUCTURAL ROOF
2501 TO 2572 WEIGHT 11.5
*
******** STRUCTURAL FLOOR
5436 TO 5466 5472 TO 5479 5483 TO 5491 5495 TO 5502 WEIGHT 14.4
*
* FOR LIVE LOAD (CONSIDERING 50% OF LIVE LOAD AS SEISMIC WEIGHT)
*
******** STRUCTURAL ROOF
2501 TO 2572 WEIGHT 5.75
*
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```
*
******** STRUCTURAL FLOOR
5436 TO 5466 5472 TO 5479 5483 TO 5491 5495 TO 5502 WEIGHT 11.55
*
******** DISCHARGE CHUTE
12798 12808 12813 12823 12965 13008 13010 13044 13087 13122 13151 13167 13169 -
13218 13230 13243 WEIGHT 30
*
*
LOAD 5 SEISMIC LOAD IN X-DIRECTION (E1)
IBC LOAD X 1
*
LOAD 6 SEISMIC LOAD IN Z-DIRECTION (E2)
IBC LOAD Z 1
*
*
**********************
LOAD 1 DEAD LOAD (D)
**********************
*
SELFWEIGHT Y -1
*
ELEMENT LOAD
* SOIL FILLING
10879 TO 10981 11198 TO 11203 11205 11206 11210 TO 11222 11225 TO 11253 11255 -
11256 TO 11259 11371 TO 11399 11487 TO 11501 11503 11511 TO 12067 PR GY -20
**
** STACKED MATERIAL (VERTICAL)
*11535 TO 11578 PR GY -96
*
* LOAD FOR STRUCTURAL ROOF
JOINT LOAD
2501 TO 2572 FY -11.5
*
* LOAD FOR STRUCTURAL FLOOR
JOINT LOAD
5436 TO 5466 5472 TO 5479 5483 TO 5491 5495 TO 5502 FY -14.4
*
* STACKED MATERIAL (HORIZONTAL)
ELEMENT LOAD
10234 TRAP JT -57.6 -57.6 -50.4 -50.4
10236 TRAP JT -57.6 -57.6 -50.4 -50.4
10238 TRAP JT -57.6 -57.6 -50.4 -50.4
10240 TRAP JT -57.6 -57.6 -50.4 -50.4
10242 TRAP JT -57.6 -57.6 -50.4 -50.4
10244 TRAP JT -57.6 -57.6 -50.4 -50.4
10246 TRAP JT -57.6 -57.6 -50.4 -50.4
10248 TRAP JT -57.6 -57.6 -50.4 -50.4
10250 TRAP JT -57.6 -57.6 -50.4 -50.4
10252 TRAP JT -57.6 -57.6 -50.4 -50.4
10254 TRAP JT -57.6 -57.6 -50.4 -50.4
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*
3449. 10256 TRAP JT -57.6 -57.6 -50.4 -50.4
3450. 10258 TRAP JT -57.6 -57.6 -50.4 -50.4
3451. 10260 TRAP JT -57.6 -57.6 -50.4 -50.4
3452. 10262 TRAP JT -57.6 -57.6 -50.4 -50.4
3453. 10264 TRAP JT -57.6 -57.6 -50.4 -50.4
3454. 10266 TRAP JT -57.6 -57.6 -50.4 -50.4
3455. 10268 TRAP JT -57.6 -57.6 -50.4 -50.4
3456. 10270 TRAP JT -57.6 -57.6 -50.4 -50.4
3457. 10272 TRAP JT -57.6 -57.6 -50.4 -50.4
3458. 10274 TRAP JT -57.6 -57.6 -50.4 -50.4
3459. 10304 TRAP JT -50.4 -50.4 -43.2 -43.2
3460. 10306 TRAP JT -50.4 -50.4 -43.2 -43.2
3461. 10308 TRAP JT -50.4 -50.4 -43.2 -43.2
3462. 10310 TRAP JT -50.4 -50.4 -43.2 -43.2
3463. 10312 TRAP JT -50.4 -50.4 -43.2 -43.2
3464. 10314 TRAP JT -50.4 -50.4 -43.2 -43.2
3465. 10316 TRAP JT -50.4 -50.4 -43.2 -43.2
3466. 10318 TRAP JT -50.4 -50.4 -43.2 -43.2
3467. 10320 TRAP JT -50.4 -50.4 -43.2 -43.2
3468. 10322 TRAP JT -50.4 -50.4 -43.2 -43.2
3469. 10324 TRAP JT -50.4 -50.4 -43.2 -43.2
3470. 10326 TRAP JT -50.4 -50.4 -43.2 -43.2
3471. 10328 TRAP JT -50.4 -50.4 -43.2 -43.2
3472. 10330 TRAP JT -50.4 -50.4 -43.2 -43.2
3473. 10332 TRAP JT -50.4 -50.4 -43.2 -43.2
3474. 10334 TRAP JT -50.4 -50.4 -43.2 -43.2
3475. 10336 TRAP JT -50.4 -50.4 -43.2 -43.2
3476. 10338 TRAP JT -50.4 -50.4 -43.2 -43.2
3477. 10340 TRAP JT -50.4 -50.4 -43.2 -43.2
3478. 10342 TRAP JT -50.4 -50.4 -43.2 -43.2
3479. 10344 TRAP JT -50.4 -50.4 -43.2 -43.2
3480. 10346 TRAP JT -50.4 -50.4 -43.2 -43.2
3481. 10376 TRAP JT -43.2 -43.2 -36 -36
3482. 10378 TRAP JT -43.2 -43.2 -36 -36
3483. 10380 TRAP JT -43.2 -43.2 -36 -36
3484. 10382 TRAP JT -43.2 -43.2 -36 -36
3485. 10384 TRAP JT -43.2 -43.2 -36 -36
3486. 10386 TRAP JT -43.2 -43.2 -36 -36
3487. 10388 TRAP JT -43.2 -43.2 -36 -36
3488. 10390 TRAP JT -43.2 -43.2 -36 -36
3489. 10392 TRAP JT -43.2 -43.2 -36 -36
3490. 10394 TRAP JT -43.2 -43.2 -36 -36
3491. 10396 TRAP JT -43.2 -43.2 -36 -36
3492. 10398 TRAP JT -43.2 -43.2 -36 -36
3493. 10400 TRAP JT -43.2 -43.2 -36 -36
3494. 10402 TRAP JT -43.2 -43.2 -36 -36
3495. 10404 TRAP JT -43.2 -43.2 -36 -36
3496. 10406 TRAP JT -43.2 -43.2 -36 -36
3497. 10408 TRAP JT -43.2 -43.2 -36 -36
3498. 10410 TRAP JT -43.2 -43.2 -36 -36
3499. 10412 TRAP JT -43.2 -43.2 -36 -36
3500. 10414 TRAP JT -43.2 -43.2 -36 -36
3501. 10416 TRAP JT -43.2 -43.2 -36 -36
3502. 10418 TRAP JT -43.2 -43.2 -36 -36
3503. 10448 TRAP JT -36 -36 -28.8 -28.8
3504. 10450 TRAP JT -36 -36 -28.8 -28.8
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*
3505. 10452 TRAP JT -36 -36 -28.8 -28.8
3506. 10454 TRAP JT -36 -36 -28.8 -28.8
3507. 10456 TRAP JT -36 -36 -28.8 -28.8
3508. 10458 TRAP JT -36 -36 -28.8 -28.8
3509. 10460 TRAP JT -36 -36 -28.8 -28.8
3510. 10462 TRAP JT -36 -36 -28.8 -28.8
3511. 10464 TRAP JT -36 -36 -28.8 -28.8
3512. 10466 TRAP JT -36 -36 -28.8 -28.8
3513. 10468 TRAP JT -36 -36 -28.8 -28.8
3514. 10470 TRAP JT -36 -36 -28.8 -28.8
3515. 10472 TRAP JT -36 -36 -28.8 -28.8
3516. 10474 TRAP JT -36 -36 -28.8 -28.8
3517. 10476 TRAP JT -36 -36 -28.8 -28.8
3518. 10478 TRAP JT -36 -36 -28.8 -28.8
3519. 10480 TRAP JT -36 -36 -28.8 -28.8
3520. 10482 TRAP JT -36 -36 -28.8 -28.8
3521. 10484 TRAP JT -36 -36 -28.8 -28.8
3522. 10486 TRAP JT -36 -36 -28.8 -28.8
3523. 10488 TRAP JT -36 -36 -28.8 -28.8
3524. 10490 TRAP JT -36 -36 -28.8 -28.8
3525. 10520 TRAP JT -28.8 -28.8 -21.6 -21.6
3526. 10522 TRAP JT -28.8 -28.8 -21.6 -21.6
3527. 10524 TRAP JT -28.8 -28.8 -21.6 -21.6
3528. 10526 TRAP JT -28.8 -28.8 -21.6 -21.6
3529. 10528 TRAP JT -28.8 -28.8 -21.6 -21.6
3530. 10530 TRAP JT -28.8 -28.8 -21.6 -21.6
3531. 10532 TRAP JT -28.8 -28.8 -21.6 -21.6
3532. 10534 TRAP JT -28.8 -28.8 -21.6 -21.6
3533. 10536 TRAP JT -28.8 -28.8 -21.6 -21.6
3534. 10538 TRAP JT -28.8 -28.8 -21.6 -21.6
3535. 10540 TRAP JT -28.8 -28.8 -21.6 -21.6
3536. 10542 TRAP JT -28.8 -28.8 -21.6 -21.6
3537. 10544 TRAP JT -28.8 -28.8 -21.6 -21.6
3538. 10546 TRAP JT -28.8 -28.8 -21.6 -21.6
3539. 10548 TRAP JT -28.8 -28.8 -21.6 -21.6
3540. 10550 TRAP JT -28.8 -28.8 -21.6 -21.6
3541. 10552 TRAP JT -28.8 -28.8 -21.6 -21.6
3542. 10554 TRAP JT -28.8 -28.8 -21.6 -21.6
3543. 10556 TRAP JT -28.8 -28.8 -21.6 -21.6
3544. 10558 TRAP JT -28.8 -28.8 -21.6 -21.6
3545. 10560 TRAP JT -28.8 -28.8 -21.6 -21.6
3546. 10562 TRAP JT -28.8 -28.8 -21.6 -21.6
3547. 10592 TRAP JT -21.6 -21.6 -14.4 -14.4
3548. 10594 TRAP JT -21.6 -21.6 -14.4 -14.4
3549. 10596 TRAP JT -21.6 -21.6 -14.4 -14.4
3550. 10598 TRAP JT -21.6 -21.6 -14.4 -14.4
3551. 10600 TRAP JT -21.6 -21.6 -14.4 -14.4
3552. 10602 TRAP JT -21.6 -21.6 -14.4 -14.4
3553. 10604 TRAP JT -21.6 -21.6 -14.4 -14.4
3554. 10606 TRAP JT -21.6 -21.6 -14.4 -14.4
3555. 10608 TRAP JT -21.6 -21.6 -14.4 -14.4
3556. 10610 TRAP JT -21.6 -21.6 -14.4 -14.4
3557. 10612 TRAP JT -21.6 -21.6 -14.4 -14.4
3558. 10614 TRAP JT -21.6 -21.6 -14.4 -14.4
3559. 10616 TRAP JT -21.6 -21.6 -14.4 -14.4
3560. 10618 TRAP JT -21.6 -21.6 -14.4 -14.4
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3561. 10620 TRAP JT -21.6 -21.6 -14.4 -14.4
3562. 10622 TRAP JT -21.6 -21.6 -14.4 -14.4
3563. 10624 TRAP JT -21.6 -21.6 -14.4 -14.4
3564. 10626 TRAP JT -21.6 -21.6 -14.4 -14.4
3565. 10628 TRAP JT -21.6 -21.6 -14.4 -14.4
3566. 10630 TRAP JT -21.6 -21.6 -14.4 -14.4
3567. 10632 TRAP JT -21.6 -21.6 -14.4 -14.4
3568. 10634 TRAP JT -21.6 -21.6 -14.4 -14.4
3569. 10664 TRAP JT -14.4 -14.4 -7.2 -7.2
3570. 10666 TRAP JT -14.4 -14.4 -7.2 -7.2
3571. 10668 TRAP JT -14.4 -14.4 -7.2 -7.2
3572. 10670 TRAP JT -14.4 -14.4 -7.2 -7.2
3573. 10672 TRAP JT -14.4 -14.4 -7.2 -7.2
3574. 10674 TRAP JT -14.4 -14.4 -7.2 -7.2
3575. 10676 TRAP JT -14.4 -14.4 -7.2 -7.2
3576. 10678 TRAP JT -14.4 -14.4 -7.2 -7.2
3577. 10680 TRAP JT -14.4 -14.4 -7.2 -7.2
3578. 10682 TRAP JT -14.4 -14.4 -7.2 -7.2
3579. 10684 TRAP JT -14.4 -14.4 -7.2 -7.2
3580. 10686 TRAP JT -14.4 -14.4 -7.2 -7.2
3581. 10688 TRAP JT -14.4 -14.4 -7.2 -7.2
3582. 10690 TRAP JT -14.4 -14.4 -7.2 -7.2
3583. 10692 TRAP JT -14.4 -14.4 -7.2 -7.2
3584. 10694 TRAP JT -14.4 -14.4 -7.2 -7.2
3585. 10696 TRAP JT -14.4 -14.4 -7.2 -7.2
3586. 10698 TRAP JT -14.4 -14.4 -7.2 -7.2
3587. 10700 TRAP JT -14.4 -14.4 -7.2 -7.2
3588. 10702 TRAP JT -14.4 -14.4 -7.2 -7.2
3589. 10704 TRAP JT -14.4 -14.4 -7.2 -7.2
3590. 10706 TRAP JT -14.4 -14.4 -7.2 -7.2
3591. 10736 TRAP JT -7.2 -7.2 0 0
3592. 10738 TRAP JT -7.2 -7.2 0 0
3593. 10740 TRAP JT -7.2 -7.2 0 0
3594. 10742 TRAP JT -7.2 -7.2 0 0
3595. 10744 TRAP JT -7.2 -7.2 0 0
3596. 10746 TRAP JT -7.2 -7.2 0 0
3597. 10748 TRAP JT -7.2 -7.2 0 0
3598. 10750 TRAP JT -7.2 -7.2 0 0
3599. 10752 TRAP JT -7.2 -7.2 0 0
3600. 10754 TRAP JT -7.2 -7.2 0 0
3601. 10756 TRAP JT -7.2 -7.2 0 0
3602. 10758 TRAP JT -7.2 -7.2 0 0
3603. 10760 TRAP JT -7.2 -7.2 0 0
3604. 10762 TRAP JT -7.2 -7.2 0 0
3605. 10764 TRAP JT -7.2 -7.2 0 0
3606. 10766 TRAP JT -7.2 -7.2 0 0
3607. 10768 TRAP JT -7.2 -7.2 0 0
3608. 10770 TRAP JT -7.2 -7.2 0 0
3609. 10772 TRAP JT -7.2 -7.2 0 0
3610. 10774 TRAP JT -7.2 -7.2 0 0
3611. 10776 TRAP JT -7.2 -7.2 0 0
3612. 10778 TRAP JT -7.2 -7.2 0 0
3613. 10232 TRAP JT -57.6 -57.6 -50.4 -50.4
3614. *
3615. *
3616. **********************
```

```
*
LOAD 2 LIVE LOAD (L)
**********************
*
* LOAD FOR STRUCTURAL ROOF
JOINT LOAD
2501 TO 2572 FY -11.5
*
* LOAD FOR STRUCTURAL FLOOR
JOINT LOAD
5436 TO 5466 5472 TO 5479 5483 TO 5491 5495 TO 5502 FY -23.1
*
* LOAD FOR MECHANICAL CHUTE
JOINT LOAD
12798 12808 12813 12823 12965 13008 13010 13044 13087 13122 13151 13167 13169 -
13218 13230 13243 FY -60
*
*
*****************************
LOAD 3 WIND LOAD ALONG X (W1)
*****************************
*
ELEMENT LOAD
* WINDWARD SIDE
10217 TO 10252 10295 TO 10324 10367 TO 10380 10382 10384 10386 10388 TO 10396 -
10439 TO 10452 10454 10456 10458 10460 TO 10468 10511 TO 10524 10526 10528 -
10530 10532 TO 10540 PR GX 0.56
10577 TO 10596 10598 10600 10602 10604 TO 10612 PR GX 0.619
10649 TO 10668 10670 10672 10674 10676 TO 10684 10721 TO 10756 PR GX 0.652
101 TO 136 201 TO 236 PR GX 0.678
301 TO 336 401 TO 436 PR GX 0.697
501 TO 536 601 TO 636 PR GX 0.717
701 TO 736 801 TO 836 PR GX 0.736
901 TO 936 1001 TO 1036 PR GX 0.756
1101 TO 1136 1201 TO 1236 PR GX 0.769
1301 TO 1336 1401 TO 1436 PR GX 0.788
1501 TO 1536 1601 TO 1636 1701 TO 1736 PR GX 0.821
1801 TO 1836 1901 TO 1936 2001 TO 2036 2101 TO 2136 PR GX 0.847
2201 TO 2236 2301 TO 2336 2401 TO 2436 PR GX 0.873
*
* LEEWARD SIDE
10253 TO 10288 10325 TO 10354 10397 TO 10404 10406 10408 10410 10412 TO 10426 -
10469 TO 10476 10478 10480 10482 10484 TO 10498 10541 TO 10548 10550 10552 -
10554 10556 TO 10570 PR GX 0.56
10613 TO 10620 10622 10624 10626 10628 TO 10648 PR GX 0.619
10685 TO 10692 10694 10696 10698 10700 TO 10720 10757 TO 10792 PR GX 0.652
137 TO 172 237 TO 272 PR GX 0.678
337 TO 372 437 TO 472 PR GX 0.697
537 TO 572 637 TO 672 PR GX 0.717
737 TO 772 837 TO 872 PR GX 0.736
937 TO 972 1037 TO 1072 PR GX 0.756
1137 TO 1172 1237 TO 1272 PR GX 0.769
1337 TO 1372 1437 TO 1472 PR GX 0.788
1537 TO 1572 1637 TO 1672 1737 TO 1772 PR GX 0.821
1837 TO 1872 1937 TO 1972 2037 TO 2072 2137 TO 2172 PR GX 0.847
2237 TO 2272 2337 TO 2372 2437 TO 2472 PR GX 0.873
*
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*
3673. *
3674. ******************************
3675. LOAD 4 WIND LOAD ALONG Z (W1B)
3676. ******************************
3677. *
3678. ELEMENT LOAD
3679. * WINDWARD SIDE
3680. 10217 10219 10221 10223 10225 10227 10229 10231 10233 10235 10237 10239 10241 -
3681. 10243 10245 10247 10249 10251 10253 10255 10257 10259 10261 10263 10265 -
3682. 10267 10269 10271 10273 10275 10277 10279 10281 10283 10285 10287 10295 -
3683. 10297 10299 10301 10303 10305 10307 10309 10311 10313 10315 10317 10319 -
3684. 10321 10323 10325 10327 10329 10331 10333 10335 10337 10339 10341 10343 -
3685. 10345 10347 10349 10351 10353 10367 10369 10371 10373 10375 10377 10379 -
3686. 10389 10391 10393 10395 10397 10399 10401 10403 10413 10415 10417 10419 -
3687. 10421 10423 10425 10439 10441 10443 10445 10447 10449 10451 10461 10463 -
3688. 10465 10467 10469 10471 10473 10475 10485 10487 10489 10491 10493 10495 -
3689. 10497 10511 10513 10515 10517 10519 10521 10523 10533 10535 10537 10539 -
3690. 10541 10543 10545 10547 10557 10559 10561 10563 10565 10567 -
3691. 10569 PR GZ 0.56
3692. 10577 10579 10581 10583 10585 10587 10589 10591 10593 10595 10605 10607 10609 -
3693. 10611 10613 10615 10617 10619 10629 10631 10633 10635 10637 10639 10641 -
3694. 10643 10645 10647 PR GZ 0.619
3695. 10649 10651 10653 10655 10657 10659 10661 10663 10665 10667 10677 10679 10681 -
3696. 10683 10685 10687 10689 10691 10701 10703 10705 10707 10709 10711 10713 -
3697. 10715 10717 10719 PR GZ 0.652
3698. 101 103 105 107 109 111 113 115 117 119 121 123 125 127 129 131 133 135 137 -
3699. 139 141 143 145 147 149 151 153 155 157 159 161 163 165 167 169 171 10721 -
3700. 10723 10725 10727 10729 10731 10733 10735 10737 10739 10741 10743 10745 -
3701. 10747 10749 10751 10753 10755 10757 10759 10761 10763 10765 10767 10769 -
3702. 10771 10773 10775 10777 10779 10781 10783 10785 10787 10789 -
3703. 10791 PR GZ 0.678
3704. 201 203 205 207 209 211 213 215 217 219 221 223 225 227 229 231 233 235 237 -
3705. 239 241 243 245 247 249 251 253 255 257 259 261 263 265 267 269 271 301 303 -
3706. 305 307 309 311 313 315 317 319 321 323 325 327 329 331 333 335 337 339 341 -
3707. 343 345 347 349 351 353 355 357 359 361 363 365 367 369 371 PR GZ 0.697
3708. 401 403 405 407 409 411 413 415 417 419 421 423 425 427 429 431 433 435 437 -
3709. 439 441 443 445 447 449 451 453 455 457 459 461 463 465 467 469 471 501 503 -
3710. 505 507 509 511 513 515 517 519 521 523 525 527 529 531 533 535 537 539 541 -
3711. 543 545 547 549 551 553 555 557 559 561 563 565 567 569 571 PR GZ 0.717
3712. 601 603 605 607 609 611 613 615 617 619 621 623 625 627 629 631 633 635 637 -
3713. 639 641 643 645 647 649 651 653 655 657 659 661 663 665 667 669 671 701 703 -
3714. 705 707 709 711 713 715 717 719 721 723 725 727 729 731 733 735 737 739 741 -
3715. 743 745 747 749 751 753 755 757 759 761 763 765 767 769 771 PR GZ 0.736
3716. 801 803 805 807 809 811 813 815 817 819 821 823 825 827 829 831 833 835 837 -
3717. 839 841 843 845 847 849 851 853 855 857 859 861 863 865 867 869 871 901 903 -
3718. 905 907 909 911 913 915 917 919 921 923 925 927 929 931 933 935 937 939 941 -
3719. 943 945 947 949 951 953 955 957 959 961 963 965 967 969 971 PR GZ 0.756
3720. 1001 1003 1005 1007 1009 1011 1013 1015 1017 1019 1021 1023 1025 1027 1029 -
3721. 1031 1033 1035 1037 1039 1041 1043 1045 1047 1049 1051 1053 1055 1057 1059 -
3722. 1061 1063 1065 1067 1069 1071 1101 1103 1105 1107 1109 1111 1113 1115 1117 -
3723. 1119 1121 1123 1125 1127 1129 1131 1133 1135 1137 1139 1141 1143 1145 1147 -
3724. 1149 1151 1153 1155 1157 1159 1161 1163 1165 1167 1169 1171 PR GZ 0.769
3725. 1201 1203 1205 1207 1209 1211 1213 1215 1217 1219 1221 1223 1225 1227 1229 -
3726. 1231 1233 1235 1237 1239 1241 1243 1245 1247 1249 1251 1253 1255 1257 1259 -
3727. 1261 1263 1265 1267 1269 1271 1301 1303 1305 1307 1309 1311 1313 1315 1317 -
3728. 1319 1321 1323 1325 1327 1329 1331 1333 1335 1337 1339 1341 1343 1345 1347 -
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3729. 1349 1351 1353 1355 1357 1359 1361 1363 1365 1367 1369 1371 PR GZ 0.788
3730. 1401 1403 1405 1407 1409 1411 1413 1415 1417 1419 1421 1423 1425 1427 1429 -
3731. 1431 1433 1435 1437 1439 1441 1443 1445 1447 1449 1451 1453 1455 1457 1459 -
3732. 1461 1463 1465 1467 1469 1471 1501 1503 1505 1507 1509 1511 1513 1515 1517 -
3733. 1519 1521 1523 1525 1527 1529 1531 1533 1535 1537 1539 1541 1543 1545 1547 -
3734. 1549 1551 1553 1555 1557 1559 1561 1563 1565 1567 1569 1571 1601 1603 1605 -
3735. 1607 1609 1611 1613 1615 1617 1619 1621 1623 1625 1627 1629 1631 1633 1635 -
3736. 1637 1639 1641 1643 1645 1647 1649 1651 1653 1655 1657 1659 1661 1663 1665 -
3737. 1667 1669 1671 PR GZ 0.821
3738. 1701 1703 1705 1707 1709 1711 1713 1715 1717 1719 1721 1723 1725 1727 1729 -
3739. 1731 1733 1735 1737 1739 1741 1743 1745 1747 1749 1751 1753 1755 1757 1759 -
3740. 1761 1763 1765 1767 1769 1771 1801 1803 1805 1807 1809 1811 1813 1815 1817 -
3741. 1819 1821 1823 1825 1827 1829 1831 1833 1835 1837 1839 1841 1843 1845 1847 -
3742. 1849 1851 1853 1855 1857 1859 1861 1863 1865 1867 1869 1871 1901 1903 1905 -
3743. 1907 1909 1911 1913 1915 1917 1919 1921 1923 1925 1927 1929 1931 1933 1935 -
3744. 1937 1939 1941 1943 1945 1947 1949 1951 1953 1955 1957 1959 1961 1963 1965 -
3745. 1967 1969 1971 2001 2003 2005 2007 2009 2011 2013 2015 2017 2019 2021 2023 -
3746. 2025 2027 2029 2031 2033 2035 2037 2039 2041 2043 2045 2047 2049 2051 2053 -
3747. 2055 2057 2059 2061 2063 2065 2067 2069 2071 2101 2103 2105 2107 2109 2111 -
3748. 2113 2115 2117 2119 2121 2123 2125 2127 2129 2131 2133 2135 2137 2139 2141 -
3749. 2143 2145 2147 2149 2151 2153 2155 2157 2159 2161 2163 2165 2167 2169 -
3750. 2171 PR GZ 0.847
3751. 2201 2203 2205 2207 2209 2211 2213 2215 2217 2219 2221 2223 2225 2227 2229 -
3752. 2231 2233 2235 2237 2239 2241 2243 2245 2247 2249 2251 2253 2255 2257 2259 -
3753. 2261 2263 2265 2267 2269 2271 2301 2303 2305 2307 2309 2311 2313 2315 2317 -
3754. 2319 2321 2323 2325 2327 2329 2331 2333 2335 2337 2339 2341 2343 2345 2347 -
3755. 2349 2351 2353 2355 2357 2359 2361 2363 2365 2367 2369 2371 2401 2403 2405 -
3756. 2407 2409 2411 2413 2415 2417 2419 2421 2423 2425 2427 2429 2431 2433 2435 -
3757. 2437 2439 2441 2443 2445 2447 2449 2451 2453 2455 2457 2459 2461 2463 2465 -
3758. 2467 2469 2471 PR GZ 0.873
3759. *
3760. * LEEWARD SIDE
3761. 10218 10220 10222 10224 10226 10228 10230 10232 10234 10236 10238 10240 10242 -
3762. 10244 10246 10248 10250 10252 10254 10256 10258 10260 10262 10264 10266 -
3763. 10268 10270 10272 10274 10276 10278 10280 10282 10284 10286 10288 10296 -
3764. 10298 10300 10302 10304 10306 10308 10310 10312 10314 10316 10318 10320 -
3765. 10322 10324 10326 10328 10330 10332 10334 10336 10338 10340 10342 10344 -
3766. 10346 10348 10350 10352 10354 10368 10370 10372 10374 10376 10378 10380 -
3767. 10382 10384 10386 10388 10390 10392 10394 10396 10398 10400 10402 10404 -
3768. 10406 10408 10410 10412 10414 10416 10418 10420 10422 10424 10426 10440 -
3769. 10442 10444 10446 10448 10450 10452 10454 10456 10458 10460 10462 10464 -
3770. 10466 10468 10470 10472 10474 10476 10478 10480 10482 10484 10486 10488 -
3771. 10490 10492 10494 10496 10498 10512 10514 10516 10518 10520 10522 10524 -
3772. 10526 10528 10530 10532 10534 10536 10538 10540 10542 10544 10546 10548 -
3773. 10550 10552 10554 10556 10558 10560 10562 10564 10566 10568 -
3774. 10570 PR GZ 0.56
3775. 10578 10580 10582 10584 10586 10588 10590 10592 10594 10596 10598 10600 10602 -
3776. 10604 10606 10608 10610 10612 10614 10616 10618 10620 10622 10624 10626 -
3777. 10628 10630 10632 10634 10636 10638 10640 10642 10644 10646 -
3778. 10648 PR GZ 0.619
3779. 10650 10652 10654 10656 10658 10660 10662 10664 10666 10668 10670 10672 10674 -
3780. 10676 10678 10680 10682 10684 10686 10688 10690 10692 10694 10696 10698 -
3781. 10700 10702 10704 10706 10708 10710 10712 10714 10716 10718 -
3782. 10720 PR GZ 0.652
3783. 102 104 106 108 110 112 114 116 118 120 122 124 126 128 130 132 134 136 138 -
3784. 140 142 144 146 148 150 152 154 156 158 160 162 164 166 168 170 172 10722 -
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3785. 10724 10726 10728 10730 10732 10734 10736 10738 10740 10742 10744 10746 -
3786. 10748 10750 10752 10754 10756 10758 10760 10762 10764 10766 10768 10770 -
3787. 10772 10774 10776 10778 10780 10782 10784 10786 10788 10790 -
3788. 10792 PR GZ 0.678
3789. 202 204 206 208 210 212 214 216 218 220 222 224 226 228 230 232 234 236 238 -
3790. 240 242 244 246 248 250 252 254 256 258 260 262 264 266 268 270 272 302 304 -
3791. 306 308 310 312 314 316 318 320 322 324 326 328 330 332 334 336 338 340 342 -
3792. 344 346 348 350 352 354 356 358 360 362 364 366 368 370 372 PR GZ 0.697
3793. 402 404 406 408 410 412 414 416 418 420 422 424 426 428 430 432 434 436 438 -
3794. 440 442 444 446 448 450 452 454 456 458 460 462 464 466 468 470 472 502 504 -
3795. 506 508 510 512 514 516 518 520 522 524 526 528 530 532 534 536 538 540 542 -
3796. 544 546 548 550 552 554 556 558 560 562 564 566 568 570 572 PR GZ 0.717
3797. 602 604 606 608 610 612 614 616 618 620 622 624 626 628 630 632 634 636 638 -
3798. 640 642 644 646 648 650 652 654 656 658 660 662 664 666 668 670 672 702 704 -
3799. 706 708 710 712 714 716 718 720 722 724 726 728 730 732 734 736 738 740 742 -
3800. 744 746 748 750 752 754 756 758 760 762 764 766 768 770 772 PR GZ 0.736
3801. 802 804 806 808 810 812 814 816 818 820 822 824 826 828 830 832 834 836 838 -
3802. 840 842 844 846 848 850 852 854 856 858 860 862 864 866 868 870 872 902 904 -
3803. 906 908 910 912 914 916 918 920 922 924 926 928 930 932 934 936 938 940 942 -
3804. 944 946 948 950 952 954 956 958 960 962 964 966 968 970 972 PR GZ 0.756
3805. 1002 1004 1006 1008 1010 1012 1014 1016 1018 1020 1022 1024 1026 1028 1030 -
3806. 1032 1034 1036 1038 1040 1042 1044 1046 1048 1050 1052 1054 1056 1058 1060 -
3807. 1062 1064 1066 1068 1070 1072 1102 1104 1106 1108 1110 1112 1114 1116 1118 -
3808. 1120 1122 1124 1126 1128 1130 1132 1134 1136 1138 1140 1142 1144 1146 1148 -
3809. 1150 1152 1154 1156 1158 1160 1162 1164 1166 1168 1170 1172 PR GZ 0.769
3810. 1202 1204 1206 1208 1210 1212 1214 1216 1218 1220 1222 1224 1226 1228 1230 -
3811. 1232 1234 1236 1238 1240 1242 1244 1246 1248 1250 1252 1254 1256 1258 1260 -
3812. 1262 1264 1266 1268 1270 1272 1302 1304 1306 1308 1310 1312 1314 1316 1318 -
3813. 1320 1322 1324 1326 1328 1330 1332 1334 1336 1338 1340 1342 1344 1346 1348 -
3814. 1350 1352 1354 1356 1358 1360 1362 1364 1366 1368 1370 1372 PR GZ 0.788
3815. 1402 1404 1406 1408 1410 1412 1414 1416 1418 1420 1422 1424 1426 1428 1430 -
3816. 1432 1434 1436 1438 1440 1442 1444 1446 1448 1450 1452 1454 1456 1458 1460 -
3817. 1462 1464 1466 1468 1470 1472 1502 1504 1506 1508 1510 1512 1514 1516 1518 -
3818. 1520 1522 1524 1526 1528 1530 1532 1534 1536 1538 1540 1542 1544 1546 1548 -
3819. 1550 1552 1554 1556 1558 1560 1562 1564 1566 1568 1570 1572 1602 1604 1606 -
3820. 1608 1610 1612 1614 1616 1618 1620 1622 1624 1626 1628 1630 1632 1634 1636 -
3821. 1638 1640 1642 1644 1646 1648 1650 1652 1654 1656 1658 1660 1662 1664 1666 -
3822. 1668 1670 1672 PR GZ 0.821
3823. 1702 1704 1706 1708 1710 1712 1714 1716 1718 1720 1722 1724 1726 1728 1730 -
3824. 1732 1734 1736 1738 1740 1742 1744 1746 1748 1750 1752 1754 1756 1758 1760 -
3825. 1762 1764 1766 1768 1770 1772 1802 1804 1806 1808 1810 1812 1814 1816 1818 -
3826. 1820 1822 1824 1826 1828 1830 1832 1834 1836 1838 1840 1842 1844 1846 1848 -
3827. 1850 1852 1854 1856 1858 1860 1862 1864 1866 1868 1870 1872 1902 1904 1906 -
3828. 1908 1910 1912 1914 1916 1918 1920 1922 1924 1926 1928 1930 1932 1934 1936 -
3829. 1938 1940 1942 1944 1946 1948 1950 1952 1954 1956 1958 1960 1962 1964 1966 -
3830. 1968 1970 1972 2002 2004 2006 2008 2010 2012 2014 2016 2018 2020 2022 2024 -
3831. 2026 2028 2030 2032 2034 2036 2038 2040 2042 2044 2046 2048 2050 2052 2054 -
3832. 2056 2058 2060 2062 2064 2066 2068 2070 2072 2102 2104 2106 2108 2110 2112 -
3833. 2114 2116 2118 2120 2122 2124 2126 2128 2130 2132 2134 2136 2138 2140 2142 -
3834. 2144 2146 2148 2150 2152 2154 2156 2158 2160 2162 2164 2166 2168 2170 -
3835. 2172 PR GZ 0.847
3836. 2202 2204 2206 2208 2210 2212 2214 2216 2218 2220 2222 2224 2226 2228 2230 -
3837. 2232 2234 2236 2238 2240 2242 2244 2246 2248 2250 2252 2254 2256 2258 2260 -
3838. 2262 2264 2266 2268 2270 2272 2302 2304 2306 2308 2310 2312 2314 2316 2318 -
3839. 2320 2322 2324 2326 2328 2330 2332 2334 2336 2338 2340 2342 2344 2346 2348 -
3840. 2350 2352 2354 2356 2358 2360 2362 2364 2366 2368 2370 2372 2402 2404 2406 -
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2408 2410 2412 2414 2416 2418 2420 2422 2424 2426 2428 2430 2432 2434 2436 -
2438 2440 2442 2444 2446 2448 2450 2452 2454 2456 2458 2460 2462 2464 2466 -
2468 2470 2472 PR GZ 0.873
*
*
*******************************************************
LOAD 11 MATERIAL FILLING WALL HORIZONTAL CASE-I (H1A)
*******************************************************
ELEMENT LOAD
2001 TO 2072 PR 18.5
1901 TO 1972 PR 34.4
1801 TO 1872 PR 48.1
1701 TO 1772 PR 60.1
1601 TO 1672 PR 70.6
1501 TO 1572 PR 80
1401 TO 1472 PR 88.3
1301 TO 1372 PR 95.7
1201 TO 1272 PR 102.3
1101 TO 1172 PR 108.4
1001 TO 1072 PR 113.8
901 TO 972 PR 118.8
801 TO 872 PR 123.3
701 TO 772 PR 127.5
601 TO 672 PR 131.3
501 TO 572 PR 134.8
401 TO 472 PR 138.1
301 TO 372 PR 141.1
201 TO 272 PR 143.8
101 TO 172 PR 146.4
*
*******************************************************
LOAD 12 MATERIAL FILLING WALL HORIZONTAL CASE-II (H1B)
*******************************************************
ELEMENT LOAD
2001 TO 2072 PR 22.6
1901 TO 1972 PR 41.2
1801 TO 1872 PR 56.8
1701 TO 1772 PR 70.1
1601 TO 1672 PR 81.5
1501 TO 1572 PR 91.3
1401 TO 1472 PR 99.9
1301 TO 1372 PR 107.4
1201 TO 1272 PR 114.1
1101 TO 1172 PR 120
1001 TO 1072 PR 125.3
901 TO 972 PR 130.1
801 TO 872 PR 134.4
701 TO 772 PR 138.3
601 TO 672 PR 141.8
501 TO 572 PR 145.1
401 TO 472 PR 148.1
301 TO 372 PR 150.8
201 TO 272 PR 153.3
101 TO 172 PR 155.7
*
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*
LOAD 13 MATERIAL FILLING WALL HORIZONTAL CASE-III (H1C)
******************************************************
ELEMENT LOAD
2001 TO 2072 PR 14.2
1901 TO 1972 PR 26.8
1801 TO 1872 PR 38.1
1701 TO 1772 PR 48.3
1601 TO 1672 PR 57.5
1501 TO 1572 PR 65.8
1401 TO 1472 PR 73.4
1301 TO 1372 PR 80.4
1201 TO 1272 PR 86.8
1101 TO 1172 PR 92.6
1001 TO 1072 PR 98
901 TO 972 PR 103
801 TO 872 PR 107.6
701 TO 772 PR 111.9
601 TO 672 PR 115.9
501 TO 572 PR 119.6
401 TO 472 PR 123.1
301 TO 372 PR 126.3
201 TO 272 PR 129.4
101 TO 172 PR 132.3
*
*
******************************************************
LOAD 14 MATERIAL EMPTYING WALL HORIZONTAL CASE-I (H2A)
******************************************************
ELEMENT LOAD
2001 TO 2072 PR 19.9
1901 TO 1972 PR 37
1801 TO 1872 PR 51.8
1701 TO 1772 PR 64.7
1601 TO 1672 PR 76.1
1501 TO 1572 PR 86.1
1401 TO 1472 PR 95.1
1301 TO 1372 PR 103.1
1201 TO 1272 PR 110.2
1101 TO 1172 PR 116.7
1001 TO 1072 PR 122.6
901 TO 972 PR 127.9
801 TO 872 PR 132.8
701 TO 772 PR 137.3
601 TO 672 PR 141.4
501 TO 572 PR 145.2
401 TO 472 PR 148.7
301 TO 372 PR 151.9
201 TO 272 PR 154.9
101 TO 172 PR 157.7
*
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******************************************************
LOAD 15 MATERIAL EMPTYING WALL HORIZONTAL CASE-II (H2B)
******************************************************
ELEMENT LOAD
2001 TO 2072 PR 24.3
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*
1901 TO 1972 PR 44.4
1801 TO 1872 PR 61.2
1701 TO 1772 PR 75.5
1601 TO 1672 PR 87.7
1501 TO 1572 PR 98.3
1401 TO 1472 PR 107.6
1301 TO 1372 PR 115.7
1201 TO 1272 PR 122.9
1101 TO 1172 PR 129.2
1001 TO 1072 PR 135
901 TO 972 PR 140.1
801 TO 872 PR 144.7
701 TO 772 PR 148.9
601 TO 672 PR 152.8
501 TO 572 PR 156.3
401 TO 472 PR 159.5
301 TO 372 PR 162.4
201 TO 272 PR 165.2
101 TO 172 PR 167.7
*
*
*********************************************************
LOAD 16 MATERIAL EMPTYING WALL HORIZONTAL CASE-III (H2C)
*********************************************************
ELEMENT LOAD
2001 TO 2072 PR 15.2
1901 TO 1972 PR 28.8
1801 TO 1872 PR 41
1701 TO 1772 PR 52
1601 TO 1672 PR 61.9
1501 TO 1572 PR 70.9
1401 TO 1472 PR 79.1
1301 TO 1372 PR 86.6
1201 TO 1272 PR 93.4
1101 TO 1172 PR 99.8
1001 TO 1072 PR 105.6
901 TO 972 PR 110.9
801 TO 872 PR 115.9
701 TO 772 PR 120.5
601 TO 672 PR 124.8
501 TO 572 PR 128.8
401 TO 472 PR 132.6
301 TO 372 PR 136.1
201 TO 272 PR 139.4
101 TO 172 PR 142.4
*
*
*****************************************************************
LOAD 21 MATERIAL WALL VERTICAL (FILLING _EMPTYING) CASE-I (H3A)
*****************************************************************
ELEMENT LOAD
2001 TO 2072 PR GY -5.9
1901 TO 1972 PR GY -10.9
1801 TO 1872 PR GY -15.3
1701 TO 1772 PR GY -19.1
1601 TO 1672 PR GY -22.5
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4009. 1501 TO 1572 PR GY -25.4
4010. 1401 TO 1472 PR GY -28.1
4011. 1301 TO 1372 PR GY -30.4
4012. 1201 TO 1272 PR GY -32.6
4013. 1101 TO 1172 PR GY -34.5
4014. 1001 TO 1072 PR GY -36.2
4015. 901 TO 972 PR GY -37.8
4016. 801 TO 872 PR GY -39.2
4017. 701 TO 772 PR GY -40.5
4018. 601 TO 672 PR GY -41.8
4019. 501 TO 572 PR GY -42.9
4020. 401 TO 472 PR GY -43.9
4021. 301 TO 372 PR GY -44.9
4022. 201 TO 272 PR GY -45.8
4023. 101 TO 172 PR GY -46.6
4024. *
4025. *
4026. ********************************************************************
4027. LOAD 22 MATERIAL WALL VERTICAL (FILLING _EMPTYING) CASE-II (H3B)
4028. ********************************************************************
4029. ELEMENT LOAD
4030. 2001 TO 2072 PR GY -6.5
4031. 1901 TO 1972 PR GY -11.9
4032. 1801 TO 1872 PR GY -16.4
4033. 1701 TO 1772 PR GY -20.3
4034. 1601 TO 1672 PR GY -23.6
4035. 1501 TO 1572 PR GY -26.4
4036. 1401 TO 1472 PR GY -28.9
4037. 1301 TO 1372 PR GY -31.1
4038. 1201 TO 1272 PR GY -33
4039. 1101 TO 1172 PR GY -34.7
4040. 1001 TO 1072 PR GY -36.2
4041. 901 TO 972 PR GY -37.6
4042. 801 TO 872 PR GY -38.9
4043. 701 TO 772 PR GY -40
4044. 601 TO 672 PR GY -41
4045. 501 TO 572 PR GY -42
4046. 401 TO 472 PR GY -42.8
4047. 301 TO 372 PR GY -43.6
4048. 201 TO 272 PR GY -44.3
4049. 101 TO 172 PR GY -45
4050. *
4051. *
4052. ********************************************************************
4053. LOAD 23 MATERIAL WALL VERTICAL (FILLING _EMPTYING) CASE-III (H3C)
4054. ********************************************************************
4055. ELEMENT LOAD
4056. 2001 TO 2072 PR GY -5.2
4057. 1901 TO 1972 PR GY -9.8
4058. 1801 TO 1872 PR GY -14
4059. 1701 TO 1772 PR GY -17.8
4060. 1601 TO 1672 PR GY -21.1
4061. 1501 TO 1572 PR GY -24.2
4062. 1401 TO 1472 PR GY -27
4063. 1301 TO 1372 PR GY -29.6
4064. 1201 TO 1272 PR GY -31.9
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1101 TO 1172 PR GY -34.1
1001 TO 1072 PR GY -36.1
901 TO 972 PR GY -37.9
801 TO 872 PR GY -39.6
701 TO 772 PR GY -41.2
601 TO 672 PR GY -42.6
501 TO 572 PR GY -44
401 TO 472 PR GY -45.3
301 TO 372 PR GY -46.5
201 TO 272 PR GY -47.6
101 TO 172 PR GY -48.7
*
*
*********************************************************************
LOAD 24 MATERIAL WALL VERTICAL (FILLING _EMPTYING) CASE-I (H4A)
*********************************************************************
ELEMENT LOAD
2001 TO 2072 PR GY -9.8
1901 TO 1972 PR GY -18.2
1801 TO 1872 PR GY -25.5
1701 TO 1772 PR GY -31.8
1601 TO 1672 PR GY -37.4
1501 TO 1572 PR GY -42.4
1401 TO 1472 PR GY -46.8
1301 TO 1372 PR GY -50.7
1201 TO 1272 PR GY -54.2
1101 TO 1172 PR GY -57.4
1001 TO 1072 PR GY -60.3
901 TO 972 PR GY -63
801 TO 872 PR GY -65.4
701 TO 772 PR GY -67.6
601 TO 672 PR GY -69.6
501 TO 572 PR GY -71.4
401 TO 472 PR GY -73.2
301 TO 372 PR GY -74.8
201 TO 272 PR GY -76.2
101 TO 172 PR GY -77.6
*
*
*********************************************************************
LOAD 25 MATERIAL WALL VERTICAL (FILLING _EMPTYING) CASE-II (H4B)
*********************************************************************
ELEMENT LOAD
2001 TO 2072 PR GY -14.9
1901 TO 1972 PR GY -27.2
1801 TO 1872 PR GY -37.5
1701 TO 1772 PR GY -46.3
1601 TO 1672 PR GY -53.8
1501 TO 1572 PR GY -60.2
1401 TO 1472 PR GY -65.9
1301 TO 1372 PR GY -70.9
1201 TO 1272 PR GY -75.3
1101 TO 1172 PR GY -79.2
1001 TO 1072 PR GY -82.7
901 TO 972 PR GY -85.8
801 TO 872 PR GY -88.7
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4121. 701 TO 772 PR GY -91.3
4122. 601 TO 672 PR GY -93.6
4123. 501 TO 572 PR GY -95.8
4124. 401 TO 472 PR GY -97.7
4125. 301 TO 372 PR GY -99.5
4126. 201 TO 272 PR GY -101.2
4127. 101 TO 172 PR GY -102.7
4128. *
4129. *
4130. ********************************************************************
4131. LOAD 26 MATERIAL WALL VERTICAL (FILLING _EMPTYING) CASE-III (H4C)
4132. ********************************************************************
4133. ELEMENT LOAD
4134. 2001 TO 2072 PR GY -9.3
4135. 1901 TO 1972 PR GY -17.7
4136. 1801 TO 1872 PR GY -25.1
4137. 1701 TO 1772 PR GY -31.9
4138. 1601 TO 1672 PR GY -37.9
4139. 1501 TO 1572 PR GY -43.4
4140. 1401 TO 1472 PR GY -48.5
4141. 1301 TO 1372 PR GY -53.1
4142. 1201 TO 1272 PR GY -57.3
4143. 1101 TO 1172 PR GY -61.1
4144. 1001 TO 1072 PR GY -64.7
4145. 901 TO 972 PR GY -68
4146. 801 TO 872 PR GY -71
4147. 701 TO 772 PR GY -73.9
4148. 601 TO 672 PR GY -76.5
4149. 501 TO 572 PR GY -78.9
4150. 401 TO 472 PR GY -81.2
4151. 301 TO 372 PR GY -83.4
4152. 201 TO 272 PR GY -85.4
4153. 101 TO 172 PR GY -87.3
4154. *
4155. *
4156. ****************************************************
4157. LOAD 31 MATERIAL VERTICAL (ON FLAT SLAB) CASE-I (H5A)
4158. ****************************************************
4159. ELEMENT LOAD
4160. 12172 TO 12787 PR GY -320.1
4161. *
4162. ****************************************************
4163. LOAD 32 MATERIAL VERTICAL (ON FLAT SLAB) CASE-II (H5B)
4164. ****************************************************
4165. ELEMENT LOAD
4166. 12172 TO 12787 PR GY -302
4167. *
4168. *****************************************************
4169. LOAD 33 MATERIAL VERTICAL (ON FLAT SLAB) CASE-III (H5C)
4170. *****************************************************
4171. ELEMENT LOAD
4172. 12172 TO 12787 PR GY -344.5
4173. *
4174. ****************************************************
4175. LOAD 34 MATERIAL VERTICAL (ON FLAT SLAB) CASE-I (H6A)
4176. ****************************************************
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4177. ELEMENT LOAD
4178. 12172 TO 12787 PR GY -247.9
4179. *
4180. ******************************************************
4181. LOAD 35 MATERIAL VERTICAL (ON FLAT SLAB) CASE-II (H6B)
4182. ******************************************************
4183. ELEMENT LOAD
4184. 12172 TO 12787 PR GY -180.3
4185. *
4186. ******************************************************
4187. LOAD 36 MATERIAL VERTICAL (ON FLAT SLAB) CASE-III (H6C)
4188. ******************************************************
4189. ELEMENT LOAD
4190. 12172 TO 12787 PR GY -233.9
4191. *
4192. *
4193. ********************
4194. * LOAD COMBINATION
4195. ********************
4196. *
4197. * FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
4198. *
4199. LOAD COMB 101 1.4X[ D ]
4200. 1 1.4
4201. *
4202. LOAD COMB 102 1.4X[ D ] + 1.7X[ L + H1A + H3A + H5A ]
4203. 1 1.4 2 1.7 11 1.7 21 1.7 31 1.7
4204. *
4205. LOAD COMB 103 1.4X[ D ] + 1.7X[ L + H1B + H3B + H5B ]
4206. 1 1.4 2 1.7 12 1.7 22 1.7 32 1.7
4207. *
4208. LOAD COMB 104 1.4X[ D ] + 1.7X[ L + H1C + H3C + H5C ]
4209. 1 1.4 2 1.7 13 1.7 23 1.7 33 1.7
4210. *
4211. LOAD COMB 105 1.4X[ D ] + 1.7X[ L + H1A + H4A + H6A ]
4212. 1 1.4 2 1.7 11 1.7 24 1.7 34 1.7
4213. *
4214. LOAD COMB 106 1.4X[ D ] + 1.7X[ L + H1B + H4B + H6B ]
4215. 1 1.4 2 1.7 12 1.7 25 1.7 35 1.7
4216. *
4217. LOAD COMB 107 1.4X[ D ] + 1.7X[ L + H1C + H4C + H6C ]
4218. 1 1.4 2 1.7 13 1.7 26 1.7 36 1.7
4219. *
4220. LOAD COMB 108 1.4X[ D ] + 1.7X[ L + H2A + H3A + H5A ]
4221. 1 1.4 2 1.7 14 1.7 21 1.7 31 1.7
4222. *
4223. LOAD COMB 109 1.4X[ D ] + 1.7X[ L + H2B + H3B + H5B ]
4224. 1 1.4 2 1.7 15 1.7 22 1.7 32 1.7
4225. *
4226. LOAD COMB 110 1.4X[ D ] + 1.7X[ L + H2C + H3C + H5C ]
4227. 1 1.4 2 1.7 16 1.7 23 1.7 33 1.7
4228. *
4229. LOAD COMB 111 1.4X[ D ] + 1.7X[ L + H2A + H4A + H6A ]
4230. 1 1.4 2 1.7 14 1.7 24 1.7 34 1.7
4231. *
4232. LOAD COMB 112 1.4X[ D ] + 1.7X[ L + H2B + H4B + H6B ]
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  *
4233. 1 1.4 2 1.7 15 1.7 25 1.7 35 1.7
4234. *
4235. LOAD COMB 113 1.4X[ D ] + 1.7X[ L + H2C + H4C + H6C ]
4236. 1 1.4 2 1.7 16 1.7 26 1.7 36 1.7
4237. *
4238. LOAD COMB 114 1.2X[ D ] + 1.0X[ L ]
4239. 1 1.2 2 1.0
4240. *
4241. LOAD COMB 115 1.2X[ D ] + 0.8X[ W1A ]
4242. 1 1.2 3 0.8
4243. *
4244. LOAD COMB 116 1.2X[ D ] + 0.8X[ W1B ]
4245. 1 1.2 4 0.8
4246. *
4247. LOAD COMB 117 1.2X[ D ] + 1.6X[ W1A ] + 1.0X[ L ]
4248. 1 1.2 3 1.6 2 1.0
4249. *
4250. LOAD COMB 118 1.2X[ D ] + 1.6X[ W1B ] + 1.0X[ L ]
4251. 1 1.2 4 1.6 2 1.0
4252. *
4253. LOAD COMB 119 1.2X[ D ] + 1.0X[ E1A ] + 1.0X[ L ]
4254. 1 1.2 5 1.0 2 1.0
4255. *
4256. LOAD COMB 120 1.2X[ D ] + 1.0X[ E1B ] + 1.0X[ L ]
4257. 1 1.2 6 1.0 2 1.0
4258. *
4259. LOAD COMB 121 0.9X[ D ] + 1.6X[ W1A + H1A + H3A + H5A ]
4260. 1 0.9 3 1.6 11 1.6 21 1.6 31 1.6
4261. *
4262. LOAD COMB 122 0.9X[ D ] + 1.6X[ W1A + H1B + H3B + H5B ]
4263. 1 0.9 3 1.6 12 1.6 22 1.6 32 1.6
4264. *
4265. LOAD COMB 123 0.9X[ D ] + 1.6X[ W1A + H1C + H3C + H5C ]
4266. 1 0.9 3 1.6 13 1.6 23 1.6 33 1.6
4267. *
4268. LOAD COMB 124 0.9X[ D ] + 1.6X[ W1A + H1A + H4A + H6A ]
4269. 1 0.9 3 1.6 11 1.6 24 1.6 34 1.6
4270. *
4271. LOAD COMB 125 0.9X[ D ] + 1.6X[ W1A + H1B + H4B + H6B ]
4272. 1 0.9 3 1.6 12 1.6 25 1.6 35 1.6
4273. *
4274. LOAD COMB 126 0.9X[ D ] + 1.6X[ W1A + H1C + H4C + H6C ]
4275. 1 0.9 3 1.6 13 1.6 26 1.6 36 1.6
4276. *
4277. LOAD COMB 127 0.9X[ D ] + 1.6X[ W1A + H2A + H3A + H5A ]
4278. 1 0.9 3 1.6 14 1.6 21 1.6 31 1.6
4279. *
4280. LOAD COMB 128 0.9X[ D ] + 1.6X[ W1A + H2B + H3B + H5B ]
4281. 1 0.9 3 1.6 15 1.6 22 1.6 32 1.6
4282. *
4283. LOAD COMB 129 0.9X[ D ] + 1.6X[ W1A + H2C + H3C + H5C ]
4284. 1 0.9 3 1.6 16 1.6 23 1.6 33 1.6
4285. *
4286. LOAD COMB 130 0.9X[ D ] + 1.6X[ W1A + H2A + H4A + H6A ]
4287. 1 0.9 3 1.6 14 1.6 24 1.6 34 1.6
4288. *
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  *
4289. LOAD COMB 131 0.9X[ D ] + 1.6X[ W1A + H2B + H4B + H6B ]
4290. 1 0.9 3 1.6 15 1.6 25 1.6 35 1.6
4291. *
4292. LOAD COMB 132 0.9X[ D ] + 1.6X[ W1A + H2C + H4C + H6C ]
4293. 1 0.9 3 1.6 16 1.6 26 1.6 36 1.6
4294. *
4295. LOAD COMB 133 0.9X[ D ] + 1.6X[ W1B + H1A + H3A + H5A ]
4296. 1 0.9 4 1.6 11 1.6 21 1.6 31 1.6
4297. *
4298. LOAD COMB 134 0.9X[ D ] + 1.6X[ W1B + H1B + H3B + H5B ]
4299. 1 0.9 4 1.6 12 1.6 22 1.6 32 1.6
4300. *
4301. LOAD COMB 135 0.9X[ D ] + 1.6X[ W1B + H1C + H3C + H5C ]
4302. 1 0.9 4 1.6 13 1.6 23 1.6 33 1.6
4303. *
4304. LOAD COMB 136 0.9X[ D ] + 1.6X[ W1B + H1A + H4A + H6A ]
4305. 1 0.9 4 1.6 11 1.6 24 1.6 34 1.6
4306. *
4307. LOAD COMB 137 0.9X[ D ] + 1.6X[ W1B + H1B + H4B + H6B ]
4308. 1 0.9 4 1.6 12 1.6 25 1.6 35 1.6
4309. *
4310. LOAD COMB 138 0.9X[ D ] + 1.6X[ W1B + H1C + H4C + H6C ]
4311. 1 0.9 4 1.6 13 1.6 26 1.6 36 1.6
4312. *
4313. LOAD COMB 139 0.9X[ D ] + 1.6X[ W1B + H2A + H3A + H5A ]
4314. 1 0.9 4 1.6 14 1.6 21 1.6 31 1.6
4315. *
4316. LOAD COMB 140 0.9X[ D ] + 1.6X[ W1B + H2B + H3B + H5B ]
4317. 1 0.9 4 1.6 15 1.6 22 1.6 32 1.6
4318. *
4319. LOAD COMB 141 0.9X[ D ] + 1.6X[ W1B + H2C + H3C + H5C ]
4320. 1 0.9 4 1.6 16 1.6 23 1.6 33 1.6
4321. *
4322. LOAD COMB 142 0.9X[ D ] + 1.6X[ W1B + H2A + H4A + H6A ]
4323. 1 0.9 4 1.6 14 1.6 24 1.6 34 1.6
4324. *
4325. LOAD COMB 143 0.9X[ D ] + 1.6X[ W1B + H2B + H4B + H6B ]
4326. 1 0.9 4 1.6 15 1.6 25 1.6 35 1.6
4327. *
4328. LOAD COMB 144 0.9X[ D ] + 1.6X[ W1B + H2C + H4C + H6C ]
4329. 1 0.9 4 1.6 16 1.6 26 1.6 36 1.6
4330. *
4331. LOAD COMB 145 0.9X[ D ] + 1.0X[ E1A ] + 1.6X 0.8X [ H1A + H3A + H5A ]
4332. 1 0.9 5 1.0 11 1.28 21 1.28 31 1.28
4333. *
4334. LOAD COMB 146 0.9X[ D ] + 1.0X[ E1A ] + 1.6X 0.8X[ H1B + H3B + H5B ]
4335. 1 0.9 5 1.0 12 1.28 22 1.28 32 1.28
4336. *
4337. LOAD COMB 147 0.9X[ D ] + 1.0X[ E1A ] + 1.6X 0.8X[ H1C + H3C + H5C ]
4338. 1 0.9 5 1.0 13 1.28 23 1.28 33 1.28
4339. *
4340. LOAD COMB 148 0.9X[ D ] + 1.0X[ E1A ] + 1.6X 0.8X[ H1A + H4A + H6A ]
4341. 1 0.9 5 1.0 11 1.28 24 1.28 34 1.28
4342. *
4343. LOAD COMB 149 0.9X[ D ] + 1.0X[ E1A ] + 1.6X 0.8X[ H1B + H4B + H6B ]
4344. 1 0.9 5 1.0 12 1.28 25 1.28 35 1.28
```

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*
4345. *
4346. LOAD COMB 150 0.9X[ D ] + 1.0X[ E1A ] + 1.6X 0.8X[ H1C + H4C + H6C ]
4347. 1 0.9 5 1.0 13 1.28 26 1.28 36 1.28
4348. *
4349. LOAD COMB 151 0.9X[ D ] + 1.0X[ E1A ] + 1.6X 0.8X[ H2A + H3A + H5A ]
4350. 1 0.9 5 1.0 14 1.28 21 1.28 31 1.28
4351. *
4352. LOAD COMB 152 0.9X[ D ] + 1.0X[ E1A ] + 1.6X 0.8X[ H2B + H3B + H5B ]
4353. 1 0.9 5 1.0 15 1.28 22 1.28 32 1.28
4354. *
4355. LOAD COMB 153 0.9X[ D ] + 1.0X[ E1A ] + 1.6X 0.8X[ H2C + H3C + H5C ]
4356. 1 0.9 5 1.0 16 1.28 23 1.28 33 1.28
4357. *
4358. LOAD COMB 154 0.9X[ D ] + 1.0X[ E1A ] + 1.6X 0.8X[ H2A + H4A + H6A ]
4359. 1 0.9 5 1.0 14 1.28 24 1.28 34 1.28
4360. *
4361. LOAD COMB 155 0.9X[ D ] + 1.0X[ E1A ] + 1.6X 0.8X[ H2B + H4B + H6B ]
4362. 1 0.9 5 1.0 15 1.28 25 1.28 35 1.28
4363. *
4364. LOAD COMB 156 0.9X[ D ] + 1.0X[ E1A ] + 1.6X 0.8X[ H2C + H4C + H6C ]
4365. 1 0.9 5 1.0 16 1.28 26 1.28 36 1.28
4366. *
4367. LOAD COMB 157 0.9X[ D ] + 1.0X[ E1B ] + 1.6X 0.8X[ H1A + H3A + H5A ]
4368. 1 0.9 6 1.0 11 1.28 21 1.28 31 1.28
4369. *
4370. LOAD COMB 158 0.9X[ D ] + 1.0X[ E1B ] + 1.6X 0.8X[ H1B + H3B + H5B ]
4371. 1 0.9 6 1.0 12 1.28 22 1.28 32 1.28
4372. *
4373. LOAD COMB 159 0.9X[ D ] + 1.0X[ E1B ] + 1.6X 0.8X[ H1C + H3C + H5C ]
4374. 1 0.9 6 1.0 13 1.28 23 1.28 33 1.28
4375. *
4376. LOAD COMB 160 0.9X[ D ] + 1.0X[ E1B ] + 1.6X 0.8X[ H1A + H4A + H6A ]
4377. 1 0.9 6 1.0 11 1.28 24 1.28 34 1.28
4378. *
4379. LOAD COMB 161 0.9X[ D ] + 1.0X[ E1B ] + 1.6X 0.8X[ H1B + H4B + H6B ]
4380. 1 0.9 6 1.0 12 1.28 25 1.28 35 1.28
4381. *
4382. LOAD COMB 162 0.9X[ D ] + 1.0X[ E1B ] + 1.6X 0.8X[ H1C + H4C + H6C ]
4383. 1 0.9 6 1.0 13 1.28 26 1.28 36 1.28
4384. *
4385. LOAD COMB 163 0.9X[ D ] + 1.0X[ E1B ] + 1.6X 0.8X[ H2A + H3A + H5A ]
4386. 1 0.9 6 1.0 14 1.28 21 1.28 31 1.28
4387. *
4388. LOAD COMB 164 0.9X[ D ] + 1.0X[ E1B ] + 1.6X 0.8X[ H2B + H3B + H5B ]
4389. 1 0.9 6 1.0 15 1.28 22 1.28 32 1.28
4390. *
4391. LOAD COMB 165 0.9X[ D ] + 1.0X[ E1B ] + 1.6X 0.8X[ H2C + H3C + H5C ]
4392. 1 0.9 6 1.0 16 1.28 23 1.28 33 1.28
4393. *
4394. LOAD COMB 166 0.9X[ D ] + 1.0X[ E1B ] + 1.6X 0.8X[ H2A + H4A + H6A ]
4395. 1 0.9 6 1.0 14 1.28 24 1.28 34 1.28
4396. *
4397. LOAD COMB 167 0.9X[ D ] + 1.0X[ E1B ] + 1.6X 0.8X[ H2B + H4B + H6B ]
4398. 1 0.9 6 1.0 15 1.28 25 1.28 35 1.28
4399. *
4400. LOAD COMB 168 0.9X[ D ] + 1.0X[ E1B ] + 1.6X 0.8X[ H2C + H4C + H6C ]
```

```
*
1 0.9 6 1.0 16 1.28 26 1.28 36 1.28
*
*
* LOAD COMBINATIONS FOR SERVICEABILITY AND ALLOWABLE STRESS DESIGN
*
LOAD COMB 501 1X[ D + H1A + H3A + H5A + L ]
1 1.0 11 1.0 21 1.0 31 1.0 2 1.0
*
LOAD COMB 502 1X[ D + H1B + H3B + H5B + L ]
1 1.0 12 1.0 22 1.0 32 1.0 2 1.0
*
LOAD COMB 503 1X[ D + H1C + H3C + H5C + L ]
1 1.0 13 1.0 23 1.0 33 1.0 2 1.0
*
LOAD COMB 504 1X[ D + H1A + H4A + H6A + L ]
1 1.0 11 1.0 24 1.0 34 1.0 2 1.0
*
LOAD COMB 505 1X[ D + H1B + H4B + H6B + L ]
1 1.0 12 1.0 25 1.0 35 1.0 2 1.0
*
LOAD COMB 506 1X[ D + H1C + H4C + H6C + L ]
1 1.0 13 1.0 26 1.0 36 1.0 2 1.0
*
LOAD COMB 507 1X[ D + H2A + H3A + H5A + L ]
1 1.0 14 1.0 21 1.0 31 1.0 2 1.0
*
LOAD COMB 508 1X[ D + H2B + H3B + H5B + L ]
1 1.0 15 1.0 22 1.0 32 1.0 2 1.0
*
LOAD COMB 509 1X[ D + H2C + H3C + H5C + L ]
1 1.0 16 1.0 23 1.0 33 1.0 2 1.0
*
LOAD COMB 510 1X[ D + H2A + H4A + H6A + L ]
1 1.0 14 1.0 24 1.0 34 1.0 2 1.0
*
LOAD COMB 511 1X[ D + H2B + H4B + H6B + L ]
1 1.0 15 1.0 25 1.0 35 1.0 2 1.0
*
LOAD COMB 512 1X[ D + H2C + H4C + H6C + L ]
1 1.0 16 1.0 26 1.0 36 1.0 2 1.0
*
LOAD COMB 513 1X[ D + H1A + H3A + H5A ]
1 1.0 11 1.0 21 1.0 31 1.0
*
LOAD COMB 514 1X[ D + H1B + H3B + H5B ]
1 1.0 12 1.0 22 1.0 32 1.0
*
LOAD COMB 515 1X[ D + H1C + H3C + H5C ]
1 1.0 13 1.0 23 1.0 33 1.0
*
LOAD COMB 516 1X[ D + H1A + H4A + H6A ]
1 1.0 11 1.0 24 1.0 34 1.0
*
LOAD COMB 517 1X[ D + H1B + H4B + H6B ]
1 1.0 12 1.0 25 1.0 35 1.0
*
```

```
*
LOAD COMB 518 1X[ D + H1C + H4C + H6C ]
1 1.0 13 1.0 26 1.0 36 1.0
*
LOAD COMB 519 1X[ D + H2A + H3A + H5A ]
1 1.0 14 1.0 21 1.0 31 1.0
*
LOAD COMB 520 1X[ D + H2B + H3B + H5B ]
1 1.0 15 1.0 22 1.0 32 1.0
*
LOAD COMB 521 1X[ D + H2C + H3C + H5C ]
1 1.0 16 1.0 23 1.0 33 1.0
*
LOAD COMB 522 1X[ D + H2A + H4A + H6A ]
1 1.0 14 1.0 24 1.0 34 1.0
*
LOAD COMB 523 1X[ D + H2B + H4B + H6B ]
1 1.0 15 1.0 25 1.0 35 1.0
*
LOAD COMB 524 1X[ D + H2C + H4C + H6C ]
1 1.0 16 1.0 26 1.0 36 1.0
*
LOAD COMB 525 1X[ D + H1A + H3A + H5A ] + 0.75X[ L ]
1 1.0 11 1.0 21 1.0 31 1.0 2 0.75
*
LOAD COMB 526 1X[ D + H1B + H3B + H5B ] + 0.75X[ L ]
1 1.0 12 1.0 22 1.0 32 1.0 2 0.75
*
LOAD COMB 527 1X[ D + H1C + H3C + H5C ] + 0.75X[ L ]
1 1.0 13 1.0 23 1.0 33 1.0 2 0.75
*
LOAD COMB 528 1X[ D + H1A + H4A + H6A ] + 0.75X[ L ]
1 1.0 11 1.0 24 1.0 34 1.0 2 0.75
*
LOAD COMB 529 1X[ D + H1B + H4B + H6B ] + 0.75X[ L ]
1 1.0 12 1.0 25 1.0 35 1.0 2 0.75
*
LOAD COMB 530 1X[ D + H1C + H4C + H6C ] + 0.75X[ L ]
1 1.0 13 1.0 26 1.0 36 1.0 2 0.75
*
LOAD COMB 531 1X[ D + H2A + H3A + H5A ] + 0.75X[ L ]
1 1.0 14 1.0 21 1.0 31 1.0 2 0.75
*
LOAD COMB 532 1X[ D + H2B + H3B + H5B ] + 0.75X[ L ]
1 1.0 15 1.0 22 1.0 32 1.0 2 0.75
*
LOAD COMB 533 1X[ D + H2C + H3C + H5C ] + 0.75X[ L ]
1 1.0 16 1.0 23 1.0 33 1.0 2 0.75
*
LOAD COMB 534 1X[ D + H2A + H4A + H6A ] + 0.75X[ L ]
1 1.0 14 1.0 24 1.0 34 1.0 2 0.75
*
LOAD COMB 535 1X[ D + H2B + H4B + H6B ] + 0.75X[ L ]
1 1.0 15 1.0 25 1.0 35 1.0 2 0.75
*
LOAD COMB 536 1X[ D + H2C + H4C + H6C ] + 0.75X[ L ]
1 1.0 16 1.0 26 1.0 36 1.0 2 0.75
```

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  *
4513. *
4514. LOAD COMB 537 1X[ D + H1A + H3A + H5A + W1A ]
4515. 1 1.0 11 1.0 21 1.0 31 1.0 3 1.0
4516. *
4517. LOAD COMB 538 1X[ D + H1B + H3B + H5B + W1A ]
4518. 1 1.0 12 1.0 22 1.0 32 1.0 3 1.0
4519. *
4520. LOAD COMB 539 1X[ D + H1C + H3C + H5C + W1A ]
4521. 1 1.0 13 1.0 23 1.0 33 1.0 3 1.0
4522. *
4523. LOAD COMB 540 1X[ D + H1A + H4A + H6A + W1A ]
4524. 1 1.0 11 1.0 24 1.0 34 1.0 3 1.0
4525. *
4526. LOAD COMB 541 1X[ D + H1B + H4B + H6B + W1A ]
4527. 1 1.0 12 1.0 25 1.0 35 1.0 3 1.0
4528. *
4529. LOAD COMB 542 1X[ D + H1C + H4C + H6C + W1A ]
4530. 1 1.0 13 1.0 26 1.0 36 1.0 3 1.0
4531. *
4532. LOAD COMB 543 1X[ D + H2A + H3A + H5A + W1A ]
4533. 1 1.0 14 1.0 21 1.0 31 1.0 3 1.0
4534. *
4535. LOAD COMB 544 1X[ D + H2B + H3B + H5B + W1A ]
4536. 1 1.0 15 1.0 22 1.0 32 1.0 3 1.0
4537. *
4538. LOAD COMB 545 1X[ D + H2C + H3C + H5C + W1A ]
4539. 1 1.0 16 1.0 23 1.0 33 1.0 3 1.0
4540. *
4541. LOAD COMB 546 1X[ D + H2A + H4A + H6A + W1A ]
4542. 1 1.0 14 1.0 24 1.0 34 1.0 3 1.0
4543. *
4544. LOAD COMB 547 1X[ D + H2B + H4B + H6B + W1A ]
4545. 1 1.0 15 1.0 25 1.0 35 1.0 3 1.0
4546. *
4547. LOAD COMB 548 1X[ D + H2C + H4C + H6C + W1A ]
4548. 1 1.0 16 1.0 26 1.0 36 1.0 3 1.0
4549. *
4550. LOAD COMB 549 1X[ D + H1A + H3A + H5A + W1B ]
4551. 1 1.0 11 1.0 21 1.0 31 1.0 4 1.0
4552. *
4553. LOAD COMB 550 1X[ D + H1B + H3B + H5B + W1B ]
4554. 1 1.0 12 1.0 22 1.0 32 1.0 4 1.0
4555. *
4556. LOAD COMB 551 1X[ D + H1C + H3C + H5C + W1B ]
4557. 1 1.0 13 1.0 23 1.0 33 1.0 4 1.0
4558. *
4559. LOAD COMB 552 1X[ D + H1A + H4A + H6A + W1B ]
4560. 1 1.0 11 1.0 24 1.0 34 1.0 4 1.0
4561. *
4562. LOAD COMB 553 1X[ D + H1B + H4B + H6B + W1B ]
4563. 1 1.0 12 1.0 25 1.0 35 1.0 4 1.0
4564. *
4565. LOAD COMB 554 1X[ D + H1C + H4C + H6C + W1B ]
4566. 1 1.0 13 1.0 26 1.0 36 1.0 4 1.0
4567. *
4568. LOAD COMB 555 1X[ D + H2A + H3A + H5A + W1B ]
```

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  *
4569. 1 1.0 14 1.0 21 1.0 31 1.0 4 1.0
4570. *
4571. LOAD COMB 556 1X[ D + H2B + H3B + H5B + W1B ]
4572. 1 1.0 15 1.0 22 1.0 32 1.0 4 1.0
4573. *
4574. LOAD COMB 557 1X[ D + H2C + H3C + H5C + W1B ]
4575. 1 1.0 16 1.0 23 1.0 33 1.0 4 1.0
4576. *
4577. LOAD COMB 558 1X[ D + H2A + H4A + H6A + W1B ]
4578. 1 1.0 14 1.0 24 1.0 34 1.0 4 1.0
4579. *
4580. LOAD COMB 559 1X[ D + H2B + H4B + H6B + W1B ]
4581. 1 1.0 15 1.0 25 1.0 35 1.0 4 1.0
4582. *
4583. LOAD COMB 560 1X[ D + H2C + H4C + H6C + W1B ]
4584. 1 1.0 16 1.0 26 1.0 36 1.0 4 1.0
4585. *
4586. LOAD COMB 561 1X[ D ] + 0.8X[ H1A + H3A + H5A ] + 0.7X[ E1A ]
4587. 1 1.0 11 0.8 21 0.8 31 0.8 5 0.7
4588. *
4589. LOAD COMB 562 1X[ D ] + 0.8X[ H1B + H3B + H5B ] + 0.7X[ E1A ]
4590. 1 1.0 12 0.8 22 0.8 32 0.8 5 0.7
4591. *
4592. LOAD COMB 563 1X[ D ] + 0.8X[ H1C + H3C + H5C ] + 0.7X[ E1A ]
4593. 1 1.0 13 0.8 23 0.8 33 0.8 5 0.7
4594. *
4595. LOAD COMB 564 1X[ D ] + 0.8X[ H1A + H4A + H6A ] + 0.7X[ E1A ]
4596. 1 1.0 11 0.8 24 0.8 34 0.8 5 0.7
4597. *
4598. LOAD COMB 565 1X[ D ] + 0.8X[ H1B + H4B + H6B ] + 0.7X[ E1A ]
4599. 1 1.0 12 0.8 25 0.8 35 0.8 5 0.7
4600. *
4601. LOAD COMB 566 1X[ D ] + 0.8X[ H1C + H4C + H6C ] + 0.7X[ E1A ]
4602. 1 1.0 13 0.8 26 0.8 36 0.8 5 0.7
4603. *
4604. LOAD COMB 567 1X[ D ] + 0.8X[ H2A + H3A + H5A ] + 0.7X[ E1A ]
4605. 1 1.0 14 0.8 21 0.8 31 0.8 5 0.7
4606. *
4607. LOAD COMB 568 1X[ D ] + 0.8X[ H2B + H3B + H5B ] + 0.7X[ E1A ]
4608. 1 1.0 15 0.8 22 0.8 32 0.8 5 0.7
4609. *
4610. LOAD COMB 569 1X[ D ] + 0.8X[ H2C + H3C + H5C ] + 0.7X[ E1A ]
4611. 1 1.0 16 0.8 23 0.8 33 0.8 5 0.7
4612. *
4613. LOAD COMB 570 1X[ D ] + 0.8X[ H2A + H4A + H6A ] + 0.7X[ E1A ]
4614. 1 1.0 14 0.8 24 0.8 34 0.8 5 0.7
4615. *
4616. LOAD COMB 571 1X[ D ] + 0.8X[ H2B + H4B + H6B ] + 0.7X[ E1A ]
4617. 1 1.0 15 0.8 25 0.8 35 0.8 5 0.7
4618. *
4619. LOAD COMB 572 1X[ D ] + 0.8X[ H2C + H4C + H6C ] + 0.7X[ E1A ]
4620. 1 1.0 16 0.8 26 0.8 36 0.8 5 0.7
4621. *
4622. LOAD COMB 573 1X[ D ] + 0.8X[ H1A + H3A + H5A ] + 0.7X[ E1B ]
4623. 1 1.0 11 0.8 21 0.8 31 0.8 6 0.7
4624. *
```

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*
LOAD COMB 574 1X[ D ] + 0.8X[ H1B + H3B + H5B ] + 0.7X[ E1B ]
1 1.0 12 0.8 22 0.8 32 0.8 6 0.7
*
LOAD COMB 575 1X[ D ] + 0.8X[ H1C + H3C + H5C ] + 0.7X[ E1B ]
1 1.0 13 0.8 23 0.8 33 0.8 6 0.7
*
LOAD COMB 576 1X[ D ] + 0.8X[ H1A + H4A + H6A ] + 0.7X[ E1B ]
1 1.0 11 0.8 24 0.8 34 0.8 6 0.7
*
LOAD COMB 577 1X[ D ] + 0.8X[ H1B + H4B + H6B ] + 0.7X[ E1B ]
1 1.0 12 0.8 25 0.8 35 0.8 6 0.7
*
LOAD COMB 578 1X[ D ] + 0.8X[ H1C + H4C + H6C ] + 0.7X[ E1B ]
1 1.0 13 0.8 26 0.8 36 0.8 6 0.7
*
LOAD COMB 579 1X[ D ] + 0.8X[ H2A + H3A + H5A ] + 0.7X[ E1B ]
1 1.0 14 0.8 21 0.8 31 0.8 6 0.7
*
LOAD COMB 580 1X[ D ] + 0.8X[ H2B + H3B + H5B ] + 0.7X[ E1B ]
1 1.0 15 0.8 22 0.8 32 0.8 6 0.7
*
LOAD COMB 581 1X[ D ] + 0.8X[ H2C + H3C + H5C ] + 0.7X[ E1B ]
1 1.0 16 0.8 23 0.8 33 0.8 6 0.7
*
LOAD COMB 582 1X[ D ] + 0.8X[ H2A + H4A + H6A ] + 0.7X[ E1B ]
1 1.0 14 0.8 24 0.8 34 0.8 6 0.7
*
LOAD COMB 583 1X[ D ] + 0.8X[ H2B + H4B + H6B ] + 0.7X[ E1B ]
1 1.0 15 0.8 25 0.8 35 0.8 6 0.7
*
LOAD COMB 584 1X[ D ] + 0.8X[ H2C + H4C + H6C ] + 0.7X[ E1B ]
1 1.0 16 0.8 26 0.8 36 0.8 6 0.7
*
LOAD COMB 585 1X[ D + H1A + H3A + H5A ] + 0.75X[ W1A + L ]
1 1.0 11 1.0 21 1.0 31 1.0 3 0.75 2 0.75
*
LOAD COMB 586 1X[ D + H1B + H3B + H5B ] + 0.75X[ W1A + L ]
1 1.0 12 1.0 22 1.0 32 1.0 3 0.75 2 0.75
*
LOAD COMB 587 1X[ D + H1C + H3C + H5C ] + 0.75X[ W1A + L ]
1 1.0 13 1.0 23 1.0 33 1.0 3 0.75 2 0.75
*
LOAD COMB 588 1X[ D + H1A + H4A + H6A ] + 0.75X[ W1A + L ]
1 1.0 11 1.0 24 1.0 34 1.0 3 0.75 2 0.75
*
LOAD COMB 589 1X[ D + H1B + H4B + H6B ] + 0.75X[ W1A + L ]
1 1.0 12 1.0 25 1.0 35 1.0 3 0.75 2 0.75
*
LOAD COMB 590 1X[ D + H1C + H4C + H6C ] + 0.75X[ W1A + L ]
1 1.0 13 1.0 26 1.0 36 1.0 3 0.75 2 0.75
*
LOAD COMB 591 1X[ D + H2A + H3A + H5A ] + 0.75X[ W1A + L ]
1 1.0 14 1.0 21 1.0 31 1.0 3 0.75 2 0.75
*
LOAD COMB 592 1X[ D + H2B + H3B + H5B ] + 0.75X[ W1A + L ]
1 1.0 15 1.0 22 1.0 32 1.0 3 0.75 2 0.75
```

```
*
4681. *
4682. LOAD COMB 593 1X[ D + H2C + H3C + H5C ] + 0.75X[ W1A + L ]
4683. 1 1.0 16 1.0 23 1.0 33 1.0 3 0.75 2 0.75
4684. *
4685. LOAD COMB 594 1X[ D + H2A + H4A + H6A ] + 0.75X[ W1A + L ]
4686. 1 1.0 14 1.0 24 1.0 34 1.0 3 0.75 2 0.75
4687. *
4688. LOAD COMB 595 1X[ D + H2B + H4B + H6B ] + 0.75X[ W1A + L ]
4689. 1 1.0 15 1.0 25 1.0 35 1.0 3 0.75 2 0.75
4690. *
4691. LOAD COMB 596 1X[ D + H2C + H4C + H6C ] + 0.75X[ W1A + L ]
4692. 1 1.0 16 1.0 26 1.0 36 1.0 3 0.75 2 0.75
4693. *
4694. LOAD COMB 597 1X[ D + H1A + H3A + H5A ] + 0.75X[ W1B + L ]
4695. 1 1.0 11 1.0 21 1.0 31 1.0 4 0.75 2 0.75
4696. *
4697. LOAD COMB 598 1X[ D + H1B + H3B + H5B ] + 0.75X[ W1B + L ]
4698. 1 1.0 12 1.0 22 1.0 32 1.0 4 0.75 2 0.75
4699. *
4700. LOAD COMB 599 1X[ D + H1C + H3C + H5C ] + 0.75X[ W1B + L ]
4701. 1 1.0 13 1.0 23 1.0 33 1.0 4 0.75 2 0.75
4702. *
4703. LOAD COMB 600 1X[ D + H1A + H4A + H6A ] + 0.75X[ W1B + L ]
4704. 1 1.0 11 1.0 24 1.0 34 1.0 4 0.75 2 0.75
4705. *
4706. LOAD COMB 601 1X[ D + H1B + H4B + H6B ] + 0.75X[ W1B + L ]
4707. 1 1.0 12 1.0 25 1.0 35 1.0 4 0.75 2 0.75
4708. *
4709. LOAD COMB 602 1X[ D + H1C + H4C + H6C ] + 0.75X[ W1B + L ]
4710. 1 1.0 13 1.0 26 1.0 36 1.0 4 0.75 2 0.75
4711. *
4712. LOAD COMB 603 1X[ D + H2A + H3A + H5A ] + 0.75X[ W1B + L ]
4713. 1 1.0 14 1.0 21 1.0 31 1.0 4 0.75 2 0.75
4714. *
4715. LOAD COMB 604 1X[ D + H2B + H3B + H5B ] + 0.75X[ W1B + L ]
4716. 1 1.0 15 1.0 22 1.0 32 1.0 4 0.75 2 0.75
4717. *
4718. LOAD COMB 605 1X[ D + H2C + H3C + H5C ] + 0.75X[ W1B + L ]
4719. 1 1.0 16 1.0 23 1.0 33 1.0 4 0.75 2 0.75
4720. *
4721. LOAD COMB 606 1X[ D + H2A + H4A + H6A ] + 0.75X[ W1B + L ]
4722. 1 1.0 14 1.0 24 1.0 34 1.0 4 0.75 2 0.75
4723. *
4724. LOAD COMB 607 1X[ D + H2B + H4B + H6B ] + 0.75X[ W1B + L ]
4725. 1 1.0 15 1.0 25 1.0 35 1.0 4 0.75 2 0.75
4726. *
4727. LOAD COMB 608 1X[ D + H2C + H4C + H6C ] + 0.75X[ W1B + L ]
4728. 1 1.0 16 1.0 26 1.0 36 1.0 4 0.75 2 0.75
4729. *
4730. LOAD COMB 609 1X[ D ] + 0.8X[ H1A + H3A + H5A ] + 0.75X0.7X[ E1A ] + 0.75X[ L ]
4731. 1 1.0 11 0.8 21 0.8 31 0.8 5 0.525 2 0.75
4732. *
4733. LOAD COMB 610 1X[ D ] + 0.8X[ H1B + H3B + H5B ] + 0.75X0.7X[ E1A ] + 0.75X[ L ]
4734. 1 1.0 12 0.8 22 0.8 32 0.8 5 0.525 2 0.75
4735. *
4736. LOAD COMB 611 1X[ D ] + 0.8X[ H1C + H3C + H5C ] + 0.75X0.7X[ E1A ] + 0.75X[ L ]
```

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  *
4737. 1 1.0 13 0.8 23 0.8 33 0.8 5 0.525 2 0.75
4738. *
4739. LOAD COMB 612 1X[ D ] + 0.8X[ H1A + H4A + H6A ] + 0.75X0.7X[ E1A ] + 0.75X[ L ]
4740. 1 1.0 11 0.8 24 0.8 34 0.8 5 0.525 2 0.75
4741. *
4742. LOAD COMB 613 1X[ D ] + 0.8X[ H1B + H4B + H6B ] + 0.75X0.7X[ E1A ] + 0.75X[ L ]
4743. 1 1.0 12 0.8 25 0.8 35 0.8 5 0.525 2 0.75
4744. *
4745. LOAD COMB 614 1X[ D ] + 0.8X[ H1C + H4C + H6C ] + 0.75X0.7X[ E1A ] + 0.75X[ L ]
4746. 1 1.0 13 0.8 26 0.8 36 0.8 5 0.525 2 0.75
4747. *
4748. LOAD COMB 615 1X[ D ] + 0.8X[ H2A + H3A + H5A ] + 0.75X0.7X[ E1A ] + 0.75X[ L ]
4749. 1 1.0 14 0.8 21 0.8 31 0.8 5 0.525 2 0.75
4750. *
4751. LOAD COMB 616 1X[ D ] + 0.8X[ H2B + H3B + H5B ] + 0.75X0.7X[ E1A ] + 0.75X[ L ]
4752. 1 1.0 15 0.8 22 0.8 32 0.8 5 0.525 2 0.75
4753. *
4754. LOAD COMB 617 1X[ D ] + 0.8X[ H2C + H3C + H5C ] + 0.75X0.7X[ E1A ] + 0.75X[ L ]
4755. 1 1.0 16 0.8 23 0.8 33 0.8 5 0.525 2 0.75
4756. *
4757. LOAD COMB 618 1X[ D ] + 0.8X[ H2A + H4A + H6A ] + 0.75X0.7X[ E1A ] + 0.75X[ L ]
4758. 1 1.0 14 0.8 24 0.8 34 0.8 5 0.525 2 0.75
4759. *
4760. LOAD COMB 619 1X[ D ] + 0.8X[ H2B + H4B + H6B ] + 0.75X0.7X[ E1A ] + 0.75X[ L ]
4761. 1 1.0 15 0.8 25 0.8 35 0.8 5 0.525 2 0.75
4762. *
4763. LOAD COMB 620 1X[ D ] + 0.8X[ H2C + H4C + H6C ] + 0.75X0.7X[ E1A ] + 0.75X[ L ]
4764. 1 1.0 16 0.8 26 0.8 36 0.8 5 0.525 2 0.75
4765. *
4766. LOAD COMB 621 1X[ D ] + 0.8X[ H1A + H3A + H5A ] + 0.75X0.7X[ E1B ] + 0.75X[ L ]
4767. 1 1.0 11 0.8 21 0.8 31 0.8 6 0.525 2 0.75
4768. *
4769. LOAD COMB 622 1X[ D ] + 0.8X[ H1B + H3B + H5B ] + 0.75X0.7X[ E1B ] + 0.75X[ L ]
4770. 1 1.0 12 0.8 22 0.8 32 0.8 6 0.525 2 0.75
4771. *
4772. LOAD COMB 623 1X[ D ] + 0.8X[ H1C + H3C + H5C ] + 0.75X0.7X[ E1B ] + 0.75X[ L ]
4773. 1 1.0 13 0.8 23 0.8 33 0.8 6 0.525 2 0.75
4774. *
4775. LOAD COMB 624 1X[ D ] + 0.8X[ H1A + H4A + H6A ] + 0.75X0.7X[ E1B ] + 0.75X[ L ]
4776. 1 1.0 11 0.8 24 0.8 34 0.8 6 0.525 2 0.75
4777. *
4778. LOAD COMB 625 1X[ D ] + 0.8X[ H1B + H4B + H6B ] + 0.75X0.7X[ E1B ] + 0.75X[ L ]
4779. 1 1.0 12 0.8 25 0.8 35 0.8 6 0.525 2 0.75
4780. *
4781. LOAD COMB 626 1X[ D ] + 0.8X[ H1C + H4C + H6C ] + 0.75X0.7X[ E1B ] + 0.75X[ L ]
4782. 1 1.0 13 0.8 26 0.8 36 0.8 6 0.525 2 0.75
4783. *
4784. LOAD COMB 627 1X[ D ] + 0.8X[ H2A + H3A + H5A ] + 0.75X0.7X[ E1B ] + 0.75X[ L ]
4785. 1 1.0 14 0.8 21 0.8 31 0.8 6 0.525 2 0.75
4786. *
4787. LOAD COMB 628 1X[ D ] + 0.8X[ H2B + H3B + H5B ] + 0.75X0.7X[ E1B ] + 0.75X[ L ]
4788. 1 1.0 15 0.8 22 0.8 32 0.8 6 0.525 2 0.75
4789. *
4790. LOAD COMB 629 1X[ D ] + 0.8X[ H2C + H3C + H5C ] + 0.75X0.7X[ E1B ] + 0.75X[ L ]
4791. 1 1.0 16 0.8 23 0.8 33 0.8 6 0.525 2 0.75
4792. *
```

```
*
LOAD COMB 630 1X[ D ] + 0.8X[ H2A + H4A + H6A ] + 0.75X0.7X[ E1B ] + 0.75X[ L ]
1 1.0 14 0.8 24 0.8 34 0.8 6 0.525 2 0.75
*
LOAD COMB 631 1X[ D ] + 0.8X[ H2B + H4B + H6B ] + 0.75X0.7X[ E1B ] + 0.75X[ L ]
1 1.0 15 0.8 25 0.8 35 0.8 6 0.525 2 0.75
*
LOAD COMB 632 1X[ D ] + 0.8X[ H2C + H4C + H6C ] + 0.75X0.7X[ E1B ] + 0.75X[ L ]
1 1.0 16 0.8 26 0.8 36 0.8 6 0.525 2 0.75
*
LOAD COMB 633 0.6X[ D ] + [ W1A + H1A + H3A + H5A ]
1 0.6 3 1.0 11 1.0 21 1.0 31 1.0
*
LOAD COMB 634 0.6X[ D ] + [ W1A + H1B + H3B + H5B ]
1 0.6 3 1.0 12 1.0 22 1.0 32 1.0
*
LOAD COMB 635 0.6X[ D ] + [ W1A + H1C + H3C + H5C ]
1 0.6 3 1.0 13 1.0 23 1.0 33 1.0
*
LOAD COMB 636 0.6X[ D ] + [ W1A + H1A + H4A + H6A ]
1 0.6 3 1.0 11 1.0 24 1.0 34 1.0
*
LOAD COMB 637 0.6X[ D ] + [ W1A + H1B + H4B + H6B ]
1 0.6 3 1.0 12 1.0 25 1.0 35 1.0
*
LOAD COMB 638 0.6X[ D ] + [ W1A + H1C + H4C + H6C ]
1 0.6 3 1.0 13 1.0 26 1.0 36 1.0
*
LOAD COMB 639 0.6X[ D ] + [ W1A + H2A + H3A + H5A ]
1 0.6 3 1.0 14 1.0 21 1.0 31 1.0
*
LOAD COMB 640 0.6X[ D ] + [ W1A + H2B + H3B + H5B ]
1 0.6 3 1.0 15 1.0 22 1.0 32 1.0
*
LOAD COMB 641 0.6X[ D ] + [ W1A + H2C + H3C + H5C ]
1 0.6 3 1.0 16 1.0 23 1.0 33 1.0
*
LOAD COMB 642 0.6X[ D ] + [ W1A + H2A + H4A + H6A ]
1 0.6 3 1.0 14 1.0 24 1.0 34 1.0
*
LOAD COMB 643 0.6X[ D ] + [ W1A + H2B + H4B + H6B ]
1 0.6 3 1.0 15 1.0 25 1.0 35 1.0
*
LOAD COMB 644 0.6X[ D ] + [ W1A + H2C + H4C + H6C ]
1 0.6 3 1.0 16 1.0 26 1.0 36 1.0
*
LOAD COMB 645 0.6X[ D ] + [ W1B + H1A + H3A + H5A ]
1 0.6 4 1.0 11 1.0 21 1.0 31 1.0
*
LOAD COMB 646 0.6X[ D ] + [ W1B + H1B + H3B + H5B ]
1 0.6 4 1.0 12 1.0 22 1.0 32 1.0
*
LOAD COMB 647 0.6X[ D ] + [ W1B + H1C + H3C + H5C ]
1 0.6 4 1.0 13 1.0 23 1.0 33 1.0
*
LOAD COMB 648 0.6X[ D ] + [ W1B + H1A + H4A + H6A ]
1 0.6 4 1.0 11 1.0 24 1.0 34 1.0
```

```
*
4849. *
4850. LOAD COMB 649 0.6X[ D ] + [ W1B + H1B + H4B + H6B ]
4851. 1 0.6 4 1.0 12 1.0 25 1.0 35 1.0
4852. *
4853. LOAD COMB 650 0.6X[ D ] + [ W1B + H1C + H4C + H6C ]
4854. 1 0.6 4 1.0 13 1.0 26 1.0 36 1.0
4855. *
4856. LOAD COMB 651 0.6X[ D ] + [ W1B + H2A + H3A + H5A ]
4857. 1 0.6 4 1.0 14 1.0 21 1.0 31 1.0
4858. *
4859. LOAD COMB 652 0.6X[ D ] + [ W1B + H2B + H3B + H5B ]
4860. 1 0.6 4 1.0 15 1.0 22 1.0 32 1.0
4861. *
4862. LOAD COMB 653 0.6X[ D ] + [ W1B + H2C + H3C + H5C ]
4863. 1 0.6 4 1.0 16 1.0 23 1.0 33 1.0
4864. *
4865. LOAD COMB 654 0.6X[ D ] + [ W1B + H2A + H4A + H6A ]
4866. 1 0.6 4 1.0 14 1.0 24 1.0 34 1.0
4867. *
4868. LOAD COMB 655 0.6X[ D ] + [ W1B + H2B + H4B + H6B ]
4869. 1 0.6 4 1.0 15 1.0 25 1.0 35 1.0
4870. *
4871. LOAD COMB 656 0.6X[ D ] + [ W1B + H2C + H4C + H6C ]
4872. 1 0.6 4 1.0 16 1.0 26 1.0 36 1.0
4873. *
4874. LOAD COMB 657 0.6X[ D ] + 0.7X[ E1A ] + 0.8X[ H1A + H3A + H5A ]
4875. 1 0.6 5 0.7 11 0.8 21 0.8 31 0.8
4876. *
4877. LOAD COMB 658 0.6X[ D ] + 0.7X[ E1A ] + 0.8X[ H1B + H3B + H5B ]
4878. 1 0.6 5 0.7 12 0.8 22 0.8 32 0.8
4879. *
4880. LOAD COMB 659 0.6X[ D ] + 0.7X[ E1A ] + 0.8X[ H1C + H3C + H5C ]
4881. 1 0.6 5 0.7 13 0.8 23 0.8 33 0.8
4882. *
4883. LOAD COMB 660 0.6X[ D ] + 0.7X[ E1A ] + 0.8X[ H1A + H4A + H6A ]
4884. 1 0.6 5 0.7 11 0.8 24 0.8 34 0.8
4885. *
4886. LOAD COMB 661 0.6X[ D ] + 0.7X[ E1A ] + 0.8X[ H1B + H4B + H6B ]
4887. 1 0.6 5 0.7 12 0.8 25 0.8 35 0.8
4888. *
4889. LOAD COMB 662 0.6X[ D ] + 0.7X[ E1A ] + 0.8X[ H1C + H4C + H6C ]
4890. 1 0.6 5 0.7 13 0.8 26 0.8 36 0.8
4891. *
4892. LOAD COMB 663 0.6X[ D ] + 0.7X[ E1A ] + 0.8X[ H2A + H3A + H5A ]
4893. 1 0.6 5 0.7 14 0.8 21 0.8 31 0.8
4894. *
4895. LOAD COMB 664 0.6X[ D ] + 0.7X[ E1A ] + 0.8X[ H2B + H3B + H5B ]
4896. 1 0.6 5 0.7 15 0.8 22 0.8 32 0.8
4897. *
4898. LOAD COMB 665 0.6X[ D ] + 0.7X[ E1A ] + 0.8X[ H2C + H3C + H5C ]
4899. 1 0.6 5 0.7 16 0.8 23 0.8 33 0.8
4900. *
4901. LOAD COMB 666 0.6X[ D ] + 0.7X[ E1A ] + 0.8X[ H2A + H4A + H6A ]
4902. 1 0.6 5 0.7 14 0.8 24 0.8 34 0.8
4903. *
4904. LOAD COMB 667 0.6X[ D ] + 0.7X[ E1A ] + 0.8X[ H2B + H4B + H6B ]
```

```
 *
1 0.6 5 0.7 15 0.8 25 0.8 35 0.8
*
LOAD COMB 668 0.6X[ D ] + 0.7X[ E1A ] + 0.8X[ H2C + H4C + H6C ]
1 0.6 5 0.7 16 0.8 26 0.8 36 0.8
*
LOAD COMB 669 0.6X[ D ] + 0.7X[ E1B ] + 0.8X[ H1A + H3A + H5A ]
1 0.6 6 0.7 11 0.8 21 0.8 31 0.8
*
LOAD COMB 670 0.6X[ D ] + 0.7X[ E1B ] + 0.8X[ H1B + H3B + H5B ]
1 0.6 6 0.7 12 0.8 22 0.8 32 0.8
*
LOAD COMB 671 0.6X[ D ] + 0.7X[ E1B ] + 0.8X[ H1C + H3C + H5C ]
1 0.6 6 0.7 13 0.8 23 0.8 33 0.8
*
LOAD COMB 672 0.6X[ D ] + 0.7X[ E1B ] + 0.8X[ H1A + H4A + H6A ]
1 0.6 6 0.7 11 0.8 24 0.8 34 0.8
*
LOAD COMB 673 0.6X[ D ] + 0.7X[ E1B ] + 0.8X[ H1B + H4B + H6B ]
1 0.6 6 0.7 12 0.8 25 0.8 35 0.8
*
LOAD COMB 674 0.6X[ D ] + 0.7X[ E1B ] + 0.8X[ H1C + H4C + H6C ]
1 0.6 6 0.7 13 0.8 26 0.8 36 0.8
*
LOAD COMB 675 0.6X[ D ] + 0.7X[ E1B ] + 0.8X[ H2A + H3A + H5A ]
1 0.6 6 0.7 14 0.8 21 0.8 31 0.8
*
LOAD COMB 676 0.6X[ D ] + 0.7X[ E1B ] + 0.8X[ H2B + H3B + H5B ]
1 0.6 6 0.7 15 0.8 22 0.8 32 0.8
*
LOAD COMB 677 0.6X[ D ] + 0.7X[ E1B ] + 0.8X[ H2C + H3C + H5C ]
1 0.6 6 0.7 16 0.8 23 0.8 33 0.8
*
LOAD COMB 678 0.6X[ D ] + 0.7X[ E1B ] + 0.8X[ H2A + H4A + H6A ]
1 0.6 6 0.7 14 0.8 24 0.8 34 0.8
*
LOAD COMB 679 0.6X[ D ] + 0.7X[ E1B ] + 0.8X[ H2B + H4B + H6B ]
1 0.6 6 0.7 15 0.8 25 0.8 35 0.8
*
LOAD COMB 680 0.6X[ D ] + 0.7X[ E1B ] + 0.8X[ H2C + H4C + H6C ]
1 0.6 6 0.7 16 0.8 26 0.8 36 0.8
*
*
PERFORM ANALYSIS PRINT STATICS CHECK
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**3888**	**NUMBER OF MEMBERS**	**24**
NUMBER OF PLATES	**3880**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**847**

*

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 3672/ 152/ 885 DOF
TOTAL PRIMARY LOAD CASES = 24, TOTAL DEGREES OF FREEDOM = 20787
TOTAL LOAD COMBINATION CASES = 248 SO FAR.
SIZE OF STIFFNESS MATRIX = 18397 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 235.2/ 82100.6 MB

*

**WARNING: IF THIS UBC/IBC ANALYSIS HAS TENSION/COMPRESSION
OR REPEAT LOAD OR RE-ANALYSIS OR SELECT OPTIMIZE, THEN EACH
UBC/IBC CASE SHOULD BE FOLLOWED BY PERFORM ANALYSIS _CHANGE.

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************************************************************
*  IBC 2003 SEISMIC LOAD ALONG X  :                        *
*     CT =  0.016 Cu =  1.700                              *
*  TIME PERIODS :                                          *
*     Ta =  1.120 T =  0.770 Tuser = 0.770                 *
*  TIME PERIOD USED (T) =  0.770                           *
*  LOAD FACTOR          =  1.000                           *
*  DESIGN BASE SHEAR    =  1.000 X  0.007 X    261428.78   *
*                       =      1765.49  KN                 *
************************************************************
```

```
************************************************************
*  IBC 2003 SEISMIC LOAD ALONG Z  :                        *
*     CT =  0.016 Cu =  1.700                              *
*  TIME PERIODS :                                          *
*     Ta =  1.120 T =  0.770 Tuser = 0.770                 *
*  TIME PERIOD USED (T) =  0.770                           *
*  LOAD FACTOR          =  1.000                           *
*  DESIGN BASE SHEAR    =  1.000 X  0.007 X    261428.78   *
*                       =      1765.49  KN                 *
************************************************************
```

*** LOADING 5 : ADDITIONAL APPLIED FORCES AND MOMENTS ABOUT ORIGIN
DUE TO SEISMIC LOADS(KN ,METE)

FORCE-X = 1765.48 FORCE-Y = 0.00 FORCE-Z = 0.00
MOM-X = 0.00 MOM-Y = 36.63 MOM-Z = -33709.41

*** LOADING 6 : ADDITIONAL APPLIED FORCES AND MOMENTS ABOUT ORIGIN
DUE TO SEISMIC LOADS(KN ,METE)

FORCE-X = 0.00 FORCE-Y = 0.00 FORCE-Z = 1765.48
MOM-X = 33709.41 MOM-Y = 0.00 MOM-Z = 0.00

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 5
SEISMIC LOAD IN X-DIRECTION (E1)

*

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = -0.212532448E-08
Y = 0.190933830E+02
Z = 0.207499186E-01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 5)
SUMMATION FORCE-X = 1765.49
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.00 MY= 36.63 MZ= -33709.23

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 5)
SUMMATION FORCE-X = -1765.49
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.00 MY= -36.63 MZ= 33709.23

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 5)

	MAXIMUMS	AT NODE
X =	9.45482E-02	2568
Y =	-2.93831E-02	2572
Z =	-1.93153E-03	2508
RX=	7.95358E-06	5358
RY=	-3.12931E-06	2537
RZ=	-3.24336E-05	5623

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 6
SEISMIC LOAD IN Z-DIRECTION (E2)

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = -0.212532448E-08
Y = 0.190933830E+02
Z = 0.207499186E-01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 6)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 1765.49

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 33709.23 MY= 0.00 MZ= 0.00

*

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 6)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -0.00
SUMMATION FORCE-Z = -1765.49

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -33709.23 MY= -0.00 MZ= 0.00

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 6)

	MAXIMUMS	AT NODE
X =	2.11854E-03	2504
Y =	-3.00804E-02	13294
Z =	9.60872E-02	2536
RX=	3.37579E-05	5466
RY=	-3.98873E-06	2522
RZ=	-6.95891E-06	5423

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 1
DEAD LOAD (D)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.384029210E-10
Y = 0.794245261E+01
Z = 0.117560242E+00

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.581540163E-09
Y = 0.298967551E+01
Z = 0.101258054E+02

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 1)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -125250.91
SUMMATION FORCE-Z = -3869.64

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 3155.55 MY= 0.00 MZ= 0.00

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 1)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 125250.91
SUMMATION FORCE-Z = 3869.64

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -3155.55 MY= -0.00 MZ= -0.00

*

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 1)

	MAXIMUMS	AT NODE
X =	2.77607E-02	2572
Y =	-4.62768E-01	2555
Z =	-5.84502E-02	2536
RX=	-1.71849E-04	12073
RY=	-3.54736E-05	2554
RZ=	1.53029E-04	12026

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 2
LIVE LOAD (L)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.000000000E+00
Y = 0.124034267E+02
Z = 0.451169626E+00

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 2)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -3081.60
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 1390.32 MY= 0.00 MZ= 0.00

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 2)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 3081.60
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -1390.32 MY= 0.00 MZ= -0.00

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 2)

	MAXIMUMS	AT NODE
X =	-4.07158E-04	13265
Y =	-1.31584E-02	13233
Z =	2.63359E-03	2536
RX=	6.97868E-06	12269
RY=	5.94603E-07	5646
RZ=	-6.26372E-06	13079

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 3
WIND LOAD ALONG X (W1)

*

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = -0.372892098E-15
Y = 0.179970026E+02
Z = 0.144637195E+00

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 3)
SUMMATION FORCE-X = 1653.44
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.00 MY= 239.15 MZ= -29756.96

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 3)
SUMMATION FORCE-X = -1653.44
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.00 MY= -239.15 MZ= 29756.96

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 3)

	MAXIMUMS	AT NODE
X =	8.43510E-02	2568
Y =	-2.62195E-02	2572
Z =	-1.80376E-03	2510
RX=	7.40436E-06	5358
RY=	2.50021E-06	5219
RZ=	-2.83808E-05	5551

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 4
WIND LOAD ALONG Z (W1B)

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = -0.154046555E-15
Y = 0.179798173E+02
Z = 0.143403596E+00

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 4)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 1667.66

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 29984.28 MY= 0.00 MZ= 0.00

*

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 4)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -0.00
SUMMATION FORCE-Z = -1667.66

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -29984.28 MY= -0.00 MZ= 0.00

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 4)

	MAXIMUMS	AT NODE
X =	1.82401E-03	2504
Y =	-2.69855E-02	13294
Z =	8.64272E-02	2536
RX=	3.02120E-05	5466
RY=	3.74549E-06	2550
RZ=	-6.57811E-06	5423

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 11
MATERIAL FILLING WALL HORIZONTAL CASE-I (H1A)

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 11)
SUMMATION FORCE-X = 2.4744445E-12
SUMMATION FORCE-Y = 0.0000000E+00
SUMMATION FORCE-Z = -3.4372086E-12

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -1.5280279E-10 MY= 1.8364427E-12 MZ= -5.0936403E-11

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 11)
SUMMATION FORCE-X = -2.4183923E-11
SUMMATION FORCE-Y = -1.1032680E-10
SUMMATION FORCE-Z = -1.4433214E-10

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -2.9007160E-06 MY= -5.4746076E-10 MZ= -1.7277576E-06

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 11)

	MAXIMUMS	AT NODE
X =	-8.25650E-02	701
Y =	-1.55491E-02	2328
Z =	8.30502E-02	736
RX=	-1.92200E-04	238
RY=	5.98608E-05	12927
RZ=	1.93796E-04	201

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 12
MATERIAL FILLING WALL HORIZONTAL CASE-II (H1B)

*

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 12)
SUMMATION FORCE-X = 3.0068168E-12
SUMMATION FORCE-Y = 0.0000000E+00
SUMMATION FORCE-Z = -2.1115128E-11

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 6.3494850E-10 MY= 1.9274356E-12 MZ= 3.1707606E-11

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 12)
SUMMATION FORCE-X = -6.2225328E-12
SUMMATION FORCE-Y = -1.1588350E-10
SUMMATION FORCE-Z = -1.4320616E-10

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -3.1107058E-06 MY= -8.1992532E-10 MZ= -1.8490623E-06

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 12)

	MAXIMUMS	AT NODE
X =	-8.93158E-02	701
Y =	-1.72031E-02	2328
Z =	8.98376E-02	736
RX=	-2.05972E-04	238
RY=	6.37960E-05	12927
RZ=	2.07659E-04	201

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 13
MATERIAL FILLING WALL HORIZONTAL CASE-III (H1C)

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 13)
SUMMATION FORCE-X = 2.4420972E-12
SUMMATION FORCE-Y = 0.0000000E+00
SUMMATION FORCE-Z = 1.7055666E-11

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -2.1340497E-10 MY= 2.2937407E-12 MZ= 3.5156458E-11

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 13)
SUMMATION FORCE-X = -1.5568185E-11
SUMMATION FORCE-Y = -9.5931147E-11
SUMMATION FORCE-Z = -1.2270934E-10

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -2.5852097E-06 MY= -3.6081129E-10 MZ= -1.5453807E-06

*

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 13)

	MAXIMUMS	AT NODE
X =	-7.27541E-02	701
Y =	-1.33401E-02	2328
Z =	7.31846E-02	736
RX=	-1.71518E-04	238
RY=	5.38729E-05	12927
RZ=	1.72974E-04	201

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 14
MATERIAL EMPTYING WALL HORIZONTAL CASE-I (H2A)

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 14)
SUMMATION FORCE-X = -1.9867658E-12
SUMMATION FORCE-Y = 0.0000000E+00
SUMMATION FORCE-Z = -7.2971544E-12

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 5.1127792E-10 MY= -2.3021451E-12 MZ= -3.4552748E-12

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 14)
SUMMATION FORCE-X = -2.3878723E-11
SUMMATION FORCE-Y = -1.2105018E-10
SUMMATION FORCE-Z = -1.6357162E-10

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -3.1237782E-06 MY= -5.5751068E-10 MZ= -1.8606168E-06

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 14)

	MAXIMUMS	AT NODE
X =	-8.89158E-02	701
Y =	-1.67450E-02	2328
Z =	8.94383E-02	736
RX=	-2.06976E-04	238
RY=	6.44707E-05	12927
RZ=	2.08696E-04	201

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 15
MATERIAL EMPTYING WALL HORIZONTAL CASE-II (H2B)

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 15)
SUMMATION FORCE-X = 2.0159031E-12
SUMMATION FORCE-Y = 0.0000000E+00
SUMMATION FORCE-Z = -2.9008856E-11

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 1.7587220E-10 MY= -1.4203134E-12 MZ= 1.8907675E-11

*

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 15)
SUMMATION FORCE-X = -4.3356238E-11
SUMMATION FORCE-Y = -1.3416577E-10
SUMMATION FORCE-Z = -1.7538950E-10

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -3.3510133E-06 MY= -5.6008127E-10 MZ= -1.9912250E-06

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 15)

	MAXIMUMS	AT NODE
X =	-9.61963E-02	701
Y =	-1.85276E-02	2328
Z =	9.67584E-02	736
RX=	-2.21862E-04	238
RY=	6.87220E-05	12927
RZ=	2.23680E-04	201

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 16
MATERIAL EMPTYING WALL HORIZONTAL CASE-III (H2C)

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 16)
SUMMATION FORCE-X = -3.4233808E-12
SUMMATION FORCE-Y = 0.0000000E+00
SUMMATION FORCE-Z = -6.8033026E-12

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -1.7854394E-11 MY= 6.1170266E-12 MZ= 2.4398865E-11

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 16)
SUMMATION FORCE-X = -2.1284025E-11
SUMMATION FORCE-Y = -1.0953022E-10
SUMMATION FORCE-Z = -1.3663843E-10

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -2.7849420E-06 MY= -5.0044727E-10 MZ= -1.6645413E-06

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 16)

	MAXIMUMS	AT NODE
X =	-7.83569E-02	701
Y =	-1.43664E-02	2328
Z =	7.88208E-02	736
RX=	-1.84752E-04	238
RY=	5.80194E-05	12927
RZ=	1.86320E-04	201

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 21
MATERIAL WALL VERTICAL (FILLING _EMPTYING) CASE-I (H3A)

*

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = -0.699704083E-16
Y = 0.159410976E+02
Z = 0.419822450E-15

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 21)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -46995.46
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -1.6576328E-11 MY= 0.0000000E+00 MZ= 2.9133811E-12

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 21)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 46995.46
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -2.2970892E-06 MY= 6.7263957E-09 MZ= 2.0128554E-06

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 21)

	MAXIMUMS	AT NODE
X =	2.34878E-02	2572
Y =	-2.19419E-01	2161
Z =	-3.51556E-02	2536
RX=	-1.79388E-04	12925
RY=	2.94277E-05	2555
RZ=	-2.08959E-04	13105

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 22
MATERIAL WALL VERTICAL (FILLING _EMPTYING) CASE-II (H3B)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = -0.804784600E-15
Y = 0.160995021E+02
Z = 0.167080281E-14

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 22)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -46988.16
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 2.3833689E-11 MY= 0.0000000E+00 MZ= 2.0829591E-11

*

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 22)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 46988.16
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -2.2981076E-06 MY= 6.9004215E-09 MZ= 2.0119286E-06

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 22)

	MAXIMUMS	AT NODE
X =	2.34791E-02	2572
Y =	-2.19854E-01	2161
Z =	-3.51540E-02	2536
RX=	-1.79362E-04	12925
RY=	2.94226E-05	2555
RZ=	-2.08925E-04	13105

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 23
MATERIAL WALL VERTICAL (FILLING _EMPTYING) CASE-III (H3C)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.699812690E-16
Y = 0.157483689E+02
Z = -0.944747131E-15

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 23)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -46988.16
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -1.4379852E-11 MY= 0.0000000E+00 MZ= 4.3046431E-12

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 23)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 46988.16
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -2.2949045E-06 MY= 6.8960847E-09 MZ= 2.0136008E-06

*

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 23)

	MAXIMUMS	AT NODE
X =	2.34899E-02	2572
Y =	-2.18815E-01	2161
Z =	-3.51458E-02	2536
RX=	-1.79359E-04	12925
RY=	2.94239E-05	2555
RZ=	-2.08929E-04	13105

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 24
MATERIAL WALL VERTICAL (FILLING _EMPTYING) CASE-I (H4A)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.126009319E-15
Y = 0.159413480E+02
Z = 0.114458464E-14

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 24)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -78286.86
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -1.9831222E-10 MY= 0.0000000E+00 MZ= -9.3558952E-12

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 24)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 78286.86
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -3.8266494E-06 MY= 1.1408309E-08 MZ= 3.3530974E-06

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 24)

	MAXIMUMS	AT NODE
X =	3.91269E-02	2572
Y =	-3.65518E-01	2161
Z =	-5.85635E-02	2536
RX=	-2.98832E-04	12925
RY=	4.90218E-05	2555
RZ=	-3.48093E-04	13105

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 25
MATERIAL WALL VERTICAL (FILLING _EMPTYING) CASE-II (H4B)

*

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.214646677E-15
Y = 0.160993050E+02
Z = -0.245310488E-15

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 25)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -107236.88
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -1.8682889E-10 MY= 0.0000000E+00 MZ= 3.5994587E-11

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 25)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 107236.88
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -5.2451536E-06 MY= 1.4732521E-08 MZ= 4.5918452E-06

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 25)

	MAXIMUMS	AT NODE
X =	5.35841E-02	2572
Y =	-5.01752E-01	2161
Z =	-8.02291E-02	2536
RX=	-4.09341E-04	12925
RY=	6.71485E-05	2555
RZ=	-4.76812E-04	13105

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 26
MATERIAL WALL VERTICAL (FILLING _EMPTYING) CASE-III (H4C)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = -0.273081973E-15
Y = 0.157499988E+02
Z = -0.212613821E-14

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 26)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -84289.85
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 4.2553187E-10 MY= 0.0000000E+00 MZ= 2.9414793E-11

*

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 26)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 84289.85
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -4.1166397E-06 MY= 1.2103327E-08 MZ= 3.6120602E-06

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 26)

	MAXIMUMS	AT NODE
X =	4.21374E-02	2572
Y =	-3.92530E-01	2161
Z =	-6.30465E-02	2536
RX=	-3.21744E-04	12925
RY=	5.27822E-05	2555
RZ=	-3.74787E-04	13105

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 31
MATERIAL VERTICAL (ON FLAT SLAB) CASE-I (H5A)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.215768958E-09
Y = 0.799999996E+01
Z = -0.100449909E-09

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 31)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -135436.45
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -2.9512414E-10 MY= 0.0000000E+00 MZ= 1.6338697E-11

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 31)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 135436.45
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 1.4924881E-06 MY= 1.0182374E-08 MZ= -1.2683328E-05

*

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 31)

	MAXIMUMS	AT NODE
X =	-5.47816E-02	13273
Y =	-1.09275E+00	13099
Z =	-5.79914E-02	13268
RX=	-1.23835E-03	13136
RY=	-4.85167E-05	5292
RZ=	1.17818E-03	13105

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 32
MATERIAL VERTICAL (ON FLAT SLAB) CASE-II (H5B)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.215768765E-09
Y = 0.799999996E+01
Z = -0.100449597E-09

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 32)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -127778.24
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -2.7580543E-10 MY= 0.0000000E+00 MZ= -1.3975238E-11

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 32)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 127778.24
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 1.4088192E-06 MY= 1.0255564E-08 MZ= -1.1966117E-05

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 32)

	MAXIMUMS	AT NODE
X =	-5.16840E-02	13273
Y =	-1.03096E+00	13099
Z =	-5.47123E-02	13268
RX=	-1.16832E-03	13136
RY=	-4.57733E-05	5292
RZ=	1.11156E-03	13105

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 33
MATERIAL VERTICAL (ON FLAT SLAB) CASE-III (H5C)

*

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.215768766E-09
Y = 0.799999996E+01
Z = -0.100450751E-09

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 33)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -145760.27
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -4.6447115E-11 MY= 0.0000000E+00 MZ= 2.6820126E-11

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 33)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 145760.27
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 1.6063381E-06 MY= 1.1509842E-08 MZ= -1.3649821E-05

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 33)

	MAXIMUMS	AT NODE
X =	-5.89574E-02	13273
Y =	-1.17604E+00	13099
Z =	-6.24119E-02	13268
RX=	-1.33274E-03	13136
RY=	-5.22149E-05	5292
RZ=	1.26799E-03	13105

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 34
MATERIAL VERTICAL (ON FLAT SLAB) CASE-I (H6A)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.215768800E-09
Y = 0.799999996E+01
Z = -0.100449777E-09

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 34)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -104888.15
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -2.1538308E-10 MY= 0.0000000E+00 MZ= -7.5784838E-12

*

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 34)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 104888.15
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 1.1565989E-06 MY= 8.3112975E-09 MZ= -9.8230454E-06

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 34)

	MAXIMUMS	AT NODE
X	-4.24254E-02	13273
Y	-8.46273E-01	13099
Z	-4.49112E-02	13268
RX	-9.59032E-04	13136
RY	-3.75735E-05	5292
RZ	9.12440E-04	13105

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 35
MATERIAL VERTICAL (ON FLAT SLAB) CASE-II (H6B)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.215768713E-09
Y = 0.799999996E+01
Z = -0.100450962E-09

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 35)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -76286.15
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -4.8502297E-11 MY= 0.0000000E+00 MZ= 1.8419569E-11

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 35)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 76286.15
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 8.4115926E-07 MY= 5.7554988E-09 MZ= -7.1439073E-06

```
 *

 MAXIMUM DISPLACEMENTS (  CM  /RADIANS) (LOADING     35)
           MAXIMUMS    AT NODE
    X = -3.08564E-02   13273
    Y = -6.15503E-01   13099
    Z = -3.26643E-02   13268
    RX= -6.97513E-04   13136
    RY= -2.73276E-05    5292
    RZ=  6.63626E-04   13105


        STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO.    36
        MATERIAL VERTICAL (ON FLAT SLAB) CASE-III (H6C)


         CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
       (FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

                     X =  0.215768517E-09
                     Y =  0.799999996E+01
                     Z = -0.100449885E-09

 ***TOTAL APPLIED LOAD ( KN   METE ) SUMMARY (LOADING    36 )
     SUMMATION FORCE-X =           0.00
     SUMMATION FORCE-Y =      -98964.65
     SUMMATION FORCE-Z =           0.00

    SUMMATION OF MOMENTS AROUND THE ORIGIN-
    MX=  6.7821008E-12  MY=  0.0000000E+00  MZ=  1.3358683E-11


 ***TOTAL REACTION LOAD( KN   METE ) SUMMARY (LOADING    36 )
     SUMMATION FORCE-X =          -0.00
     SUMMATION FORCE-Y =       98964.65
     SUMMATION FORCE-Z =           0.00

    SUMMATION OF MOMENTS AROUND THE ORIGIN-
    MX=  1.0905348E-06  MY=  8.0940776E-09  MZ= -9.2678944E-06


 MAXIMUM DISPLACEMENTS (  CM  /RADIANS) (LOADING     36)
           MAXIMUMS    AT NODE
    X = -4.00294E-02   13273
    Y = -7.98480E-01   13099
    Z = -4.23749E-02   13268
    RX= -9.04871E-04   13136
    RY= -3.54516E-05    5292
    RZ=  8.60910E-04   13105



 ************ END OF DATA FROM INTERNAL STORAGE ************


4948. *
4949. LOAD LIST 101 TO 168
4950. *
4951. START CONCRETE DESIGN
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*
CODE ACI
FYMAIN 420000 MEMB 101 TO 172 201 TO 272 301 TO 372 401 TO 472 501 TO 572 -
601 TO 672 701 TO 772 801 TO 872 901 TO 972 1001 TO 1072 1101 TO 1172 1201 -
1202 TO 1272 1301 TO 1372 1401 TO 1472 1501 TO 1572 1601 TO 1672 1701 TO 1772 -
1801 TO 1872 1901 TO 1972 2001 TO 2072 2101 TO 2172 2201 TO 2272 -
2301 TO 2372 2401 TO 2472 10001 TO 10288 10295 TO 10354 10367 TO 10380 10382 -
10384 10386 10388 TO 10404 10406 10408 10410 10412 TO 10426 10439 TO 10452 -
10454 10456 10458 10460 TO 10476 10478 10480 10482 10484 TO 10498 -
10511 TO 10524 10526 10528 10530 10532 TO 10548 10550 10552 10554 -
10556 TO 10570 10577 TO 10596 10598 10600 10602 10604 TO 10620 10622 10624 -
10626 10628 TO 10668 10670 10672 10674 10676 TO 10692 10694 10696 10698 -
10700 TO 10792 10799 10802 10803 10808 10879 TO 10981 11198 TO 11203 11205 -
11206 11210 TO 11222 11225 TO 11253 11255 TO 11259 11371 TO 11399 -
11487 TO 11501 11503 11511 TO 12067 12070 12071 12074 12075 12869 TO 12956
MINSEC 10 ALL
FYSEC 420000 ALL
CLS 0.06 ALL
CLB 0.06 MEMB 101 TO 172 201 TO 272 301 TO 372 401 TO 472 501 TO 572 -
601 TO 672 701 TO 772 801 TO 872 901 TO 972 1001 TO 1072 1101 TO 1172 1201 -
1202 TO 1272 1301 TO 1372 1401 TO 1472 1501 TO 1572 1601 TO 1672 1701 TO 1772 -
1801 TO 1872 1901 TO 1972 2001 TO 2072 2101 TO 2172 2201 TO 2272 -
2301 TO 2372 2401 TO 2472 10001 TO 10288 10295 TO 10354 10367 TO 10380 10382 -
10384 10386 10388 TO 10404 10406 10408 10410 10412 TO 10426 10439 TO 10452 -
10454 10456 10458 10460 TO 10476 10478 10480 10482 10484 TO 10498 -
10511 TO 10524 10526 10528 10530 10532 TO 10548 10550 10552 10554 -
10556 TO 10570 10577 TO 10596 10598 10600 10602 10604 TO 10620 10622 10624 -
10626 10628 TO 10668 10670 10672 10674 10676 TO 10692 10694 10696 10698 -
10700 TO 10792 10799 10802 10803 10808 10879 TO 10981 11198 TO 11203 11205 -
11206 11210 TO 11222 11225 TO 11253 11255 TO 11259 11371 TO 11399 -
11487 TO 11501 11503 11511 TO 12067 12070 12071 12074 12075 12869 TO 12956
CLT 0.06 MEMB 101 TO 172 201 TO 272 301 TO 372 401 TO 472 501 TO 572 -
601 TO 672 701 TO 772 801 TO 872 901 TO 972 1001 TO 1072 1101 TO 1172 1201 -
1202 TO 1272 1301 TO 1372 1401 TO 1472 1501 TO 1572 1601 TO 1672 1701 TO 1772 -
1801 TO 1872 1901 TO 1972 2001 TO 2072 2101 TO 2172 2201 TO 2272 -
2301 TO 2372 2401 TO 2472 10001 TO 10288 10295 TO 10354 10367 TO 10380 10382 -
10384 10386 10388 TO 10404 10406 10408 10410 10412 TO 10426 10439 TO 10452 -
10454 10456 10458 10460 TO 10476 10478 10480 10482 10484 TO 10498 -
10511 TO 10524 10526 10528 10530 10532 TO 10548 10550 10552 10554 -
10556 TO 10570 10577 TO 10596 10598 10600 10602 10604 TO 10620 10622 10624 -
10626 10628 TO 10668 10670 10672 10674 10676 TO 10692 10694 10696 10698 -
10700 TO 10792 10799 10802 10803 10808 10879 TO 10981 11198 TO 11203 11205 -
11206 11210 TO 11222 11225 TO 11253 11255 TO 11259 11371 TO 11399 -
11487 TO 11501 11503 11511 TO 12067 12070 12071 12074 12075 12869 TO 12956
MAXMAIN 32 ALL
MINMAIN 32 ALL
FC 35000 MEMB 101 TO 172 201 TO 272 301 TO 372 401 TO 472 501 TO 572 -
601 TO 672 701 TO 772 801 TO 872 901 TO 972 1001 TO 1072 1101 TO 1172 1201 -
1202 TO 1272 1301 TO 1372 1401 TO 1472 1501 TO 1572 1601 TO 1672 1701 TO 1772 -
1801 TO 1872 1901 TO 1972 2001 TO 2072 2101 TO 2172 2201 TO 2272 -
2301 TO 2372 2401 TO 2472 10001 TO 10288 10295 TO 10354 10367 TO 10380 10382 -
10384 10386 10388 TO 10404 10406 10408 10410 10412 TO 10426 10439 TO 10452 -
10454 10456 10458 10460 TO 10476 10478 10480 10482 10484 TO 10498 -
10511 TO 10524 10526 10528 10530 10532 TO 10548 10550 10552 10554 -
10556 TO 10570 10577 TO 10596 10598 10600 10602 10604 TO 10620 10622 10624 -
10626 10628 TO 10668 10670 10672 10674 10676 TO 10692 10694 10696 10698 -
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*
5007. 10700 TO 10792 10799 10802 10803 10808 10879 TO 10981 11198 TO 11203 11205 -
5008. 11206 11210 TO 11222 11225 TO 11253 11255 TO 11259 11371 TO 11399 -
5009. 11487 TO 11501 11503 11511 TO 12067 12070 12071 12074 12075 12869 TO 12956
5010. TRACK 0 ALL
5011. DESIGN COLUMN 10799 10802 10803 10808 12070 12071 12074 12075 12869 TO 12884
```

ACI 318-11 COLUMN NO. 10799 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE -1000.0 X1500.0 MMS, TIED
AREA OF STEEL REQUIRED = 16575.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
24 - 32 MM	1.287	137	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 10802 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE -1250.0 X2000.0 MMS, TIED
AREA OF STEEL REQUIRED = 25875.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
36 - 32 MM	1.158	116	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 10803 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE -1250.0 X2000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 25000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI

*
32 - 32 MM 1.029 101 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 10808 DESIGN RESULTS
==========

FY - 420.0 FC - 35.0 MPA, RECT SIZE -1000.0 X1500.0 MMS, TIED
AREA OF STEEL REQUIRED = 16470.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
24 - 32 MM	1.287	160	STA	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 12070 DESIGN RESULTS
==========

FY - 420.0 FC - 35.0 MPA, CIRC SIZE 1400.0 MMS DIAMETER TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 15393.8 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 32 MM	1.045	101	END	0.650

(EQUALLY SPACED)
TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 12071 DESIGN RESULTS
==========

FY - 420.0 FC - 35.0 MPA, CIRC SIZE 1400.0 MMS DIAMETER TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 15393.8 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 32 MM	1.045	101	END	0.650

(EQUALLY SPACED)
TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 12074 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, CIRC SIZE 1400.0 MMS DIAMETER TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 15393.8 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 32 MM	1.045	101	END	0.650

(EQUALLY SPACED)
TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 12075 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, CIRC SIZE 1400.0 MMS DIAMETER TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 15393.8 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 32 MM	1.045	101	END	0.650

(EQUALLY SPACED)
TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 12869 DESIGN RESULTS

*
FY - 420.0 FC - 35.0 MPA, RECT SIZE -1000.0 X1500.0 MMS, TIED
AREA OF STEEL REQUIRED = 16575.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
24 - 32 MM	1.287	137	STA	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 12870 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE -1250.0 X2000.0 MMS, TIED
AREA OF STEEL REQUIRED = 25875.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
36 - 32 MM	1.158	116	STA	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 12871 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE -1250.0 X2000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 25000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 32 MM	1.029	101	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 12872 DESIGN RESULTS
==========

FY - 420.0 FC - 35.0 MPA, RECT SIZE -1000.0 X1500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 15000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 32 MM	1.072	101	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 12873 DESIGN RESULTS
==========

FY - 420.0 FC - 35.0 MPA, CIRC SIZE 1400.0 MMS DIAMETER TIED
AREA OF STEEL REQUIRED = 16255.9 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
21 - 32 MM	1.097	139	END	0.650

(EQUALLY SPACED)
TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 12874 DESIGN RESULTS
==========

FY - 420.0 FC - 35.0 MPA, CIRC SIZE 1400.0 MMS DIAMETER TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 15393.8 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 32 MM	1.045	101	END	0.650

(EQUALLY SPACED)
TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 12875 DESIGN RESULTS
==========

FY - 420.0 FC - 35.0 MPA, CIRC SIZE 1400.0 MMS DIAMETER TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 15393.8 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 32 MM	1.045	101	END	0.650

(EQUALLY SPACED)
TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 12876 DESIGN RESULTS
==========

FY - 420.0 FC - 35.0 MPA, CIRC SIZE 1400.0 MMS DIAMETER TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 15393.8 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 32 MM	1.045	101	END	0.650

(EQUALLY SPACED)
TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 12877 DESIGN RESULTS
==========

FY - 420.0 FC - 35.0 MPA, RECT SIZE -1000.0 X1500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 15000.0 SQ. MM

*

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 32 MM	1.072	101	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 12878 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE -1250.0 X2000.0 MMS, TIED
AREA OF STEEL REQUIRED = 25875.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
36 - 32 MM	1.158	120	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 12879 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE -1250.0 X2000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 25000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 32 MM	1.029	101	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 12880 DESIGN RESULTS

*

FY - 420.0 FC - 35.0 MPA, RECT SIZE -1000.0 X1500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 15000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 32 MM	1.072	101	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 12881 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, CIRC SIZE 1400.0 MMS DIAMETER TIED
AREA OF STEEL REQUIRED = 16255.9 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
21 - 32 MM	1.097	139	STA	0.650

(EQUALLY SPACED)
TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 12882 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, CIRC SIZE 1400.0 MMS DIAMETER TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 15393.8 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 32 MM	1.045	101	END	0.650

(EQUALLY SPACED)
TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 12883 DESIGN RESULTS
==========

FY - 420.0 FC - 35.0 MPA, CIRC SIZE 1400.0 MMS DIAMETER TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 15393.8 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 32 MM	1.045	101	END	0.650

(EQUALLY SPACED)
TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 12884 DESIGN RESULTS
==========

FY - 420.0 FC - 35.0 MPA, CIRC SIZE 1400.0 MMS DIAMETER TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 15393.8 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 32 MM	1.045	101	END	0.650

(EQUALLY SPACED)
TIE BAR NUMBER 10 SPACING 480.00 MM

**********END OF COLUMN DESIGN RESULTS**********

```
5012. *DESIGN ELEMENT 101 TO 172 201 TO 272 301 TO 372 401 TO 472 501 TO 572 601 -
5013. *602 TO 672 701 TO 772 801 TO 872 901 TO 972 1001 TO 1072 1101 TO 1172 1201 -
5014. *1202 TO 1272 1301 TO 1372 1401 TO 1472 1501 TO 1572 1601 TO 1672 1701 TO 1772
5015. *1801 TO 1872 1901 TO 1972 2001 TO 2072 2101 TO 2172 2201 TO 2272 -
5016. *2301 TO 2372 2401 TO 2472 10001 TO 10288 10295 TO 10354 10367 TO 10380 10382 -
5017. *10384 10386 10388 TO 10404 10406 10408 10410 10412 TO 10426 10439 TO 10452 -
5018. *10454 10456 10458 10460 TO 10476 10478 10480 10482 10484 TO 10498 -
5019. *10511 TO 10524 10526 10528 10530 10532 TO 10548 10550 10552 10554 -
5020. *10556 TO 10570 10577 TO 10596 10598 10600 10602 10604 TO 10620 10622 10624 -
5021. *10626 10628 TO 10668 10670 10672 10674 10676 TO 10692 10694 10696 10698 -
5022. *10700 TO 10792 10879 TO 10981 11198 TO 11203 11205 11206 11210 TO 11222 -
```

```
  *
5023. *11225 TO 11253 11255 TO 11259 11371 TO 11399 11487 TO 11501 11503 -
5024. *11511 TO 12067 12885 TO 12956
5025. END CONCRETE DESIGN
5026. FINISH


     ********** END OF THE STAAD.Pro RUN **********

       **** DATE= JUN 24,2020   TIME= 16:32:57 ****

    ************************************************************
    *   For technical assistance on STAAD.Pro, please visit    *
    *   http://selectservices.bentley.com/en-US/               *
    *                                                          *
    *   Details about additional assistance from               *
    *   Bentley and Partners can be found at program menu      *
    *   Help->Technical Support                                *
    *                                                          *
    *      Copyright (c) 1997-2016 Bentley Systems, Inc.       *
    *                http://www.bentley.com                    *
    ************************************************************
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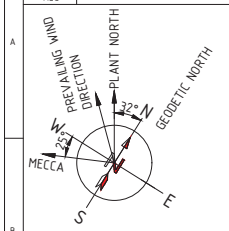
ATTACHMENT 3 MECHANICAL INPUT DRAWINGS

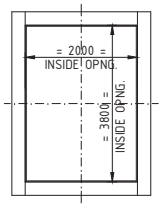
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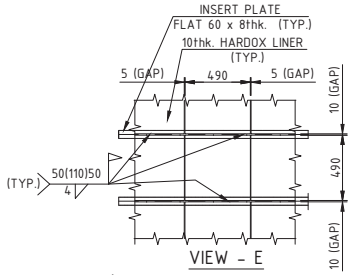
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DRAWING NUMBER MD-216-3210-EG-ME-ELV-1009 REV A02

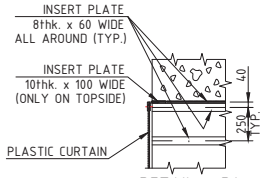




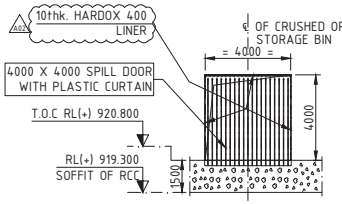
VIEW C-C
SCALE 1:40



VIEW - E
(TYP. LINER FIXING DETAILS
ON VERTICAL RCC WALL)
SCALE 1:20



DETAIL - D4
SCALE 1:20



VIEW - G
SCALE 1:125

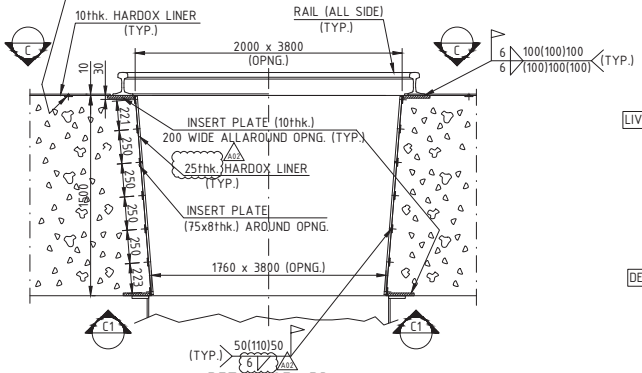
ABBREVIATION

W.P.:	WORKING POINT
℄:	CENTER LINE
DN:	DOWNWARD
UP:	UPWARD
CRS.:	CENTERS
thk.:	THICKNESS
RCC:	REINFORCED CEMENT CONCRETE
deg.:	DEGREE
B.O.M.R:	BOTTOM OF MONORAIL
OPNG.:	OPENING
TR:	TRESTLE
TYP.:	TYPICAL
PROJ:	PROJECTED
W:	WIDTH
H:	HEIGHT

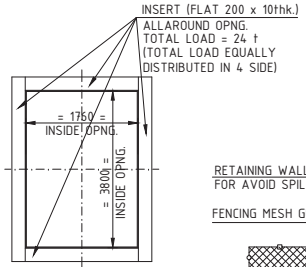
STORAGE MATERIAL CHARACTERISTICS

PARTICULARS	SPECIFICATION
STORAGE MATERIAL	CRUSHED GOLD ORE
LUMP SIZE (mm)	(-)10
BULK DENSITY (t/m³)	1.6 (FOR VOLUME CALCULATION)
	1.9 (FOR STRUCTURAL CALCULATION)
MOISTURE CONTENT (%)	2
ANGLE OF REPOSE (deg.)	37
STORAGE PERIOD (day)	1
STORAGE CAPACITY (m³)	~9600 (INCLUDING DEAD VOLUME)
CHEMICAL PROPERTIES OF GOLD ORE :	REFER PROCESS DESIGN CRITERIA DOC. NO. MD-216-3000-PR-CRT-0001
DEAD VOLUME TO BE CLEARED PERIODICALLY TO AVOID CAKE FORMATION	

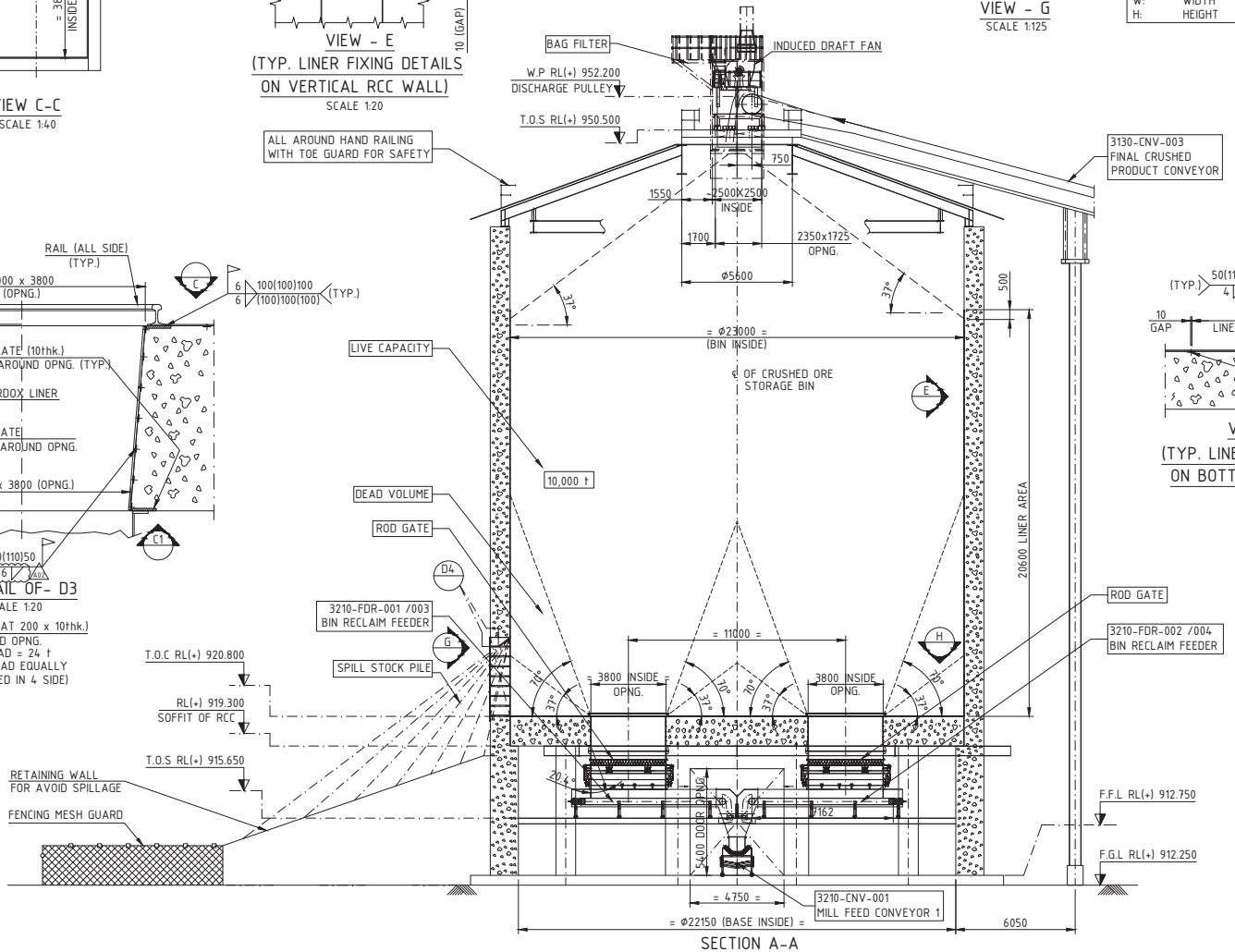
NOTE:
1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+)913.000 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.500 m
4. F.F.L, F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN TENTATIVE.



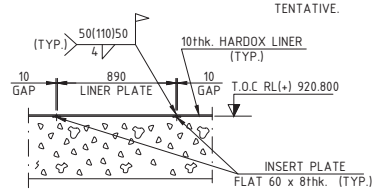
DETAIL OF- D3
SCALE 1:20



VIEW C1-C1
SCALE 1:40



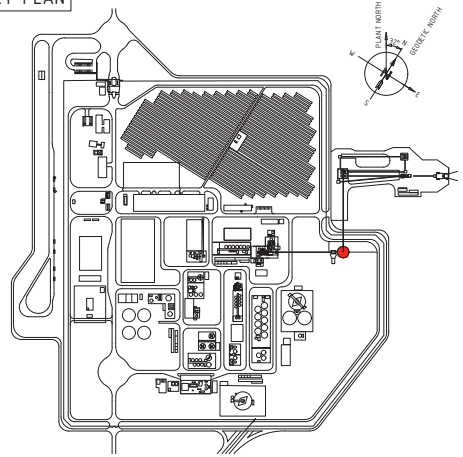
SECTION A-A
SCALE 1:125



VIEW - H
(TYP. LINER FIXING DETAILS
ON BOTTOM OF RCC BIN)
SCALE 1:20

CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

KEY PLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3210-EG-ME-LPX-1006	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN								
MD-216-3210-EG-ME-LPX-1005	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN								
MD-216-3210-EG-ME-LPX-1004	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN	A02	10/04/2020	ISSUED FOR APPROVAL & REVISED AS PER EMPLOYER	HKD	BRD	BD	BGR	
MD-216-3210-EG-ME-ELV-1010	CRUSHED ORE STORAGE BIN & TRANSFER - SECTION			COMMENTS DT. 20.03.2020 & REVISED AS MARKED A02					
MD-216-3210-EG-ME-LPX-1003	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN	A01	21/02/2020	ISSUED FOR APPROVAL	HKD	BRD	BD	BGR	

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Gold Arabian Mining Company

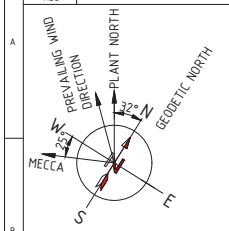
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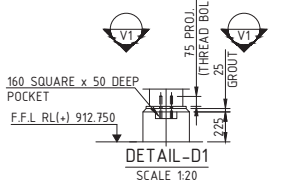
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DRAWN	HKD	18/02/2020
CHECKED	BRD	20/02/2020
ENGINEER	BD	20/02/2020
APPROVED	BGR	21/02/2020
EMPLOYER		
SCALE	1:125, 1:40, 1:20	A1

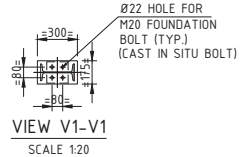
MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
GRINDING & CLASSIFICATION
CRUSHED ORE STORAGE BIN & TRANSFER - ELEVATION

DRAWING NUMBER: MD-216-3210-EG-ME-ELV-1009 — REV: A02

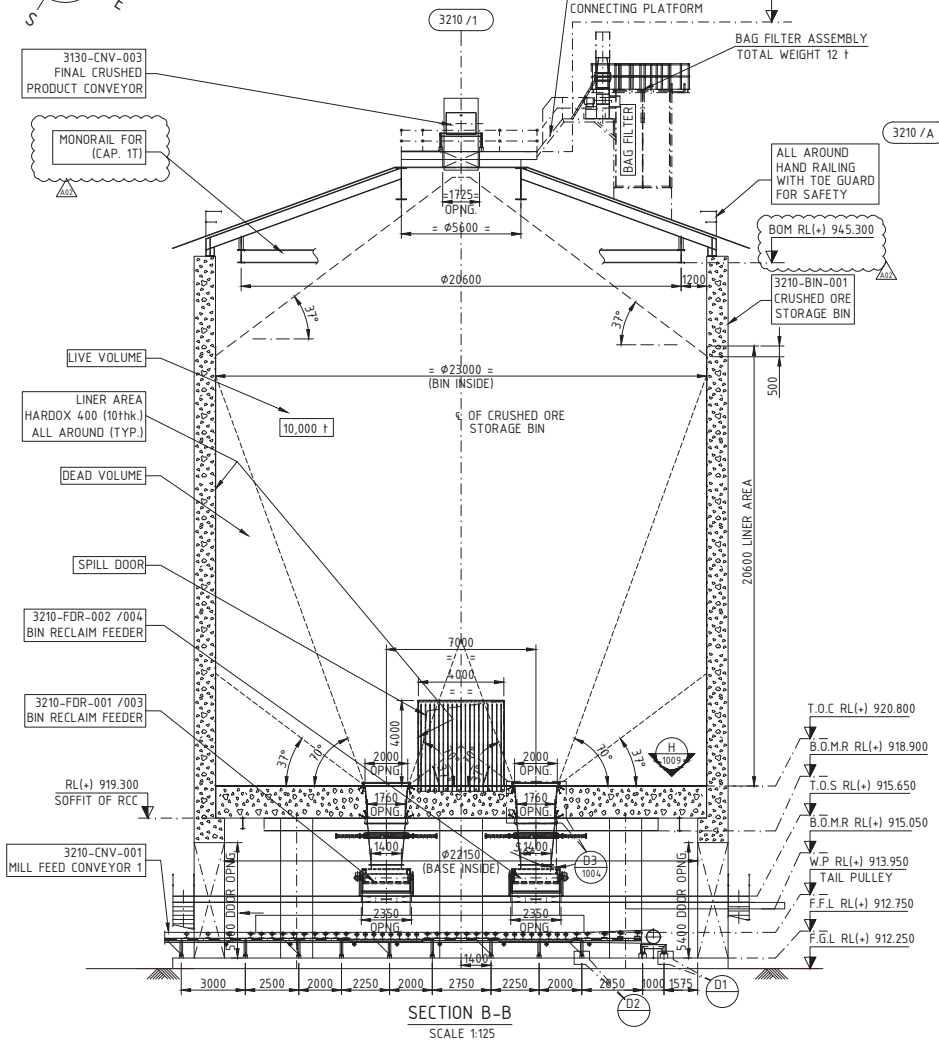




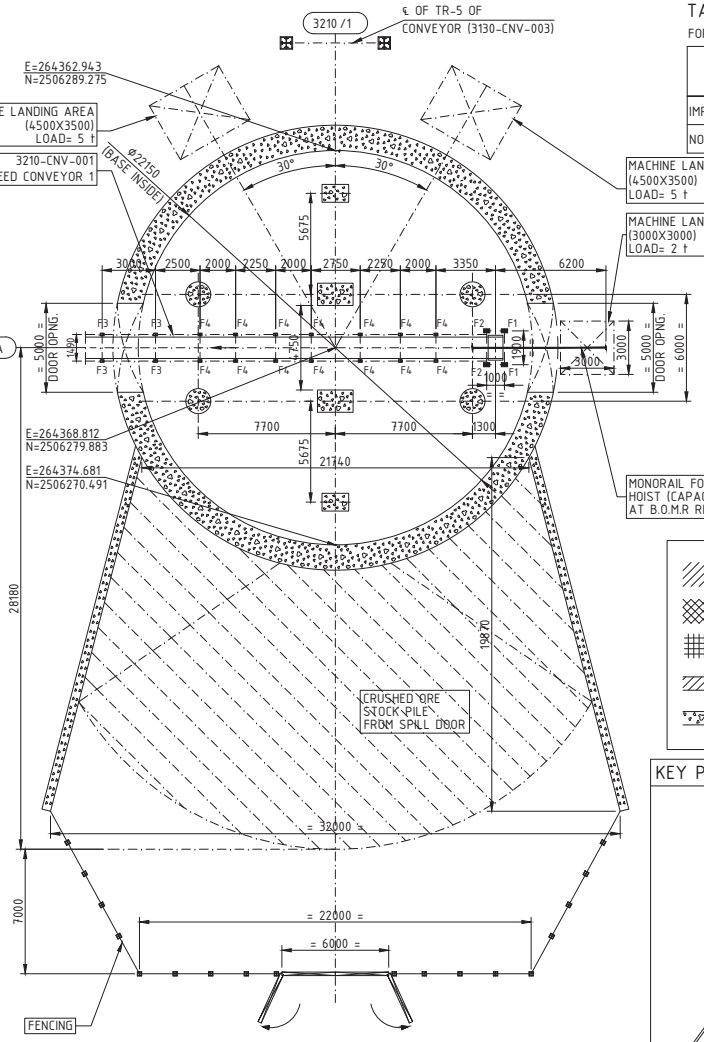
DETAIL-D1
SCALE 1:20



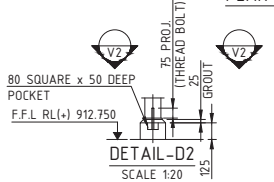
VIEW V1-V1
SCALE 1:20



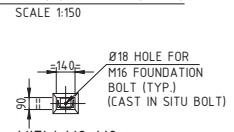
SECTION B-B
SCALE 1:125



PLAN AT RL(+) 912.750 (F.F.L)
SCALE 1:150



DETAIL-D2
SCALE 1:20



VIEW V2-V2
SCALE 1:20

TABLE- 2 : LOAD TABLE FOR SHORT POST

FOR CONV. 3210-CNV-001

ZONE	POINT	VERTICAL LOAD (IN "t")		HORIZONTAL LOAD (IN "t")	
		DEAD LOAD	LIVE LOAD	ALONG	ACROSS
IMPACT ZONE	F4	(+)1.2	(+)0.55	(±)0.17	(±)0.09
NORMAL ZONE	F3	(+)0.98	(+)0.55	(±)0.17	(±)0.09

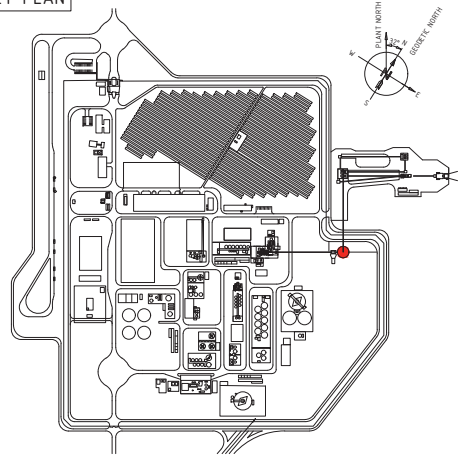
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CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

ABBREVIATION

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RCC:	REINFORCED CEMENT CONCRETE
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B.O.M.R:	BOTTOM OF MONORAIL
OPNG.:	OPENING
TR:	TRESTLE
TYP.:	TYPICAL
PROJ:	PROJECTED
W:	WIDTH
H:	HEIGHT

KEY PLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3210-EG-ME-LPX-1006	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN								
MD-216-3210-EG-ME-LPX-1005	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN								
MD-216-3210-EG-ME-LPX-1004	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN	A02	10/04/2020	ISSUED FOR APPROVAL & REVISED AS PER EMPLOYER	HKD	BRD	BD	BGR	
MD-216-3210-EG-ME-ELV-1009	CRUSHED ORE STORAGE BIN & TRANSFER - ELEVATION			COMMENTS DT. 20.03.2020 & REVISED AS MARKED A03					
MD-216-3210-EG-ME-LPX-1003	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN	A01	21/02/2020	ISSUED FOR APPROVAL	HKD	BRD	BD	BGR	

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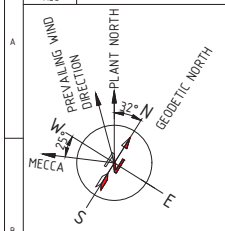
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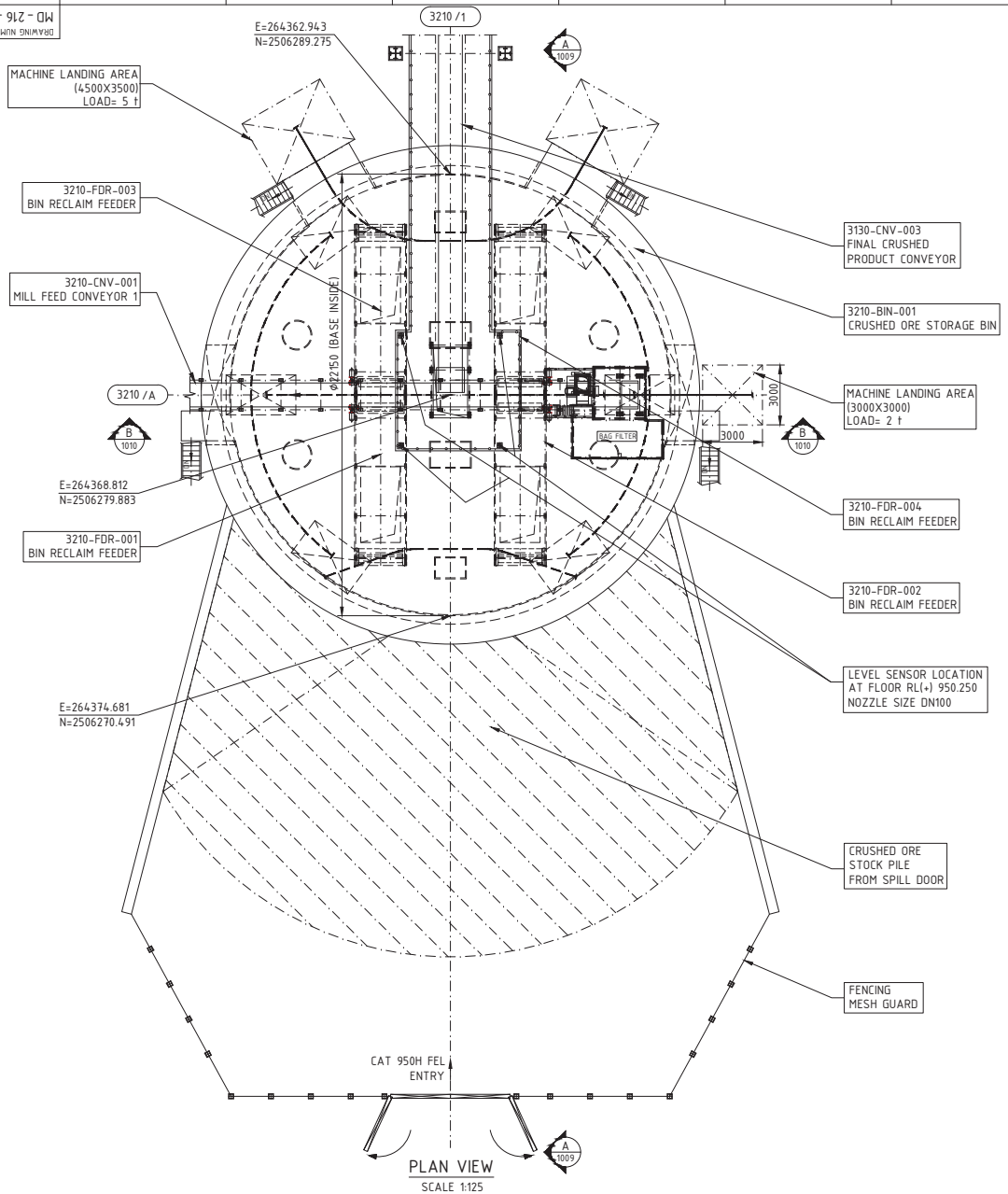
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OUTOTEC DRAWING NUMBER: 900707__GAU01__ZC01__US01__32115 — DOCUMENT ID: OU100669134 — REVISION: 1

DRAWN	HKD	18/02/2020
CHECKED	BRD	20/02/2020
ENGINEER	BD	20/02/2020
APPROVED	BGR	21/02/2020
EMPLOYER		
SCALE	1:150, 1:125, 1:20	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
GRINDING & CLASSIFICATION
CRUSHED ORE STORAGE BIN & TRANSFER - SECTION

DRAWING NUMBER: MD-216-3210-EG-ME-ELV-1010 — REV: A02





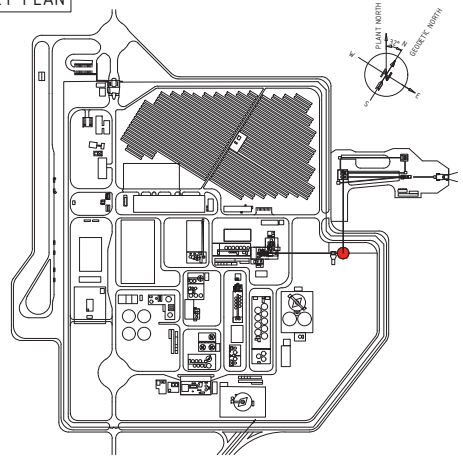
PLAN VIEW
SCALE 1:125

NOTE:
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CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

KEY PLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3210-EG-ME-LPX-1006	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN								
MD-216-3210-EG-ME-LPX-1005	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN								
MD-216-3210-EG-ME-LPX-1004	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN	A02	10/04/2020	ISSUED FOR APPROVAL & REVISED AS PER EMPLOYER	HKD	BRD	BD	BGR	
MD-216-3210-EG-ME-ELV-1010	CRUSHED ORE STORAGE BIN & TRANSFER - SECTION			COMMENTS DT. 20.03.2020					
MD-216-3210-EG-ME-ELV-1009	CRUSHED ORE STORAGE BIN & TRANSFER - ELEVATION	A01	21/02/2020	ISSUED FOR APPROVAL	HKD	BRD	BD	BGR	

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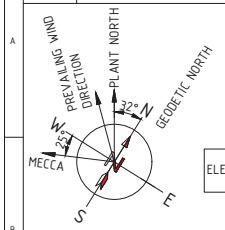
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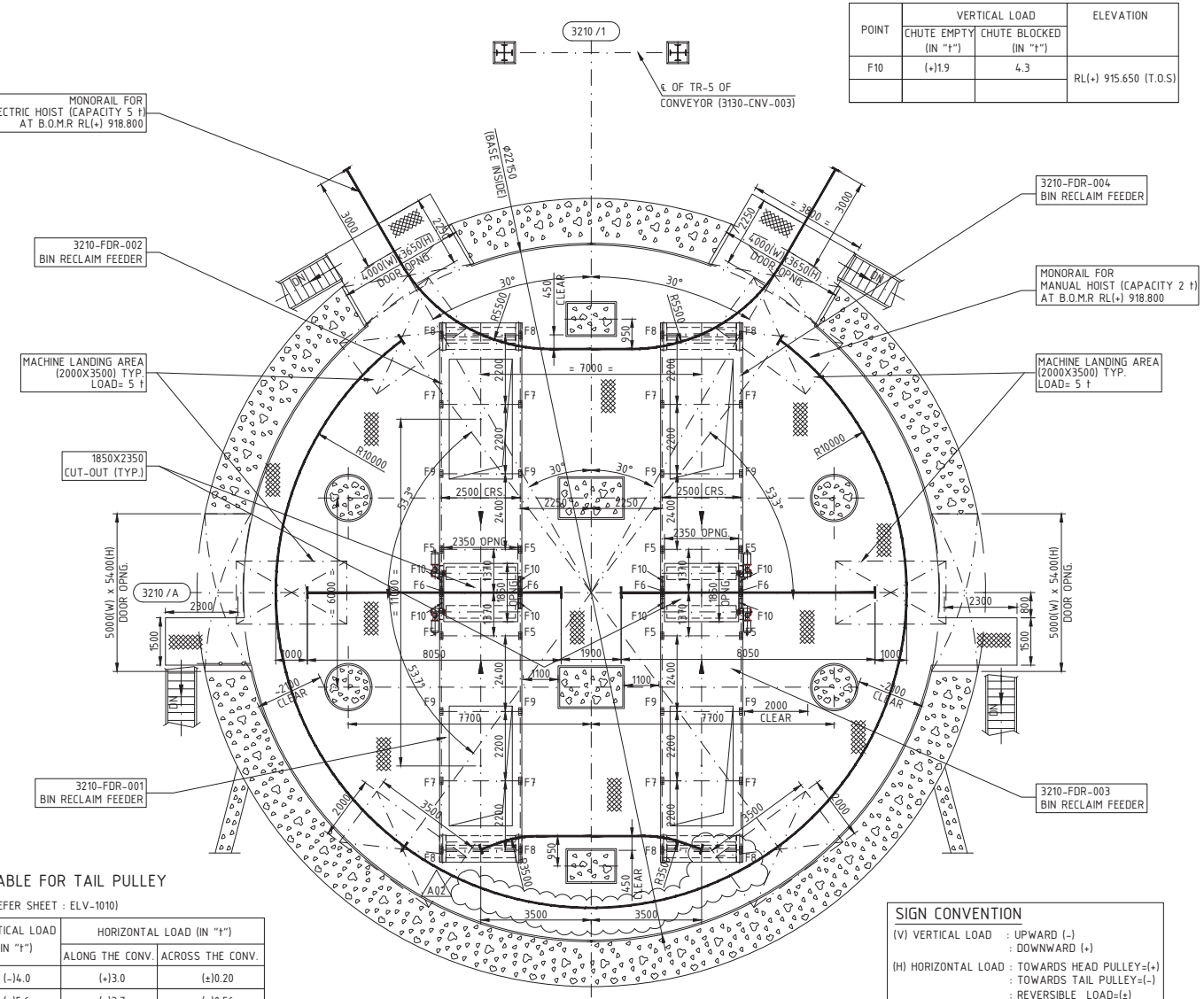
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LARSEN & TOUBRO DRAWING NUMBER	REVISION
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OUTOTEC DRAWING NUMBER	DOCUMENT ID / REVISION
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DRAWN	HKD	18/02/2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	BRD	20/02/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	BD	20/02/2020	GRINDING & CLASSIFICATION
APPROVED	BGR	21/02/2020	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN
EMPLOYER			
SCALE 1:125	A1		DRAWING NUMBER: MD-216-3210-EG-ME-LPX-1003 REV: A02

DRAWING NUMBER: MD-216-3210-EG-ME-LPX-1004 REV: A02





PLAN AT RL(+)915.650 (T.O.S)

SCALE 1:75

LOAD TABLE-4 (LOAD TABLE FOR CHUTE SADDLE)

FOR CONV. 3210-FDR-001, 3210-FDR-002, 3210-FDR-003,3210-FDR-004

POINT	VERTICAL LOAD		ELEVATION
	CHUTE EMPTY (IN "t")	CHUTE BLOCKED (IN "t")	
F10	(+)1.9	4.3	RL(+) 915.650 (T.O.S)

NOTE:

1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+)913.000 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.500 m
4. F.F.L, F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN TENTATIVE.

LOAD TABLE-3 : LOAD TABLE FOR BIN RECLAIM FEEDER

3210-520 HOLD

FOR CONV. 3210-FDR-001, 3210-FDR-002, 3210-FDR-003,3210-FDR-004

POINT	ELEVATION (IN "m")	DEAD LOAD		DEAD LOAD + MATERIAL LOAD	
		V (IN "t")	H (IN "t")	V (IN "t")	H (IN "t")
F5	RL(+) 915.650	1.9	0.19	3.8	0.38
F6		1.9	0.19	3.8	0.38
F7		1.9	0.19	4.5	0.45
F8		1.9	0.19	4.5	0.45
F9		1.9	0.19	6.5	0.65

TABLE-1 : LOAD TABLE FOR TAIL PULLEY

FOR CONV. 3210-CNV-001 (REFER SHEET : ELV-1010)

POINT	ELEVATION (IN "m")	VERTICAL LOAD (IN "t")	HORIZONTAL LOAD (IN "t")	
			ALONG THE CONV.	ACROSS THE CONV.
F1	RL(+) 912.750	(-)4.0	(+)3.0	(±)0.20
F2		(+)5.6	(+)3.7	(±)0.56

SIGN CONVENTION

(V) VERTICAL LOAD : UPWARD (-)
: DOWNWARD (+)

(H) HORIZONTAL LOAD : TOWARDS HEAD PULLEY=(+)
: TOWARDS TAIL PULLEY=(-)
: REVERSIBLE LOAD=(±)

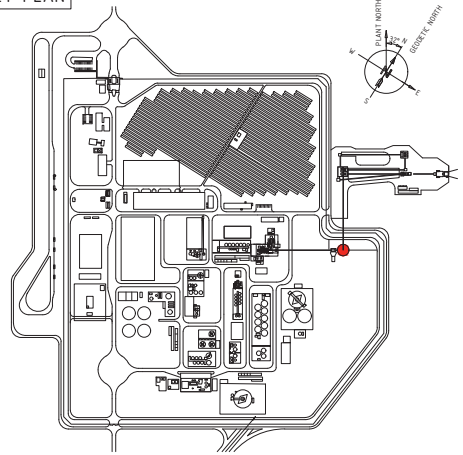
Note: Wind load & Seismic load to be considered during civil & structural design

ABBREVIATION

W.P.:	WORKING POINT
℄:	CENTER LINE
DN:	DOWNWARD
UP:	UPWARD
CRS.:	CENTERS
thk.:	THICKNESS
RCC:	REINFORCED CEMENT CONCRETE
deg.:	DEGREE
B.O.M.R:	BOTTOM OF MONORAIL
OPNG.:	OPENING
TR:	TRESTLE
TYP.:	TYPICAL
PROJ:	PROJECTED
W:	WIDTH
H:	HEIGHT

CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

KEY PLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3210-EG-ME-LPX-1006	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN								
MD-216-3210-EG-ME-LPX-1005	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN								
MD-216-3210-EG-ME-LPX-1010	CRUSHED ORE STORAGE BIN & TRANSFER - SECTION	A02	10/04/2020	ISSUED FOR APPROVAL & REVISED AS PER EMPLOYER	HKD	BRD	BD	BGR	
MD-216-3210-EG-ME-ELV-1009	CRUSHED ORE STORAGE BIN & TRANSFER - ELEVATION			COMMENTS DT. 20.03.2020 & REVISED AS MARKED A02					
MD-216-3210-EG-ME-LPX-1003	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN	A01	21/02/2020	ISSUED FOR APPROVAL	HKD	BRD	BD	BGR	

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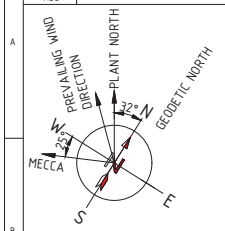
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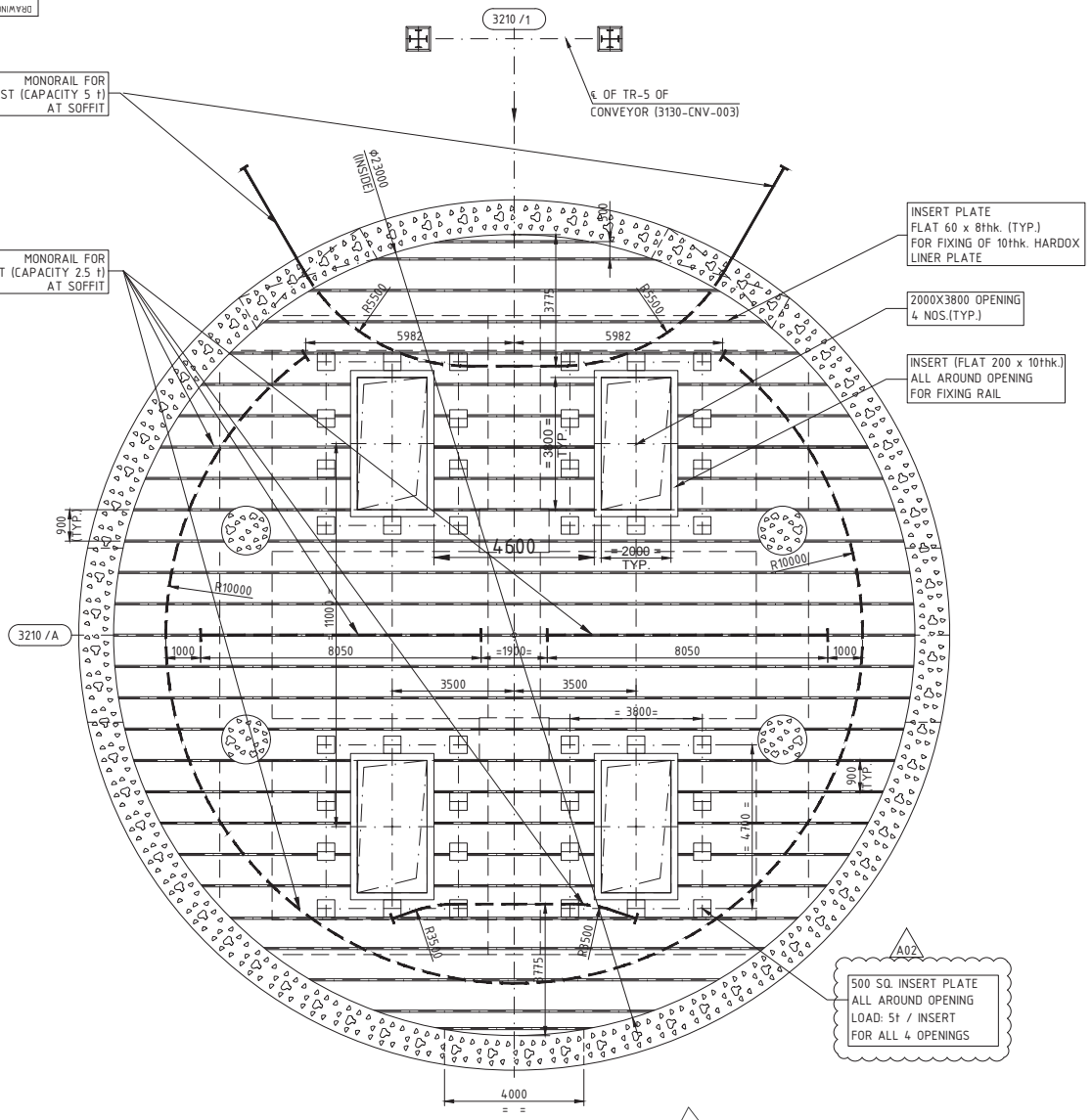
DRAWN	HKD	18/02/2020
CHECKED	BRD	20/02/2020
ENGINEER	BD	20/02/2020
APPROVED	BGR	21/02/2020
EMPLOYER		
SCALE	1:75	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
GRINDING & CLASSIFICATION
CRUSHED ORE STORAGE BIN & TRANSFER - PLAN

DRAWING NUMBER: MD-216-3210-EG-ME-LPX-1004 — REV: A02

DRAWING NUMBER: MD-216-3210-EG-ME-LPX-1005 REV: A02





PLAN AT RL(+) 920.800 (SOFFIT) A02
SCALE 1:75

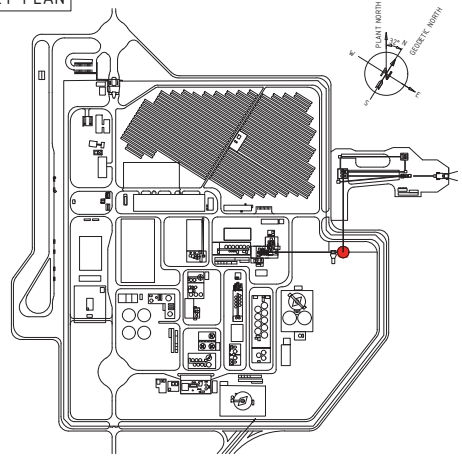
NOTE:
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ABBREVIATION

W.P.:	WORKING POINT
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deg.:	DEGREE
B.O.M.R:	BOTTOM OF MONORAIL
OPNG.:	OPENING
TR:	TRESTLE
TYP.:	TYPICAL
PROJ:	PROJECTED
W:	WIDTH
H:	HEIGHT

CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

KEY PLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3210-EG-ME-LPX-1006	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN								
MD-216-3210-EG-ME-LPX-1004	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN								
MD-216-3210-EG-ME-LPX-1010	CRUSHED ORE STORAGE BIN & TRANSFER - SECTION	A02	10/04/2020	ISSUED FOR APPROVAL & REVISED AS PER EMPLOYER	HKD	BRD	BD	BGR	
MD-216-3210-EG-ME-ELV-1009	CRUSHED ORE STORAGE BIN & TRANSFER - ELEVATION			COMMENTS DT. 20.03.2020 & REVISED AS MARKED A02					
MD-216-3210-EG-ME-LPX-1003	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN	A01	21/02/2020	ISSUED FOR APPROVAL	HKD	BRD	BD	BGR	

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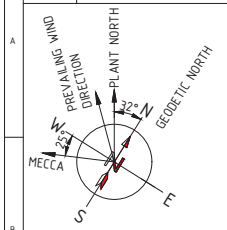
LARSEN & TOUBRO DRAWING NUMBER: 019045_100_H_G_16126_005 — REVISION: 1
OUTOTEC DRAWING NUMBER: 900707__GAU01__US01__MTL01__32105 — DOCUMENT ID: OU100669119 — REVISION: 1

DRAWN	HKD	18/02/2020
CHECKED	BRD	20/02/2020
ENGINEER	BD	20/02/2020
APPROVED	BGR	21/02/2020
EMPLOYER		
SCALE	1:75	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
GRINDING & CLASSIFICATION
CRUSHED ORE STORAGE BIN & TRANSFER - PLAN

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DRAWING NUMBER: MD-216-3210-EG-ME-LPX-1006 REV: A02



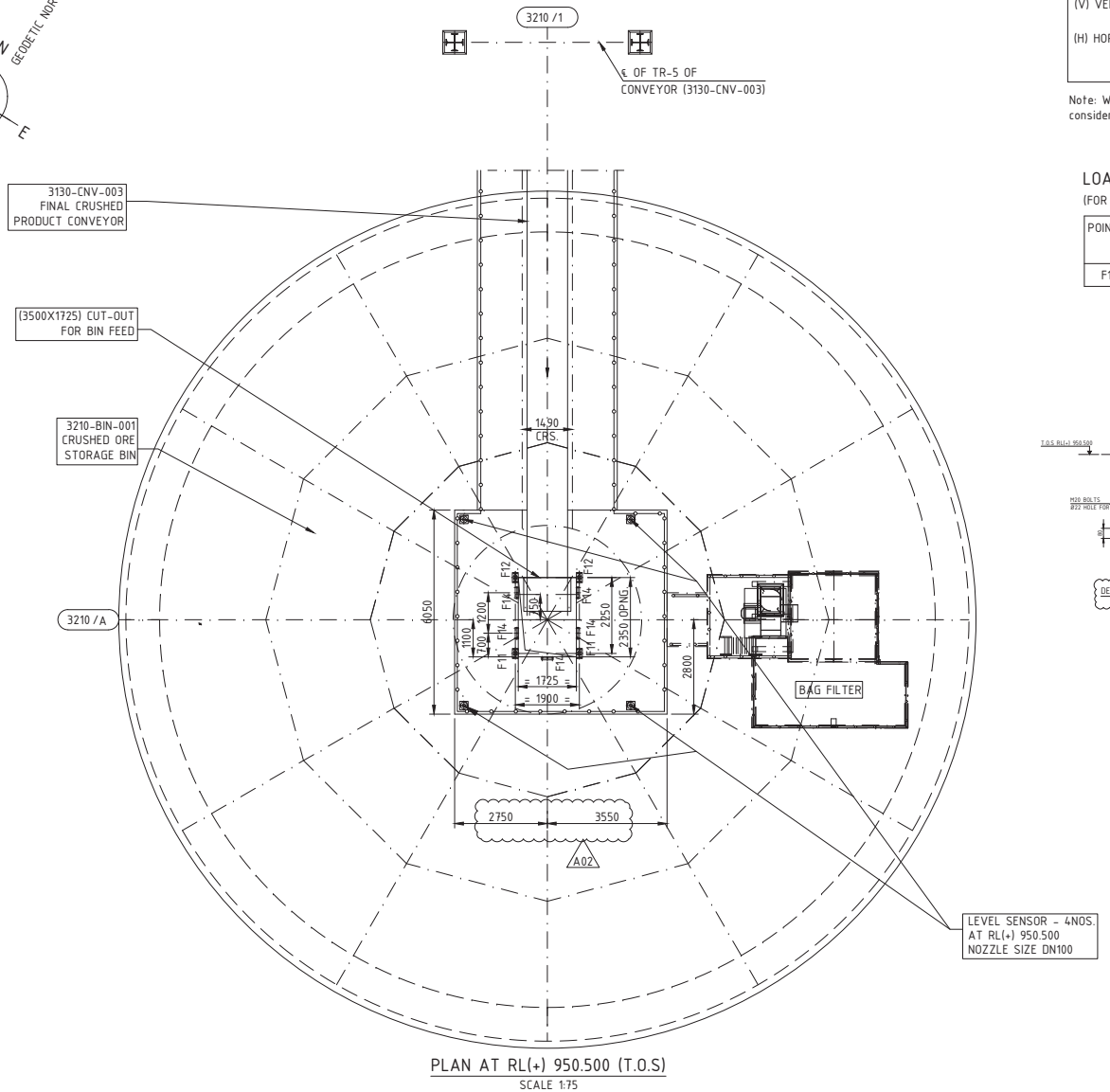
SIGN CONVENTION

(V) VERTICAL LOAD : UPWARD (–)
: DOWNWARD (+)

(H) HORIZONTAL LOAD : TOWARDS HEAD PULLEY=(+)
: TOWARDS TAIL PULLEY=(–)
: REVERSIBLE LOAD=(±)

Note: Wind load & Seismic load to be considered during civil & structural design

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PLAN AT RL(+) 950.500 (T.O.S)
SCALE 1:75

LOAD TABLE -5

(FOR LOAD POINT AT CHUTE SADDLE)

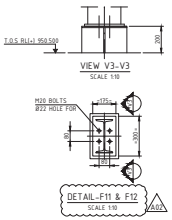
POINTS	ELEVATIONS IN "m"	V (IN "t")
F14	(+)950.250	(+)1.5

LOAD TABLE-6 : LOAD TABLE FOR HEAD FRAME

FOR CONV. 3130-CNV-003

POINT	LOAD CASE I			LOAD CASE II		
	V (IN "t")	H (IN "t")	ACROSS	V (IN "t")	H (IN "t")	ACROSS
F11	(–)4.2	(–)7.1	(±)0.5	(–)9.2	(–)10.4	(±)0.9
F12	(+)13.8	(–)7.8	(±)1.4	(+)17.1	(–)10.0	(±)1.7

NOTE:CASE I :CONV. NORMAL RUNNING+CHUTE BLOCKED+DEAD
CASE II :CONV. STARTING +CHUTE EMPTY+DEAD

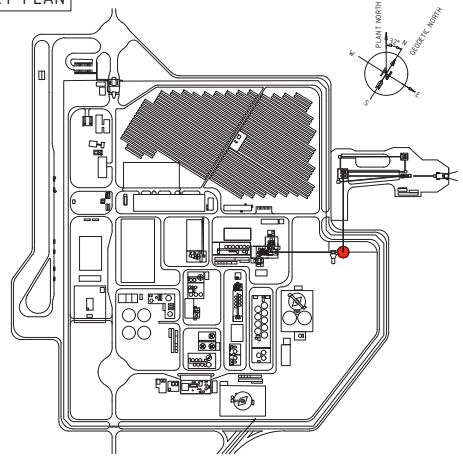


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OPNG.:	OPENING
TR:	TRESTLE
TYP.:	TYPICAL
PROJ:	PROJECTED
W:	WIDTH
H:	HEIGHT

CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

KEY PLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3210-EG-ME-LPX-1005	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN								
MD-216-3210-EG-ME-LPX-1004	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN								
MD-216-3210-EG-ME-ELV-1010	CRUSHED ORE STORAGE BIN & TRANSFER - SECTION	A02	10/04/2020	ISSUED FOR APPROVAL & REVISED AS PER EMPLOYER	HKD	BRD	BD	BGR	
MD-216-3210-EG-ME-ELV-1009	CRUSHED ORE STORAGE BIN & TRANSFER - ELEVATION			COMMENTS DT. 20.03.2020 & REVISED AS MARKED A01					
MD-216-3210-EG-ME-LPX-1003	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN	A01	21/02/2020	ISSUED FOR APPROVAL	HKD	BRD	BD	BGR	

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OUTOTEC DRAWING NUMBER: 900707__GAU01__US01__MTL01__32106 — DOCUMENT ID: OU100669120 — REVISION: 1

DRAWN	HKD	18/02/2020
CHECKED	BRD	20/02/2020
ENGINEER	BD	20/02/2020
APPROVED	BGR	21/02/2020
EMPLOYER		
SCALE	1:75	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
GRINDING & CLASSIFICATION
CRUSHED ORE STORAGE BIN & TRANSFER - PLAN

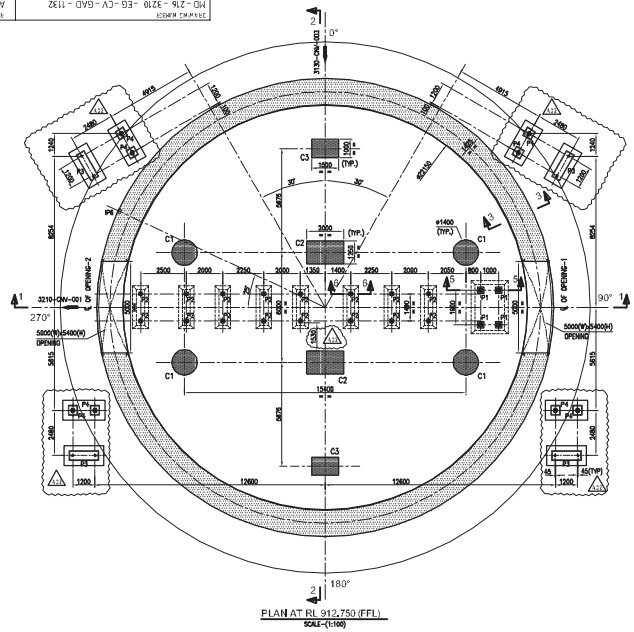
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ATTACHMENT 4 CIVIL DRAWINGS

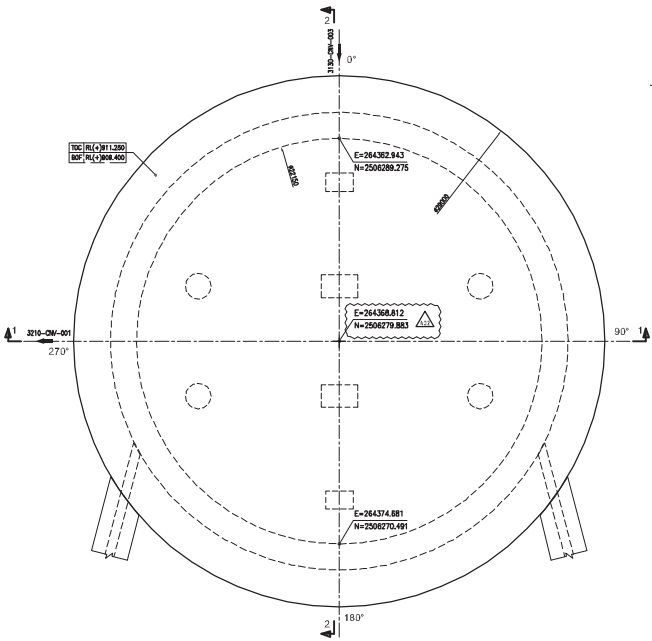
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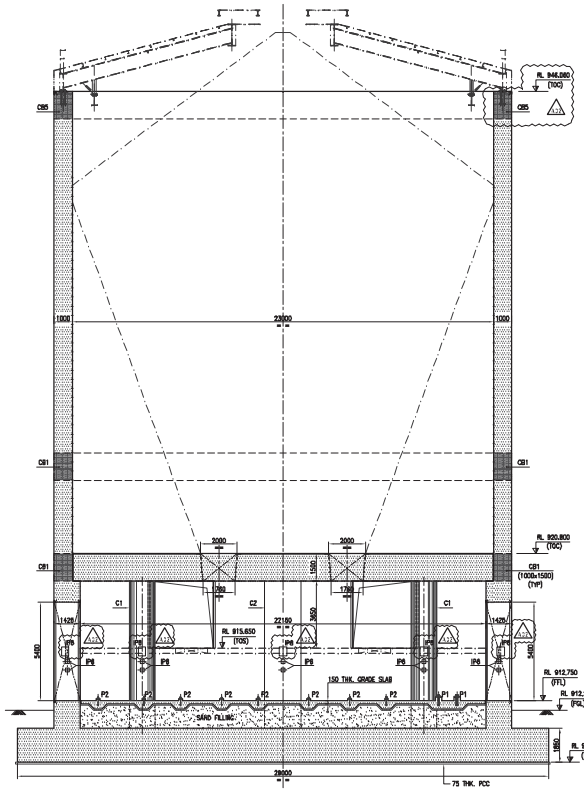
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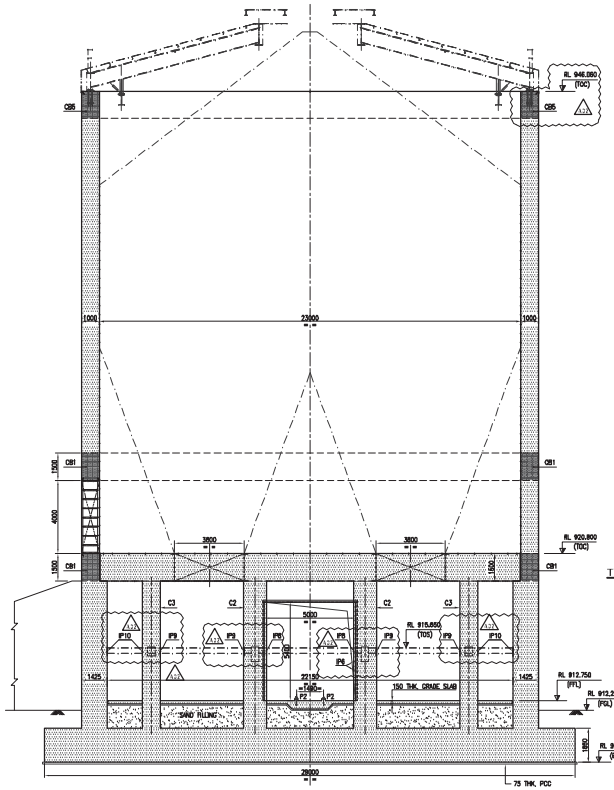
PLAN AT RL 912.750 (FFL)
SCALE-(1:100)



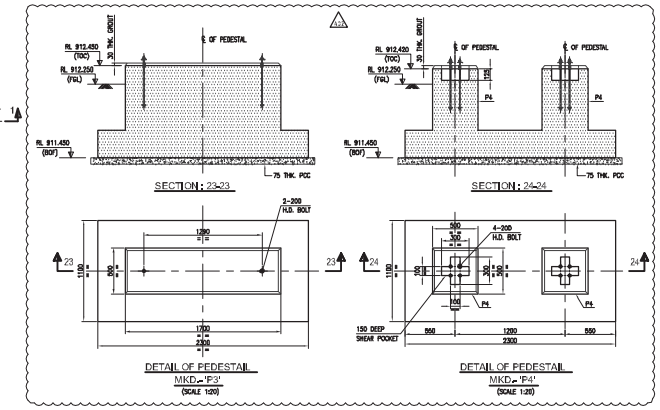
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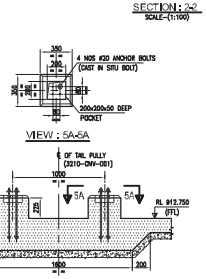


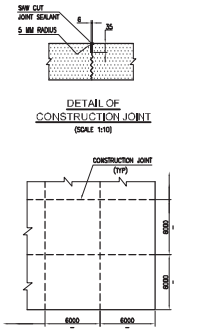
SECTION : 1-1
SCALE-(1:100)



SECTION : 2-2
SCALE-(1:100)







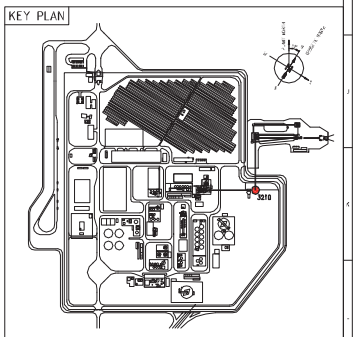
LEGEND :-

TOC = TOP OF CONCRETE
BOC = BOTTOM OF CONCRETE
TOS = TOP OF STRUCTURE
FGL = FINISHED GROUND LEVEL
BOF = BOTTOM OF FOOTING
BOM = BOTTOM OF MONORAIL
TYP = TYPICAL
MKD = MARKED
PCC = PLAIN CEMENT CONCRETE
FL = FLAT
IP = INSERT PLATE
TOI = TOP OF INSERT
MR = MONORAIL
W = WIDTH
H = HEIGHT
℄ = CENTRE LINE
THK = THICKNESS
Ø = DIAMETER
R = RADIUS

NOTES :-

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND CO-ORDINATES ARE IN METRES, UNLESS OTHERWISE STATED.
2. THIS DRAWING SHALL BE READ IN CONJUNCTION WITH DRG. NO. MD-216-3210-EG-CV-GAD-1133 & 1134.
3. FOR ALL OTHER NOTES REFER DRAWING NO. MD-216-1000-EG-CV-DED-1001, 1003 & 1004.

KEY PLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRW	CHK	DES	APP	DEP
MD-216-3210-EG-MC-CVA-1002	CRUSHED ORE BIN SECTION								
MD-216-3210-EG-MC-CVA-1003	CRUSHED ORE BIN ELEVATION								
MD-216-3210-EG-MC-CPA-1001	CRUSHED ORE BIN PLAN								
MD-216-3210-EG-MC-CPA-1002	CRUSHED ORE BIN PLAN	A02	23/01/2023	REVISED AS PER DEPARTMENTAL COMMENTS	ASH	JR	SVK	AG	
MD-216-3210-EG-MC-CPA-1004	CRUSHED ORE BIN PLAN			DW 24/01/2023 & REVISED AS MARKED					
MD-216-3210-EG-MC-CPA-1003	CRUSHED ORE BIN PLAN	A01	20/01/2023	ISSUED FOR REFERENCE	ASH	JR	SVK	AG	
DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRW	CHK	DES	APP	DEP

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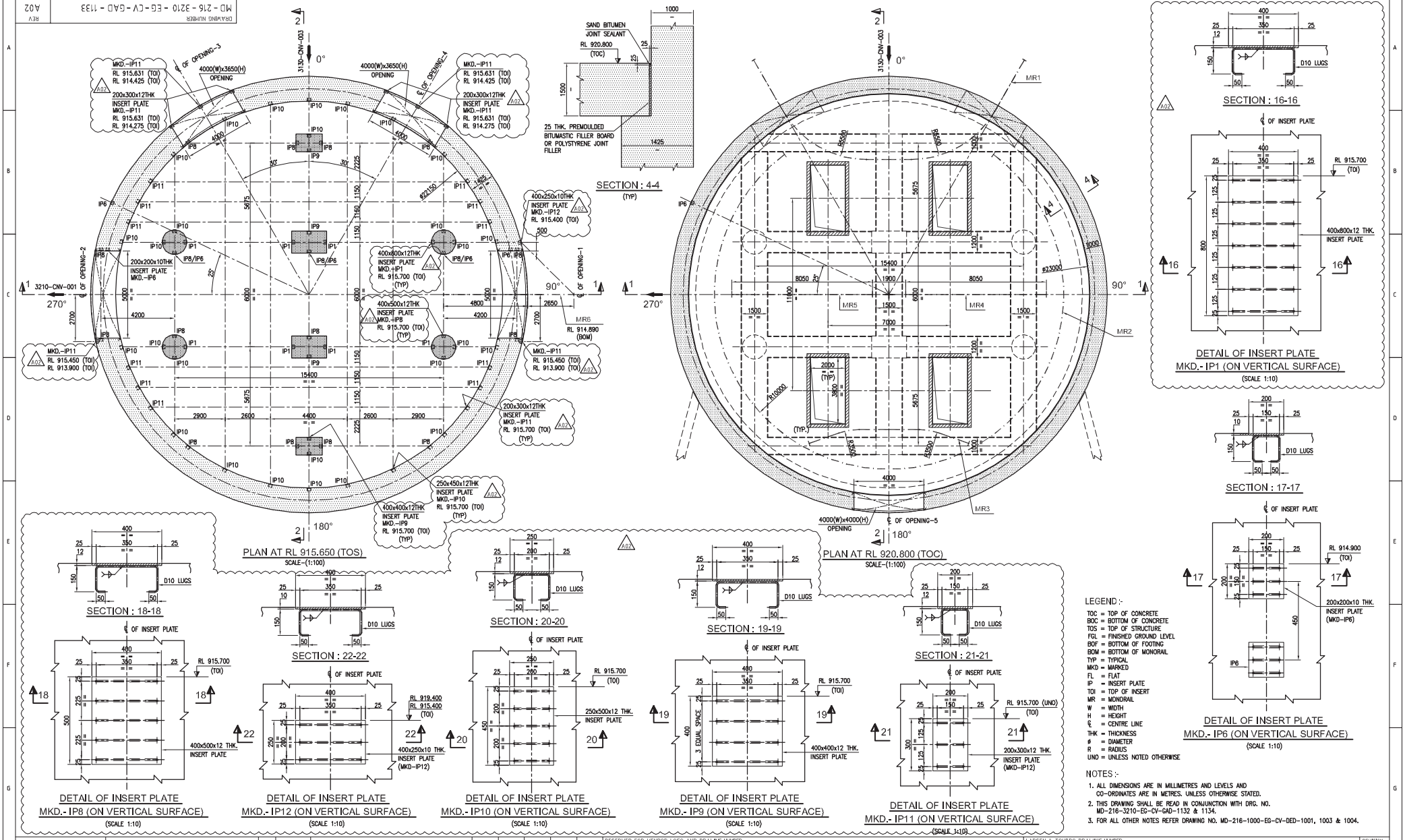
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019045_A50_E_G_24071_001		
SUBSIDIARY DRAWING NUMBER	DOCUMENT ID	REVISION 1
900707__GAU01__US01__MTL01__32121	OU100673540	

DRAWN	ASH	20/01/2023
CHECKED	JR	24/01/2023
ENGINEER	SVK	20/01/2023
APPROVED	AG	20/01/2023
TRACEABILITY		
SCALE	1:100	A0

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
GRINDING & CLASSIFICATION
CRUSHED ORE STORAGE BIN - LAYOUT & DETAIL

DRAWING NUMBER: MD-216-3210-EG-CV-GAD-1132
REV: A02



LEGEND :-

TOC = TOP OF CONCRETE
BOC = BOTTOM OF CONCRETE
TOS = TOP OF STRUCTURE
FGL = FINISHED GROUND LEVEL
BOF = BOTTOM OF FOOTING
BOM = BOTTOM OF MONORAIL
TYP = TYPICAL
MKD = MARKED
FL = FLAT
IP = INSERT PLATE
TOI = TOP OF INSERT
MR = MONORAIL
W = WIDTH
H = HEIGHT
℄ = CENTRE LINE
THK = THICKNESS
ø = DIAMETER
R = RADIUS
UNO = UNLESS NOTED OTHERWISE

NOTES :-

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND CO-ORDINATES ARE IN METRES, UNLESS OTHERWISE STATED.
2. THIS DRAWING SHALL BE READ IN CONJUNCTION WITH DRG. NO. MD-216-3210-EG-CV-GAD-1132 & 1134.
3. FOR ALL OTHER NOTES REFER DRAWING NO. MD-216-1000-EG-CV-DED-1001, 1003 & 1004.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3210-EG-ME-ELV-1010	CRUSHED ORE BIN-SECTION								
MD-216-3210-EG-ME-ELV-1009	CRUSHED ORE BIN-ELEVATION								
MD-216-3210-EG-ME-LPX-1006	CRUSHED ORE BIN-PLAN								
MD-216-3210-EG-ME-LPX-1005	CRUSHED ORE BIN-PLAN	A02	30/04/2020	REVISED AS PER EMPLOYER COMMENTS	ABH	UR	SVK	AG	
MD-216-3210-EG-ME-LPX-1004	CRUSHED ORE BIN-PLAN			ON 04/04/2020 & REVISED AS MARKED A02					
MD-216-3210-EG-ME-LPX-1003	CRUSHED ORE BIN-PLAN	A01	05/03/2020	ISSUED FOR REFERENCE	ABH	UR	SVK	AG	

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Saudi Arabian Mining Company

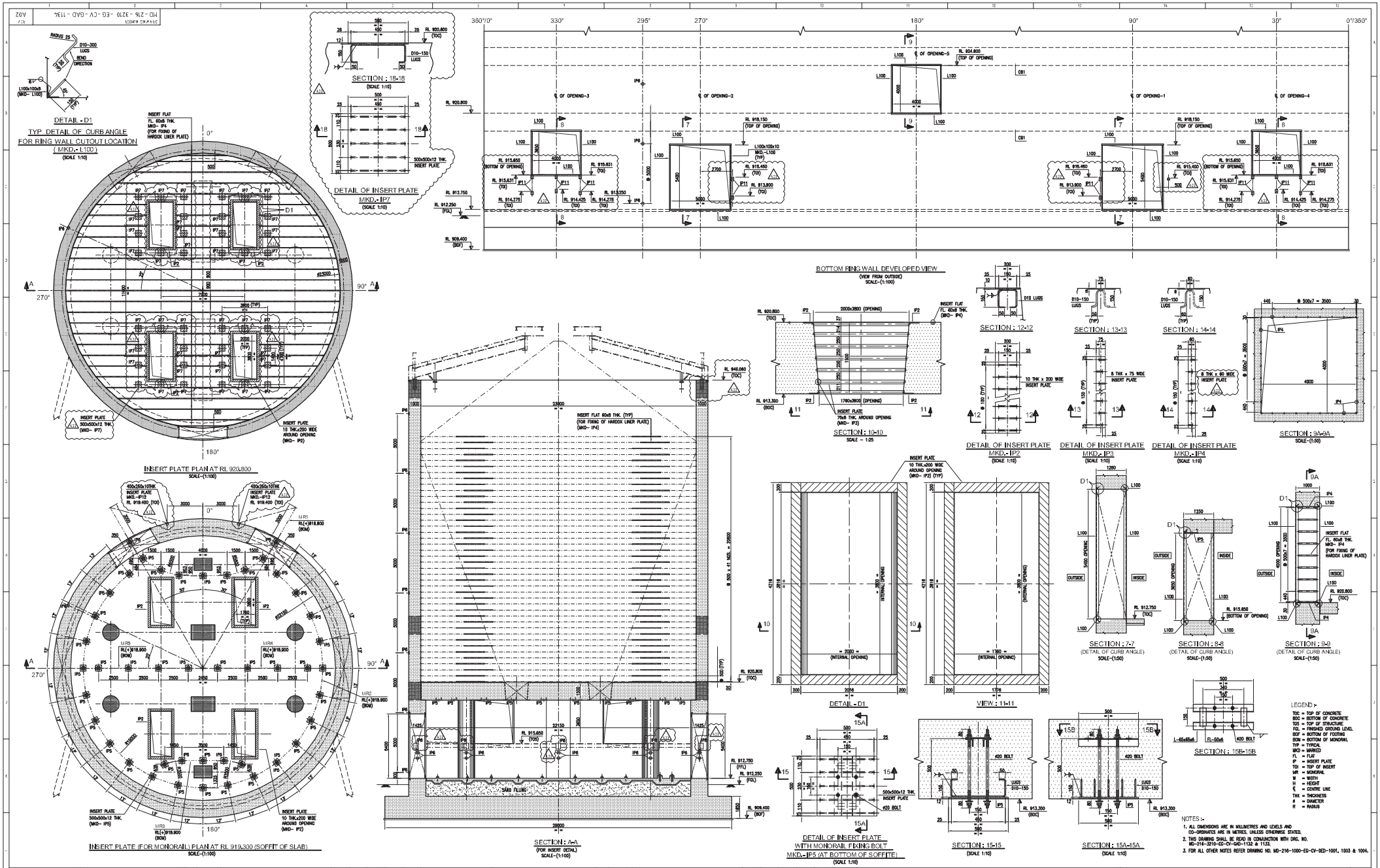
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LARSEN & TOUBRO DRAWING NUMBER: 019045_450_C_G_24071_002 — REVISION: 1
OUTOTEC DRAWING NUMBER: 900707__GAU01__US01__MTL01__32122 — DOCUMENT ID: OU100673541 — REVISION: 1

DRAWN	ABH	03/03/2020
CHECKED	UR	04/03/2020
ENGINEER	SVK	05/03/2020
APPROVED	AG	05/03/2020
EMPLOYER		
SCALE	1:100	A1

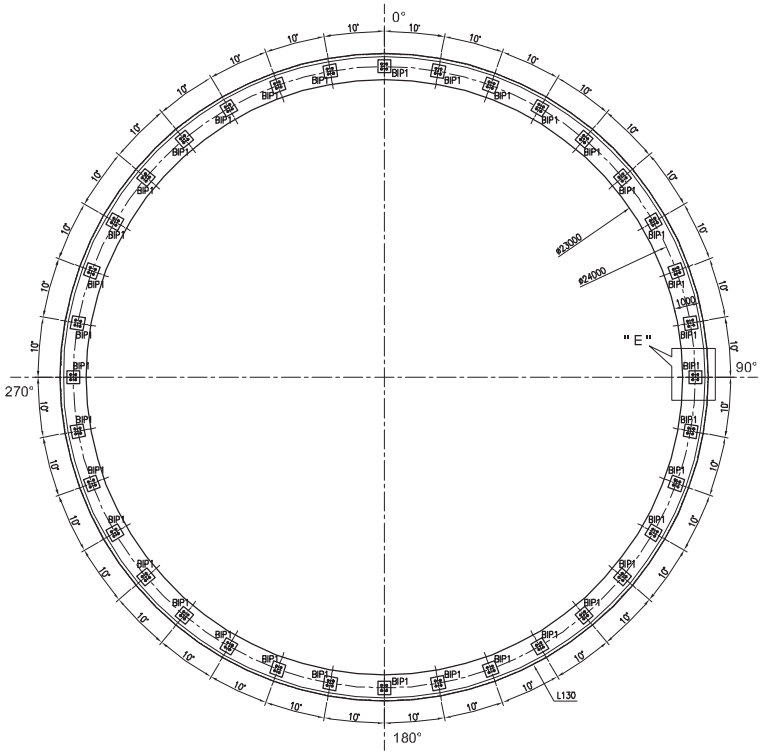
MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
GRINDING & CLASSIFICATION
CRUSHED ORE STORAGE BIN - LAYOUT & DETAIL

DRAWING NUMBER: MD - 216 - 3210 - EG - CV - GAD - 1133 — REV: A02

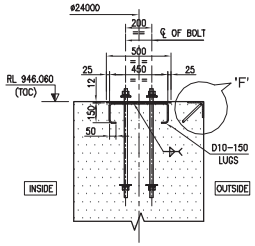

DETAIL - D1
TYP. DETAIL OF CURB ANGLE FOR RING WALL CUTOUT LOCATION
(MKD.- L100)
SECTION : 18-18
DETAIL OF INSERT PLATE MKD.- IP7
INSERT PLATE PLAN AT RL 920.800
INSERT PLATE (FOR MONORAIL) PLAN AT RL 919.300 (SOFFIT OF SLAB)
SECTION : A-A
BOTTOM RING WALL DEVELOPED VIEW
SECTION : 10-10
SECTION : 12-12
SECTION : 13-13
SECTION : 14-14
DETAIL OF INSERT PLATE MKD.- IP2
DETAIL OF INSERT PLATE MKD.- IP3
DETAIL OF INSERT PLATE MKD.- IP4
SECTION : 9A-9A
DETAIL - D1
VIEW : 11-11
SECTION : 7-7
SECTION : 8-8
SECTION : 9-9
DETAIL OF INSERT PLATE WITH MONORAIL FIXING BOLT MKD.- IP5 (AT BOTTOM OF SOFFITE)
SECTION : 15-15
SECTION : 15A-15A
SECTION : 15B-15B
LEGEND :
NOTES :-
1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND CO-ORDINATES ARE IN METRES, UNLESS OTHERWISE STATED.
2. THIS DRAWING SHALL BE READ IN CONJUNCTION WITH DRG. NO. MD-216-3210-EG-CV-GAD-1132 & 1133.
3. FOR ALL OTHER NOTES REFER DRAWING NO. MD-216-1000-EG-CV-SED-1001, 1003 & 1004.

| | | |

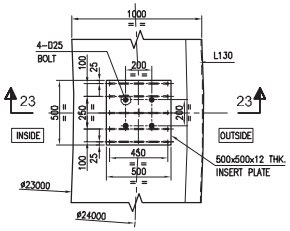
DRAWING NUMBER: MD - 216 - 3210 - EG - CV - GAD - 1135 REV A01



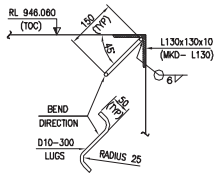
PLAN AT RL 946.060 (TOC)
SCALE-(1:100)



SECTION : 23-23
(SCALE 1:20)



DETAIL - E
(MKD.- BIP1)
(SCALE 1:20)



DETAIL - F
TYP DETAIL OF CURB ANGLE
(MKD.- L125)
(SCALE 1:10)

LEGEND :-

TOC = TOP OF CONCRETE
PCC = PLAIN CEMENT CONCRETE
TOB = TOP OF BEAM
BOC = BOTTOM OF CONCRETE
TOS = TOP OF STRUCTURE
FGL = FINISHED GROUND LEVEL
FFL = FINISH FLOOR LEVEL
BOF = BOTTOM OF FOOTING
BOM = BOTTOM OF MONORAIL
TYP = TYPICAL
MKD = MARKED
FL = FLAT
IP = INSERT PLATE
TOI = TOP OF INSERT
MR = MONORAIL
W = WIDTH
H = HEIGHT
℄ = CENTRE LINE
THK = THICKNESS
ø = DIAMETER
R = RADIUS

NOTES :-

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND CO-ORDINATES ARE IN METRES. UNLESS OTHERWISE STATED.
2. THIS DRAWING SHALL BE READ IN CONJUNCTION WITH DRG. NO. MD-216-3210-EG-CV-GAD-1132, 1133 & 1134.
3. FOR ALL OTHER NOTES REFER DRAWING NO. MD-216-1000-EG-CV-DED-1001, 1003 & 1004.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3210-EG-ME-ELV-1010	CRUSHED ORE BIN-SECTION								
MD-216-3210-EG-ME-ELV-1009	CRUSHED ORE BIN-ELEVATION								
MD-216-3210-EG-ME-LPX-1006	CRUSHED ORE BIN-PLAN								
MD-216-3210-EG-ME-LPX-1005	CRUSHED ORE BIN-PLAN								
MD-216-3210-EG-ME-LPX-1004	CRUSHED ORE BIN-PLAN								
MD-216-3210-EG-ME-LPX-1003	CRUSHED ORE BIN-PLAN	A01	23/04/2020	ISSUED FOR APPROVAL	ABH	UR	SVK	AG	

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Saudi Arabian Mining Company

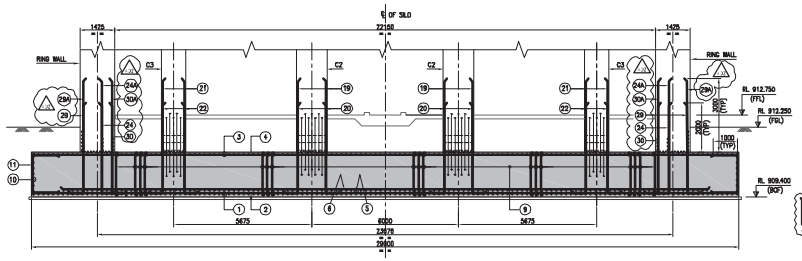
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LARSEN & TOUBRO DRAWING NUMBER: 019045_450_C_G_24071_004 — REVISION: 0
OUTOTEC DRAWING NUMBER: 900707 _GAU01 _US01 _MTL01 _32124 — DOCUMENT ID: OU100673543 — REVISION: 0

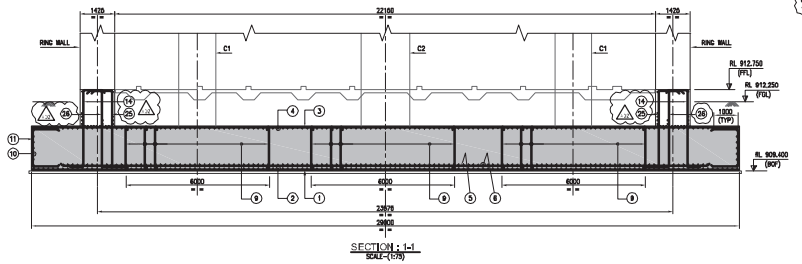
DRAWN	ABH	23/04/2020
CHECKED	UR	24/04/2020
ENGINEER	SVK	23/04/2020
APPROVED	AG	23/04/2020
EMPLOYER		
SCALE 1:100	A1	

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
GRINDING & CLASSIFICATION
CRUSHED ORE STORAGE BIN - LAYOUT & DETAIL

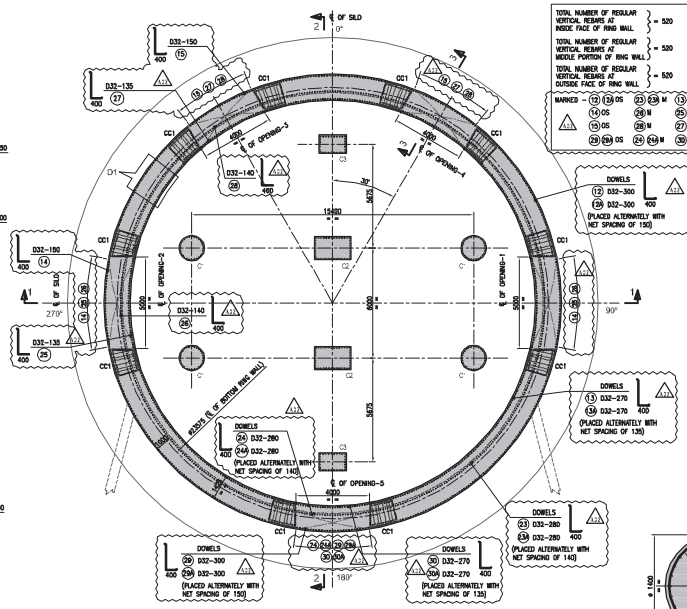
DRAWING NUMBER: MD - 216 - 3210 - EG - CV - GAD - 1135 — REV: A01



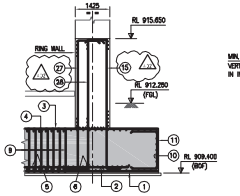
SECTION : 2-2
SCALE-(1:75)



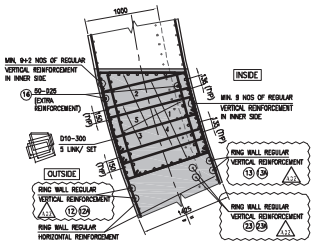
SECTION : 1-1
SCALE-(1:75)



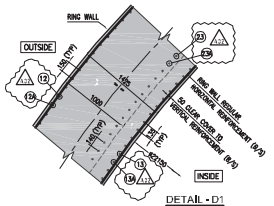
PLAN SHOWING DOWELS FOR RING WALL AND INTERNAL COLUMN REINFORCEMENT
SCALE-(1:100)



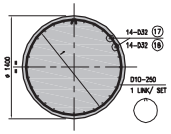
SECTION : 3-3
SCALE-(1:75)



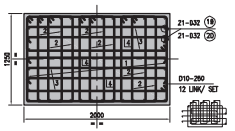
SECTIONAL DETAIL OF COLUMN (MKD. CC1)
("a"-EXTRA VERTICAL REINFORCEMENT AROUND OPENING)
("-"-REGULAR VERTICAL REINFORCEMENT)
SCALE 1:20



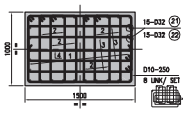
DETAIL - D1
SCALE - (1:25)



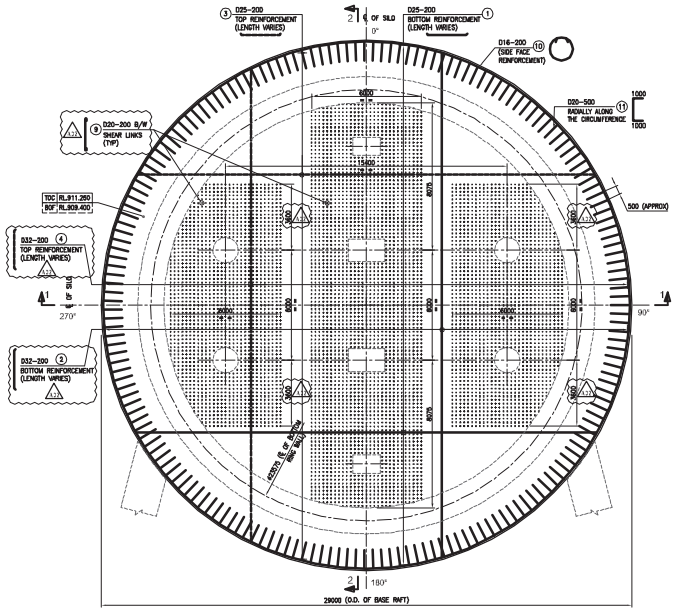
SECTIONAL DETAIL OF COLUMN (MKD. C1)
SCALE 1:25



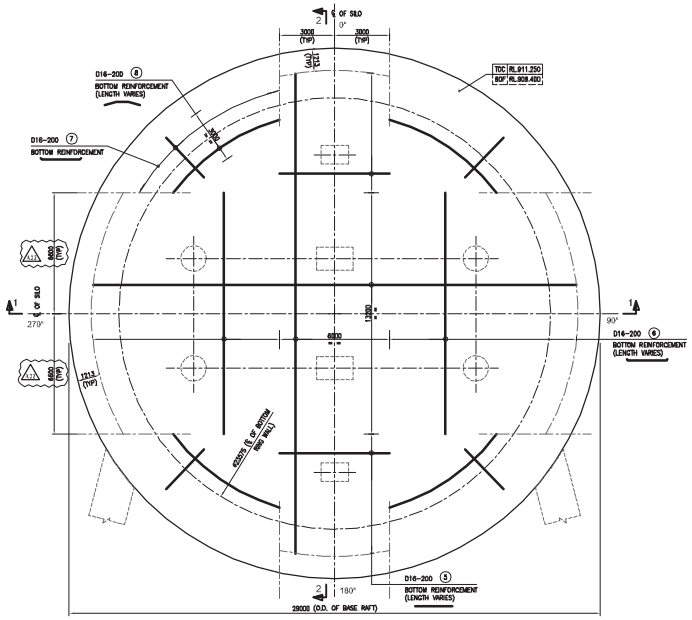
SECTIONAL DETAIL OF COLUMN (MKD. C2)
SCALE 1:25



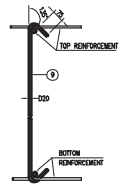
SECTIONAL DETAIL OF COLUMN (MKD. C3)
SCALE 1:25



PLAN AT RL 909.400 (BOF)
(SHOWING TOP AND BOTTOM REINFORCEMENT)
SCALE-(1:100)



PLAN AT RL 909.400 (BOF)
(SHOWING BOTTOM EXTRA REINFORCEMENT)
SCALE-(1:100)



TYP DETAIL OF VERTICAL SHEAR LINK (IN RAFT)
(ELEVATION)

LEGEND :-
TOC = TOP OF CONCRETE
FGL = FINISHED GROUND LEVEL
BOF = BOTTOM OF FOOTING
TYP = TYPICAL
MKD = MARKED
℄ = CENTRE LINE
ø = DIAMETER
R = RADIUS
B/S = BOTH SIDE
OS = OUT SIDE
M = MIDDLE
I/S = IN SIDE
= TOP REINFORCEMENT
= BOTTOM REINFORCEMENT

NOTES :-
1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND CO-ORDINATES ARE IN METRES. UNLESS OTHERWISE STATED.
2. THIS DRAWING SHALL BE READ IN CONJUNCTION WITH DRAWING NUMBER : MD-216-3210-EG-CV-DAD-1132, 1133 & 1134.
3. FOR ALL OTHER NOTES REFER DRAWING NO. MD-216-1000-EG-CV-DED-1001, 1003 & 1004.
4. GRADE OF CONCRETE SHALL BE C35 AND GRADE OF CONCRETE SHALL BE C30 FOR GRADE SLAB.
5. GRADE OF REINFORCEMENT SHALL BE DEFORMED BAR ASTM A615-GRADE 60 (fy=420 MPa)
6. GRADE OF WELDED WIRE FABRIC – ASTM A185(PLAIN) / A497(DEFORMED).
7. NET SAFE BEARING CAPACITY OF SOIL AT BOTTOM OF FOUNDATION CONSIDER 600KN/ SQ. M
8. IF ANY PART OF FOUNDATION COMES ON BACKFILL ZONE OF ADJACENT FOUNDATION OR ANY LOOSE SOIL ARE ENCOUNTERED BELOW FOUNDATION LEVEL THE SAME SHALL BE REMOVED AND FILLED WITH PCC.
9. REINFORCEMENT SHALL BE SUITABLY ADJUSTED AT SITE TO CLEAR POCKETS, BOLTS ETC.
10. SUFFICIENT AND SUITABLE NOS OF CHAIRS SHALL BE PROVIDED AT APPROPRIATE LOCATIONS TO MAINTAIN CORRECT POSITION OF REINFORCEMENT.
11. CLEAR COVER TO MAIN REINFORCEMENT SHALL BE AS FOLLOWS :
    a) RAFT : 75 mm (BOTTOM & SIDE)
    b) RAFT : 00 mm (TOP)
    c) WALL : 50 mm
    d) SUB : 40 mm
    e) PEDESTAL & COLUMN : 50 mm
12. MINIMUM LAP LENGTHS OF REINFORCEMENT SHALL BE 40xBAR DIAMETER.
13. THIS DRAWING SHALL BE READ IN CONJUNCTION WITH DRG. NO. MD-216-3210-EG-CV-DED-1137, 1138 & 1139.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRW	CHK	DES	APP	DMP
		A01	22/06/2022	ISSUED FOR REFERENCE	ASH	JR	SJK	AD	
		A02	22/09/2022	REVISED AS PER EMPLOYER COMMENTS SW 24/09/2022 & REVISED AS MARKED	ASH	JR	SJK	AD	

ISSUED FOR VENDOR / DOC. NO. DRAWING NUMBER

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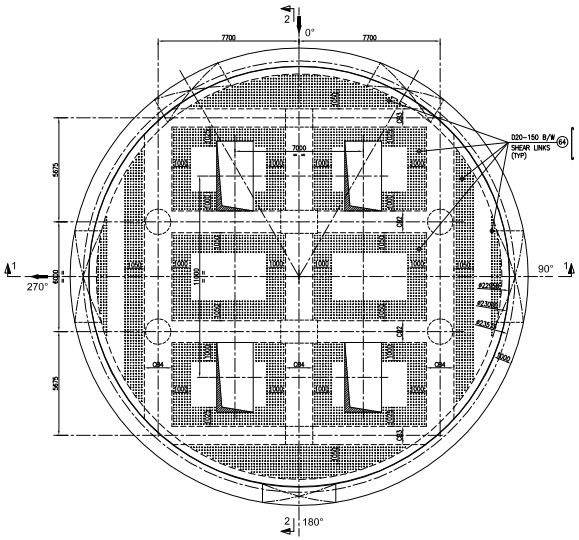
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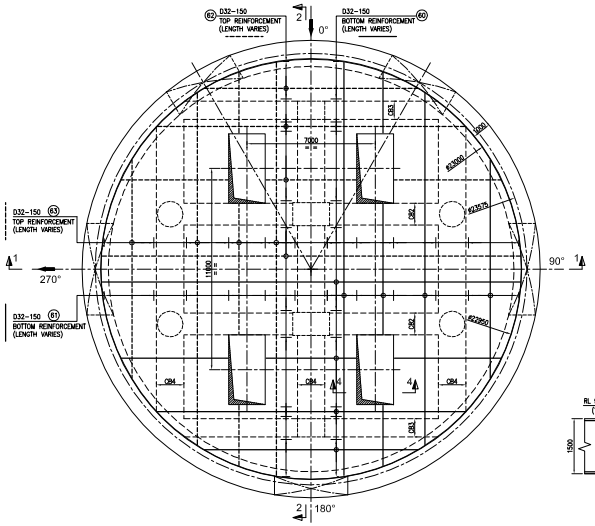
LARSEN & TOUBRO DRAWING NUMBER	019045_A50_C_D_24072_001	REVISION 1
SUBSIDIARY DRAWING NUMBER	900707__GA001__US01__CTB17__32101	DOCUMENT ID OU100673544 REVISION 1
DRAWN	ASH	22/03/2022
CHECKED	JR	24/03/2022
ENGINEER	SJK	22/03/2022
APPROVED	AD	22/03/2022
FINAL VER.		
SCALE	1:100	A0

MANSOURAH – MASSARAH GOLD PROJECT
DETAIL DRAWING
GRINDING & CLASSIFICATION
CRUSHED ORE STORAGE BIN – R.C. DETAIL

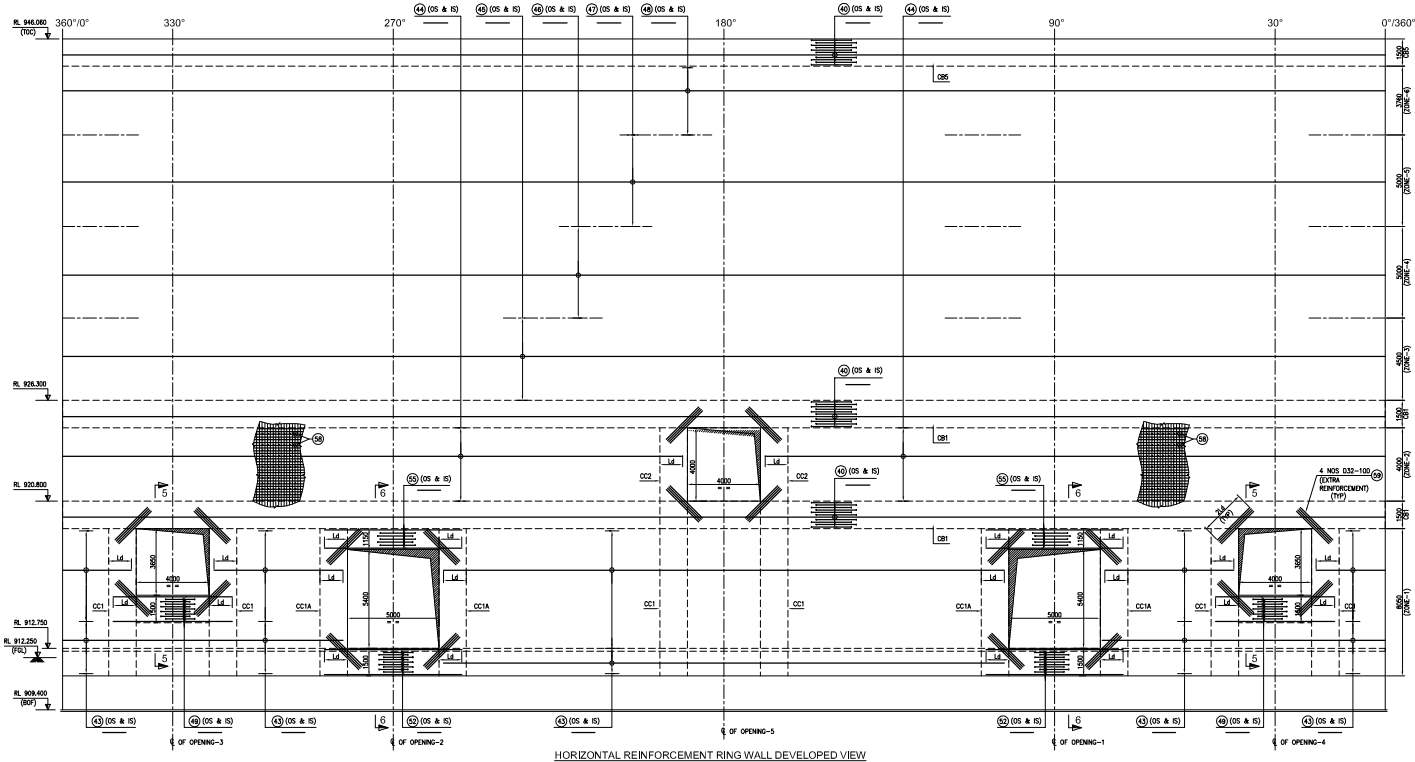
DRAWING NUMBER: MD - 216 - 3210 - EG - CV - DED - 1136
REV: A02



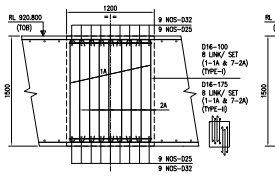
PLAN AT RL 920.800 (TOC)
(SLAB REINFORCEMENT SHOWING ONLY SHEAR LINKS)
SCALE-(1:100)



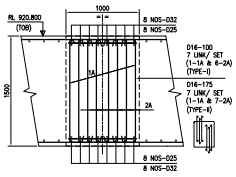
PLAN AT RL 920.800 (TOC)
(SLAB REINFORCEMENT SHOWING ONLY MAIN BAR, TOP & BOTTOM)
SCALE-(1:100)



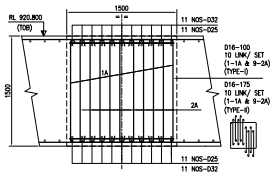
HORIZONTAL REINFORCEMENT RING WALL DEVELOPED VIEW
(VIEW FROM OUTSIDE)
SCALE-(1:100)



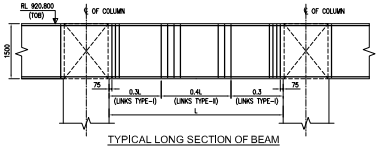
SECTIONAL DETAIL OF BEAM
(MKD. CB2)
SCALE 1:25



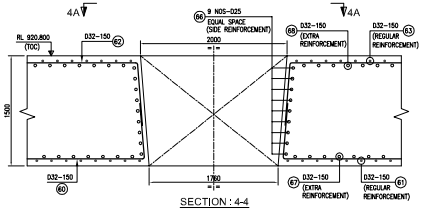
SECTIONAL DETAIL OF BEAM
(MKD. CB3)
SCALE 1:25



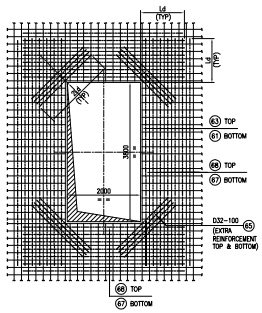
SECTIONAL DETAIL OF BEAM
(MKD. CB4)
SCALE 1:25



TYPICAL LONG SECTION OF BEAM
(SHOWING ONLY LINKS ARRANGEMENT)
SCALE 1:50



SECTION : 4-4
("e"-EXTRA REINFORCEMENT AROUND OPENING)
("r"-REGULAR REINFORCEMENT)
SCALE 1:25



SECTION : 4A-4A
SCALE 1:50

LEGEND:-

TOC = TOP OF CONCRETE
PCC = PLAIN CEMENT CONCRETE
TOB = TOP OF BEAM
FGL = FINISHED GROUND LEVEL
BOF = BOTTOM OF FOOTING
FFL = FINISH FLOOR LEVEL
TYP = TYPICAL
MKD = MARKED
℄ = CENTRE LINE
ø = DIAMETER
R = RADIUS
B/S = BOTH SIDE
OS = OUT SIDE
M = MIDDLE
IS = IN SIDE
RL = REDUCED LEVEL

_____ = TOP REINFORCEMENT
_____ = BOTTOM REINFORCEMENT
_ _ _ _ = EXTRA REINFORCEMENT

NOTES:-

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND CO-ORDINATES ARE IN METRES. UNLESS OTHERWISE STATED.
2. THIS DRAWING SHALL BE READ IN CONJUNCTION WITH DRAWING NUMBER : MD-216-3210-EG-CV-GAD-1132, 1133 & 1134.
3. ALL OTHER NOTES REFER DRAWING NO. MD-216-1000-EG-CV-DED-1001, 1003, 1004 & 1005.
4. GRADE OF CONCRETE SHALL BE C30 (30MPa) ONLY FOR GRADE SLAB AND C35 (35MPa) FOR OTHERS.
5. GRADE OF REINFORCEMENT SHALL BE DEFORMED BAR 420 MPa (GRADE 60) – ASTM A615/A615M.
6. GRADE OF WELDED WIRE FABRIC – ASTM A185(PLAIN) / A497(DEFORMED).
7. NET SAFE BEARING CAPACITY OF SOIL AT BOTTOM OF FOUNDATION IS CONSIDERED AS 600KN/ SQ. M
8. IF ANY PART OF FOUNDATION COMES ON BACKFILL ZONE OF ADJACENT FOUNDATION OR ANY LOOSE SOIL ARE ENCOUNTERED BELOW FOUNDATION LEVEL THE SAME SHALL BE REMOVED AND FILLED WITH PCC.
9. REINFORCEMENT SHALL BE SUITABLY ADJUSTED AT SITE TO CLEAR POCKETS, BOLTS ETC.
10. SUITABLE ARRANGEMENT TO BE PROVIDED AT SITE DURING EXECUTION, TO MAINTAIN CORRECT POSITION OF REINFORCEMENTS & CORRECT DISTANCE BETWEEN TOP & BOTTOM BARS.
11. CLEAR COVER TO MAIN REINFORCEMENT SHALL BE AS FOLLOWS :
    a) RAFT : 75 mm (BOTTOM & SIDE)
    b) RAFT : 50 mm (TOP)
    c) WALL : 50 mm
    d) SLAB : 40 mm
    e) PEDESTAL & COLUMN : 50 mm
12. MINIMUM LAP LENGTHS OF REINFORCEMENT SHALL BE AS PER GENERAL NOTES FOR CIVIL WORKS DRAWING NO : MD-216-1000-EG-CV-DED-1001.
13. THIS DRAWING SHALL BE READ IN CONJUNCTION WITH DRAWING NO. MD-216-3210-EG-CV-DED-1136, 1138 & 1139.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		D01	02/11/2020	AS BUILT	ABH	UR	SVK	AG	
		C01	23/07/2020	ISSUED FOR CONSTRUCTION	ABH	UR	SVK	AG	
		A02	25/06/2020	REVISED AS PER EMPLOYER COMMENTS ON 26/05/2020 & REVISED AS PAINTED	ABH	UR	SVK	AG	
		A01	30/04/2020	ISSUED FOR REFERENCE	ABH	UR	SVK	AG	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO

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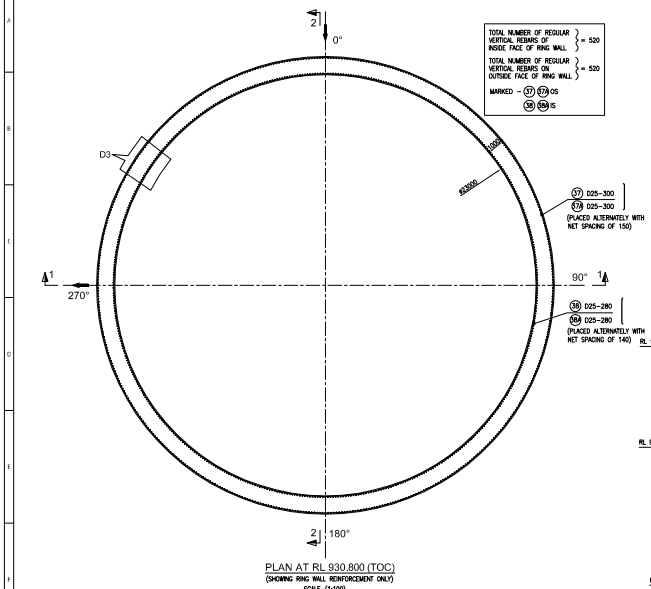
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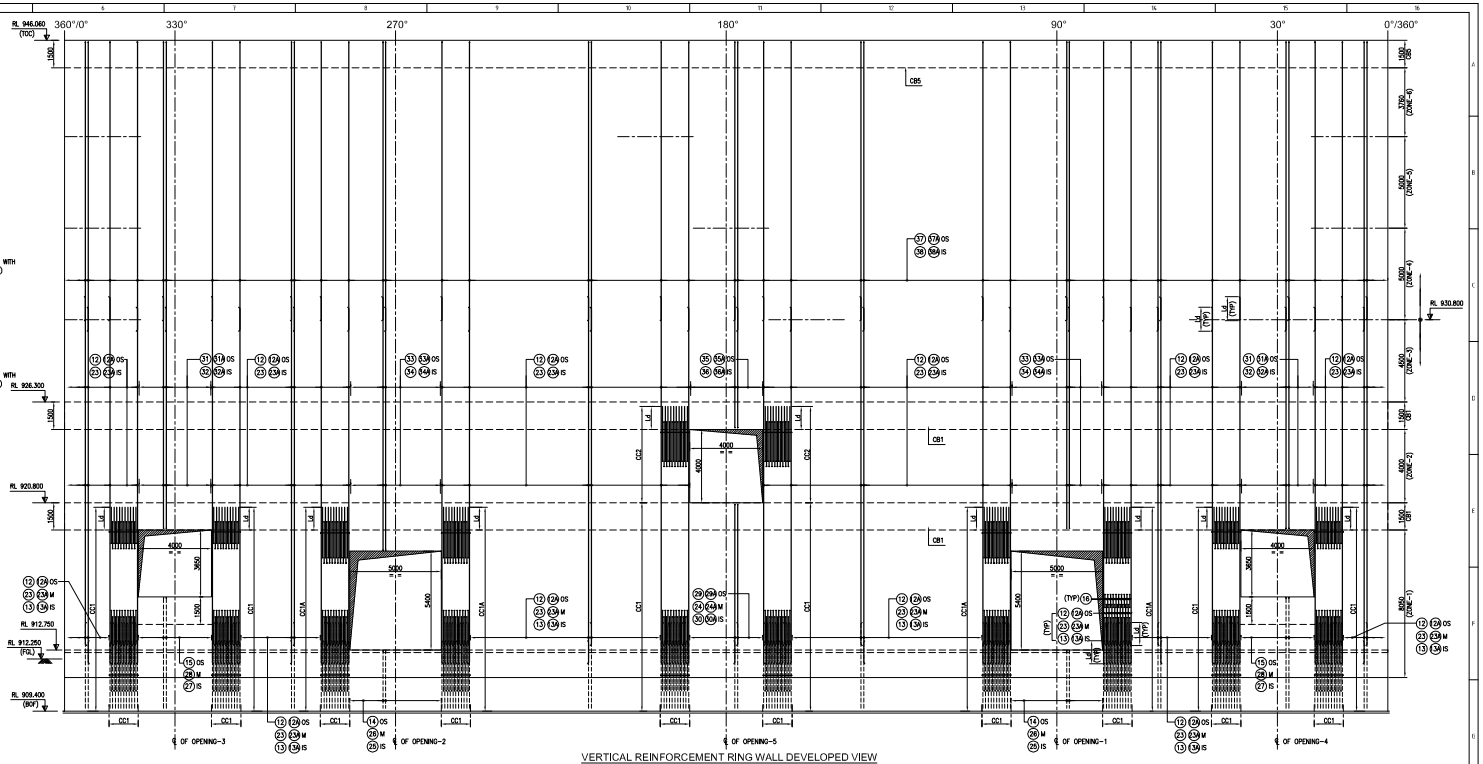
MA'ADEN
Saudi Arabian Mining Company

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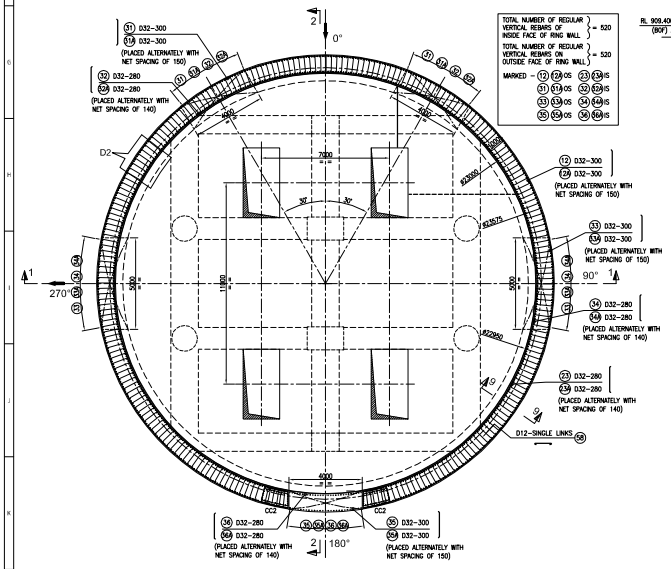
LARSEN & TOUBRO DRAWING NUMBER		REVISION
D19045_450_C_D_24072_002		3
OUTOTEC DRAWING NUMBER	DOCUMENT ID	REVISION
900707__GAU01__US01__CTB17__32102	OU100673545	3
DRAWN: ABH 24/04/2020	MANSOURAH – MASSARAH GOLD PROJECT	
CHECKED: UR 29/04/2020	DETAIL DRAWING	
ENGINEER: SVK 24/04/2020	GRINDING & CLASSIFICATION	
APPROVED: AG 30/04/2020	CRUSHED ORE STORAGE BIN – R.C. DETAIL	
EMPLOYER:		
SCALE: 1:100	A0	DRAWING NUMBER: MD-216-3210-EG-CV-DED-1137 REV: D01



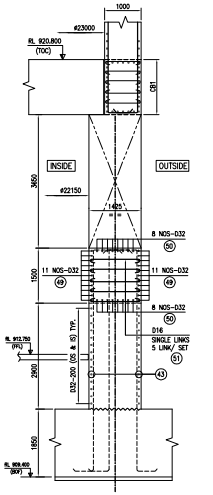
PLAN AT RL 930.800 (TOC)
(SHOWING RING WALL REINFORCEMENT ONLY)
SCALE-(1:100)



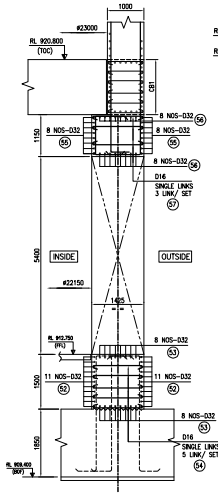
VERTICAL REINFORCEMENT RING WALL DEVELOPED VIEW
(VIEW FROM OUTSIDE)
SCALE-(1:100)



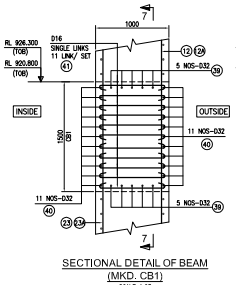
PLAN AT RL 920.800 (TOC)
(SHOWING RING WALL REINFORCEMENT WITH LINKS ONLY)
SCALE-(1:100)



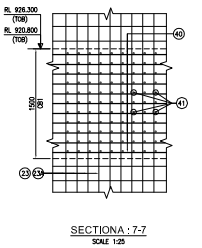
SECTION : 5-5
SCALE 1:50



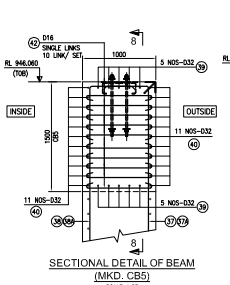
SECTION : 6-6
SCALE 1:50



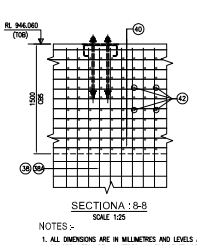
SECTIONAL DETAIL OF BEAM
(MKD. CB1)
SCALE 1:25



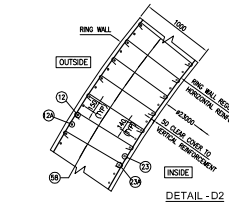
SECTIONA : 7-7
SCALE 1:25



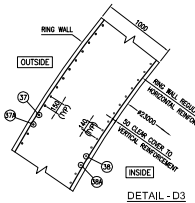
SECTIONAL DETAIL OF BEAM
(MKD. CB5)
SCALE 1:25



SECTIONA : 8-8
SCALE 1:25



DETAIL - D2
SCALE - (1:25)



DETAIL - D3
SCALE - (1:25)

LEGEND :-
TOC = TOP OF CONCRETE
PCC = PLAIN CEMENT CONCRETE
TOB = TOP OF BEAM
FGL = FINISHED GROUND LEVEL
BOF = BOTTOM OF FOOTING
FFL = FINISH FLOOR LEVEL
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MKD = MARKED
℄ = CENTRE LINE
ø = DIAMETER
R = RADIUS
B/S = BOTH SIDE
OS = OUT SIDE
M = MIDDLE
IS = IN SIDE
RL = REDUCED LEVEL
= TOP REINFORCEMENT
= BOTTOM REINFORCEMENT
= EXTRA REINFORCEMENT

NOTES :-
1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND CO-ORDINATES ARE IN METRES. UNLESS OTHERWISE STATED.
2. THIS DRAWING SHALL BE READ IN CONJUNCTION WITH DRAWING NUMBER : MD-216-3210-EG-CV-DAD-1132, 1133 & 1134.
3. ALL OTHER NOTES REFER DRAWING NO. MD-216-1000-EG-CV-DED-1001, 1003, 1004 & 1005.
4. GRADE OF CONCRETE SHALL BE C30 (30MPa) ONLY FOR GRADE SLAB AND C35 (35MPa) FOR OTHERS.
5. GRADE OF REINFORCEMENT SHALL BE DEFORMED BAR 420 MPa (GRADE 60) – ASTM A615/A615M.
6. GRADE OF WELDED WIRE FABRIC – ASTM A185(PLAIN) / A497(DEFORMED).
7. NET SAFE BEARING CAPACITY OF SOIL AT BOTTOM OF FOUNDATION IS CONSIDERED AS 600KN/ SQ. M
8. IF ANY PART OF FOUNDATION COMES ON BACKFILL ZONE OF ADJACENT FOUNDATION OR ANY LOOSE SOIL ARE ENCOUNTERED BELOW FOUNDATION LEVEL THE SAME SHALL BE REMOVED AND FILLED WITH PCC.
9. REINFORCEMENT SHALL BE SUITABLY ADJUSTED AT SITE TO CLEAR POCKETS, BOLTS ETC.
10. SUITABLE ARRANGEMENT TO BE PROVIDED AT SITE DURING EXECUTION, TO MAINTAIN CORRECT POSITION OF REINFORCEMENTS & CORRECT DISTANCE BETWEEN TOP & BOTTOM BARS.
11. CLEAR COVER TO MAIN REINFORCEMENT SHALL BE AS FOLLOWS :
    a) RAFT : 75 mm (BOTTOM & SIDE)
    b) RAFT : 50 mm (TOP)
    c) WALL : 50 mm
    d) SLAB : 40 mm
    e) PEDESTAL & COLUMN : 50 mm
12. MINIMUM LAP LENGTHS OF REINFORCEMENT SHALL BE AS PER GENERAL NOTES FOR CIVIL WORKS DRAWING NO : MD-216-1000-EG-CV-DED-1001.
13. THIS DRAWING SHALL BE READ IN CONJUNCTION WITH DRAWING NO. MD-216-3210-EG-CV-DED-1136, 1137 & 1139.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EPP
		A01	03/12/2020	AS BUILT	ABH	UR	SVK	AG	
		C01	23/09/2020	ISSUED FOR CONSTRUCTION	ABH	UR	SVK	AG	
		A02	25/06/2020	REVISED AS PER EMPLOYER COMMENTS ON 26/05/2020 & REVISED AS MARKED	ABH	UR	SVK	AG	
		A01	30/04/2020	ISSUED FOR REFERENCE	ABH	UR	SVK	AG	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO

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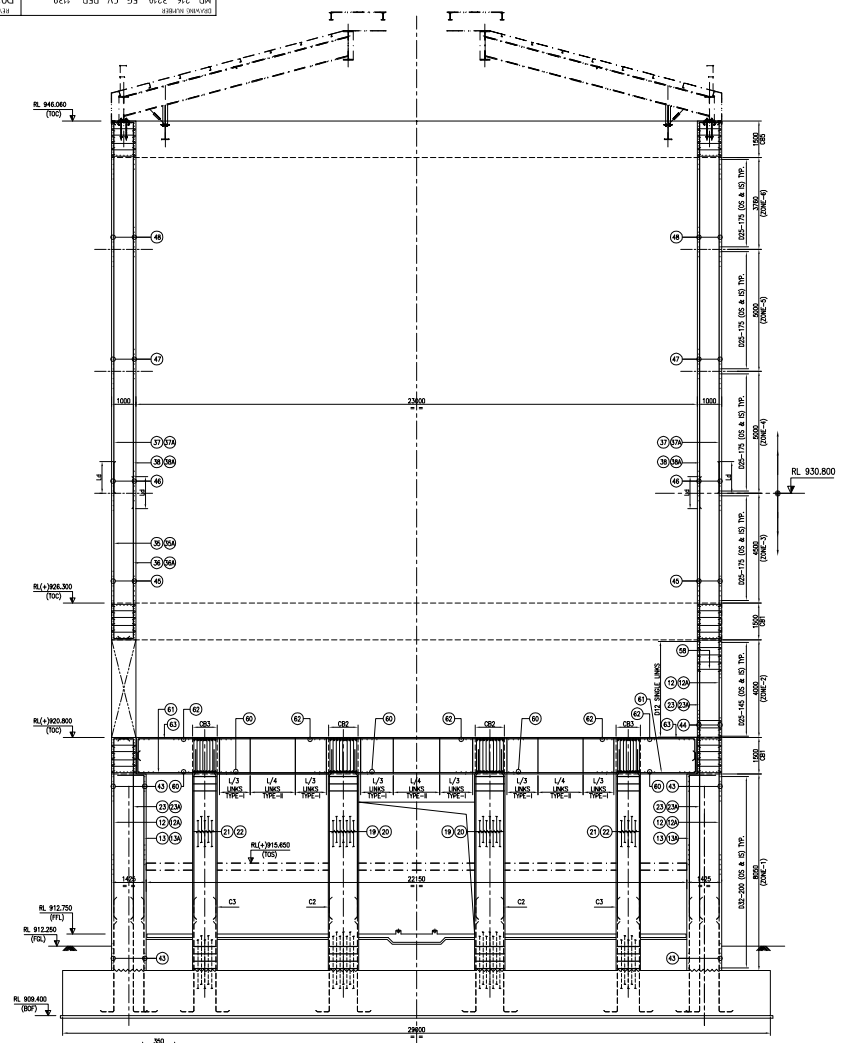
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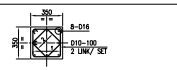
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LARSEN & TOUBRO DRAWING NUMBER	D19045_450_C_D_24072_003		REVISION 3
OUTOTEC DRAWING NUMBER	900707__GAU01__US01__CTB17__32103	DOCUMENT ID OU100673546	REVISION 3
DRAWN	ABH	24/04/2020	MANSOURAH – MASSARAH GOLD PROJECT
CHECKED	UR	29/04/2020	DETAIL DRAWING
ENGINEER	SVK	24/04/2020	GRINDING & CLASSIFICATION
APPROVED	AG	30/04/2020	CRUSHED ORE STORAGE BIN – R.C. DETAIL
EMPLOYER			
SCALE 1:100	A0	DRAWING NUMBER MD-216-3210-EG-CV-DED-1138	REV D01



SECTION : 2-2
SCALE-(1:75)

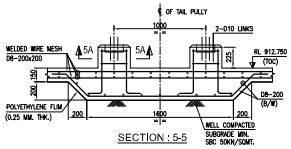
SECTION : 1-1
SCALE-(1:75)



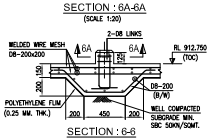
SECTION : 5A-5A
(SCALE 1:20)



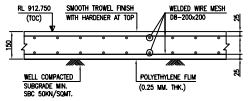
SECTION : 6A-6A
(SCALE 1:20)



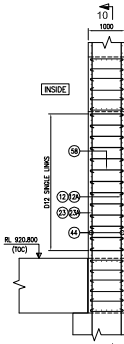
SECTION : 5-5
DETAIL OF PEDESTAL
MKD.- 'P1'
(SCALE 1:20)



SECTION : 6-6
DETAIL OF PEDESTAL
MKD.- 'P2'
(SCALE 1:20)

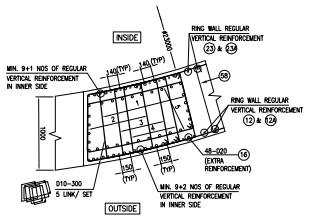


TYPICAL DETAIL OF 150 THK GRADE SLAB
HEAVY DUTY
(SCALE 1:10)



SECTION : 9-9
(SECTIONAL DETAIL FOR ZONE-2)
SCALE 1:50

SECTION : 10-10
SCALE 1:50



SECTIONAL DETAIL OF COLUMN
(MKD. CC2)
('o'-EXTRA VERTICAL REINFORCEMENT AROUND OPENING)
('x'-REGULAR VERTICAL REINFORCEMENT)
SCALE 1:25

LEGEND :-

TOC = TOP OF CONCRETE
PCC = PLAIN CEMENT CONCRETE
TOB = TOP OF BEAM
FGL = FINISHED GROUND LEVEL
BOF = BOTTOM OF FOOTING
FFL = FINISH FLOOR LEVEL
TYP. = TYPICAL
MKD = MARKED
℄ = CENTRE LINE
ø = DIAMETER
R = RADIUS
B/S = BOTH SIDE
OS = OUT SIDE
M = MIDDLE
IS = IN SIDE
RL = REDUCED LEVEL
B/W = BOTH WAY
THK = THICK
MIN. = MINIMUM
SBC = SAFE BEARING CAPACITY
_ _ _ _ = TOP REINFORCEMENT
———— = BOTTOM REINFORCEMENT
———— = EXTRA REINFORCEMENT

NOTES :-

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND CO-ORDINATES ARE IN METRES. UNLESS OTHERWISE STATED.
2. THIS DRAWING SHALL BE READ IN CONJUNCTION WITH DRAWING NUMBER : MD-216-3210-EG-CV-GAD-1132, 1133 & 1134.
3. ALL OTHER NOTES REFER DRAWING NO. MD-216-1000-EG-CV-DED-1001, 1003, 1004 & 1005.
4. GRADE OF CONCRETE SHALL BE C30 (30MPa) ONLY FOR GRADE SLAB AND C35 (35MPa) FOR OTHERS.
5. GRADE OF REINFORCEMENT SHALL BE DEFORMED BAR 420 MPa (GRADE 60) – ASTM A615/A615M.
6. GRADE OF WELDED WIRE FABRIC – ASTM A185(PLAIN) / A497(DEFORMED).
7. NET SAFE BEARING CAPACITY OF SOIL AT BOTTOM OF FOUNDATION IS CONSIDERED AS 600KN/ SQ. M
8. IF ANY PART OF FOUNDATION COMES ON BACKFILL ZONE OF ADJACENT FOUNDATION OR ANY LOOSE SOIL ARE ENCOUNTERED BELOW FOUNDATION LEVEL THE SAME SHALL BE REMOVED AND FILLED WITH PCC.
9. REINFORCEMENT SHALL BE SUITABLY ADJUSTED AT SITE TO CLEAR POCKETS, BOLTS ETC.
10. SUITABLE ARRANGEMENT TO BE PROVIDED AT SITE DURING EXECUTION, TO MAINTAIN CORRECT POSITION OF REINFORCEMENTS & CORRECT DISTANCE BETWEEN TOP & BOTTOM BARS.
11. CLEAR COVER TO MAIN REINFORCEMENT SHALL BE AS FOLLOWS :
    a) RAFT : 75 mm (BOTTOM & SIDE)
    b) RAFT : 50 mm (TOP)
    c) WALL : 50 mm
    d) SLAB : 40 mm
    e) PEDESTAL & COLUMN : 50 mm
12. MINIMUM LAP LENGTHS OF REINFORCEMENT SHALL BE AS PER GENERAL NOTES FOR CIVIL WORKS DRAWING NO : MD-216-1000-EG-CV-DED-1001.
13. DETAIL OF GRADE SLAB REFER DRAWING NO – MD-216-1000-EG-CV-DED-1002.
14. THIS DRAWING SHALL BE READ IN CONJUNCTION WITH DRAWING NO. MD-216-3210-EG-CV-DED-1136, 1137 & 1138.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
D01	B01000			AS BUILT	ABH	UR	SVK	AG	
C01	23/07/2020			ISSUED FOR CONSTRUCTION	ABH	UR	SVK	AG	
A02	25/06/2020			REVISED AS PER EMPLOYER COMMENTS ON 26/05/2020 & REVISED AS PAIRED	ABH	UR	SVK	AG	
A01	30/04/2020			ISSUED FOR REFERENCE	ABH	UR	SVK	AG	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO

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L&T/VENDOR DRAWING NUMBER			REVISION
019045_450_C_D_24072_004			3
OUTOTEC DRAWING NUMBER		DOCUMENT ID	REVISION
900707__GA001__US01__CTB17__32104		OU100673547	3
DRAWN	ABH	24/04/2020	MANSOURAH – MASSARAH GOLD PROJECT
CHECKED	UR	29/04/2020	DETAIL DRAWING
ENGINEER	SVK	24/04/2020	GRINDING & CLASSIFICATION
APPROVED	AG	30/04/2020	CRUSHED ORE STORAGE BIN – R.C. DETAIL
EMPLOYER			
SCALE 1:100	A0	DRAWING NUMBER MD-216-3210-EG-CV-DED-1139	REV D01